

IMAGINARIES OF IMMUTABLE TRUTH

**THE CULTURAL
POLITICS OF
BLOCKCHAIN**

INTE GLOERICH

Colophon

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Imaginaries of Immutable Truth: The Cultural Politics of Blockchain

Author: Inte Gloerich

Edited by: Bresser-Chapple

Production: Inte Gloerich

Cover design: Katja van Stiphout

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Contact

Institute of Network Cultures

Email: info@networkcultures.org

Web: <http://www.networkcultures.org>

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COMPUTATIONAL RATIONALISM AND THE ELEMENT OF THE IMAGINATION IN BLOCKCHAIN CULTURE

This book is about what happens when a complex technology catches the collective imagination as the harbinger of a revolution. The technology I refer to is blockchain and what is revolutionised is the truth—or so the story goes. The imaginary of ‘blockchain as a truth machine’ presents the technology as being capable of producing a new and better record of the truth than was possible prior to its invention. On the surface, the idea of a ‘truth machine’ evokes a rationalist vision of logically calculated, universally relevant truths produced by an indiscriminate, emotionless mechanical tool. However, my analyses show that this ‘truth machine’s’ reliance on rationality is always accompanied by different forms of what I call ‘more-than-rational’ belief. ‘Truth’ has many different connotations—from the philosophical to the logical and the spiritual—each of which will be untangled in the coming chapters to develop a more nuanced understanding of the worlds this imaginary performs.

While, fundamentally, I understand imaginaries as collective visions about the future of socio-cultural life in the context of emerging technologies that are enacted materially and discursively, throughout this book I finetune the concept to respond to the complexities of blockchain culture. I warn against the colonial logics of imaginaries that appear as if from nowhere and make their situated explicit to understand the social and cultural implications of the worlds that they envision. Truth imaginaries not only perform consequential idealisations of societal power dynamics, but also impact how people understand their relationships to others on an everyday level and their capacity to imagine these relations otherwise. Throughout this book, I reveal how different blockchain truth imaginaries impact power relations, knowledge politics, and—as the idea of a blockchain revolution promises—the technology’s capacity of playing a role in radical change.

The coexistence of the rational and more-than-rational already featured in the Bitcoin whitepaper¹—the text in which pseudonymous blockchain inventor Satoshi Nakamoto first publicly described the details of the technology.² Satoshi proposes blockchain technology as a solution to the power of corruptible human institutions such as banks and governments, instead offering a decentralised, networked system of exchange in which no such centralised points of power can exist. The whitepaper is structured like a technical manual, featuring abstract drawings of the proposed system’s architecture and descriptions of its algorithmic

1 Nakamoto, Satoshi, ‘Bitcoin: A Peer-to-peer Electronic Cash System’, *Cryptography mailinglist*, 2008, <https://bitcoin.org/bitcoin.pdf>.

2 The Bitcoin white paper is written in first person plural and signed by Satoshi Nakamoto. Since Satoshi’s identity remains unknown, I will use they/them pronouns and the first name shorthand that is commonly used in the blockchain community.

decision-making processes. Reading the whitepaper, it is easy to forget that what it describes is the messy social phenomenon of an economy—a phenomenon that inherently features hopes and dreams, desperation and speculation, risk-taking and the hedging of bets. However, presenting the economy as if it consists merely of sterile computational processes does not relieve it of its more-than-rational factors. The idea of a purely rational economy made possible by blockchain technology is itself part of a techno-utopian belief system. In other words, the belief in the virtue and feasibility of pure rationality is itself not purely rational.

Over the years, Satoshi has acquired almost divine status, functioning like a supreme being at the centre of the culture of belief in blockchain technology, with self-described evangelists declaring their eternal dedication. For example, an essay in online magazine *The Bitcoin Times* starts as follows:

As always, we commence by sending thanks and good tidings to Satoshi Nakamoto—The Creator and First of the Believers, General of the Byzantines, Breaker of Banks and Fighter of Fiat—and to Hal Finney and The Apostles and The Disciples thereafter, and to The Stoic and Patient True Believers, who keep their tithe holy, and stack sats for their salvation. Oh you who believe, fear the day of economic reckoning, and do not face the Angel of Hyperbitcoinization as a nocoiner, for punishment awaits them in the hereafter.³

Beyond this rather literal prayer to Satoshi and the logics of Bitcoin—which I will dissect in detail in chapter 3—the mix of rationality with more-than-rational forms of belief is often much more subtle, or even unintentional, in blockchain culture. It features in claims that blockchain technology represents a societal revolution, imbuing the technology with far more than simply technical capacities. It is also recognisable in the ritualised practices that cryptocurrency traders perform in their attempts to summon good luck. In this book, I focus on the paradoxical yet inseparable mix of, on the one hand, the superior rationality with which blockchain technology was first presented and is usually still described, and the processes of mystification, spiritualisation, and speculation that have emerged around it on the other. In the next section, I expand on the role of power in the intertwined histories of the rational and the more-than-rational in order to be able to draw connections and distinctions with dynamic in the contemporary blockchain truth imaginaries discussed throughout this book.

Rationalism in History and Contemporary Technology Culture

The history of Western philosophy provides an important context within which to understand the mix of the rational and the more-than-rational in contemporary technology culture. Two related forms of belief should be highlighted in this history: The first rose to prominence during the Enlightenment and consists of the belief in rationality as a superior approach to knowledge

3 Hass McCook, 'The Passion of the Believers', *The Bitcoin Times*, 2019, <https://bitcointimes.io/the-passion-of-the-believers/>. Many religious references that feature in this quote—e.g., the mention of apostles, holiness, and a creator—are clear without further explanation of the Bitcoin-specific terminology. I will go into more depth about its meaning in chapter 3.

that can produce universal truths that reveal the order that structures the world. The second became increasingly influential from the start of the industrial revolution onwards and is the belief that mathematic and algorithmic processes are capable of achieving pure rationality and therefore their use inherently signifies progress. Additionally, complex technologies⁴ themselves are inherently entangled with belief purely because they are considered difficult to comprehend.⁵ Later in this book—particularly in chapters 2 and 3—I lay the foundations for my claim that rationalism in blockchain culture has its roots firmly anchored in this long history, while also introducing medium-specific shifts.

Rationalism's Self-Proclaimed Superiority

It is useful to distinguish between the descriptive use of rational and rationality and the broader philosophical system referred to as rationalism. The former generally denote the quality of being guided by logic or reason, or the capacity to exercise such reasoning. Rationalism, by contrast, represents a philosophical and epistemological stance: a structured belief that reason constitutes the highest or most reliable source of knowledge. Over time, this commitment has gained dominance through the intertwining of epistemic hierarchies—where rationality is positioned as superior to other modes of knowing—and the geopolitical power structures that have reinforced and legitimised that supremacy. In this book, I focus on the element of power that is associated with this hierarchisation between rationalism and the other ways of knowing it aims to supersede, particularly phenomena that are difficult to pin down in abstract formulas and systematic universals, such as belief, speculation, and the imagination.

Because rationalism is built atop and around rationality, I will first unpack that term in more detail. This is not as straightforward as it may seem, since the specific meanings attached to rationality have often shifted and sometimes contradicted each other over time. Rather than attempting to create a precise definition of rationality to use throughout this book, I allow the concept to shift in both meaning and connotation from chapter to chapter, dependent on the way it is used by the communities under analysis. Each of the uses highlighted below will resurface in subsequent chapters.

Thinking of rationality as 'the application of logic, of pure thought untempered by emotions'⁶ highlights its association with the sense that embodiment and worldly particulars are distractions from the truths that exist in the world of the mind. For this kind of rationality, the most certain way of establishing truth is through abstract logical statements and reason rather than the senses and observation: here, rational approaches to knowledge are strictly separate from empiricism. Plato, for example, argued that knowledge is achieved through reason

4 'Complex' is a relative term, e.g., at different moments in history different technologies were considered hard to grasp and bewildering.

5 Science fiction author Arthur C. Clarke's famous adage conveys a similar idea: 'Any sufficiently advanced technology is indistinguishable from magic'. See Arthur C. Clarke, *Profiles of the Future: An Inquiry Into the Limits of the Possible*, New York: Harper & Row, 1973 [1962], 36.

6 David Graeber, *The Utopia of Rules: On Technology, Stupidity, and the Secret Joys of Bureaucracy*, New York: Melville House, 2015, 167.

rather than humanity's deceivable senses.⁷ A related understanding is associated with René Descartes' dualism between body and mind, in which he argued that the pure use of the mind was the closest humanity could possibly get to all-encompassing and ultimately truthful godly knowledge.⁸

Other uses of the term 'rationality' do allow for a form of empiricism, despite the fact that these philosophical strands are often presented as mutually exclusive. For example, Carl Linnaeus, most famous for creating taxonomies of natural phenomena,⁹ used abstract reason to create an understanding that captured nature's essence. Instead of being interested in the particulars of individual flowers and the specifics of a particular specimen of a natural phenomenon as it exists in its uniqueness in reality, Linnaeus and naturalists like him were interested in establishing something more akin to 'truth-to-nature':¹⁰ generalised, universal, abstract, and reasoned truths like those of a given taxonomy, which are nevertheless also based on observation.¹¹

After the industrial revolution, uses of the term 'rational' travelled into the fields of economics and sociology, gaining its connotations of technical efficiency and capitalist sensibility. Sociologist Max Weber¹² described the changes to the way in which work and production processes were organised that characterise the industrial revolution of the nineteenth century as the rationalisation of labour. These changes meant that previously holistic production practices became isolated elements that could be routinised and managed. In the economic

7 Peter Markie and M. Folescu, 'Rationalism Vs. Empiricism', in Edward N. Zalta and Uri Nodelman (eds) *The Stanford Encyclopedia of Philosophy*, Stanford: Metaphysics Research Lab, Stanford University, 2023, n.p. <https://plato.stanford.edu/archives/spr2023/entries/rationalism-empiricism/>.

8 David F. Noble, *The Religion of Technology: The Divinity of Man and the Spirit of Invention*, New York: Penguin Group, 1997.

9 Linnaeus' nested categories, which position different species in relation to each other, were instrumental in understanding evolution and his taxonomies continue to be widely used to this day. By naming species foreign to European colonists, these taxonomies supported the logic of ownership and the business of cultivation for profit that featured so prominently in the colonial capitalism that emerged during his time. See Londa L. Schiebinger, *Plants and Empire: Colonial Bioprospecting in the Atlantic World*, Cambridge, MA: Harvard University Press, 2004; M. Ideland, 'Science, Coloniality, and 'the Great Rationality Divide': How Practices, Places, and Persons Are Culturally Attached to One Another in Science Education', *Science & Education* 27 (October, 2018): 783-803. Besides plants and non-human animals, Linnaeus is also a key figure in the institution of the still influential yet erroneous idea of the existence of different human races. See Ideland, 'Science, Coloniality, and 'the Great Rationality Divide''. Linnaeus' eighteenth-century descriptions of various of these supposed races are all too recognisable in today's racist stereotypes—a terrible legacy if there ever was one.

10 Lorraine Daston and Peter Galison, *Objectivity*, New York: Zone Books, 2010, p. 59.

11 Although often used interchangeably, truth and objectivity each have their own scientific histories and meanings. While the search for truth connotes abstracted universals and characterises histories of scientific rationality—but also, for instance, religion—objectivity as it is understood today only became a virtue in the sciences in the nineteenth century. This is a shift that cannot be seen separately from technological developments such as the invention of photography, which made it possible to create mechanical reproductions of 'reality' for its scientific study. The scientific virtue of objectivity and the similar-sounding concept of objectivism—i.e., the Cartesian distinction between subjects and objects—should also not be confused. See Daston and Galison, *Objectivity*.

12 Max Weber, *The Protestant Ethic and the Spirit of Capitalism*, London: Routledge, 2005 [1904].

sciences that emerged in the nineteenth century, the notion of the free market became characterised by the purely rational and self-interested decisions of homo economicus—an understanding that, although also criticised, still lingers in contemporary discussions of the market and individual decision-making.¹³

As a scientific and philosophical system that considers itself superior to others, rationalism also dates back to antiquity.¹⁴ However, the element of power becomes particularly relevant in the modern era, when rationalism, capitalism, and colonialism all start to develop in relation to each other.¹⁵ When the European colonisation of the Americas gathered pace in the sixteenth and seventeenth century, and indigenous Americans and African people were displaced and eradicated or enslaved and forced to work on the plantations, a global plantation-based system of production emerged. This moment when labour and the economy began to be organised internationally is inseparable from the scientific, political, and economic changes of the Renaissance, such as the scientific revolution, the development of sovereign states, and the birth of mercantilism as a precursor to capitalism. All these processes demanded the increasingly streamlined management of nature and people. Rationalism provided the necessary logic to form what became a global production machine of colonial capitalism, eradicating indigenous practices that were perceived as non-rational in the process.¹⁶

During European colonisation, belief in the supremacy of rationalism produced concrete effects far beyond Europe. Decolonial theorist Enrique Dussel describes this dynamic as the ‘irrational myth’ underpinning modernity’s moral order.¹⁷ Modernity signified a shift from the religious logics of the European Middle Ages in which humanity was thought of as subject to god’s will and too much intervention with the world as it was divinely designed as bad. Instead, Europeans increasingly thought of humans as having free will, of the future as malleable by humanity, and of societal and technological progress as important goals for which to strive. Europeans constructed themselves as the embodiment of civilisation and progress, a self-image that justified what they perceived as a moral duty to civilise those they labelled primitive or savage.¹⁸ This narrative was reinforced by emerging scientific “truths”—new taxonomies and classificatory systems that framed racialised colonial Others as irrational, less than human, and thus in need of enlightenment. It also drew strength from technological innovations that

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- 13 Galit Ailon, ‘The Phenomenology of Homo Economicus’, *Sociological Theory* 38.1 (2020): 36-50; Didem Özkiziltan, ‘Goodbye Labouring Man, Long Live Homo Economicus: The New Precarity in the World of Work’, *Globalizations* 18.4 (2021): 499-515.
 - 14 Graeber, *The Utopia of Rules*; David Golumbia, *The Cultural Logic of Computation*, Cambridge, MA: Harvard University Press, 2009.
 - 15 I discuss this history briefly in the paragraphs that follow, and return to the topic in relation to environmentalist blockchain projects in chapter 2.
 - 16 Sylvia Wynter, ‘Unsettling the Coloniality of Being/ Power/ Truth/ Freedom: Towards the Human, After Man, Its Overrepresentation—an Argument’, *The New Centennial Review* 3.3 (2003): 257-337. <https://doi.org/10.2307/41949874>; Anibal Quijano, ‘Coloniality and Modernity/rationality’, *Cultural Studies* 21.2–3 (2007): 168-78.
 - 17 Enrique Dussel, ‘Eurocentrism and Modernity (Introduction to the Frankfurt Lectures)’, *Boundary 2* 20.3 (1993), 66.
 - 18 Wynter, ‘Unsettling the Coloniality of Being/ Power/ Truth/ Freedom’; Quijano, ‘Coloniality and Modernity/rationality’.

offered new tools of domination as well as novel insights into nature.¹⁹ Convinced of their moral and rational superiority, Europeans positioned themselves as masters over both nature and humanity, legitimising the use of violence as a necessary means toward the supposedly emancipatory goal of progress.²⁰

The Perceived Primacy of Algorithmic Logic

Over time, the idea of rationality grew to rely increasingly on calculated rather than reasoned logic and was meant to be able to ‘account for every part of the material world’²¹ in even more abstraction. In striving for objectivity, calculated logic started to replace natural language—which is always at risk of being caught in ambiguities and subjectivity—with the symbols and variables of mathematics.²² In this process of abstraction and formalisation, the validity of a decision became increasingly disconnected from its meaning or morality.²³ Instead, rationality increasingly served functionality in the globalised machine of industrial capitalism that was taking shape from the late eighteenth century onwards.²⁴

The move from reason to calculation was a gradual and partial shift rather than a clear break, but one that would prove influential for the automated logic at work in programming languages and algorithms today.²⁵ For instance, the focus on calculation is already recognisable in the work of Enlightenment philosopher Gottfried Wilhelm Leibniz,²⁶ who considered human cognition to work in logical processes, which could be represented by symbols and formal rules. He dreamt of a universal language of calculation that could discover and express truths about the world logically. This is a task later elaborated upon by those who would develop modern logic in the late eighteenth and nineteenth century. Of particular significance to the history of computation is George Boole,²⁷ whose work—together with Leibniz’s earlier ideas—laid the foundations of binary logic.²⁸ His formulation of Boolean operations, based on true and false values, remains central to digital computation today.

19 Peter Wagner, ‘Progress and Modernity: The Problem With Autonomy’, *Sociología Histórica* 7 (2017): 71–94.; Scott McQuire, ‘Technology’, *Theory, Culture & Society* 23.2–3 (2006): 253–65. <https://doi.org/10.1177/0263276406062681>; Dussel, ‘Eurocentrism and Modernity’.

20 Dussel, ‘Eurocentrism and Modernity’.

21 Golumbia, *The Cultural Logic of Computation*, 1.

22 Daston and Galison, *Objectivity*.

23 Golumbia, *The Cultural Logic of Computation*; Achille Mbembe, *The Earthly Community: Reflections on the Last Utopia*, Rotterdam: V2_Publishing, 2022.

24 Christopher Miles, ‘The Combine Will Tell the Truth: On Precision Agriculture and Algorithmic Rationality’, *Big Data & Society* 6.1 (2019): 1–12.; Paul Erickson et al., *How Reason Almost Lost Its Mind: The Strange Career of Cold War Rationality*, Chicago: University of Chicago Press, 2013; Max Horkheimer, *Critique of Instrumental Reason*, London: Verso Books, 2014 [1947].

25 Daston and Galison, *Objectivity*; Lloyd Strickland and Harry Lewis, *Leibniz on Binary: The Invention of Computer Arithmetic*, Cambridge, MA: MIT Press, 2022; Golumbia, *The Cultural Logic of Computation*.

26 Gottfried Wilhelm Leibniz, ‘Explanation of Binary Arithmetic, Which Uses Only the Characters 0 and 1, With Some Remarks on Its Usefulness, and on the Light it Throws on the Ancient Chinese Figures of Fuxi’, (1703):

27 George Boole, *An Investigation of the Laws of Thought*, London: Project Gutenberg, 2005 [1894].

28 Earlier, Leibniz also worked on binary arithmetic, although his work would not be widely available until much later. See Volker Peckhaus, ‘Leibniz’s Influence on 19th Century Logic’, in Edward N. Zalta and

Another genealogy that is important to highlight with respect to the use of algorithms today—especially in the context of the rise of data collection and analysis—is that of statistics. Although it developed at the same time as modern logic and is also concerned with mathematical calculations as a way to understand reality, there is an important difference between the two. In striving for objectivity as a rationalist system of thought, modern logic tried to eliminate ties to the specifics of the real world, hoping rather to achieve universal, abstract true statements.²⁹ The field of statistics is instead rooted in positivism and is more empirical, applying the methods of mathematics to make sense of an observable yet messy social world. Nevertheless, the process of quantifying the world and readying it for statistical analysis, was similarly part of the pursuit of objectivity.³⁰

As with rationalism, the element of power is palpable in statistics. For example, administration and statistics support states: the state both governs its population by acquiring knowledge about it through statistics, but also derives its existence from the legibility of populations in the form of statistics.³¹ States still use statistics to understand the makeup of their population and base policies and spending strategies on this knowledge. How exactly it is decided who counts as part of a given population, and what insight individuals have in the way data about them is used, are questions of power.³²

Such questions have become increasingly pressing in the past two decades, during which statistical methods have been applied to many aspects of life, by more and more public and private actors, and in increasingly consequential and opaque ways through the production of (big) data.³³ This is where the genealogies of statistics and rationalism merge, as the statistics of '[d]ata science and data practices reincarnate rationalism in many forms,' for example through its emphasis on categorisation and order, its reinforcement of binaries, its idealisation

Uri Nodelman (eds) *The Stanford Encyclopedia of Philosophy*, Stanford: Metaphysics Research Lab, Stanford University, 2024, <https://plato.stanford.edu/archives/spr2024/entries/leibniz-logic-influence/>. Leibniz likely built his binary arithmetic on the foundational I Ching binary system, which was used in ancient China in divination practices and as a philosophy of nature. However, dismissing such practices, Leibniz thought a European rationalist reading of the I Ching's code would be superior and reveal its true meaning: 'The Chinese lost the meaning of [the origin of the I Ching], perhaps more than a thousand years ago, and they have written commentaries on the subject in which they have sought I know not what far out meanings, so that their true explanation now has to come from Europeans'. See Evelyn Wan, *Decolonising the Binary Arithmetic*, 2024. <https://youtu.be/jjP2xb9z7hM?feature=shared>; also Leibniz, 'Explanation of Binary Arithmetic'.

29 Daston and Galison, *Objectivity*.

30 Theodore M. Porter, *The Rise of Statistical Thinking, 1820–1900*, Princeton: Princeton University Press, 1986.

31 Michel Foucault, *The Essential Foucault: Selections From Essential Works of Foucault, 1954–1984*, New York: The New Press, 2003.

32 Porter, *The Rise of Statistical Thinking*; Theodore M. Porter, *Trust in Numbers: The Pursuit of Objectivity in Science and Public Life*, Princeton: Princeton University Press, 1995.

33 danah boyd and Kate Crawford, 'Critical Questions for Big Data: Provocations for a Cultural, Technological, and Scholarly Phenomenon', *Information, Communication & Society* 15.5 (2012): 662–79. <https://doi.org/10.1080/1369118X.2012.678878>; José van Dijck, 'Datafication, Dataism and Dataveillance: Big Data Between Scientific Paradigm and Ideology', *Surveillance & Society* 12.2 (2014): 197–208.

of disembodied knowledge processes, and its claims to universal applicability and objectivity without regard for the power it exerts.³⁴

The late David Golumbia coined the term computationalism to describe how contemporary digital culture continues to embody the intellectual legacy of rationalism. Four aspects of this connection are important to highlight here: First, like the logics of rationality, computation functions through abstraction, quantification, and mathematical formulas. These abstractions allow for computational processes to be automated and applied in many different contexts.

Second, and building on from the first, computation is imagined as a system capable of modelling and accounting for every dimension of the social and natural world through the operations of logical calculation.³⁵ Leibniz' dream of a universal language is recognisable in more recent ideas such as the Turing Machine, generative artificial intelligence, and blockchain as a general-purpose technology.³⁶

Third, because of this presumed quantitative capacity to 'account for every part of the material world'³⁷ computation is frequently positioned as exceeding human cognition. Although it is assumed to operate according to similar principles of logic, it is valued for eliminating what are perceived as the limitations of the human mind—emotion, embodiment, and mistakes. At the same time, its speed and scale vastly expand the volume of logical operations that can be performed within any given timeframe. Echoing earlier hierarchies between the rational mind of the civilised European and the irrational mind of the colonial Other, algorithmic decision-making is increasingly believed to be superior to embodied human decision-making.³⁸ As a result, computational and algorithmic processes now permeate domains ranging from governance to business and everyday life. This diffusion has fostered a pervasive culture of 'computational thinking'³⁹—the conviction that 'any given problem can be solved by the application of computation'.⁴⁰ Examples include the allocation of social benefits, the deployment of police in cities, the targeting of advertisements, or the invitation of candidates for job interviews.

Fourth, power is embedded at the core of computationalism, just as it was in rationalism. The conviction that computation is inherently superior, along with its expanding reach across social domains, serves the interests of those who design and govern its architectures—the

34 Abeba Birhane, 'Algorithmic Injustice: A Relational Ethics Approach', *Patterns* 2.2 (2021), 3.

35 Golumbia, *The Cultural Logic of Computation*.

36 Martin Davis, *The Universal Computer: The Road From Leibniz to Turing*, Boca Raton: CRC Press, 2018; Buterin, Vitalik, 'Visions, Part 1: The Value of Blockchain Technology', *Ethereum Foundation Blog*, 13 april 2015, <https://blog.ethereum.org/2015/04/13/visions-part-1-the-value-of-blockchain-technology>.

37 Golumbia, *The Cultural Logic of Computation*, 1.

38 Golumbia, *The Cultural Logic of Computation*; Paolo Totaro and Domenico Ninno, 'Algorithms and the Practical World', *Theory, Culture & Society* 33.1 (2016): 139-52. <https://doi.org/10.1177/0263276415620284>; Miles, 'The Combine Will Tell the Truth'; Paolo Totaro and Domenico Ninno, 'The Concept of Algorithm as an Interpretative Key of Modern Rationality', *Theory, Culture & Society* 31.4 (2014): 29-49.

39 James Bridle, *New Dark Age: Technology and the End of the Future*, London: Verso Books, 2018, 4.

40 Bridle, *New Dark Age*, 4.

actors who define categories, manage data flows, and control algorithmic operations.⁴¹ However, no technology is neutral, and the history of rationalism shows the dangers of the overlaps between systems of knowing and concentrations of power.⁴² Computation is employed by particular actors or institutions to achieve particular goals, even if those intentions are frequently concealed by the layers of abstraction that characterise complex technological systems.⁴³

Computation is thus used to institute a particular version of socio-cultural reality while also abstracting it. In the words of Ed Finn—a researcher in innovation culture and media technology—computation and algorithms are ‘culture machines’⁴⁴ that always balance between rational calculation based on abstraction and generalisation on the one hand, and the messy reality of culture and the more-than-rational narratives that are attached to make sense of such complex technologies on the other. Finn proposes that understanding such computational culture machines ‘requires the critical methods of the humanities’⁴⁵ to address the processes through which computational knowledge is ordered, the abstractions that determine what is included and obscured in computational systems of knowledge, the contexts of implementation in which the gaps between abstraction and socio-cultural reality become tangible, and the forms of imagination and future projection that knowing through these systems makes possible. In section 4, I detail how I render these concerns for both the technical processes and abstractions as well as the contexts and socio-cultural impacts for my own analyses of the culture machine that is blockchain.

Addressing the ‘More-than-Rational’

My use of words such as ‘myth,’ ‘belief,’ and ‘godly knowledge’ in the two previous sections indicate that there is more to rationalism and its legacy in the contemporary culture of computationalism than just pure logic and objective calculation. At the most basic level, it is important to recognise that rationalism and computationalism are belief systems that consist of the conviction that their way of knowing is better than others. As belief systems the logical practices of which are suited to efficient operations and automation, rationalism and computationalism permeate contemporary capitalist cultures and as a result are incentivised and capable of supplanting other ways of knowing.⁴⁶

Additionally, this belief system is supplemented by the modernist belief in the promise that technoscientific development leads to societal progress. For example, the industrial revolution was in part powered by the ‘myth[s] that technology equals progress’ and that

41 Columbia, *The Cultural Logic of Computation*; Porter, *Trust in Numbers*.

42 Foucault, *The Essential Foucault*; Quijano, ‘Coloniality and Modernity/rationality’.

43 Columbia, *The Cultural Logic of Computation*; Evgeny Morozov, *To Save Everything, Click Here: The Folly of Technological Solutionism*, New York: PublicAffairs, 2014; Wendy Hui Kyong Chun, ‘On ‘Sourcery,’ or Code as Fetish’, *Configurations* 16.3 (2008): 299–324.

44 Ed Finn, *What Algorithms Want: Imagination in the Age of Computing*, Cambridge, MA: The MIT Press, 2017, 35.

45 Finn, *What Algorithms Want*, 2.

46 Columbia, *The Cultural Logic of Computation*.

‘technology is infallible’ as the best operationalisation of rationality.⁴⁷ Not only does this mean that continuous technological innovation is understood as inherently good for societies, but also, increasingly, the conviction that social issues should ideally be addressed through technologies rather than the practice of politics, because they supposedly offer efficient, objective, and neutral alternatives to it that signify progress in and of themselves.⁴⁸ I will discuss the religious dimension of the promise of progress that is attributed to technoscientific development in more detail in chapter 3.

Finally, there is something inherently mystical or magical about technologies that are deemed difficult to understand—whether one lives at a time when the physics at work in telescopes or in lightbulbs exceeds common levels of knowledge, or at a time when the workings of the cryptographic and algorithmic logics of blockchain technology seem to always remain just beyond one’s grasp. As a result, throughout history, new technologies have been imbued with certain mystical, divine, magical, or otherwise more-than-rational capacities that are beyond our own human reach.⁴⁹ Their continuous development is supported in part by the idea that, because of this more-than-rational touch of brilliance, technologies provide a better understanding of how the world works than humans can attain.⁵⁰

A key point I make in the chapters that follow is that complex, digital technologies are considered to contribute to progress and a common ‘good’ because of their attainment of the purest form of rationality and, simultaneously, their inherent more-than-rational characteristics. The more-than-rational always accompanies the rational in contemporary technological culture, despite—or rather, because of—the culture’s continuation of rationalism.

This simultaneous existence of rationalism and more-than-rational beliefs is the complexity I address in my analyses of blockchain culture. In blockchain culture, the mix of rationalism and more-than-rational beliefs connects to the history of knowledge and power discussed above, but it is also rendered in new, medium-specific ways. This mix does not emerge solely out of blockchain’s code, nor is it located entirely in the culture that surrounds that code. Rather, it is situated in the continuous interplay between the evolving technology and the evolving set of ideas that is projected onto it. The imaginary of blockchain as a truth machine relies on a new form of epistemology—one made possible by the technology itself. This has been called ‘cryptological positivism’⁵¹ and rests on logical, cryptographic processes and assumptions

47 William A. Stahl, *God and the Chip: Religion and the Culture of Technology*, Waterloo: Wilfrid Laurier University Press, 1999, 32–33.

48 Noble, *The Religion of Technology*; Stahl, *God and the Chip*; Morozov, *To Save Everything, Click Here*.

49 Addressing the continuing relevance of this connection between rationalism and the more-than-rational in relation to technology culture, Ian Bogost states that contemporary ‘algorithmic culture is not a material phenomenon so much as a devotional one, a supplication made to the computers people have allowed to replace gods in their minds, even as they simultaneously claim that science has made us impervious to religion’. See Ian Bogost, ‘The Cathedral of Computation’, *The Atlantic*, 2015, <https://www.theatlantic.com/technology/archive/2015/01/the-cathedral-of-computation/384300/>.

50 Lee Worth Bailey, *The Enchantments of Technology*, Urbana: University of Illinois Press, 2005; Noble, *The Religion of Technology*; Morozov, *To Save Everything, Click Here*.

51 Pedro Jacobetty and Kate Orton-Johnson, ‘Blockchain Imaginaries and Their Metaphors: Organising Principles in Decentralised Digital Technologies’, *Social Epistemology* 37.1 (2023), 11.

about their capacity to produce truthful statements.⁵² This new way of knowing inspires a belief in a truth that is ‘more reliable than any truth we’ve ever seen’,⁵³ made possible by blockchain. Versions of the imaginary of blockchain as a truth machine that centre rationalism spread their version of knowing into ever newer domains and reinforce the superiority of its form of knowledge production. Because of this entanglement with power, I critique the imaginary of blockchain as a truth machine through questions regarding the assumptions and norms carried forward by blockchain-based truth claims.

Understanding ‘Blockchain’

‘Blockchain’ as a phenomenon does not begin and end where its code does, but rather should be understood as an assemblage that continuously changes due to technological updates, new ideas about its future potential, regulations, major movements of financial markets, and many other large and small shifts. Whenever I use the term ‘blockchain’ by itself in this book, I mean to address this complexity. Moreover, I avoid using ‘the blockchain’ in its general sense, even if it is often used in this way by others within blockchain culture. The reason for this is that this phrasing can make it seem like there is one underlying or overarching blockchain to which all blockchain projects are connected. This is not the case, nor is there a single piece of code that defines what would constitute ‘the blockchain’ as a general technological category. After all, the core building blocks of blockchain technology are open source and can be updated, amended, or altered by anyone. Consequently, many different versions of blockchain technology exist and there is no central instance or standardised technical phenomenon that can be meaningfully understood as the blockchain.⁵⁴

In what follows, I provide a partial history of the way in which this evolving blockchain assemblage has been pulled in different directions through social, technical, and cultural forces. This history is laced with more-than-rational portrayals of blockchain as the universal solution to society’s ills through its unprecedented capacity for neutral scientific rationality. Despite these claims, suggested applications of blockchain technology are inherently political in their visions for a changed future. This section serves as a background to the analyses in upcoming chapters and contextualises the communities and phenomena that feature there in the broader history of blockchain.

52 Jillian Crandall, ‘Blockchains and the ‘Chains of Empire’: Contextualizing Blockchain, Cryptocurrency, and Neoliberalism in Puerto Rico’, *Design and Culture* 11.3 (2019): 279-300. <https://doi.org/10.1080/17547075.2019.1673989>.

53 Paul Vigna and Michael J. Casey, *The Truth Machine: The Blockchain and the Future of Everything*, New York: St. Martin’s Press, 2018, 20.

54 Additionally, to address (the logics of) blockchain’s software code more specifically, I will use ‘blockchain technology,’ while I reserve ‘a blockchain’ or ‘the blockchain’ for references to a specific instance of the technology.

Approaching Blockchain as an Assemblage

Since its early years, public discussions on blockchain have often struggled with the perceived complexity of the technology. Countless explainer articles and videos circulate on YouTube, technology blogs, and the websites of blockchain projects. Yet, the technology still seems difficult to grasp for many people. As a case in point that illustrates both the ubiquity of blockchain in popular culture and its continued elusiveness, Dutch punk band Hang Youth—whose songs often feature witty, activist takes on current affairs in their lyrics—released a song in 2021 making fun of the way in which blockchain always remains just out of reach. The song features lyrics such as: ‘Have they ever actually used blockchain for anything yet? [...] Why don’t we, for example, build a democracy on the blockchain? Or perhaps I still don’t understand exactly how it works?’.⁵⁵

In part, this difficulty in understanding blockchain technology can be explained by its continuously changing technical features. The Internet Policy Review’s Glossary of Decentralized Techno-Social Systems states that ‘there is neither a formal definition nor a common understanding of what blockchain-based technologies means’.⁵⁶ In the early days of the technology, blockchain might have still been rather straightforwardly understood based on Satoshi’s technical descriptions in the Bitcoin whitepaper, personal experiences with using Bitcoin, or through the slight alterations offered by other cryptocurrencies in their own whitepapers. However, in 2013, the launch of Ethereum—another blockchain that would grow into a platform for the technology’s continued development—expanded the technology’s applicability. Blockchain became increasingly framed as a general-purpose technology with new features proposed by a plethora of new blockchain projects, each with their own distinct use-case.⁵⁷

The proliferation of ideas about what blockchain is capable of doing also has an economic dimension. In part due to the perceived complexity of blockchain technology, the culture around it is characterised by the proliferation of texts explaining what blockchain is believed to be able to do, rather than the actual widespread deployment of robust technological systems. Many blockchain projects evoke what they frame as the technology’s ‘disruptive’ potential

55 The original lyrics read: ‘Hebben ze nou al een keer serieus wat gedaan met de blockchain? [...] Kunnen we niet bijvoorbeeld een democratie op de blockchain doen? Of snap ik dan toch niet precies hoe het werkt?’ See Hang Youth, *Hebben Ze Nou Al Eens Een Keer Serieus Wat Gedaan Met De Blockchain*, 2021. <https://open.spotify.com/track/0UZOJl8yXF4zrLZmZrsWlq?si=6d9f0e77933142e3>. Translation by me. The name Hang Youth is a pun on the Dutch term ‘hangjeugd,’ which is used to refer to young people that hang out in public space (seemingly) without purpose and are perceived by others as a nuisance that is to be eliminated. Despite the general reference to youth, the term ‘hangjeugd’ is often used in a racially coded way to mean young people of colour.

56 María-Cruz Valiente and Florian Tschorsch, ‘Blockchain-Based Technologies_’, in Valeria Ferrari et al. (eds) *Log Out_ A Glossary of Technological Resistance and Decentralization_*, Amsterdam: Institute of Network Cultures, 2023, 35, emphasis in original.

57 Paul Dylan-Ennis, Donncha Kavanagh, and Luis Araujo, ‘The Dynamic Imaginaries of the Ethereum Project’, *Economy and Society* 52.1 (2022): 1–23. <https://doi.org/10.1080/03085147.2022.2131280>. The idea of blockchain being a general-purpose technology is exemplified by references to Ethereum as a ‘world computer’ because it can serve as a technical foundation to run other applications on. See Dylan-Ennis, Kavanagh, and Araujo, ‘The Dynamic Imaginaries of the Ethereum Project,’ 1.

through enticing visions and stories well before building systems that people can actually test and experience. This means that, over the years, blockchain technology has inspired ‘a wholesale of imaginaries, claims, visions, and story-telling and yet very few real-life implementations that have gone beyond Proof-of-Concept’.⁵⁸ Coding such systems takes time and money, and, lacking usable systems, projects in beta or the early stages of pitching instead rely on speculative narratives about the disruptive potential of future applications in the hope of accumulating venture capital investments.

As capital investment cycles push speculation rather than locking down ambitions, technological development can seem stuck in a perpetual beta stage, making blockchain a remarkably idea-driven and continuously changing phenomenon. The combination of slick stories meant to convince rather than provide nuance or contribute to deep understanding and venture capitalist culture forms a self-fulfilling feedback loop between technology and hypes based on superficial engagement. Speculative narratives often feature ill-defined concepts such as decentralisation, transparency or equality, which function like floating signifiers that are able to stand in for generalised ideas about progress or innovation purportedly achievable through the application of the technology while at the same time obscuring the precise logics at work.⁵⁹ Blockchain’s social, technical, and cultural complexity and the lack of a stable definition of the technology afford⁶⁰ the proliferation of multiple visions for blockchain’s use that can equally reinforce or be in conflict with each other.⁶¹

Blockchain continues to be perceived as complex because, rather than just being made up of complicated technical procedures and infrastructures, blockchain is also a sprawling phenomenon that consists of many different and continuously changing elements beyond mere technology. Rather than defining specific technical features that will inevitably change, it is more fruitful to think of blockchain as an assemblage that ranges from apps and platforms such as wallets and exchanges, to stories about blockchain that influence how its functionalities are understood; and from algorithmic elements such as smart contracts, DAOs, and oracles to the actors engaging with them—the day traders, venture capitalist investors, blockchain-based companies, regulators, to name but a few.⁶²

58 Karim Jabbar and Pernille Bjørn, ‘Blockchain Assemblages: Whiteboxing Technology and Transforming Infrastructural Imaginaries’, *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, Glasgow, UK, 4–9 May, 2019. <https://doi.org/10.1145/3290605.3300496>.

59 Silvia Semenzin, *Blockchain and Data Justice: The Political Culture of Technology*, PhD diss., Università degli Studi di Milano & Università degli Studi di Torino, Milan & Turin, 2021.

60 Affordance theory is a key element in my approach to the study of blockchain imaginaries and I will elaborate on my understanding and operationalisation of the term later in this chapter.

61 Silvia Semenzin, David Rozas, and Samer Hassan, ‘Blockchain-Based Application At a Governmental Level: Disruption or Illusion? The Case of Estonia’, *Policy and Society* 41.3 (2022): 386–401.

62 See e.g. the use of the term ‘blockchain assemblage’ by Jaya Klara Brekke, *Disassembling the Trust Machine*, PhD diss., Durham University, Durham, 2019.

Claims of Neutral Revolutions with Blockchain

A core element of blockchain's initial appeal is the belief that fallible, corruptible, and partisan human decision-making can be improved and supplanted by what are thought of as blockchain's neutral mathematical, computational, and algorithmic processes. The timing of blockchain's first announcement and who it was published by are significant in relation to these a-political aims. The Bitcoin whitepaper⁶³ was published on 1 November 2008 on a cryptography-themed mailing list by a person (or several people) going by the pseudonym Satoshi Nakamoto, whose true identity remains unknown to this day. The Bitcoin whitepaper was published during some of the most unstable months of the global financial crisis of 2007–2008—a time during which faith in the banking system was at an all-time low. Bitcoin was designed expressly to route around the incumbent financial institutions that make up the banking sector. It would do so by replacing the trust needed in such (ultimately corruptible) intermediaries with what Satoshi deemed to be 'trustless' cryptographic proof.⁶⁴ Whether intended as such or not, the fact that Satoshi's identity remains unknown enhances the anti-institutional rhetoric of the Bitcoin whitepaper and its associated enthusiasm for supposedly a-political computational systems to facilitate socio-economic processes. In a god-like move, Satoshi created and then disappeared, keeping Bitcoin's seemingly immaculate conception free from the personal politics that any known human developer would bring. This vision of Bitcoin and its underlying blockchain technology, as neutral tools outside of politics, remains present in the community to this day⁶⁵ and has contributed to a culture of solutionist optimism and utopianism around the technology.⁶⁶

In the whitepaper, Satoshi significantly described Bitcoin as 'a peer-to-peer electronic cash system'.⁶⁷ The idea of digital cash is different from the digital money that already existed when the whitepaper was released, for example the digital form of money that is used in internet banking. Physical cash has particular properties that are unavailable in these earlier forms of digital money. Particularly, the privacy associated with cash transactions was fundamental to people working on developing its digital equivalent, many of whom were part of the cypherpunk community of the 1980s and 1990s.⁶⁸ Many members of this community believed Bitcoin

63 Nakamoto, 'Bitcoin'.

64 Nakamoto, 'Bitcoin'.

65 A recent addition to this line of thought features in the New York Times bestseller written by Chris Dixon, *Read Write Own: Building the Next Era of the Internet*, New York: Random House, 2024. Dixon is a general partner at venture capitalist firm Andreessen Horowitz, which funds many blockchain projects, particularly through its subsidiary a16z crypto.

66 Morozov, *To Save Everything, Click Here*; Olivier Jutel, 'Blockchain Imperialism in the Pacific', *Big Data & Society* 8.1 (2021): 1–14.

67 Nakamoto, 'Bitcoin,' 1. Although it was not the first experiment with digital cash by any means, it would soon become the most well-known one. Computer scientist and cryptographer David Chaum is considered the first to write about the idea of digital cash in 1983. See David Chaum, 'Blind Signatures for Untraceable Payments', in D. Chaum, R.L. Rivest, and A.T. Sherman (eds) *Advances in Cryptology*, Boston: Springer, 1983, 199–203. DigiCash (1989–1998) and e-gold (1996–2008) are generally considered to be the most developed experiments with digital cash that predate Bitcoin.

68 Finn Brunton, *Digital Cash: The Unknown History of the Anarchists, Utopians, and Technologists Who Created Cryptocurrency*, Princeton: Princeton University Press, 2020.

solved key issues that prevented previous versions of digital cash from taking off, and after its launch, the Bitcoin user-base grew continuously, breaking out of its subcultural niche. Its price started a volatile yet often upward journey, reaching dramatic proportions from the mid-2010s onward—a phenomenon that was often discussed in financial, business, and technology-themed news outlets at the time.

Nevertheless, it was not until a few years later that the blockchain assemblage produced what seemed like a Cambrian explosion of diverse revolutionary claims. A key reason for this second hype cycle was the 2013 introduction of the Ethereum blockchain,⁶⁹ heralding a period that is referred to by some as blockchain 2.0.⁷⁰ While Satoshi's blockchain design is particularly geared towards monetary assets and led to the launch of a host of cryptocurrencies, Ethereum transformed the technology into a platform for a much wider range of applications. Its new technological capacities—such as smart contracts and DAOs⁷¹—led to a flurry of blockchain-based start-ups claiming to disrupt one sector after the other, from finance to health care and journalism. Blockchain technology became so hyped and its proponents so convinced and vocal about its societal benefits that they were sometimes referred to as blockchain evangelists.⁷² Between 2016 and 2018, it was not uncommon to see articles in technology journalism, blog posts, and even complete books⁷³ proclaiming that blockchain would soon change the world in fundamental and revolutionary ways.

However, any revolution bankrolled by venture capital is not incentivised to undo the power dynamics that make asymmetrical wealth accumulation possible in the first place, and the limitations and biases hidden underneath 'universalist claims'⁷⁴ of revolution need to be critically examined. The utopian moment in the mid-2010s of an imminent blockchain revolution was inherently tied to the status quo through its dependence on venture capital.⁷⁵ Blockchain was often portrayed as either a way to make existing businesses more efficient or as a technology that would make new kinds of profit streams possible. Paradoxically, but also perhaps predictably, the incumbent financial institutions that the initial invention of blockchain aimed to circumvent also started to dedicate teams and task forces to finding ways to exploit the technology for their own financial gain.

69 Vitalik Buterin, 'Ethereum Whitepaper', *Ethereum*, 2013, <https://ethereum.org/en/whitepaper>.

70 Just as Web 2.0 did not efface the static websites that characterise Web 1, this new version of blockchain did not mean the end of earlier versions of the technology.

71 For now, it is not necessary to understand the meaning of these technical phenomena. I will explain them later in this dissertation, at moments in which a more detailed understanding is necessary in order to follow the rest of my discussion of the technology.

72 see e.g. Rachel O'Dwyer, 'Blockchains and Their Pitfalls', in Trebor Scholz and Nathan Schneider (eds) *Ours to Hack and Own: The Rise of Platform Cooperatives a New Vision for the Future of Work and a Fairer Internet*, New York: OR Books, 2016, 228-32.

73 See e.g., Don Tapscott and Alex Tapscott, *Blockchain Revolution: How the Technology Behind Bitcoin and Other Cryptocurrencies is Changing the World*, London: Penguin, 2016.

74 Sally Wyatt, 'Danger! Metaphors At Work in Economics, Geophysics, and the Internet', *Science, Technology, and Human Values* 29.2 (2004), 257.

75 For example, in 2018, blockchain start-ups collectively amassed more than 4 billion US dollars in investment funding. See Statista, <https://www.statista.com/statistics/621207/worldwide-blockchain-startup-financing-history/>.

Blockchain projects that portray fictional futures in which society has changed for the better due to the use of their systems still often envision this as an a-political endeavour. However, this necessarily relies on a narrow understanding of the word ‘political’: what is meant in such framings is that change is envisioned to emerge through non-governmental initiatives and in self-organised ways. Under the banner of Blockchain for Good, countless initiatives promise social and environmental benefits through technological innovation.⁷⁶ Yet these claims rarely specify what “good” entails or how it will be realised in practice. The result is that genuine social and ecological effects are often concealed behind persuasive slogans and glossy marketing campaigns.⁷⁷ Instead of using the narrow understanding of politics used by many blockchain projects, I follow a broader understanding that emerges from cultural studies to do justice to these deeper implications. I understand politics as ‘involving all those processes whereby power relationships are implemented, maintained, challenged, or altered in any sphere of activity whatsoever’.⁷⁸ With this definition of politics, all blockchain projects can in fact be understood to be political, opening them up for analysis of their entanglement with power.

Blockchain technology has not stopped evolving since Ethereum’s first major innovations, leading to ever newer visions for the technology’s future sociocultural potential. In recent years, DAOs and Non-Fungible Tokens (NFTs) have accumulated their own, sometimes overlapping crowds of enthusiasts. Just like Satoshi’s announcement of the Bitcoin whitepaper during the financial crisis of 2007–2008, NFTs rose to fame in another global moment of crisis: the COVID-19 pandemic. NFTs enable the trading of unique digital certificates of ownership, representing, among others, artworks, physical objects, or digital collectibles. Digital art, which was previously notoriously hard to sell because of its easy and cheap reproducibility, became a key focus for the NFT hype that raged in 2021–2022, perhaps also stimulated by the fact that COVID-19 lockdowns complicated people’s normal way of engaging with art in galleries and museums. Because of NFTs, many artists were able to sell their work for the first time, and they inspired hopes of a more equitable art world. Nevertheless, many others also lost

76 Such aims may be financially motivated because framing a project as contributing to a common societal or environmental good can play a significant role in its success at accumulating funding, whether that is from governmental subsidies or venture capital.

77 Glenn Parry and John Collomosse, ‘Perspectives on ‘Good’ in Blockchain for Good’, *Frontiers in Blockchain* 3.1 (2021): 1–5. <https://doi.org/10.3389/fbloc.2020.609136>; Silvia Semenzin, ‘Blockchain for Good’: Exploring the Notion of Social Good Inside the Blockchain Scene’, *Big Data & Society* 10.2 (2023): 1–14. The ‘Blockchain for Good’ sector is discussed in various parts of this thesis, but takes centre stage specifically in chapter 2. Here, blockchain can be seen as a new addition in a long line of administrative and informational technologies that support control, from punch cards to logbooks and algorithms (see O’Dwyer, ‘Blockchains and Their Pitfalls’), the socio-political nature of which has been described in many different contexts. See e.g.: Geoffrey C. Bowker and Susan Leigh Star, *Sorting Things Out: Classification and Its Consequences*, Cambridge, MA: MIT Press, 2000; Jacqueline Wernimont, *Numbered Lives: Life and Death in Quantum Media*, Cambridge, MA: MIT Press, 2018; Safiya Umoja Noble, *Algorithms of Oppression: How Search Engines Reinforce Racism*, New York: New York University Press, 2018.

78 Jeremy Gilbert, *Anticapitalism and Culture: Radical Theory and Popular Politics*, Oxford: Berg Publishers, 2008, 8.

money due to high transfer fees or the shady actors that were attracted by the financial value that suddenly accumulated in the emerging scene.

While NFTs fundamentally represented a new way to pay for and own (digital) objects, around the same time—and sometimes connected in the same overarching project—DAO experiments emerged that focussed on social organisation via blockchain technology. Essentially, a DAO can be used to collectively manage funds in a community-based or commercial project. This can be a highly automated process, or it can take place mostly off-chain, such as in chat groups or in-person meetings—or anything in between these extremes. DAOs will feature prominently in chapter 5, where I will show that certain artistic and activist DAO projects contribute to countercultural imaginaries, featuring new forms of sociality and knowledge practices.

This brief history shows that the utopian and a-political narratives that surround blockchain technology are, in fact, untenable. Blockchain projects, whether for the good and/or for the money, are thoroughly political in their aim to effect change⁷⁹ because all change is political: it has different effects for and on different people and thus requires prioritisation between their needs. Blockchain imaginaries have also evolved throughout the history of blockchain itself, and demand ongoing critical analyses that deal with the changing shape of the ideals they perpetuate. I address three key dimensions of blockchain imaginaries that took shape throughout this history in the following section: ideals of self-sovereignty, practices of algorithmic social organisation, and belief in the social and environmental ‘good’ of technology.

Imagining Futures with Blockchain

The diverse hopes for, narratives about, iterations on and uses of blockchain technology discussed above have been the subject of researchers studying imaginaries. Before presenting the concept of imaginaries in depth and reworking it in the context of my specific project in section 3.2, I will first use the general understanding that technology imaginaries are ‘collective beliefs about how society functions’⁸⁰ in relation to a technology, and that the future-oriented narratives around emerging technologies are particularly insightful places to research them. Below, I present three sociotechnical blockchain imaginaries identified by other researchers that are relevant to this book because rationalism plays an important role in them. I will highlight this role and detail how my own analyses connect to these imaginaries in my discussion of each. I will continue this practice of making explicit the connection between established insights and concepts and their relevance for upcoming chapters in the rest of the current chapter. However, it should be noted that these brief discussions do not constitute a full overview of the book—which will instead feature in section 5 below.

79 See e.g.: Primavera de Filippi and Benjamin Loveluck, ‘The Invisible Politics of Bitcoin: Governance Crisis of a Decentralized Infrastructure’, *Internet Policy Review* 5.3 (2016): 1–28.; O’Dwyer, ‘Blockchains and Their Pitfalls’; David Golumbia, *The Politics of Bitcoin*, Minneapolis: University of Minnesota Press, 2016.

80 Sheila Jasanoff, ‘Future Imperfect: Science, Technology, and the Imaginations of Modernity’, in Sang-Hyun Kim and Sheila Jasanoff (eds) *Dreamscapes of Modernity: Sociotechnical Imaginaries and the Fabrication of Power*, Chicago: University of Chicago Press, 2015, 5.

Importantly, the imaginaries discussed in this section are not an exhaustive list, but rather serve to highlight how the study of sociotechnical imaginaries often picks up on the importance of rationalism in them, while their reliance on the more-than-rational remains obscured. To address the rational and the more-than-rational simultaneously, I expand the concept of sociotechnical imaginaries in the second part of this section into socio-techno-cultural imaginaries.

Three Sociotechnical Blockchain Imaginaries

The first blockchain imaginary that is relevant to my research frames technological rationality as a virtue in economic systems because it is thought to align with and facilitate libertarianism and other right-wing political ideologies, especially as they intersect with ideals of self-sovereignty. While this imaginary is particularly well-represented in the Bitcoin community, it also resonates outside of it in the broader blockchain community. It is characterised by shared ideals with regard to the evils of government oversight, central banking, and monetary policy on the one hand, and reverence for computational knowledge processes on the other.⁸¹ Important precursors to these views are found in the cypherpunk community, the Austrian School of Economics, and in gold-rush-era pioneers, particularly with regard to the importance of privacy and self-sovereignty,⁸² a shared sense of techno-optimism, and the economic logics of individualism, freedom, private property, metallism,⁸³ and decentralised free trade.⁸⁴ This imaginary seeks independence from state control by establishing a new mode of value production grounded in automated, mathematical processes that operate beyond governmental oversight.⁸⁵

Particularly, the use of Bitcoin is seen as a financial speculation on a 'crypto-anarchist future'⁸⁶ where it will become ever more vital to have access to value outside of that which crumbling states can (be trusted to) secure or control. Projects such as Liberland—an unrecognised micronation between Croatia and Serbia—which fall under broader umbrella-term of 'Network States'⁸⁷ use blockchain as a way to govern an autonomous society. Such blockchain-based

81 Golumbia, *The Politics of Bitcoin*.

82 Although the community emphasises ideals of self-sovereignty, it is important to point out that Bitcoin relies on existing infrastructures and is embedded in their—often more centralised—systems and socio-political contexts, which is also true for blockchain technology more generally Henrik Karlstrøm, 'Do Libertarians Dream of Electric Coins? The Material Embeddedness of Bitcoin', *Distinktion: Scandinavian Journal of Social Theory* 15.1 (2014): 23–36.; de Filippi and Loveluck, 'The Invisible Politics of Bitcoin'.

83 I.e., the idea that the value of money should be 'backed by a commodity like gold, which derives its intrinsic value from the market'. See Lana Swartz, 'What Was Bitcoin, What Will it be? The Techno-Economic Imaginaries of a New Money Technology', *Cultural Studies* 32.4 (2018), 630.

84 Swartz, 'What Was Bitcoin, What Will it be?'; John Flood and Lachlan Robb, 'Trust, Anarcho-Capitalism, Blockchain and Initial Coin Offerings', *SSRN Electronic Journal* (November, 2017): 1–27. <https://doi.org/10.2139/ssrn.3074263>; Bill Maurer and Lana Swartz, 'Wild, Wild West: A View From Two Californian Schoolmarms', in Geert Lovink, Nathaniel Tkacz, and Patricia De Vries (eds) *Moneylab Reader: An Intervention in Digital Economy*, Amsterdam: Institute of Network Cultures, 2015, 222–29.

85 Jacobetty and Orton-Johnson, 'Blockchain Imaginaries and Their Metaphors'.

86 Swartz, 'What Was Bitcoin, What Will it be?', 631.

87 Balaji Srinivasan, *The Network State*, self-published, 2022.

state-creation envisions its autonomy as emerging in (virtual) societies free from punitive law and regulation, and instead featuring quantified reputational systems tied to the distribution of rights and access.⁸⁸

These developments can be understood as blockchain-based examples of 'libertarian 'exit' imaginaries'.⁸⁹ This imaginary highlights the connection between libertarian politics and optimism about automation and technological decision-making. The knowledge politics involved in this connection has a broader sociocultural history in colonial power relations and contemporary neoliberal social relations.

The second blockchain imaginary that is relevant to my own analyses features visions of distributed social organisation at scale, especially as they rely on idealisations of technological rationalism and algorithmic governance. Here, ideals of efficiency, trust in computational and automated processes for decision-making and distributed governance, and peer-to-peer connections are key ingredients.⁹⁰ A central feature in this imaginary is the influence of informational infrastructures on the kinds of futures that could be achievable.⁹¹ This imaginary is particularly prominent in DAO-based future visions, in which automation and self-organisation are central.⁹² For example, the idea of 'blockchain cities' in which rights, access, energy, transportation, etc. are algorithmically governed via blockchain systems is part of this imaginary.⁹³ The networked architecture of blockchain technology and the peer-to-peer connections that constitute it feature in visions where trust between humans, and between humans and societal institutions, is replaced with what is thought of as trustless computational processes.

Yet, in this context, 'trustless' does not imply free from trust, but rather replaces trust in humans and their institutions with trust in code, computation, and the logics of the network.⁹⁴ Algorithmic decision-making processes feature in the imaginary as an idealised deliberation process that does not allow for ambiguity and is considered to be 'highly technocratic'.⁹⁵ As such, this blockchain imaginary not only features in corporate and community projects, but

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- 88 Inês Faria, 'Trust, Reputation and Ambiguous Freedoms: Financial Institutions and Subversive Libertarians Navigating Blockchain, Markets, and Regulation', *Journal of Cultural Economy* 12.2 (2019): 119-32.; Amaël Cattaruzza, 'Where is Liberland? Ideology and Power Beyond Territory', *Dialogues in Human Geography* 12.2 (2022): 307-11.
 - 89 Casey R Lynch and Alex Muñoz-Viso, 'Blockchain Urbanism: Evolving Geographies of Libertarian Exit and Technopolitical Failure', *Progress in Human Geography* 48.1 (2023), 1; Raymond Craib, *Adventure Capitalism: A History of Libertarian Exit, From the Era of Decolonization to the Digital Age*, Oakland: PM Press, 2022.
 - 90 Caitlin Lustig, 'Intersecting Imaginaries: Visions of Decentralized Autonomous Systems', *Proceedings of the ACM on Human-Computer Interaction* 3 (2019): 1-27.
 - 91 Swartz, 'What Was Bitcoin, What Will it be?.'
 - 92 Lustig, 'Intersecting Imaginaries'.
 - 93 Chris Elsdén et al., 'Making the Blockchain Civic', *Interactions* 26.2 (2019): 60-65. <https://doi.org/10.1145/3305364>; Charles Shen and Feniosky Pena-Mora, 'Blockchain for Cities—a Systematic Literature Review', *IEEE Access* 6 (2018): 76787-819.
 - 94 Jacobetty and Orton-Johnson, 'Blockchain Imaginaries and Their Metaphors'; Gili Vidan and Vili Lehdonvirta, 'Mine the Gap: Bitcoin and the Maintenance of Trustlessness', *New Media & Society* 21.1 (2019): 42-59.
 - 95 de Filippi and Loveluck, 'The Invisible Politics of Bitcoin,' 1.

also in proposals and experiments by (non-) governmental organisations.⁹⁶ This imaginary shows that, besides economic and financial uses, blockchain technology is often imagined in terms of its role in new forms of social organisation. The organisational dimension of blockchain will feature in several different upcoming chapters, sometimes critiquing the way in which an imaginary reproduces the social hierarchies and power dynamics that already characterise the status quo, other times expanding on the ways in which blockchain inspires forms of organisation that create fundamental breaks from it instead.

The third blockchain imaginary that is important to highlight here is closely connected to the previous one, but here particularly skewed towards social and environmental goals. This ‘blockchain for good’ imaginary applies the perceived benefits of rationalism to ideals that are societal in nature, for example the UN Sustainable Development Goals,⁹⁷ rather than the ideals of unbridled free trade that dominate other corners of blockchain culture. A particularly well-established imaginary exists around the idea of applying blockchain technology to secure common resources, for example building systems to share solar energy between members of a community depending on shifting needs and availabilities.⁹⁸ Some envision the possibility of a ‘technological commonwealth’,⁹⁹ while others look to blockchain technology in order to build commonly owned and governed alternatives to the proprietary platforms that dominate digital society.¹⁰⁰ In the context of heightening awareness of the climate crisis, blockchain technology is increasingly used to imagine ways to prevent its collapse through automatic data-based decision-making.¹⁰¹ Although the ‘blockchain for good’ imaginary focusses on futures that are inspired by ideals such as equality and sustainability, it is important to point out that these projects are also embedded in established contexts and sociotechnical logics, and often become ‘a tool for reviving the neoliberal vision’.¹⁰² This imaginary shows that, despite its claims of neutrality, blockchain culture features ideals that are inherently political, although the political influence of blockchain’s knowledge processes often remains unaddressed.

The three imaginaries outlined above weave through my own analyses in upcoming chapters. In chapter 2, I dissect the knowledge politics of blockchain projects that aim to contribute

96 Inês Faria, ‘Blockchain in the EU: Transforming Imaginaries and the Social Making of Financial Regulation’, *Anthropology Today* 39.4 (2023): 21–23.; Semenzin, Rozas, and Hassan, ‘Blockchain-Based Applications At the Governmental Level’.

97 See e.g., Beth Kewell, Richard Adams, and Glenn Parry, ‘Blockchain for Good’, *Strategic Change* 26.5 (2017): 429–37. <https://doi.org/10.1002/jsc.2143>.

98 See e.g., David Rozas et al., ‘When Ostrom Meets Blockchain: Exploring the Potentials of Blockchain for Commons Governance’, *SSRN Electronic Journal*, (2021): 1–14.; Nazli Cila et al., ‘The Blockchain and the Commons: Dilemmas in the Design of Local Platforms’, *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, Honolulu, HI, April 25–30, New York, 2020. <https://doi.org/10.1145/3313831.3376660>.

99 Sarah Manski, ‘Building the Blockchain World: Technological Commonwealth or Just More of the Same’, *Strategic Change* 26.5 (2017): <https://doi.org/10.1002/jsc.2151>, 511.

100 Alex Pazaitis, Primavera de Filippi, and Vasilis Kostakis, ‘Blockchain and Value Systems in the Sharing Economy: The Illustrative Case of Backfeed’, *Technological Forecasting and Social Change* 125 (December, 2017): 105–15. <https://doi.org/10.1016/j.techfore.2017.05.025>.

101 Malcolm Campbell-Verduyn, ‘Conjuring a Cooler World? Imaginaries of Improvement in Blockchain Climate Finance Experiments’, *Environment and Planning C: Politics and Space* 42.5 (2023): 1–18.

102 Semenzin, *Blockchain and Data Justice*, 145.

to the fight against climate change by putting the logics of tokenisation to use. I will argue that blockchain's capacity for storing data immutably and creating from it some universal truth reproduces assumptions about knowledge established throughout modernity and also reproduces the colonial relations associated with them. In chapter 3, I will expand on the religious and mystical dimensions of blockchain culture and how it relates to ideas surrounding the agency of individuals and communities to organise societies. I argue that these dimensions naturalise the computational knowledge processes and organisational capacities of blockchains to such a degree that they appear divinely ordained or otherwise beyond human control. In chapter 4, I show the limits of rationalist logics in the context of the fundamental uncertainty that characterises the cryptocurrency market. Instead of universal and immutable truths, here blockchain features in an imaginary that doubles down on the performativity of information. Although it is concerned with how society functions and is upheld, this is a profoundly individualist vision for society. In chapter 5, experiments with alternative forms of social organisation through the use of DAOs are key in imaginaries that break with dominant versions of 'blockchain as a truth machine.' Instead of universal and immutable truths, here situated and mutating understandings of shared knowledge processes are key. Furthermore, cases discussed in this chapter break with the tendencies of the 'blockchain for good' project by expressing their aims in overtly political terms.

Conceptualising Socio-Techno-Cultural Imaginaries

I will now introduce the concept of imaginaries in more detail and explain my own iterations on it. Imaginaries have been studied in the past century to understand the shared sense of identity, reality, norms, and/or purpose that groups of people who do not know each other personally collectively establish, and which in effect has an influence on the future through its own performative reproduction.¹⁰³ Émile Durkheim¹⁰⁴ is often cited as one of the forebears of the notion of imaginaries, particularly with regard to his understanding of the collective conscience or consciousness as the shared ideals, beliefs, and norms that form a society. From here, the idea has seen several iterations: Jacques Lacan's 'imaginary order'¹⁰⁵ addresses how people understand themselves and others through images in the mind; Benedict Anderson¹⁰⁶ addressed the way in which citizens imagine their nation collectively; Cornelius Castoriadis¹⁰⁷ wrote on the social imaginary and its role in instituting society (I will return in more depth to Castoriadis' influential ideas in chapter 4); Charles Taylor¹⁰⁸ thinks about the establishment

103 This performativity has not always been explicitly outlined in more historical uses of the concept, although imaginaries can retroactively be thought of as inherently performative. Performativity is a key element of my approach to the study of socio-techno-cultural imaginaries, and I will explore it in more detail in the next section.

104 Emile Durkheim, *The Division of Labor in Society*, Glencoe: The Free Press, 1960 [1893].

105 Jacques Lacan, *The Seminars of Jacques Lacan, Book 1: Freud's Papers on Technique, 1953–1954*, New York: W.W. Norton, 1991 [1988].

106 Benedict Anderson, *Imagined Communities: Reflections on the Origin and Spread of Nationalism*, London: Verso, 1983.

107 Cornelius Castoriadis, *The Imaginary Institution of Society*, Cambridge, MA: MIT Press, 1998 [1975].

108 Charles Taylor, *Modern Social Imaginaries*, Durham: Duke University Press, 2004.

and shifts of paradigms like modernity; and Arjun Appadurai¹⁰⁹ writes about how the social imagination changes under the influence of globalisation.

Within the context of digital technology, Vincent Mosco, Patrick Flichy, and Taina Bucher are important contributors to the study of imaginaries. Whereas Mosco¹¹⁰ and Flichy¹¹¹ present theories on what can be understood as the origins of internet imaginaries using semiotics and the myth structure in making sense of the future, Bucher¹¹² brings the concept of imaginaries to bear on the way in which users make sense of the blackboxed workings of algorithmic technologies.

However, the most comprehensive conceptualisation that is targeted to the study of (digital) technologies comes from Sheila Jasanoff, who writes that 'sociotechnical imaginaries [are] collectively held, institutionally stabilized, and publicly performed visions of desirable futures, animated by shared understandings of forms of social life and social order attainable through, and supportive of, advances in science and technology'.¹¹³ Sociotechnical imaginaries consist of the interplay between both scientific and technological developments, and the social forms and practices those developments are imagined to make possible. Studying sociotechnical imaginaries requires an interdisciplinary approach that bridges the social sciences and cultural theory to become attuned to the way in which future societies are envisioned to be organised with the use of technologies, the meanings attached to those technologies and the practices that emerge around them, and is capable of critically assessing the power relations upon which such futures are built and hence reinforce.¹¹⁴

Despite this interdisciplinarity, research on sociotechnical imaginaries does tend to skew towards the social, political, and economic, rather than the cultural. Jasanoff's own involvement in such fields as law and policy likely influence her framing of the concept, and most of the studies she and her co-editor Sang-Hyun Kim select to illustrate it in their edited book on the topic address imaginaries at the level of nation states, politics, and governance.¹¹⁵

Likewise, while the previous section shows that significant work has been done with regard to the study of blockchain imaginaries, such contributions also reveal that the sociotechnical focus of the concept is not fully adequate to understand the more-than-rational elements in blockchain culture. Whereas important aspects of the rationalist dimension of blockchain imaginaries become immediately apparent in a sociotechnical analysis that focusses on how society is imagined to be reorganised through technological innovation, the more-than-

109 Arjun Appadurai, 'Disjuncture and Difference in the Global Cultural Economy', *Theory, Culture & Society* 7.2–3 (1990): 295–310. <https://doi.org/10.1177/026327690007002017>.

110 Vincent Mosco, *The Digital Sublime: Myth, Power, and Cyberspace*, Cambridge, MA: MIT Press, 2005.

111 Patrice Flichy, *The Internet Imaginaire*, Cambridge, MA: MIT Press, 2001.

112 Taina Bucher, 'The Algorithmic Imaginary: Exploring the Ordinary Affects of Facebook Algorithms', *Information, Communication & Society* 20.1 (2017): 30–44.

113 Jasanoff, 'Future Imperfect', 4.

114 Jasanoff, 'Future Imperfect'.

115 Sheila Jasanoff and Sang-Hyun Kim, *Dreamscapes of Modernity: Sociotechnical Imaginaries and the Fabrication of Power*, Chicago: University of Chicago Press, 2015.

rational is relegated to the background in such studies. The mix of the rational and the more-than-rational functions socially and culturally: it shapes the kinds of institutions that exist, it features in social debates, and it can be recognised in meanings attached to the technology and in the everyday practices that emerge around it.

To address rationalism and the more-than-rational in blockchain culture, I modify Jasanoff's concept: socio-techno-cultural imaginaries¹¹⁶ are collective future-oriented visions of sociocultural life—its organisation, politics, meanings, practices, etc.—made possible by technological developments. These visions influence the realisation of such potentials because of their power, both in terms of the key societal actors they involve, and their spread into culture and everyday practices. The interdisciplinary approach that is necessary to analyse socio-techno-cultural blockchain imaginaries features methods from media studies and cultural analysis, and needs to be combined with insights from critical theory, critical data studies, and science and technology studies to address the sociocultural implications of the intersection of technology and ideas about its sociocultural potential. As an emerging technology that is embedded in constantly evolving narratives and practices, I argue that blockchain imaginaries should be understood in relation to those cultural processes of meaning-making and bring into focus the norms, values, and ideologies that influence their visions of the future.

To be clear, I do not propose to abandon the social dimension that Jasanoff emphasises, but rather to integrate the analysis of imagined social worlds with the cultural implications with which such visions are entangled. While Jasanoff's concept itself has room for a cultural approach, in practice this possibility is not explored to its fullest potential. Socio-techno-cultural imaginaries are a modified approach to the study of the imaginaries that surround technologies that combine an imagined social order, cultural norms, and values.

To illustrate the limitations of a narrower sociotechnical approach to blockchain imaginaries, and what it means to rebalance the approach to the socio-techno-cultural, I now briefly present the example of blockchain-based consent apps. Although these apps amount to little more than a footnote in blockchain history and will not appear in the rest of this book, these apps offer an informative example via which to explain my perspective on socio-techno-cultural blockchain imaginaries.

In 2018, at the height of the #MeToo movement, which created public awareness about the ubiquity of sexual abuse and harassment in contemporary societies, a Dutch blockchain project called LegalFling attempted to establish itself. It envisioned an improved future in which such aggressions would be drastically reduced through consent contracts that would be logged immutably onto a blockchain. Such consent contracts would detail the kinds of sexual acts one consents to on a particular occasion with a particular person and should be assembled and signed anew before any subsequent sexual encounter. By logging consent immutably onto a blockchain, transgressions would be easier to prove, they argued. However, the project had a major conceptual flaw: logging consent immutably only serves to prove

116 I will discuss below what I mean to address specifically by the addition of 'cultural.'

that at one point in time, consent was given. It does not show whether this consent was later rescinded off-chain or even given under duress in the first place. Blockchain consent apps thus run the terrible risk of becoming a tool for those committing acts of sexual violence rather than their victims. LegalFling was not a one-off case either: Loly and SaSie were blockchain projects launched around the same time encompassing fundamentally similar concepts, with fundamentally similar limitations. Perhaps unsurprisingly, blockchain consent apps have not stood the test of time: their operationalisations of consent revealed a meagre understanding of the issue at hand, and as a result the projects were often subject to public ridicule and critique, and were soon abandoned by their developers.

The vision of a future with reduced instances of sexual aggression through the application of blockchain technology could be studied in the vein of Jasanoff's concept of sociotechnical imaginaries. Such a study might describe how blockchain consent systems are imagined to interface with existing law institutions and police work. Alternatively, it might explore how such institutions would be overhauled completely and how social order might alternatively take shape through blockchain technology. It could also look into how social hierarchies in the workplace—as the site of much harassment—are shifted as a consequence, or perhaps how institutions such as marriage might evolve in the context of blockchain consent apps. These are all interesting possible research endeavours, but they engage little with culture and as such leave unacknowledged a large part of the performative power of the imaginary of blockchain consent apps to enact the futures they envision.

A socio-techno-cultural approach to the study of imaginaries is necessary to address the connections between (perceived) technological capacities of blockchain, ideas about how societies (should, could) function with the technology, and what it means to coexist together with others in these future worlds. Culture provides this focus because it encompasses both the 'ordinary' and 'a whole way of life,' both big, shared ideals and small, situated meaning-making practices.¹¹⁷ Such an appreciation of the ordinary allows for a focus on objects that often get overlooked in research on imaginaries,¹¹⁸ and is especially relevant in the context of digital technologies with which many people gain personal experiences, rather than for example less accessible technologies such as space travel. In the context of widespread and popular use of digital technologies that are continuously evolving, imaginaries may be plural and in contestation with each other¹¹⁹—something towards which a cultural approach is critically attuned.

It is notoriously difficult to define what 'culture' is.¹²⁰ Rather than defining the dimensions of a particular culture, an approach based in cultural analysis attempts to understand how diverse

117 Raymond Williams, *Resources of Hope: Culture, Democracy, Socialism*, London: Verso Books, 2016 [1989], 4.

118 c.f. Mieke Bal, *Double Exposures: The Subject of Cultural Analysis*, New York: Routledge, 1996.

119 Astrid Mager and Christian Katzenbach, 'Future Imaginaries in the Making and Governing of Digital Technology: Multiple, Contested, Commodified', *New Media & Society* 23.2 (2021): 223–36.

120 Raymond Williams, *Keywords: A Vocabulary of Culture and Society*, Oxford: Oxford University Press, 2015 [1976].

phenomena contribute to cultures in flux,¹²¹ like the way blockchain culture is in a continuous state of flux, influenced by both technology and ideas. Cultural analysis allows for this focus, as it studies its objects 'in view of their existence in culture'¹²² in the 'now',¹²³ aiming to understand 'how [they] contribut[e] to cultural debates',¹²⁴ rather than viewing them as representative of it. Thus, rather than untangling what is or is not blockchain culture, the continued interplay between technical blockchain developments and their cultural appropriation through practices, debates, and objects is key in socio-techno-cultural blockchain imaginaries. These imaginaries take shape in the values and norms that are attached to a given technology, and the meaning attributed to it in the context of wider ideas about life, nature, and society. Furthermore, a socio-techno-cultural approach has the potential to yield findings that support a broader understanding of the influence of imaginaries to bring new realities into being, because they engage with the performativity of everyday practices, the performativity of ideas and values, and the performativity of institutionalisation and power.

If we return to the example of blockchain consent apps based on this understanding of a socio-techno-cultural analysis of their imaginary, different questions might be asked. For example, with this expanded focus, research could be done with regard to how blockchain consent apps affect the meaning that is attached to consensual and nonconsensual sex as interpersonal experiences, in addition to its institutional ramifications. Research might inquire into how that meaning relates to dominant or countercultural norms and values. It might explore the everyday practices that accompany sexual encounters and how they differ between communities in relation to different ideals and sexual identities. It might look to popular or subcultural representations to find answers to such questions, rather than just the 'languages of power'¹²⁵ such as policy documents and political speeches, to which sociotechnical imaginaries are attuned. As such, socio-techno-cultural imaginaries become indicative of particular, situated visions of future life with technology. Socio-techno-cultural imaginaries can be used to reveal in even more detail 'the complex topographies of power and morality'¹²⁶ that influence the ways in which technology is imagined, developed, and used.

Examples do exist of the study of socio-techno-cultural blockchain imaginaries, for example, studies that explore how 'trust' is understood in blockchain communities, projects, and technological structures;¹²⁷ how a value such as belonging is imagined in communities that

121 Mieke Bal, *Travelling Concepts in the Humanities: A Rough Guide*, Toronto: University of Toronto Press, 2002.

122 Bal, *Travelling Concepts in the Humanities*, 9, emphasis in original.

123 Bal, *Double Exposures*.

124 Bal, *Travelling Concepts in the Humanities*, 9.

125 Jasanoff, 'Future Imperfect,' 25.

126 Jasanoff, 'Future Imperfect,' 22.

127 See J. Z. Garrod, 'The Real World of the Decentralized Autonomous Society', *tripleC: Communication, Capitalism & Critique* 14.1 (2016), 67; Gianluca Miscione and Donncha Kavanagh, 'Bitcoin and the Blockchain: A Coup D'État Through Digital Heterotopia', *SSRN* (2015): 1-27; Wessel Reijers, Fiachra O'Brolcháin, and Paul Haynes, 'Governance in Blockchain Technologies & Social Contract Theories', *Ledger* 1 (2016): 134-51. <https://doi.org/10.5195/LEDGER.2016.62>; Vidan, and Lehdonvirta, 'Mine the Gap'; Taylor C Nelms et al., 'Social Payments: Innovation, Trust, Bitcoin, and the Sharing Economy', *Theory, Culture & Society* 35.3 (2018): 13-33.; Sarah Pink, Debora Lanzani, and Heather Horst, 'Data

are organised via DAOs;¹²⁸ or how political dissent might be thought of and put into practice differently with blockchain technology.¹²⁹ All of these projects combine a vision of the social (e.g., social organisation and its discontents) and the cultural (e.g., the values that shape what trust, belonging, or dissent mean in blockchain cultures and how blockchain, in turn, influences cultural norms about what can be expected around such ideas). This book sits at exactly this intersection as well.

In the course of subsequent chapters, I will continue to add nuance to the concept of socio-techno-cultural imaginaries in order to be able to respond to the needs of particular case studies. In chapters 2 and 3, I show that, although imaginaries are always situated in particular temporal and cultural contexts, this situatedness is often hidden in favour of a universal appearance. In chapters 4 and 5, I focus on overtly situated imaginaries, and their complicated entanglement with rationalism. In chapter 4, I deal with the incalculability that permeates the cryptocurrency market and use Castoriadis' theory on the imagination¹³⁰ to illuminate how unexpected worlds may emerge from a given status quo. This insight is then rendered through the activist practices that feature in chapter 5 to reveal how imaginaries can evolve from an analytical concept into a deliberate tactic and, in doing so, performatively institute radically different sociocultural worlds.

The Significance of Affordances and Performativity in Imaginaries

At the core of the imaginary of blockchain as a truth machine lies a claim that blockchain has a particularly strong capacity to produce truths—i.e., that it affords establishing the truth. Unpacking such a claim, which seems to be taken at face value in many parts of blockchain culture, and addressing how truth affordances are performed in blockchain culture, requires a multi-faceted approach leveraging concepts that bridge technology and the socio-cultural meaning given to it. The concept of affordances brings into view fields of possibilities for the use of technologies that may be based in material and imagined characteristics of those technologies. The concept of performativity explains how language, concepts, and ideas have the capacity to do things, to make a difference in the world, and to bring about certain realities. Significantly, performativity highlights the way in which power relations are normatively reproduced through imaginaries.

The imaginary of blockchain as a truth machine brings about a reality in which blockchain's logics structure what can be understood as the 'truth' and, as such, requires critical analysis. In the context of complex technologies such as blockchain, beliefs, expectations, and imaginations that are performed discursively around them are particularly important and shape the ongoing development of such technologies. At the same time, the concept of imaginary

Anxieties: Finding Trust in Everyday Digital Mess', *Big Data & Society* 5.1 (2018): 1-14.

128 Martin Zeilinger, *Structures of Belonging*, Ljubljana: Aksioma – Institute for Contemporary Art, 2023. This is a topic to which I return in chapter 5.

129 Barbara Cueto Gutierrez, *White Papers on Dissent: Politics and Poetics of Blockchain*, PhD diss., University of Amsterdam, Amsterdam, 2022.

130 Castoriadis, *The Imaginary Institution of Society*.

affordances also opens the door to a countercultural engagement with imaginaries in which different relations are performed. This is an aspect that I will briefly discuss below, before expanding upon it in more detail in chapter 5.

How Affordances of Complex Technologies are Imagined and Performed

The concept of affordances was first developed by psychologist James Gibson to describe the way in which environments shape one's potential for different actions: 'The affordances of the environment are what it offers the animal, what it provides or furnishes, either for good or ill'.¹³¹ An affordance is thus a characteristic of an object (or objects) in an environment, but it also depends on the actions of an agent to be actualised: 'It is equally a fact of the environment and a fact of behaviour. It is both physical and psychical, yet, neither. An affordance points both ways, to the environment and to the observer'.¹³² Therefore, different people with different bodies, capacities, needs, or cultural systems of meaning may realise different affordances in relation to the same environment. Understood in this way, affordances are relational and context-dependent. They are 'the primary way in which [agents] become enmeshed with their surroundings'.¹³³ Although the concept of affordances soon began to travel to other disciplinary contexts—gaining new meanings and applications along the way before it was applied to digital technologies—Gibson's early conceptualisation already makes clear that affordances occupy the space between structure and agency, between the individual and the collective, and between materiality and imagination, all of which are so relevant to understanding socio-techno-cultural imaginaries.

The concept of affordances travelled through design research before it reached digital media studies. Design researcher Don Norman's¹³⁴ reworking of the concept has been particularly influential, although it disregards key elements of Gibson's conceptualisation. Instead of Gibson's relational and context-dependent understanding, Norman thinks of affordances as a design tool, arguing that objects should be consciously designed so that their intended affordances are perceptible and stable, universally understood and useable in the same way by a diverse user-base. This shift impacts the way in which affordances intersect with power: rather than the shifting power relations that agents and environments call into being in particular moments and constellations, 'Norman's account hinges on the designer's instrumentalization of the affordance concept and, therefore, on ideologically-laden interventions that tend to close down, rather than broaden, the interactional horizon of an object or environment'.¹³⁵ Although Gibson's understanding is stronger analytically, Norman's application of affordances to designed objects and technologies popularised the term. It is through this history that the concept travelled to media studies, where it particularly gained traction in the study of digital technologies.¹³⁶

131 James Gibson, *The Ecological Approach to Visual Perception*, Boston: Houghton Mifflin Harcourt, 1979, 127, emphasis in original.

132 Gibson, *The Ecological Approach to Visual Perception*, 129.

133 Ashley Scarlett and Martin Zeilinger, 'Rethinking Affordance', *Media Theory* 3.1 (2019), 5.

134 Donald Norman, *The Psychology of Everyday Things*, New York: Basic Books, 1988.

135 Scarlett and Zeilinger, 'Rethinking Affordance', 12.

136 Scarlett and Zeilinger, 'Rethinking Affordance'.

The translation of the concept of affordances to the analysis of digital technologies necessitates significant shifts. First, digital technologies are complex and opaque in ways that complicate the relationship between environment and agent. Gibson understood this as the relational space in which affordances are actualised.¹³⁷ As I argued above about blockchain technology specifically, algorithmic digital technologies more generally are complex and continuously shifting assemblages comprised of many different actors, including those with commercial or political interests.¹³⁸ The affordances of such technologies are rendered through ‘complex constellations of intersecting feedback loops, driven in large part by the solicitation, aggregation, operationalization, and calculated implementation of data’.¹³⁹ An individual interacting with a digital technology may activate obscured layers of algorithmic affordances within this complex assemblage without being aware of it. Furthermore, with each interaction, such a technology’s affordances change along with the evolving relational constellations that are addressed. This means that the affordances of digital technologies are often not readily perceivable.

However, the imperceptibility of certain affordances does not stop users from interacting with them—i.e., attempting to realise the potential of an affordance they do perceive.¹⁴⁰ Many designers and users interact with complex technologies in ways that involve beliefs, expectations, and imagination about their affordances rather than basing their interactions on a rational understanding of technological processes.¹⁴¹ In the context of users that attempt to make sense of social media algorithms, researchers have approached the phenomenon of imperceptible affordances with concepts such as folk theories¹⁴² and algorithmic imaginaries¹⁴³ that account for the guess-work, belief, and ritualisation that occur in interactions with such complex technological assemblages. Furthermore, the complexity of digital technological assemblages not only make it difficult for users to perceive certain affordances, but also for their developers to design for particular interactions, problematising Norman’s¹⁴⁴ ideal of intentionally designed affordances.

As such, complex digital technologies may have imagined affordances that are performatively enacted, particularly in the context of the feedback loops of algorithmic processes.¹⁴⁵

137 Scarlett and Zeilinger, ‘Rethinking Affordance’.

138 Nancy Ettlinger, ‘Algorithmic Affordances for Productive Resistance’, *Big Data & Society* 5.1 (2018): 1-13.

139 Scarlett and Zeilinger, ‘Rethinking Affordance,’ 13.

140 Ettlinger, ‘Algorithmic Affordances for Productive Resistance’; Scarlett and Zeilinger, ‘Rethinking Affordance’; Taina Bucher and Anne Helmond, ‘The Affordances of Social Media Platforms’, in Jean Burgess, Thomas Poell, and Alice E. Marwick (eds) *The Sage Handbook of Social Media*, Thousand Oaks: Sage Publications Ltd., 2017, 233-53.

141 Peter Nagy and Gina Neff, ‘Imagined Affordance: Reconstructing a Keyword for Communication Theory’, *Social Media + Society* 1.2 (2015): 1-9.

142 Motahhare Eslami et al., ‘First I ‘Like’ it, Then I Hide it: Folk Theories of Social Feeds’, *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, San Jose, USA, 7-12 May 2016. <https://doi.org/10.1145/2858036.2858494>.

143 Bucher, ‘The Algorithmic Imaginary’.

144 Norman, *The Psychology of Everyday Things*.

145 Bucher and Helmond, ‘The Affordances of Social Media Platforms’.

Imagined affordances may range from mundane ideas such as the ability to influence the advertisements with which one is presented on social media by talking about related topics near one's phone, to the literally otherworldly space-colonisation affordances that are imagined around Elon Musk's SpaceX rockets. As users act upon imagined affordances, they are involved in the technology's continuous shaping and reshaping.¹⁴⁶

The concept of performativity, on the other hand, refers to the capacity of language to have an effect in the world and highlights how affordances derive power, not just from their relation to materiality, but also, significantly, from their relation to discourse and the enactment of sociocultural ideas about a technology's potential in interactions with it. The concept of performativity makes sense of the sociocultural embeddedness and the power of words. Initially, as a concept in linguistics, performativity was linked to specific kinds of words—verbs such as promising, ordering, or declaring—that perform an act in the utterance itself, rather than describing, for example, a situation in the world. Unlike other forms of speech, the meaning of a such a performative utterance is detached from the intention of the speaker, its effectiveness instead dependent on social conventions and rules that differ between culturally specific discourses.¹⁴⁷

Language gets its performative capacity as part of just such a social system of conventions through repetition and iteration: discourse does not exist in isolation, but emerges out of the status quo. Individuals cannot express themselves without referencing, reiterating, and reflecting on expressions made previously—a phenomenon Deleuze and Guattari refer to as the 'indirectness' of discourse¹⁴⁸—and so, enunciation is not individual, but always already social.

Through this self-referentiality, expressions carry forward 'implicit presuppositions'¹⁴⁹—e.g., assumptions about norms and ideals—that make language function as a system that structures and orders the world. Each individual speech act exists as part of, is the result of, and performs a 'collective assemblage of enunciation',¹⁵⁰ thus reproducing the social world in particular dimensions based on particular shared assumptions. As a result, discourse defines and establishes boundaries, for example of acceptability, normalcy, or even what is considered to be natural. Such normative discourse is heavily shaped and regulated through powerful actors that define what is considered acceptable or alternatively, what has to be fixed, punished, or prevented. Norms and their performative expression thus co-constitute each other, but this process nevertheless also allows for shifts, evolutions, and the creation of new norms.¹⁵¹

146 Nagy and Neff, 'Imagined Affordance,' 5.

147 John Langshaw Austin, *How to Do Things With Words*, Oxford: Oxford University Press, 1962; Stanley Fish, *Is There a Text in This Class? The Authority of Interpretive Communities*, Cambridge, MA: Harvard University Press, 1980; James Loxley, *Performativity*, Oxon: Routledge, 2007.

148 Gilles Deleuze and Félix Guattari, *A Thousand Plateaus*, London: Bloomsbury Academic, 2021 [1980].

149 Deleuze and Guattari, *A Thousand Plateaus*, 91.

150 Deleuze and Guattari, *A Thousand Plateaus*, 91.

151 Judith Butler, *Bodies That Matter: On the Discursive Limits of Sex*, London: Routledge, 1993; Jon McKenzie, *Perform or Else: From Discipline to Performance*, London: Routledge, 2002. I will expand on the notion of performativity in relation to memetic online cultural objects in chapter 4.

Affordances and performativity are key points of access to understanding how imaginaries emerge, develop, and spread. When affordances become actualised because people engage with a given technology, they contribute to the reproduction of the power relations that led to their development. Similarly, when affordances are imagined and performed through discourse, they reproduce shared ideals about what social relations should be like.¹⁵² Below, I detail how I will apply this framework to blockchain culture and the widespread idea of the technology's affordances of truth.

Studying the Imaginary of Blockchain as a Truth Machine

Socio-techno-cultural imaginaries have an effect based on the performative relationship between technology and ideas about technology. Jasanoff¹⁵³ studies sociotechnical imaginaries through normative and well-established discourses, such as 'languages of power, especially the official discourses of the state' expressed by 'actors with authority' (25). Indeed, discourses are an important site of analysis in order to gain an understanding both of what affordances blockchain is understood to have, and how they intersect with power. However, Jasanoff's approach does not reflect the fact that any discourse carries with it performative power.¹⁵⁴ As a complex technology, blockchain is particularly idea-driven, making material-discursive meaning-making practices, including in everyday and subcultural ones, key sites from which to study socio-techno-cultural imaginaries that emerge around it.

Imaginaries are negotiated between various actors with varying degrees of agency and take shape in a continuous process of iterative expressions about what blockchain technology could be.¹⁵⁵ The different communities involved in imagining and working towards blockchain futures collectively go through moments of development, negotiation, and crisis, during which its imaginaries can multiply and become more precise.¹⁵⁶ For example, a hard fork is a very explicit version of evolving and multiplying blockchain imaginaries that can be understood through the performativity of imagined affordances: a hard fork is a moment in which differences of opinion about future implications of the technology are deemed unsolvable and the community and the technology diverge into different directions to pursue different affordances. These are points at which cracks within imaginaries appear and different imaginaries can begin to emerge.

152 Jasanoff, 'Future Imperfect'.

153 Jasanoff, 'Future Imperfect'.

154 Jacques Derrida, *Limited Inc*, Evanston: Northwestern University Press, 1988.

155 Yong Ming Kow and Caitlin Lustig, 'Imaginaries and Crystallization Processes in Bitcoin Infrastructuring', *Computer Supported Cooperative Work* 27.2 (2018): 209-32. <https://doi.org/10.1007/s10606-017-9300-2>; Laura J Neumann and Susan Leigh Star, 'Making Infrastructure: The Dream of a Common Language', *Proceedings of the 4th Biennial Participatory Design Conference*, Cambridge, MA, 13-16 November, 1996. <http://ojs.ruc.dk/index.php/pdc/article/view/153>.

156 Kow and Lustig, 'Imaginaries and Crystallization Processes in Bitcoin Infrastructuring'; Susan Leigh Star and James R. Griesemer, 'Institutional Ecology, 'translations' and Boundary Objects: Amateurs and Professionals in Berkeley's Museum of Vertebrate Zoology, 1907-39', *Social Studies of Science* 19.3 (1989): 387-420.

Affordance analysis, critical discourse analysis, and performativity theory combine into a methodological approach that allows me to ‘bring the material and the imaginary onto one analytical plane’¹⁵⁷ in order to address how they continuously influence each other as part of the imaginary of blockchain as a truth machine. As do imaginaries, affordance theory offers an approach to technology that ‘cuts through the binary of structure and agency’,¹⁵⁸ bridging the technical and sociocultural forces that co-construct a technology’s potential. Such a cut does justice both to the ways in which technologies and infrastructures exert power on sociocultural life, and the ways in which individuals and groups are able to tactically repurpose technologies in specific situations. It critically addresses the power dynamics and knowledge politics that are performed within and by imaginaries. Critical discourse analysis ‘is not an analysis of discourse in itself,’ instead approaching language and ideas as embedded in social, cultural, and material conditions that are not neutral.¹⁵⁹ By applying performativity theory, I centre the capacity of discourses and affordances to shape the future and address the normativity that is involved in this performative process.

My analyses of blockchain culture in subsequent chapters activate the concepts of affordances and performativity in different ways. In chapter 2, I start with a practical case study in which belief in blockchain’s truth affordances stands at the base of various new organisations. By analysing their systems and documentation, I show what truth affordances consist of technically and discursively, and how this reproduces particular power relations. In chapter 3, I move my attention to the cultural narratives, systems of belief, and ritualised practices that give meaning to the idea of truth affordances in three influential blockchain subcultures. The idea that blockchain affords the establishment of truth normalises the technology as a force for good without regard for its entanglement with power. In chapter 4, I show what happens when real experiences with blockchain in the cryptocurrency market cause friction with (or outright negate) the idea that the technology affords truth. Here, performativity—rather than truth—takes centre stage as the logic that decides what is important. In chapter 5, an active engagement with the imagined affordances of complex technologies is presented as a way in which what these technologies are and what they could become in the future can be reimagined in an activist and artistic manner.

Chapter Overview

In the following chapters, I analyse four case studies that each represent a different position in relation to three key characteristics. First, the case studies cover diverse ideas of truth that represent differing positions between rationalism and the more-than rational. Second, the communities and individual actors that play key roles in the establishment of a chapter’s

157 Aris Komporozos-Athanasiou, *Speculative Communities: Living With Uncertainty in a Financialized World*, Chicago: The University of Chicago Press, 2022, 28. Although Komporozos-Athanasiou does not address these concepts specifically here, his reflections on the way in which communities engage with technologies in ways that are more-than-rational resonate with the key points about affordance theory that I described above. I will return to Komporozos-Athanasiou’s theory in more depth in chapter 4.

158 Jasanoff, ‘Future Imperfect,’ 24.

159 Norman Fairclough, *Critical Discourse Analysis: The Critical Study of Language*, London: Routledge, 2013, 4.

imaginary each have different forms and degrees of access to power. Third, the futures that are envisioned in these imaginaries assume various standpoints between the reinforcement or expansion of elements of the status quo and a radical reimagination of contemporary ways of life. Below, I will explain the case study upon which each chapter focusses, how this case study relates to the above characteristics, and highlight the chapter's role in the wider book.

In chapter 2, I focus on concrete projects that implement blockchain because they understand it as capable of producing truths. Specifically, I analyse how the idea that blockchain affords truth is materially-discursively performed in blockchain forest protection projects. These projects render trees and forests through blockchain technology—i.e., they 'tokenise' them—in order to protect them against deforestation and present this process as a tool that can be used in the fight against climate change. The materials I analyse consist of documents produced by these projects—whitepapers, websites, social media posts, and FAQs—and contextual public documents, such as journalistic articles and judicial documents. Not only is the notion of truth that features in this case study heavily influenced by ideals of rationality derived from Western modernity, but it also reproduces the power imbalances that characterise this history. On the one hand, the tokenisation of nature affords extractivism by appropriating land and trees into Western systems of ownership and ways of knowing. On the other hand, it affords the establishment of universal and immutable categories regarding what is and what is not understood to be nature, and how those categories are inherently financialised while creating a more-than-rational myth of tokenised nature as nature itself. The blockchain imaginary of a thriving tokenised natural world that these projects perform is based on a colonial understanding of truth that is universalising, immutable, and financial, while obscuring other (ways of knowing) nature in the process.

In chapter 3, I contextualise these concrete projects by taking a wider perspective on blockchain culture and how it makes sense of the notion of truth, both socially and culturally. I analyse materials produced at three significant moments in blockchain's history: the inception of blockchain's central algorithm; the moment when blockchain becomes more than just a ledger for cryptocurrencies; and the response to this broadening of the technology by a subculture that remains committed to Bitcoin above all else. Religious narratives and technological mysticism are important factors that shape the kinds of truth that blockchain is thought to afford. Religious narratives imbue blockchain-based truths with a sense of predestination, universality, and timelessness. Like technological mysticism, it presents blockchain as if lying outside of humanity's capacity to understand or influence it. In narratives of blockchain's truth affordances, rationalism and religious and mystical forms of belief reinforce each other because they share a preoccupation with perfection and both function through obfuscation. Rather than showing the opposite of the previous chapter, the blockchain imaginary of religious and mystical truthfulness reveals how rationalism and the more-than-rational are inherently intertwined.

This point signals a shift in the book: I move from imaginaries that are presented as if they are universally relevant and applicable to imaginaries that overtly emerge from the lived experiences of specific communities. In chapter 4, I focus on the cryptocurrency market and the imaginaries of blockchain's affordance for truth that are established by crypto traders

in relation to its profound volatility. First, I analyse several interviews with a prominent crypto astrologer who analyses the movements of the planets to predict which cryptocurrency trades will be beneficial to make at a given moment. In the context of the unpredictability of the market, the charts and mathematics of astrology offer an unexpected, pseudo-rational road back to rational calculability through the 'predestined' movements of the planets. The second case study in chapter 4 takes cryptocurrency memes as its starting point. These memes perform an imaginary that revels in truthlessness and speculation as the only way to survive in the post-truth, financialised society. What is important here is not the lasting truthfulness of blockchain-based data, but rather the operability of information on a blockchain amidst a widespread lack of believable, stable, shared truth. Instead of a truth machine, blockchain is imagined as an uncertainty machine that produces volatile waves to surf by speculating. The blockchain imaginary of uncertainty highlights that the cryptocurrency market lives in a continuous 'now' in which radical rethinking is practically impossible.

In chapter 5, I continue the situated perspective set out in the previous chapter, but introduce another shift as well: I move from using the concept of imaginaries as a purely analytical framework to understanding it as a tool that may be used in activist practices. Here, I zoom in on artistic-activist engagement with DAOs. I start from the imaginative work of an edited book titled *Radical Friends: Decentralised Autonomous Organisations and the Arts* that frames blockchain in terms of social belonging rather than financial exchange. DAO practices that are discussed in the book and that emerge in the community around it can be understood as prefigurative practices that live out their visions for future social organisation in the here and now, and consist of what I propose are radically reimagined affordances. Furthermore, I show that, when engaging more directly with the notion of truth, artistic-activist DAOs perform work that opens up the imaginary of blockchain as a truth machine to radically different ways of understanding knowledge and the practices that constitute knowing. The blockchain imaginary performed in these countercultural communities is one of mutable and situated truths, and the possibility to relate differently to others by knowing together with them in new ways.

In the final chapter, I formulate conclusions on three different levels. First, I reflect on the contribution of my approach to imaginaries and formulate the conceptual insights following from my analyses. I highlight the importance of emphasising cultural situatedness in order to gain a better understanding of the performativity of imaginaries. I also show that understanding imaginaries as an artistic-activist tool opens up new avenues for engagement with emerging technologies to support the reimagining of what life itself could mean and constitute. Second, I draw the findings of the previous chapters together to present my conclusions with regard to the rational and the more-than-rational elements at play in the imaginary of blockchain as a truth machine. Despite their seeming oppositions, the rational and the more-than-rational in imaginaries, apparently from nowhere, reinforce fundamentally the same universalist, extractivist, and controlling vision of truth. When imaginaries are more clearly situated, rationalism starts to be problematised—e.g., in the context of the profound uncertainty of many aspects of contemporary life. In the end, the notion of a 'truth machine' might start to take on radical, emancipatory, and decolonial meanings through the conscious use of the notion of imaginaries. Finally, I take a step back to discuss how the findings of this book can be gen-

eralised to encompass wider technology culture and serve to find ways to probe emerging technologies in the future in radically imaginative ways.

TRUE TREES AND EFFECTIVE FORESTS: THE MYTH OF TOKENISED NATURE AS NATURE ITSELF

Recent years have seen a proliferation of blockchain-based projects that aim to contribute to the fight against climate change. Many projects do this by tokenising nature and bringing it into the economy in the form of datafied yet living organisms, rather than selling it in the form of physical, harvested natural resources. EcoMatcher, for example, specialises in the planting and tokenisation of trees and boast about the unprecedented level of detail of their interface on nature. On their website, they state: ‘Through our blockchain-based platform, you will know everything about every tree planted’,¹ arguing that this new depth of knowledge will support positive outcomes for the protection of trees and forests.

EcoMatcher and projects like it present the tokenisation of nature as a way to provide access to knowledge that supports sustainable decision-making. Such tokenisation is inspired by a rationalist belief in the possibility and virtue of establishing universal truths about nature. The meticulous and exact knowledge about trees that blockchain forest protection projects like EcoMatcher imagine, is made possible by tokenising—i.e. logging—them on a blockchain and continuously updating the ledger with new status reports on, for example, the health and growth of individual trees. Here, tokenisation is presented as a new, fundamentally better way to know nature. Being structured by the logics of blockchain, this system of knowledge is intimately connected to ideas of value and processes of financialisation. As with any other system of knowledge, the epistemology made possible by EcoMatcher and similar initiatives is far from neutral. Indeed, in the same manner as rationalism has done historically, such processes of tokenisation create a myth of universality to hide their partisanship.

My aim in this chapter is to show the power dynamics implicated in the rationalist myth of tokenised nature as nature itself that is afforded by these processes of tokenisation. The blockchain projects I analyse here present an imaginary of blockchain as a truth machine that works through this myth. Their imaginary establishes blockchain as *the* solution to societal problems while impeding the deeper understanding of the biases involved in their operations. Rationalist claims of the virtue of tokenisation are characterised by belief in the superiority of their blockchain-based epistemology over other ways of knowing and echo key aspects of the way rationalism has been implicated historically in the eradication of other ways of knowing.²

In my analysis, I bring decolonial theory to bear on blockchain technology and culture to demonstrate how their reliance on rationalism affords the reproduction of coloniality—i.e., the persistence of colonial relations and logics after the period of historical European colonisation

1 EcoMatcher, <https://www.ecomatcher.com/>.

2 Boaventura de Sousa Santos, *Epistemologies of the South: Justice Against Epistemicide*, London: Routledge, 2014.

ended. From a decolonial perspective, rationalism has to be understood as inherently entangled with the power relations, social organisation, and knowledge politics of the colonialist histories that enabled Europe's political, economic, and scientific dominance and continue to shape global relations.³ I argue that tokenisation plays a key role in affording such coloniality because of its supposedly pure rationalism. Where blockchain forest protection projects actually 'touch the ground' rather than stay speculative project pitches—e.g., when they take ownership of land, implement tokenisation procedures, or sell tokens connected to nature—I reveal the lived colonial implications they embody.

In this chapter, I make three claims to support this argument. First, I claim that the rationalist myth of a blockchain-based revolution founded on its supposed capacity to create new and better kinds of truth hides how tokenisation affords the centralisation of power along familiar colonial imbalances. The tokenisation of nature is used to create a myth of redistributed ownership and to imply the power of governance to token-holders. However, power is instead concentrated within the projects themselves. Tokenisation in effect functions to dispossess those deemed not 'developed' enough to care for forests and trees, of the ownership, access, and control they previously held. Here, a rhetoric of modernity accompanies tokenisation, which is meant to establish a 'civilised' and rationalist upgrade to life among the trees. Blockchain forest protection projects impose their systems upon communities living in rainforests with the goal of emancipating them from their (supposed) economic dependence on logging. While it may seem that the abstractions of tokenisation are without sacrifice—they keep the trees' roots in the ground rather than selling them as logs—in reality, the projects reproduce colonial power dynamics that historically lurk beneath the charitable surface of Western environmentalism.⁴

The second claim I make in this chapter is that the rationalist logics of tokenisation afford the reproduction of the coloniality of knowledge⁵ and attempt to position tokenisation as superior over other ways of knowing. Tokenisation continues the logics of the colonial epistemologies that datafication made possible,⁶ but enhances them in medium-specific ways. I lay my foundation for this claim in two steps: First, I show that the verification and validation processes of blockchains connect truth claims to financialised logics. This creates a feedback loop that empowers tokenised nature over other kinds of nature. By creating this verified and validated nature, these blockchain forest protection projects also create less-than-true and less-than-valid nature. As such, these projects establish token-based epistemologies that reproduce colonial categories and relations that support dominant Western systems of power. Second, I show that tokenisation ideally functions as a totalising system and how the

3 Walter Mignolo, 'Delinking: The Rhetoric of Modernity, the Logic of Coloniality and the Grammar of De-Coloniality', *Cultural studies* 21.2–3 (2007): 449–514; Walter Mignolo, *The Darker Side of Western Modernity: Global Future, Decolonial Options*, Durham: Duke University Press, 2011.

4 Malcom Ferdinand, *Decolonial Ecology: Thinking From the Caribbean World*, Cambridge: Polity Press, 2022.

5 Mignolo, 'Delinking'; Walter Mignolo, 'The Geopolitics of knowledge and the Colonial Difference', *The South Atlantic Quarterly* 101.1 (2002): 57–96.

6 Paola Ricaurte, 'Data Epistemologies, the Coloniality of Power, and Resistance', *Television & New Media* 20.4 (2019): 350–65.

tokenisation of nature aims for the naturalisation of tokenised nature as nature itself. Although isolated, tokenised bits of nature might thrive if these projects fulfil their promises, on the flipside, nature as a wild and messy ecology may equally wither in such a totalising system of financialised knowledge.

Finally, I bring the findings of this chapter together to claim that the tokenisation of nature performs a colonial onto-epistemology supported by the myth of rationalism's neutrality. Not only are 'knowing and being [...] intertwined'⁷ through financialised blockchains, but via their totalising affordances and their colonial power dynamics and knowledge politics, tokenised nature also proposes a theory of nature that produces the conditions for that nature to exist while other understandings of nature are deprioritised. The myth of pure, tokenised rationality hides the involvement of tokenisation in the establishment of new forms and clusters of power behind a layer of financialised abstraction that onto-epistemologically reinforces itself. The myth of tokenised rationality suggests financialised truths are innocent and neutral, thus hiding the affordance of coloniality inherent within such tokenisation.

The Case for the Tokenisation of Nature

Superficially, the imaginary of blockchain as a truth machine of nature envisions technology as being able to produce sustainability. This environmentalist imaginary relies heavily on rationalist ideals of verifiable and universal truths. Particularly, projects that tokenise nature operationalise these ideals through processes of atomisation and disintermediation. These processes are meant to make nature accessible to be acted upon more directly and with more precision in service of countering climate change. While there are many different ways in which blockchain projects address climate-related issues, I focus specifically on the tokenisation of physical and chemical elements—carbon sequestered in trees, the individual trees themselves, or entire forests—in order to address the imaginary of blockchain as a truth machine as it is applied to nature. By representing these ecologically entangled phenomena as isolated units on a blockchain, these projects aim to make the truth about nature accessible, incontrovertible, and actionable for sustainable decision-making.

Three Kinds of Tokenised Granularity

Tokenisation is the mechanism through which blockchain is imagined to render tangible truths about the world—in this case, in the context of sustainable activity. Tokenisation is a central feature of blockchain technology referring to the process of creating on-chain data. Generally speaking, tokens may be fungible or not, meaning that they function similar to interchangeable coins or unique ID tags respectively.⁸ Any kind of digital data can be tokenised and stored

7 Chiara Carboni et al., 'Eye for an AI: More-Than-seeing, Fauxtimation, and the Enactment of Uncertain Data in Digital Pathology', *Social Studies of Science* 53.5 (2023), 714; Karen Barad, *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning*, Durham: Duke University Press, 2007.

8 Although there are more detailed terms for the different functionalities tokens have—such as utility tokens, security tokens, or governance tokens—the projects discussed here generally do not use these terms. The tokenisation that occurs in most projects can be understood as consisting of a combination

immutably on a blockchain ledger. In the blockchain forest protection projects I analyse here, the following kinds of data are present: 1) a measure of the carbon sequestered by a forest or tree; 2) a living plant to be tracked, preferably in (close to) real-time and with much detail; or 3) a plot of land (usually forested) to be owned, protected, and/or governed. Projects may focus on one or more of these kinds of tokenisation. Below, I outline key characteristics of each kind of tokenisation of nature and elaborate on the environmentalist truth imaginary they establish.

The blockchain forest protection projects Single.Earth, Carbon Collectible NFT (ccNFT), MCO2 Token, and Corcocooin are examples of the first kind of tokenisation, namely, sequestered carbon. On July 7, 2021, Single.Earth's founder Merit Valdsalu announced that the project raised 7.9 million USD 'to tokenize nature'.⁹ To do this, Single.Earth creates 'a digital twin of Earth: We combine satellite data, big data analysis, and machine learning to create a digital representation of how the world's nature [...] sequesters carbon and holds biodiversity'.¹⁰ Based on this digital twin, and all the data it is built on, the system then mints a 'MERIT token for every 100kg of CO2 captured in biodiverse nature in real time',¹¹ in effect making carbon sequestration and biodiversity tradable on a blockchain.

Essentially, this kind of project participates in the wider carbon credit phenomenon, particularly in the voluntary carbon market. While the compliance carbon market is regulated and governmental bodies such as the European Union mandate that certain polluters buy these carbon credits in an effort to meet carbon reduction targets, the voluntary market operates differently.¹² The most important difference is that they are not regulated and as a result present issues with regard to the additionality of the credits, the claims of sustainability made based on their purchase, and the fragmented market possibly leading to unfair pricing, double sales, and conflicts of interest.¹³

Nevertheless, this lack of regulation is turned into an asset by a project such as ccNFT, which claims that profits can be harvested as soon as five months after registering a piece of land

of utility tokens (where nature is tokenised) and governance tokens (where ownership of tokenised nature is connected to voting rights in the management of forests).

9 Merit Valdsalu, 'Single.earth Raises \$7.9m Led By Eqt Ventures to Tokenize Nature', *Single.Earth*, 2021, <https://www.single.earth/blog/joining-eqt-ventures>.

10 Valdsalu, 'Single.earth Raises \$7.9m Led By Eqt Ventures to Tokenize Nature'.

11 Valdsalu, 'Single.earth Raises \$7.9m Led By Eqt Ventures to Tokenize Nature'.

12 The regulated compliance market system is criticised as a vehicle for so-called 'greenwashing' because it does not require polluters to alter their polluting production processes while allowing them to claim that their products are sustainable, for example, by suggesting their production is carbon neutral. The voluntary market is generally even more open to such criticism because of the lack of regulation, in particular with regard to 'additionality,' meaning that voluntary carbon credits may be sold for forests that are under no threat to be cut down and thus the sale of carbon credits does not represent additional carbon sequestration. This criticism also holds true for certain blockchain carbon credit projects. Single.Earth, for example, writes that '[I]andowners are not required to take additional action to improve the provision of the services (e.g., improve CO2 sequestration) beyond business as usual.' See Single.Earth, 'Lightpaper: An Introduction to Single Earth', *Single.Earth*, 2022, https://assets.single.earth/documents/Single.Earth_Lightpaper_August_2022.pdf, 5.

13 Gerko Wessel and Remco De Boer, *Voluntary Carbon Markets: Supervisory Issues*, Amsterdam: Autoriteit Financiële Markten, 2023.

in their system, while certifying it in the compliance market may take up to five years. ccNFT estimates the compliance market certification process, which includes site visits by experts, can cost landowners up to 725,000 USD. This means it excludes small- and low-income communities—a problem ccNFT claims to provide a solution for. CcNFT replaces compliance certification process with an automated combination of satellite data, artificial intelligence, and machine learning, which ‘reduces the cost of the NFT impact certificates toward \$0’.¹⁴ The immutability and transparency of the blockchain ledger is then thought to increase the trustworthiness of the credits by making accessible the quantities of removed carbon that are assigned to each NFT and giving their owners access to view the plot of land linked to their NFT via satellite imagery.¹⁵



Fig. 1. Screenshot of ccNFT's carbon map. ccNFT sells NFTs that represent 1 hectare of forest. Using a combination of metrics, different rates of carbon sequestration are then attached to each NFT.

The second kind of tokenisation is present in a project called EcoMatcher, which is essentially a tool for buying trees to be planted in forests around the world. Each tree is represented by a unique token, which includes all sorts of information about its species, its geolocation, the current weather conditions where it is located, the date it was planted, who it was planted by (including a picture of the planter and the tree itself), its height, and how much CO₂ it is estimated to have already sequestered (see Figure 2). A further example is Open Forest Protocol, which aims to establish a universal protocol for forest measurement, reporting, and verifying via blockchain technology. It logs updates about the growth of individual trees in forest protection projects and makes this database accessible online (see Figure 3). Veritree—as the name suggests—verifies the planting of trees by uploading metadata and proof of trees planted onto their blockchain in a way similar to EcoMatcher.

14 Carbon Collectible NFTs, 'Our Theory of Change for Carbon Collectible NFTs', *Carbon Collectible NFT*, 2022, <https://carboncollectiblenfts.com/Downloadables/toc.pdf>, 4.

15 Carbon Collectible NFTs, 'Our Theory of Change for Carbon Collectible NFTs'.



Fig. 2. Screenshot of a sample tree in the EcoMatcher TreeTracker tool.

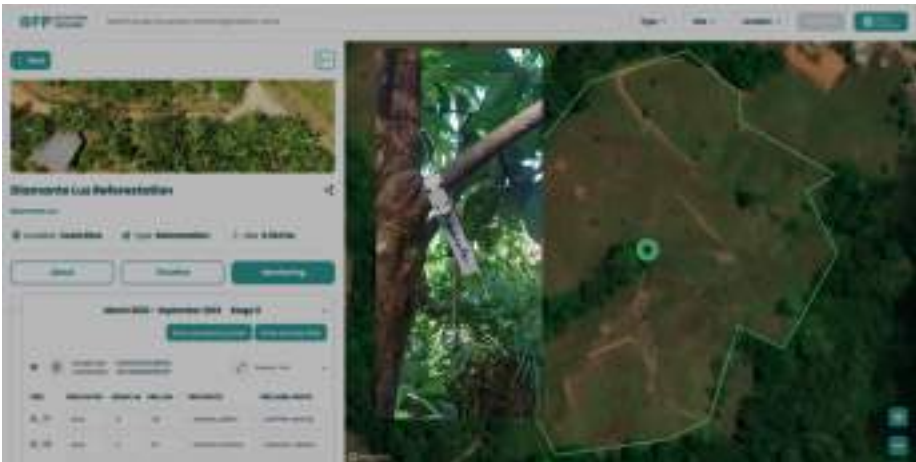


Fig. 3. Screenshot of Open Forest Protocol's dashboard.

The third form of tokenised nature occurs at the level of plots of land. This usually involves the traditional—i.e., non-blockchain—purchase of a piece of land and its subsequent tokenisation into smaller plots linked to individual NFTs. For example, Nemus tokenised a plot of land in the Amazon rainforest and added rarity features, including noteworthy flora and fauna that exist there, to each NFT. NFT-holders are called ‘Guardians,’ and they can navigate to their plot via a game-like map of the region. They can then add their own avatar to the NFT as if they are planting a flag in their acquired plot (see Figure 4). Moss’ Amazon NFT—a sister organisation of MCO2 Token mentioned above—operates in a comparable, albeit less granular manner. Instead of the different sizes that Nemus offers, the Amazon NFTs each represent one

hectare of rainforest. The organisation promises that ownership of an NFT represents *actual* legal ownership of the plot of land.¹⁶ The tokenised plots of land are situated in such a way that, together, they build a ‘green wall’ that protects the deeper forest from deforestation.¹⁷ Both Amazon NFT and Nemus envision a collective form of governance via a Decentralised Autonomous Organisation (DAO) that would be populated (primarily) by the NFT-holders. The DAO should then make decisions about the use and protection of the forest as well as how the new owners of the plots of land should engage with local inhabitants.



Fig. 4. Screenshot of Nemus' Guardian app (accessed 19 October 2023).

The Imaginary of Blockchain as a Truth Machine of Nature

Taken together, these blockchain forest protection projects establish an environmentalist imaginary in which blockchain's atomising and disintermediating affordances are central. Projects that tokenise carbon say they improve the status quo—the highly centralised, slow, and abstract compliance market and the untrustworthy non-blockchain voluntary market—by disintermediating the relationship between the person buying carbon credits and the trees that do the sequestering. Credits are often tied to particular trees or plots of land that are visible from afar, providing buyers with the capacity to continuously verify the status of their purchases, while on the compliance market, credits are reduced to abstracted values, detached from specific locales and trees.

16 These claims are highly questionable, as no notary seems to be involved in the sale of these NFTs.

17 Meaning that the tokenised plots are situated between land that is accessible by road traffic, and land that is not. The idea is that by making this green wall where no roads may be built, the inaccessible land will remain forested.

Projects that tokenise individual trees establish direct links between tokens and physical reality in which precision and atomisation are key. The real-time progress of such initiatives is made traceable through databases that are accessible online. Instead of trusting centralised authorisation institutions, these projects imagine an informed customer that bases their decisions and repeated purchases on a continuous and fine-grained flow of disintermediated data. Information goes straight from the planter to the buyer, who can base their decision to stop or continue purchasing tokenised nature on the verified, objective reality at precise locations across the world.

Projects that tokenise plots of land envision blockchain as a governance tool that overcomes sluggish governmental policy, corrupt politicians and corporations, or lack of influence by people that want to 'do good' in localities across the world. The disintermediation and new forms of trust that these blockchain-based systems offer, present an environmentalism that claims to give individuals that might be as far away as half-way across the world new sorts of agency to overcome decades of political inaction and criminal deforestation. Although NFTs are not indicative of individual trees, NFT-holders can still get to know their plot of land with a level of detail and a sense of ownership. The DAO-based governance of Nemus and Amazon NFT introduce a new layer of power and governance to the plots of land.

These atomising and disintermediating affordances of tokenisation play a role on three levels: inclusion, influence, and information. First, blockchain climate projects include different kinds of actors in their systems than more traditional climate projects. Their imaginary rests on assumptions about the inefficiency or corruption of traditional climate action as enacted by governments and corporations. The argument is that by disintermediating processes of onboarding and certifying forest protection projects—made possible by the trustless data infrastructures of blockchain—more nature can be preserved. Blockchain's distributed systems are imagined to enable a new level of trust and precision previously unachievable by governments and corporations.

Second, influence and governance are atomised and disintermediated in this environmentalist imaginary. Individuals and businesses across the world, it is claimed, are afforded the agency to establish zones of protection in remote areas and are sometimes even imagined to govern on-the-ground proceedings at these locations via the distributed governance infrastructures of DAOs. This governance-from-afar is made possible by the establishment of direct ownership and isolated informational links between tokenised abstraction and physical reality.

At the third level, the atomisation and disintermediation of information blockchain forest protection projects are imagined to offer superior systems in support of nature, biodiversity, and forestation because of the isolated precision of the link between token and physical reality, and the accessible, continuous flow of granular and verified data. Without these new capacities of environmental information, and the presumed objectivity and verifiability it provides, the imagined trustlessness and distributed governance would not be possible.

The Entangled Histories of Environmentalism, Coloniality, and Digital Technology

Environmentalism has historically functioned along colonial logics,¹⁸ and it is important to discuss the environmentalist imaginary of blockchain as a truth machine of nature side by side with its potential for the affordance of coloniality if we are to understand the sociocultural implications of the future it aims to bring about. Below, a historical overview of the entanglement of nature management with technological and scientific methods reveals the ways in which tokenisation continues legacies of techno-scientific knowledge production about nature. Historically, such methods have been used in the establishment of colonial relations, warranting critical analysis of their continuing influence. Digital technologies building on this history afford the reproduction of colonial logics and dynamics of power and knowledge, and need to be understood 'within the historical processes of domination, extraction, exploitation, and oppression that are central to the modern world'.¹⁹ The overview below lays the foundation for my later claim that the rationalism of the tokenisation of nature hides its continuation of colonial environmentalism.

From Scientific Forestry to Precision Agriculture and the Tokenisation of Nature

As liberal democracies and nation states started to take shape across Europe in late modernity, nature increasingly became subject to the governance of distant centres of power. As part of this shift, nature needed to be simplified and made legible from afar. As a result, scientific methods were used to turn forests into 'administrative entities' subject to nation-wide control.²⁰ In late eighteenth-century Prussia, scientific forestry emerged to improve and stabilise the yield of timber and to make long-term planning possible. Instead of harvesting whatever a forest had to offer in a particular year, forests were remodelled to become significantly less diverse monocultures targeted to particularly useful types of wood production. Forests became legible through numbers and statistics, making optimised extraction and centralised control possible. This new method of forest management turned unpredictable and heterogeneous ancient forests into dependable modern log machines. Significantly, the scalability made possible by technological and scientific methods that turn nature into natural resources was a prerequisite for the continued growth of capitalism.²¹

Although the detrimental impact of scientific forestry on the health and resilience of forests as well as the multi-species ecologies of which they are part—including local human populations—soon became apparent,²² its influence can still be recognised today. Precision agricul-

18 Ferdinand, *Decolonial Ecology*.

19 Stefania Milan and Emiliano Treré, 'Big Data From the South(s): Beyond Data Universalism', *Television & New Media* 20.4 (2019), 326.

20 Jennifer Gabrys et al., 'Reworking the Political in Digital Forests: The Cosmopolitics of Socio-Technical Worlds', *Progress in Environmental Geography* 1.1–4 (2022), 60.

21 James C Scott, *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed*, New Haven: Yale University Press, 1998.

22 Ferdinand, *Decolonial Ecology*.

ture and smart forestry are examples of its evolution into new domains. These phenomena use digital technologies while building on the logics of scientific management in service of a 'bottom line' and centralised control.²³ The use of data to represent nature reduces its many forms to 'nonsocial landscape elements' to allow for a perspective on nature that is alienated from lived reality and optimised for growing scales and governance from afar.²⁴ Reducing nature to standardised elements provides the backbone for continued scalability and expansionism.

Contemporary environmentalism is under increasing pressure to be accountable and auditable, as well as to function to the logics of the market. As such, digital forest management technologies that promise transparency and efficiency have gained increasing popularity within environmental initiatives.²⁵ The tokenisation of nature purports to empower all these developments, presenting increased capacities for detailed and real-time monitoring (atomisation) and transparent processes and informational systems (disintermediation) in combination with inherent financialisation through the inclusion on blockchains.

Colonial Logics in the History of Environmentalism

From a Western perspective, modernity is often thought of as 'an essentially or exclusively European phenomenon'.²⁶ However, from a decolonial perspective, it must be understood as the other side of the coin of coloniality: the power relations, social organisation, and knowledge politics that powered Europe's political, economic, and scientific dominance. This 'coloniality' emerged as an integral part of modernity and thus neither can be understood without the other. To signify this entanglement, the phenomenon is referred to by decolonial scholars as modernity/coloniality.²⁷

Modernity/coloniality is characterised by scientific developments that purport to be neutral, while in reality, they facilitate partisan practices of domination and dispossession. For example, based on philosophical and scientific knowledge produced during modernity, the world was often understood to be comprised of hierarchical parts, with the West righteously in control of other parts of the world. Western Man was conceived as inherently politically, culturally, and scientifically superior to (colonial) Others and animals, and thus thought of as the rightful keeper of the natural world. Such conceptions, presented as truths, naturalised Western

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- 23 Christopher Miles, 'The Combine Will Tell the Truth: On Precision Agriculture and Algorithmic Rationality', *Big Data & Society* 6.1 (2019): 1-12; Jennifer Gabrys, 'Smart Forests and Data Practices: From the Internet of Trees to Planetary Governance', *Big Data & Society* 7.1 (2020): 1-10; Gabrys, 'Reworking the Political in Digital Forests'.
 - 24 Anna Lowenhaupt Tsing, 'On Nonscalability: The Living World is Not Amenable to Precision-Nested Scales', *Common knowledge* 18.3 (2012), 508.
 - 25 Kate Lewis Hood and Jennifer Gabrys, 'Keeping Time With Digital Technologies: From Real-Time Environments to Forest Futurisms', *Environment and Planning D: Society and Space* 42.2 (2024): 254-74; see also Mirca Madianou, 'Technocolonialism: Digital Innovation and Data Practices in the Humanitarian Response to Refugee Crises', *Social media+ society* 5.3 (2019): 1-13.
 - 26 Enrique Dussel, 'Eurocentrism and Modernity (Introduction to the Frankfurt Lectures)', *Boundary 2* 20.3 (1993), 65.
 - 27 Mignolo, 'Delinking'; Aníbal Quijano, 'Coloniality of Power and Eurocentrism in Latin America', *International sociology* 15.2 (2000): 215-32; Mignolo, *The Darker Side of Western Modernity*.

domination, exploitation, and extractivism through the rhetoric of modernity/coloniality.²⁸ Hence, modernity conceptualised the colonial Other as inherently uncivilised and in need of development. At the same, this rhetoric served to conceal the violence of colonialism that came with this enforced 'progress' behind supposed scientific objectivity and neutrality.²⁹ This 'irrational myth' ensured modernity remained morally 'clean' for centuries.³⁰

European colonialism, while involved in the destruction of natural habitats for the establishment of plantations, is equally implicated in the emergence of environmentalism. Environmentalist concerns go back at least three centuries and would not have emerged as they did without colonialism, capitalism, and the scientific development of modernity. Globalised production processes destroyed ecologies in the colonies, but also provided an overview of the world and the changes in yield after such destruction that was not previously accessible. Those that drove colonial extractivist processes gained a new understanding of the capacity of human activity to affect the climate and, as such, laid the foundations for the concerns that characterise contemporary environmentalism.³¹

The colonies provided a place on the periphery of Western life in which to experiment with environmentalist practices. This environmentalism was itself colonial: it did not have regard for the welfare of the people it dominated, but rather rethought its engagement with nature in order to sustain colonial and economic power. Decolonial scholar Malcom Ferdinand calls this 'Modernity's colonial and environmental double fracture,' which 'separates the colonial history of the world from its environmental history,' leading to disconnected movements for either cause.³² Environmentalism generally has little regard for coloniality, racism, and indigenous struggles, while civil rights, anti-racist, and feminist movements often do not engage with their intersection with environmental destruction and climate-related issues.³³ Many environmentalist movements today continue to envision better futures through Western-centric colonial logics.³⁴

28 Mignolo, 'Delinking'; Quijano, 'Coloniality of Power and Eurocentrism in Latin America'; Mignolo, *The Darker Side of Western Modernity*; Sylvia Wynter, 'Unsettling the Coloniality of Being/ Power/ Truth/ Freedom: Towards the Human, After Man, Its Overrepresentation—an Argument', *The New Centennial Review* 3.3 (2003): 257-337. <https://doi.org/10.2307/41949874>.

29 Walter Mignolo, *Local Histories/global Designs: Coloniality, Subaltern Knowledges, and Border Thinking*, Princeton: Princeton University Press, 2000; Quijano, 'Coloniality of Power and Eurocentrism in Latin America'.

30 Dussel, 'Eurocentrism and Modernity,' 66.

31 Richard Grove, 'Climatic Fears: Colonialism and the History of Environmentalism', *Harvard International Review* 23.4 (2002): 50-55.

32 Ferdinand, *Decolonial Ecology*, 3, emphasis in original.

33 Ferdinand, *Decolonial Ecology*. This, as Ferdinand explains, despite the fact that movements that bridge the 'double fracture' and perform a decolonial ecology also exist, and have done so for centuries.

34 Bruce Erickson, 'Anthropocene Futures: Linking Colonialism and Environmentalism in an Age of Crisis', *Environment and Planning D: Society and Space* 38.1 (2020): 111-28; Joe Curnow and Anjali Helferty, 'Contradictions of Solidarity: Whiteness, Settler Coloniality, and the Mainstream Environmental Movement', *Environment and Society* 9.1 (2018): 145-63; Ferdinand, *Decolonial Ecology*.

Although they are consistently portrayed as politically neutral, the technological and scientific methods used in environmentally motivated forestry and agriculture are inherently political and have been since they emerged as scientific forestry.³⁵ Contemporary digital nature management systems build on the colonial logics that featured in scientific forestry, which were used to turn colonised places into streamlined production centres for the West.³⁶ The exploitation of enslaved people in colonised territories in turn funded the technological innovations that enhanced the control of nature, agricultural crops, and people.³⁷ Through the abstractions of contemporary digital nature management, new concentrations of power emerge that align contemporary digital environmentalism with extractive colonial dynamics and 'reinvigorate and rework colonial relationships of dependency'³⁸ and demand urgent critical attention.

Ongoing Coloniality and the Role of Digital Technologies

Coloniality can be recognised in many different realms of global relations, including the coloniality of power (such as in politics and economics) and the coloniality of knowledge (through the epistemology of modernity).³⁹ The coloniality of power relies on the racial hierarchies that took shape during modernity to justify exploitation, violence, and the Eurocentrism of the global power dynamic.⁴⁰ It features, for example, in the rhetoric of progress, emancipation, and development that establish the West as civilised and authoritative while framing those in other parts of the world as uncivilised and in need of discipline toward emancipation.⁴¹ The coloniality of knowledge denies the geopolitics of knowledge and the way it structures relations between people in service of existing power relations.⁴² To underscore this point,

35 Susan Paulson, 'World-Making Technology Entangled With Coloniality, Race and Gender: Ecomodernist and Degrowth Perspectives', *Environmental Values* 33.1 (2024): 71–89.

36 Gabrys, 'Reworking the Political in Digital Forests'; Nancy Lee Peluso and Peter Vandergeest, 'Genealogies of the Political Forest and Customary Rights in Indonesia, Malaysia, and Thailand', *The Journal of Asian Studies* 60.3 (2001): 761–812; Ferdinand, *Decolonial Ecology*.

37 Alf Hornborg, *Global Magic: Technologies of Appropriation From Ancient Rome to Wall Street*, Basingstoke: Palgrave Macmillan, 2016.

38 Mirca Madianou, 'Technological Futures as Colonial Debris: 'Tech-for-Good' as Technocolonialism', in Joanna Zylińska (ed) *The Future of Media*, London: Goldsmiths Press, 2022, 10.

39 Besides these two, often mentioned is the coloniality of being, for example, in terms of gender, sexuality, and subjectivity. See Mignolo, 'Delinking'.

40 Aníbal Quijano, 'Coloniality of Power, Eurocentrism, and Latin America', *Neplanta: Views from South* 1.3 (2000): 533–80.; Mignolo, 'Delinking'.

41 Dussel, 'Eurocentrism and Modernity'; Aníbal Quijano, 'Coloniality and Modernity/Rationality', *Cultural Studies* 21.2–3 (2007): 168–78.

42 This geopolitics of knowledge bears a resemblance to Michel Foucault's theory of power/knowledge and regimes of truth. Foucault's work similarly highlights the interaction between power and knowledge: power shapes knowledge and knowledge supports power. What is considered knowledge is structured by regimes of truth. See Michel Foucault, *Power/knowledge: Selected Interviews and Other Writings 1972–1977*, New York: Pantheon Books, 1980; Michel Foucault, 'The Political Function of the Intellectual', *Radical Philosophy* 17. Summer (1977): 12–14. Foucault writes: 'Each society has its regime of truth, its 'general politics' of truth: that is, the types of discourse it harbours and causes to function as true; the mechanisms and instances which enable one to distinguish true from false statements, the way in which each is sanctioned; the techniques and procedures which are valorised for obtaining truth; the status of those who are charged with saying what counts as true'. See Foucault,

decolonial theorist Sylvia Wynter uses the phrase ‘truth-for’⁴³ (270) to indicate the political purposes of scientific truths. The coloniality of knowledge establishes Europe as the centre of the world, universalising its own way of knowing—the rational scientific method—while eradicating others.⁴⁴

Although European colonialism in the form of territorial expansion has ended, the logics of coloniality that supported it continue in contemporary societies. For example, after independence, formerly colonised territories remain dependent on Western states through the power relations of the global economy, which demand the ongoing export of produce rather than shifting to prioritise the sustenance of local populations. As such, the coloniality of power reproduces the political and economic North-South relations that emerged during colonial occupations.⁴⁵ Similarly, the scientific racism and eugenics that the Modernist sciences produced, which were used to substantiate the superiority of White people over racialised Others and justify their exploitation, lives on today. While cultural racism is more common now than biological racism, both rest on essentialised understandings of the inherent inferiority of racialised groups of people. Cultural racism is used politically to blame the ‘uncivilised’ or ‘underdeveloped’ culture of racialised people for the high poverty rates in their communities, rather than institutional racism and continued logics of coloniality.⁴⁶

Digital technologies, and the culture of capitalist digital economy around them, afford the ongoing coloniality of power, and render it in medium-specific ways. Systems that function through the datafication of everyday life—social media, self-trackers, public transport cards, etc.—produce ‘a new type of social relation’ that is structured by big tech corporations who have privileged access to the power and economic value that can be gained from them.⁴⁷ Digital systems afford the financialisation of life while simultaneously denying people access to this newly articulated value.⁴⁸ Interactions that take place through digital systems are struc-

‘The Political Function of the Intellectual,’ 13. As such, tokenisation could also be analysed as a regime of truth. However, decolonial theorists point out that Foucault’s work is lacking in its illumination of the racist and colonial dimensions of the geopolitics of knowledge and is still too embedded within Western histories of knowledge. See Wynter, ‘Unsettling the Coloniality of Being/ Power/ Truth/ Freedom’; Denise Ferreira da Silva, ‘Before Man: Sylvia Wynter’s Rewriting of the Modern Episteme’, in Katherine McKittrick (ed) *Sylvia Wynter: On Being Human as Praxis*, Durham: Duke University Press, 2015, 90–105.; Mignolo, ‘Delinking’. To analyse how tokenisation perpetuates the colonial ramifications of the geopolitics of knowledge, I build my argument using decolonial theory.

43 Wynter, ‘Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,’ 270.

44 Mignolo, ‘Delinking’; Mignolo, ‘The Geopolitics of Knowledge and the Colonial Difference’.

45 Quijano, ‘Coloniality of Power and Eurocentrism in Latin America’.

46 Ramón Grosfoguel, ‘Colonial Difference, Geopolitics of Knowledge, and Global Coloniality in the Modern/colonial Capitalist World-System’, *Utopian Thinking* 25.3 (2002): 203–24. <https://doi.org/10.2307/40241548>.

47 Nick Couldry and Ulises A. Mejias, *The Costs of Connection: How Data is Colonizing Human Life and Appropriating it for Capitalism*, Stanford: Stanford University Press, 2020, 12.

48 Jonathan Beller, *The Message is Murder: Substrates of Computational Capital*, London: Pluto Press, 2018. Beller ends his book on this topic with a subsection about blockchain. To him, it is clear that the socio-technical phenomena of Bitcoin and other mainstream cryptocurrencies do not fulfil any anti-capitalist hopes and rather continue earlier forms of the financialisation of the social. At the time of his writing, the programmability of the Ethereum blockchain was the cause of much hype, and

tured in significant ways by the demands of the corporate and usually Western accumulation of wealth and power.

Besides the power and wealth accumulation that data-based resource extraction afford, Densua Mumford argues for a potentially more fundamental critique of the way datafication also affords the coloniality of knowledge.⁴⁹ Digital culture reinforces the myth of modernity—that knowledge is unpolitical and science is neutral—while expanding the reach of rationalism in automated and technologised ways.⁵⁰ The ways in which data is structured to make life knowable are heavily informed by the demands of extractivist economic practices⁵¹ and in turn ‘life as seen by capital [is reduced] to what can be recorded and exchanged as digital data’.⁵² Data-based systems order life to produce a quantified view from afar that facilitates new and expanded forms of power that are centralised and wielded by those in control of the infrastructures that create them.⁵³ Data epistemologies are used to monitor, surveil, and control those that produce data, but have no agency over its use.⁵⁴

Through this combination of power and knowledge, data-based coloniality functions onto-epistemologically. Datafication operationalises its epistemic logics to produce its own realm of possibilities. Epistemologically, data-based systems determine what knowledge entails along rationalist lines that are essentialised through the logics of automation and algorithmisation, while ontologically, these ways of knowing produce powerful hierarchies that financially favour particular ways of being and make invisible or eradicate others.⁵⁵

Beller is suspicious of any partnerships between blockchain projects and big tech. Nevertheless, Beller expresses hope for the emergence of a different, anti-capitalist use of this new kind of blockchain, which aligns with his own non-academic work at the Economic Space Agency—a group that experiments with distributed ledger technologies in order to facilitate post-capitalist economies.

- 49 Densua Mumford, ‘Data Colonialism: Compelling and Useful, But Whither Epistemes’, *Information, Communication & Society* 25.10 (2022): 1511–16. Mumford is critical of Couldry and Mejias’ focus on data-based resource extraction over an engagement with the concept of the coloniality of data-based ways of knowing and argues this limits the strength of their concept of data colonialism. Critiquing the social quantification sector for its continuation of colonial relations in service of wealth accumulation is valid, she argues, but uses a profoundly Western system of thought—in this case, Marxism—to do so. This excludes the fundamental critique that modernity/coloniality offers. Additionally, Stefano Calzati points out that Couldry and Mejias build on Western conceptions of the autonomous self that are not universal and should themselves be decolonised. See Stefano Calzati, ‘Decolonising ‘Data Colonialism’ Propositions for Investigating the Realpolitik of Today’s Networked Ecology’, *Television & New Media* 22.8 (2021): 914–29. These critiques make clear that Couldry and Mejias’ data colonialism theory should be applied carefully. I operationalise their theories so they can function as medium-specific insights on how coloniality functions through datafication, rather than making their concept itself central to my analysis.
- 50 Ricaurte, ‘Data Epistemologies, the Coloniality of Power, and Resistance’.
- 51 Milan and Tréré, ‘Big Data From the South(s)’.
- 52 Jim Thatcher, David O’Sullivan, and Dillon Mahmoudi, ‘Data Colonialism Through Accumulation By Dispossession: New Metaphors for Daily Data’, *Environment and Planning D: Society and Space* 34.6 (2016), 1000.
- 53 Couldry and Mejias, *The Cost of Connection*.
- 54 Ricaurte, ‘Data Epistemologies, the Coloniality of Power, and Resistance’.
- 55 Nelson Maldonado-Torres, ‘On the Coloniality of Being: Contributions to the Development of a Concept’, *Cultural Studies* 21.2–3 (2007): 240–70.; Quijano, ‘Coloniality and Modernity/rationality’; Mignolo, *Local*

Blockchain and the Coloniality of Power

Blockchain constitutes an iteration of, rather than a break with, previous digital technologies and the way in which they afford the coloniality of power. Often, this takes shape in new, medium-specific ways and needs to be understood through blockchain's specific social, cultural, and technical dimensions. Blockchain affords the perpetuation of the coloniality of power on several levels, including the economic and political realms. While these levels are interconnected, I will highlight several relevant concrete manifestations of each below to show that coloniality can be recognised in phenomena such as the frontierism, extractivism, and imperialism present in blockchain culture.⁵⁶ Finally, I argue below that tokenisation plays a significant yet under-explored role in blockchain's affordances to reproduce the coloniality of power.

Economic Coloniality in Blockchain

A first dimension of economic coloniality is exemplified by blockchain frontierism and the exploitation of regulatory weaknesses in the Global South with the aim of maximising profit. Discursively, frontierism features in the analogy between the language used in historical settler colonialism and in the blockchain sector operating in the Global South. For example, 'a group of primarily white males from the United States referred to as 'Puertopians'' settled on the island of Puerto Rico, which they imagine as a 'blank slate' ready for the 'crypto-utopia' they aim to establish there.⁵⁷ The myth of (still-)to-be-colonised spaces as blank slates, history-less and culture-less, conceptualises them as available to own and dominate.⁵⁸ This discourse is co-constructed with the (lack of) legal frameworks that support frontierism. Although blockchain is often understood as a radical break with the past, as a culture it in fact functions largely through a continuation of the (venture) capitalist logics that structure the broader technology sector and feature partnerships between blockchain projects and established big tech actors. Technological innovation—including through blockchains—is used to continuously occupy not-yet (thoroughly) regulated localities in order to gain a competitive share in ever newer markets.⁵⁹ Examples include the settlement of blockchain entrepreneurs—mostly

Histories/global Designs.

- 56 In literature on these phenomena, authors use terms such as cryptocolonialism and blockchain colonialism to identify cases in which blockchain is used to establish colonial extractions of value. My analysis also features such cases. However, the broader aim of this chapter is to establish how blockchain supports coloniality through tokenisation, including the structural potentialities of the power dynamics and knowledge politics at play. To address this level of abstraction, I use the concept of coloniality.
- 57 Jillian Crandall, 'Blockchains and the 'Chains of Empire': Contextualizing Blockchain, Cryptocurrency, and Neoliberalism in Puerto Rico', *Design and Culture* 11.3 (2019): 287-88. <https://doi.org/10.1080/17547075.2019.1673989>.
- 58 Lyman Tower Sargent, *Utopianism: A Very Short Introduction*, Oxford: Oxford University Press, 2010. It is evident in such an example that the coloniality of knowledge that reproduces such categories as 'blank slates' is entangled with the coloniality of power through which such occupations materialise.
- 59 Beller, *The Message is Murder*; Anna Lowenhaupt Tsing, *Friction: An Ethnography of Global Connection*, Princeton: Princeton University Press, 2005; Peter Howson, 'Climate Crises and Crypto-Colonialism: Conjuring Value on the Blockchain Frontiers of the Global South', *Frontiers in Blockchain* 3.1 (2020): 1-6.

focussed on cryptocurrencies—on, for example, Puerto Rico, El Salvador, and Honduras, while making use of favourable tax schemes and cheap local resources.⁶⁰ Often, such localities embrace technological innovation as a way to advance their economies and implement weak regulatory frameworks for the business sectors around new technologies in order to attract investors and entrepreneurs.⁶¹ This proliferation of blockchain entrepreneurship, designed specifically with particular regulatory frameworks in mind, is a remarkable shift from the fiercely anti-regulation stances that dominated the early Bitcoin and cryptocurrency community.⁶² The coloniality of power and its reinforcement of global inequalities featuring in these developments is exemplified by Puerto Rico, which has seen rising energy prices due to the energy consumption of crypto mining and increasing rents due to the space occupied by the blockchain sector.⁶³

A second dimension of economic coloniality is exemplified by blockchain-based extractivism through the implementation of capitalist property logics and a Modernist rhetoric of emancipation. It is closely related to the previous dimension, as blockchain frontierism has ‘developing world resources as its prized objects’ and extracts them to the Global North.⁶⁴ For example, in Fiji, indigenous communities have largely succeeded in keeping out Western property logics, although this is now under threat due to the blockchain-based land registry that Western actors are trying to implement on the islands. These systems veil ‘the calculus of computational capitalism’ and its ‘neoliberal development model of property rights’ under the Modernist rhetoric of development aid and environmentalism.⁶⁵ Similar processes are taking shape in Papua New Guinea.⁶⁶ Favourable legal frameworks go hand in hand with ‘the rhetoric of tech-transformation and experimentation’⁶⁷ that brings tech entrepreneurs to the developing world. Even though many blockchain projects ‘do not demonstrate proof of concept, there is a rhetorical success’ in the hyped narratives that are spread about the technology’s capacities for social transformation that serves their finances.⁶⁸ The narratives and business strategies that emerge from ‘an assemblage of development agencies, foreign governments and platforms’ present blockchain technology as the key toward changing societies for the better and lays the groundwork for the continued involvement of the West in localities in the Global South.⁶⁹

<https://doi.org/10.3389/fbloc.2020.00022>.

- 60 Jose Atiles, ‘The Paradise Performs: Blockchain, Cryptocurrencies, and the Puerto Rican Tax Haven’, *South Atlantic Quarterly* 121.3 (2022): 612–27.; Olivier Jutel, ‘Blockchain Imperialism in the Pacific’, *Big Data & Society* 8.1 (2021): 1–14.
- 61 Olivier Jutel, ‘Blockchain Humanitarianism and Crypto-Colonialism’, *Patterns* 3.1 (2021): 1–8. <https://doi.org/10.1016/j.patter.2021.100422>.
- 62 Crandall, ‘Blockchains and the ‘Chains of Empire’’; Atiles, ‘The Paradise Performs’.
- 63 Crandall, ‘Blockchains and the ‘Chains of Empire’’.
- 64 Jutel, ‘Blockchain Imperialism in the Pacific,’ 2–3.
- 65 Jutel, ‘Blockchain Imperialism in the Pacific,’ 2.
- 66 Jutel, ‘Blockchain Imperialism in the Pacific’.
- 67 Jutel, ‘Blockchain Imperialism in the Pacific,’ 6.
- 68 Jutel, ‘Blockchain Imperialism in the Pacific,’ 11.
- 69 Jutel, ‘Blockchain Imperialism in the Pacific,’ 2.

This potent mix of technological solutionism, capitalist opportunism, colonial conceptualisations of emancipation, and global inequalities in the economic coloniality that blockchain affords ‘exacerbates already existing population vulnerabilities, socioeconomic and financial crises, and the climate crisis.’⁷⁰ Economic coloniality in the use of blockchain concentrates power in corporate and entrepreneurial players, and should thus be understood through the logics of neoliberalism.

Political Coloniality in Blockchain

In terms of political coloniality, blockchain perpetuates imperialist relations and power dynamics. Western actors apply technological innovation to humanitarian or environmental goals and as a result implement systems that establish dependencies in the Global South. However, such innovations are often part of a larger political scheme, such as in the case of ‘the US State Department’s evangelism,’ which implements soft power strategies—for example, funding and organising technology events in the Pacific—in order to extend geopolitical influence to these locales.⁷¹ As a new form of platform imperialism,⁷² blockchain imperialism attempts to implement control over the social realm through computation, while ‘extend[ing] both the cultural and economic power of Silicon Valley and American geopolitical interests’ through humanitarian rhetoric.⁷³ The state occupies a contradictory role in this imperialism, as it is seen by ‘blockchain enthusiasts as an entity that can be sold governance solutions, superseded by innovation or confer legitimacy upon blockchain’.⁷⁴ Ultimately, blockchain imperialism offers ‘a means to inculcate a market logic, Eurocentric techno-rationality’ and to reinforce the dependency of developing countries on the West while ‘oscillating between the objectives of state, capital and the solutionist prospecting of developers’.⁷⁵ For example, the Fiji-based tuna supply chain blockchain pilot—led by the World Wide Fund for Nature and developed in the context of US embassy-funded tech events—was envisioned as a way to make sustainably-sourced tuna traceable by logging it on a blockchain. However, the project disregarded local concerns regarding the availability of the Radio Frequency Identification (RFID) technology needed as well as brushing over the costs of these and other technologies to be used in the system, which would have to be shouldered by the local fisheries themselves. In effect, the project exacerbated North-South power imbalances with profits going to Northern technology companies supported by, and reinforcing, US-based geopolitical power, and the costs shouldered by Southern fisheries wishing to enter the global market.⁷⁶ The political coloniality of blockchain-based imperialism highlights the use of so-called ‘soft power’ to reinforce geopolitical power imbalances and dependencies and to continue to extract value from the Global South to benefit the Global North.

70 Jose Atilas, ‘Introduction: Decoding Crypto-Paradises: Fraud, Crypto-Colonialism, Climate Crisis, and Dispossession in the Global South’, *South Atlantic Quarterly* 121.3 (2022), 594.

71 Jutel, ‘Blockchain Imperialism in the Pacific,’ 8.

72 Dal Yong Jin, *Digital Platforms, Imperialism and Political Culture*, New York: Routledge, 2015.

73 Jutel, ‘Blockchain Imperialism in the Pacific,’ 2.

74 Jutel, ‘Blockchain Imperialism in the Pacific,’ 2.

75 Jutel, ‘Blockchain Imperialism in the Pacific,’ 2, 11.

76 Jutel, ‘Blockchain Imperialism in the Pacific’.

A second dimension of political coloniality resides in the asymmetrical power dynamics that blockchain-based computation represents. For example, blockchain-based identity systems run the risk of becoming ‘a technology of the state’ when mixed with governmental decision-making systems for the distribution of social benefits or border control.⁷⁷ Blockchain technology is involved in the construction of everyday rights and access in different ways for different people. For example, the UN World Food Programme’s Building Blocks project facilitates the distribution of aid to refugees, making sure individuals receive a share of the funds available through various organisations that is commensurate with their individual needs. To do this, refugees are given access to their funds by scanning their irises in shops and logging personal details on the Ethereum blockchain. The immutable exchange of personal information for immediate survival necessities extends the vulnerability of these groups into the future, where the data logged on the blockchain may be used in unforeseeable ways, potentially including decisions on housing allocation and insurances.⁷⁸ Other projects code the exclusion of those that are not deemed wealthy or trustworthy enough from participating into their systems.⁷⁹ Computational logics are a key process through which blockchain supports the reproduction of colonial power dynamics.

Tokenisation and the Myth of Emancipation in the Colonial Off-World

Blockchain researcher Olivier Jutel writes that blockchain-based datafication establishes new ‘cartographies of control’⁸⁰ that use ‘code to lock in proprietary relations’ (2). Although, as the previous section demonstrated, the existence of the coloniality of power in blockchain’s socio-political landscape is well-established, what remains under-evaluated is the instrumental role played by tokenisation, as well as the intersection of coloniality with blockchain-based environmentalism. In this section, I show that the tokenisation of nature affords the translation of power imbalances into infrastructural dictates that regulate life among that tokenised nature in ways that reproduce the coloniality of power. This is supported by the tokenised operationalisation of the myth of modernity that hides these colonial effects.

Below, I focus on how tokenisation supports a contemporary iteration of what decolonial scholar Malcom Ferdinand calls the ‘off-world’,⁸¹ in which colonial Others are barred from visibility and participation in the world as it is constituted by those in power. The colonial logics of environmentalism make forests ready for their dispossession from local communities by framing those communities as the cause of deforestation. In this process, ‘peasant farmers literally become the *targets* of the technical ‘solutions’ that are provided’ by environmentalism.⁸² The environmentalism of blockchain forest protection projects promotes a rhetoric of modernity that frames those in the Global South as underdeveloped and in need of emancipation by

77 Jutel, ‘Blockchain Imperialism in the Pacific,’ 9.

78 Howson, ‘Climate Crises and Crypto-Colonialism’.

79 Howson, ‘Climate Crises and Crypto-Colonialism’.

80 Jutel, ‘Blockchain Imperialism in the Pacific,’ 9.

81 Ferdinand, *Decolonial Ecology*, 138.

82 Ferdinand, *Decolonial Ecology*, 91, emphasis in original.

presenting their ways of life as unsustainable rather than addressing the global capitalist logics that demand resource depletion.

The Case of Nemus

The blockchain forest protection project Nemus exemplifies this phenomenon. The project creates NFTs of variously sized plots of the Amazon rainforest. With the money raised from the sale of these NFTs, the plots are intended to be protected against logging and other forms of deforestation. However, the power dynamics of this protection constitute colonial environmentalism. A professionally produced documentary-style promotional video for the project, their FAQ, and public documents about the project give more insight as to how this colonial environmentalism takes shape. The video follows an indigenous community that lives in the rainforest as it goes through the process of legally changing the name of its local area into ‘Non-Fungible Territory.’ Figure 5 shows a representative of the community placing a thumbprint on the legal document to sign it.

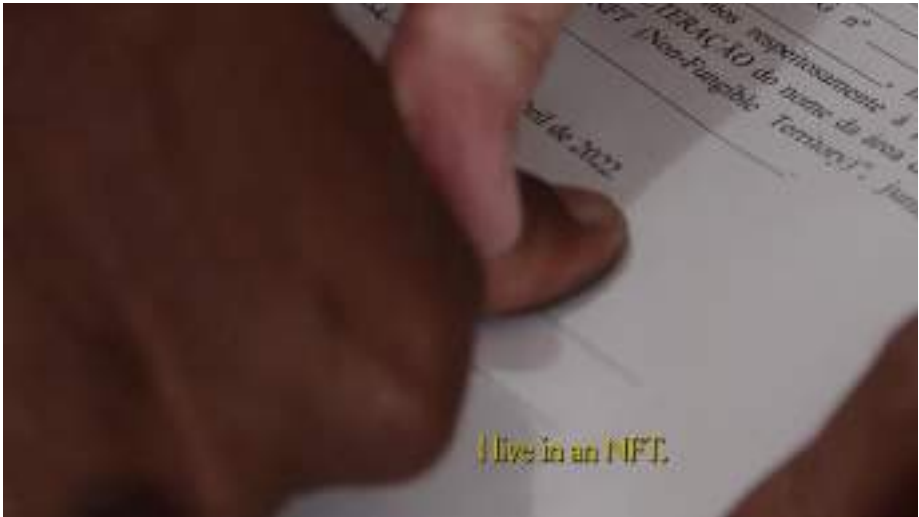


Fig. 5. Still from ‘Non-Fungible Territory (NFT Short Film) - Nemus’.⁸³

While this renaming may be a gimmick to call attention to what the video describes as ‘the most non-fungible thing for the planet’⁸⁴—i.e., the rainforest—much less fanfare is created about the rather more impactful legal change of ownership that the forest supposedly went through as well. To be able to secure the protection of the plots, the project’s FAQ states that the Nemus Foundation officially bought the land and is now registered as its owner in Brazil’s land registry.⁸⁵ The Nemus Foundation will act on the instructions of NFT-holders—Nemus

83 Nemus, *Non-Fungible Territory (NFT Short Film) - Nemus*, 2022. <https://www.youtube.com/watch?v=HIVD4qmowql>.

84 Nemus, *Non-Fungible Territory (NFT Short Film) - Nemus*.

85 Nemus, ‘Faq’, *Nemus Docs*, n.d., <https://docs.nemus.earth/nemus-docs/resources/faq>.

calls them ‘Guardians’—that are organised in the Nemus DAO in managing the property.⁸⁶ Local users of the forest—i.e., indigenous communities and local farmers—are imagined to effectively live under this new rule. While local farmers may become Guardians ‘on a case-by-case basis,’ no such option is explicitly mentioned for the indigenous community.⁸⁷ Nonetheless, Nemus blames the farmers for the deforestation and sees their own system as a way to emancipate them and encourage ‘better’ ways of living: ‘[W]hether this means working to rewild/reforest the land they’ve destroyed, or working to help monitor and protect the remaining forest, Nemus will create opportunities for them to earn a living by helping the rainforest, without having to abandon their homestead’.⁸⁸

This distinction between the different stakeholders was drawn into even sharper focus when the Federal Public Ministry of Brazil questioned whether Nemus had lawfully acquired the land at all, stating that it is public land that is reserved for use by indigenous communities and should remain so, especially when, as they claim, the indigenous community was not properly informed about the contents of the documents they were signing.⁸⁹ A member of the indigenous community is elsewhere quoted, stating:

(Nemus) said they would buy the land and give it to us [...]. They say they want to buy the land to preserve the forest. But we are preservers, we have been here for a long time and it was never deforested.⁹⁰

On December 29, 2022, the Ministry recommended that Nemus stop selling NFTs related to the area and act in accordance with regulations set by the International Labour Organisation of the United Nations in their contact with indigenous communities.⁹¹ Nemus’ online activities dropped significantly in the period following the recommendation, however, on May 8, 2023, they announced plans to move the project to a different plot of land, which is again an area in which indigenous communities live.⁹²

86 Nemus, ‘Nemus Litepaper’, *Nemus Docs*, n.d., <https://docs.nemus.earth/nemus-docs/nemus-litepaper/the-nemus-ecosystem/participants>.

87 Nemus, ‘Nemus Drop’, *Nemus Docs*, n.d., <https://docs.nemus.earth/nemus-docs/nemus-drops/drop-one-the-genesis-drop/initiatives-on-the-land/social-and-employment-opportunities/squatters>.

88 Nemus, ‘Nemus Drop’.

89 Ministério Público Federal, ‘MPF Aciona Empresa Que Vende Ativos Digitais (NFTs) De Áreas Da Amazônia’, *Ministério Público Federal*, 2022, <https://www.mpf.mp.br/am/sala-de-imprensa/noticias-am/mpf-aciona-empresa-que-vende-ativos-digitais-nfts-de-areas-da-amazonia>. The Federal Public Ministry claims on behalf of the indigenous community that even though literacy is low in the community, the documents they were presented with were not explained to them and included text in English. Furthermore, the community was not provided with a copy of the document.

90 Andre Cabette Fabio and Avi Asher-Schapiro, ‘Brazilian Prosecutors Probe Company Over Amazon Forest NFT Sale’, *Context*, 2022, <https://www.context.news/big-tech/brazilian-prosecutors-probe-company-over-amazon-forest-nft-sale>.

91 Ministério Público Federal, ‘MPF Recomenda Que Empresa Interrompa Venda De Ativos Digitais De Territórios Indígenas E Tradicionais Em Pauini (Am)’, *Ministério Público Federal*, 2022, <https://www.mpf.mp.br/am/sala-de-imprensa/noticias-am/mpf-recomenda-que-empresa-interrompa-venda-de-ativos-digitais-de-territorios-indigenas-e-tradicionais-em-pauini-am>.

92 Nemus, ‘EN: The Future of Nemus: An Update on Our Mission’, *Medium*, 2023, <https://medium.com/@Nemus.Earth/en-the-future-of-nemus-an-update-on-our-mission-fcfa3e6e52b>.

Instances such as those outlined above exemplify the kinds of asymmetrical data ownership, access, and agency that the tokenisation of nature can create, and show that processes of tokenisation—like datafication⁹³—do not have universally equal consequences. While the Nemus DAO might give a wider variety of individuals access to rainforest ownership and governance, it reproduces existing forms of inequality in the distribution of these capacities as it excludes indigenous communities while offering limited opportunities to others, such as local farmers. Most importantly, the Nemus DAO privileges those in the blockchain community who are sufficiently wealthy to purchase NFTs, wherever they might be located in the world. Furthermore, while the data represented in the NFTs—for example, features present on the individual plots of land, such as caves or waterfalls—are publicly available through the Nemus website, based on the indigenous community's lack of internet connection,⁹⁴ they are unlikely to be able to access it, let alone purchase one for a minimum of approximately 250 USD per NFT.⁹⁵ The actual legal ownership of the land—which makes possible the claim of DAO-based governance in support of the forest's protection—is now private and centralised where once it was public, at least in the way Nemus presents it. The complexity of blockchain, and the difficulty people have in understanding what it is capable of doing, afford the colonial accumulation of power, and straightforward explanations like the following are of great importance:

If [Nemus] owns the land, the ultimate power and control goes to them [...]. At the end of the day, they can just shut the DAO down. They own the land. They can either stay true to their word or get rid of it.⁹⁶

Thus, the project's claim that the new forms of ownership that its technology offers benefit the preservation of the rainforest through the emancipation of local communities stands in stark contrast with the actual affordances of its tokenisation of nature which, in effect, merely functions as a barrier between those that get to participate and those that do not. By tokenising nature and making it inaccessible to local communities, Nemus is involved in a form of colonial environmentalism that shallowly reproduces the myth of emancipation while creating an off-world in which extraction and dispossession can persist unabatedly.

Tokenisation and the Coloniality of Power in the Off-World

Although extreme, Nemus' case does not stand in isolation. Moss Amazon NFT, for example, makes similar promises when they state that '[t]his is NOT just another game card or vehicle for raising money for conservation. None of that 'buy this art to protect the forest' stuff. This NFT carries rights related to the actual real estate'.⁹⁷ However, just like Nemus, Moss retains

93 Milan and Tréré, 'Big Data From the South(s)'.

94 Nemus, 'Nemus Drop'.

95 Cabette Fabio and Asher-Schapiro, 'Brazilian Prosecutors Probe Company Over Amazon Forest NFT Sale'.

96 Crandall qtd. in Diane Jeantet, 'An NFT Company is Trying to Buy the Amazon to Save it. It's Not Going Well', *Vice*, 2022, <https://www.vice.com/en/article/z349be/an-nft-company-is-trying-to-buy-the-amazon-to-save-it-its-not-going-well>.

97 Moss, 'Moss Amazon Forest NFT – White Paper and FAQ', n.d., <https://moss-2.gitbook.io/moss-amazon-forest-nft-faq/>.

the legal ownership of their plots of Amazon rainforest in a foundation and does not officially transfer it to the NFT holders.⁹⁸ Another example is the now defunct The Hague-based Impact Earth, which made promises toward the empowerment of local communities in the Kariba conservation area in Zimbabwe through their Earth Token, but excluded ‘investors from “high-risk” jurisdictions,’ including Zimbabweans themselves.⁹⁹ Meanwhile, IBM partner organisation Infinite Earth, which aims to establish ‘for-profit conservation’ by selling tokens representing carbon sequestered in the Rimba Raya forest in Indonesia,¹⁰⁰ failed to inform local people of the details of this appropriation of the virtual value of their resources and left them powerless to have any say in its processes and logics.¹⁰¹

Through the tokenisation of nature, blockchain forest protection projects simultaneously capture a ‘last remaining ‘outside’ for capitalist extraction¹⁰² and define a new outside from which participation in this new economy is not possible. This new outside is like the ‘off-world’ of colonial plantations, without which the global colonial economy could not function, but which were kept outside of the view of the West.¹⁰³ Europeans were able to envision themselves as enlightened and moral humanists exactly because the space of violence and exploitation was confined to the off-world locale. Those in the off-world existed without humanity and power, ‘offside to a set of institutions, public and political arenas where the world is built and organized’.¹⁰⁴ The tokenisation of nature represents a new off-world where local communities are simultaneously exploited and excised from its systems.

The idea of a ‘last remaining outside’ relates to the ongoing process of accumulation by dispossession¹⁰⁵ that builds on, appropriates, and reaffirms established racist and colonial hierarchies.¹⁰⁶ The centralised accumulation of (tokenised) data as a new outside that is captured dispossesses people of a degree of control over their lives while commercial actors gain new forms of control. Those that produce data have no real access to it, ownership of it, or agency to act upon it, while those that own the system increase their access, ownership, and agency.¹⁰⁷ These examples show that the tokenisation of nature can dispossess indigenous communities and marginalised people in the Global South of both the agency to determine their own ways of being, and their relationship with nature.

98 Moss, ‘Moss Amazon Forest NFT.’

99 Howson, ‘Climate Crises and Crypto-Colonialism,’ 3.

100 Infinite Earth, *Rimba Raya - What We Do*, n.d. <https://vimeo.com/130461017>.

101 Howson, ‘Climate Crises and Crypto-Colonialism’.

102 Couldry and Mejias, *The Cost of Connection*, 6.

103 Ferdinand, *Decolonial Ecology*, 137.

104 Ferdinand, *Decolonial Ecology*, 138.

105 See e.g., David Harvey, ‘The ‘New’ Imperialism: Accumulation By Dispossession’, *Socialist Register* 40.1 (2004): 63-87.; Silvia Federici, *Re-Enchanting the World: Feminism and the Politics of the Commons*, Oakland: PM Press, 2018; Rosa Luxemburg, *The Accumulation of Capital*, London: Routledge, 1913.

106 Paula Chakravartty and Denise Ferreira da Silva, ‘Accumulation, Dispossession, and Debt: The Racial Logic of Global Capitalism—an Introduction’, *American Quarterly* 64.3 (2012): 361-85. <https://doi.org/10.2307/23273527>.

107 Couldry and Mejias, *The Cost of Connection*.

In these projects, the tokenisation of nature facilitates the coloniality of power. It promotes the dispossession of resources and agency from people that are already historically dispossessed. While blockchain forest protection projects structure the relationships between various communities—i.e., the indigenous or other local communities that inhabit the physical space targeted by the project on the one hand, and virtual communities of crypto- or NFT-holders and/or DAO participants who (it has to be assumed) aim to have some kind of positive environmental impact through their acquisitions on the other¹⁰⁸—they fail to decentralise in terms of ownership and agency to either of them. Rather, they act as centralising and intermediating forces that have the power to decide who gets to participate, how data is collected, stored, and made available, and what ownership entails. These projects insert themselves into existing relations between humans and forests, centralising ownership and agency within their own ranks.

The off-worlds in which nature is tokenised become ‘sacrifice zones,’ which ‘to their extractors, somehow don’t count and therefore can be [...] destroyed’¹⁰⁹ in service of some greater good defined by the extractors. While blockchain forest protection projects purport to capture the value of nature in a datafied rather than physical form, aiming to protect nature against functioning like a sacrifice zone for capitalism, instead, life around that which is tokenised constitutes the off-world and is sacrificed. Here, as it has done historically, ‘environmentalism embodies a *colonial ecology*’¹¹⁰ that reinforces colonial power relations under the guise of sustainability. These blockchain forest protection projects build on the idea that local communities cannot be trusted to care for the forest. They instead produce a tokenised view of the world that promotes the centralisation of power and the performance of an environmentalism that is disconnected from lived experiences and real engagement. As Ferdinand writes, they constitute ‘reforestation *without the world*’.¹¹¹

Truthful and Effective Tokens that Perpetuate the Coloniality of Knowledge

Tokenisation supports the coloniality of knowledge—which denies the geopolitics of knowledge and universalises Western epistemologies¹¹²—through the expansion of rationalism into new domains, exercising its supposed qualities of ‘immutable truth and pure mediation’.¹¹³ Whereas the previous section centres on the role of tokenisation in the reproduction of colonial power relations, I argue in this section and the next that it is imperative to understand tokenised ways of knowing in terms of their power to reproduce colonial epistemologies as well. While the effects of such colonial epistemologies may be less directly visible, ways of knowing that are imbued with economic, political, and technological power have very real ongoing consequences in structuring the world in seemingly neutral ways.

108 As shown above, often these communities do not overlap by design.

109 Naomi Klein, *This Changes Everything: Capitalism Vs. The Climate*, New York: Simon & Schuster, 2014, 148.

110 Ferdinand, *Decolonial Ecology*, 115 emphasis in original.

111 Ferdinand, *Decolonial Ecology*, 89, emphasis in original.

112 Mignolo, ‘Delinking’; Mignolo, ‘The Geopolitics of Knowledge and the Colonial Difference’.

113 Jutel, ‘Blockchain Imperialism in the Pacific,’ 4.

Furthermore, tokenisation perpetuates the rationalist myth of objectivity and universality, while the abstracted and partial truths of the tokenisation of nature serve the colonial geopolitics of knowledge. Although the abstractions of (tokenised) datafication always involve the translation of one thing into a generalised or simplified other, data is often presented as a natural by-product of life. This myth makes it seem as if those creating datafying systems simply build the right devices to listen to, and capture, real-world phenomena losslessly. However, much is indeed lost in this abstraction, on the one hand with regard to that which is rendered invisible—i.e., elements of life that do not fit financialised categories—and on the other hand with regard to the capacity to understand, reflect on, or object to the way in which datafication contributes to coloniality.¹¹⁴

My analysis is centred around two key verbs—namely, to verify and to validate—that are each discussed in a separate subsection below. Both verbs are linked to rationalism and are also often used to explain why blockchain technology is both trustworthy and effective.¹¹⁵ I focus on projects that use tokenisation to establish truths about trees and forests through the process of logging minute data about them—including their height, circumference, species, geolocation, information about their planters, pictures of the actual trees, local weather, and estimations of sequestered carbon—and combine media theory with decolonial theory to uncover the coloniality of knowledge that this rationalist tokenisation supports.

Verifying Nature

Open Forest Protocol (OPF) is an exemplary project with regard to the rationalism that characterises the use of fine-grained tokenised data about nature in blockchain-based environmentalism. In their introductory video, OPF envisions a measuring system for individual trees that is accessible to anyone and applicable anywhere. The video presents this in a profoundly romanticised way, following its protagonist through a forest, cycling and walking around, touching tree trunks, and measuring the circumference of a sapling (see Figure 6). In a meditative, quasi-philosophical style, the voice-over of the video asks: ‘why is what we measure what we value?’ The voice-over goes on to explain: ‘we measure trees, and we value a moment from the moment of truth [sic]. The location of a tree is permanent. Data reporting a tree is a permanent truth’.¹¹⁶

114 Couldry and Mejias, *The Cost of Connection*; Ricaurte, ‘Data Epistemologies, the Coloniality of Power, and Resistance’.

115 I use the word ‘effective’ here in a way that includes an etymological history that has persisted less in English than in Dutch: effective derives from the Latin effectus, meaning accomplishment or performance. In Dutch effecten are both consequences and financial stocks. In English the sense of ownership is carried in effects in the sense of ‘personal belongings,’ but the sense of the potential financial performance and accomplished gains is lost.

116 Open Forest Protocol, *Welcome to the Open Forest Protocol*, 2022. <https://www.youtube.com/watch?v=QB-e4Xv-9kE>.



Fig. 6. Still from 'Welcome to the Open Forest Protocol'.¹¹⁷

This translation, from off-chain reality (i.e., the untokenised tree as it exists in the world) to an on-chain tokenised 'permanent truth' about reality (i.e., the token that represents the tree immutably on a blockchain), necessarily involves both a process of simplification and the exercising of power. A tree's circumference, height, and carbon sequestration rate do not define its existence, but rather form a standardised set of metrics for its approximation. Datafication and tokenisation ready the world for abstracted commodification. This transformation of life into something else, is stabilised by the way 'data shadows—and eventually stands in for—the very thing it is supposed to measure'.¹¹⁸ The parts of life that are datafied 'remain, much of the time, irrelevant to this process of knowledge extraction'.¹¹⁹ Life itself is obscured by the data that is supposed to represent it. OPF's processes of tokenisation do not support understanding the tree itself, but rather establish a sufficiently detailed abstraction to sustain its commodification in a quantified way. By abstracting, quantifying, and categorising, processes of tokenisation normalise and reinforce token-based knowledge as objective and standardised ways in which life can be known. In effect, by continuously reaffirming itself over other ways of knowing the world, data turns into truth and, on the blockchain, that truth becomes 'permanent'.¹²⁰

The idea of truthful knowledge about nature in support of sustainable activities features in many blockchain forest protection projects. Some emphasise the importance of making

117 Open Forest Protocol, *Welcome to the Open Forest Protocol*.

118 Couldry and Mejias, *The Cost of Connection*, 68.

119 Couldry and Mejias, *The Cost of Connection*, 131.

120 With regard to these processes, Helen Kennedy writes of a 'desire for numbers' that reinforces itself. These projects show how this may be further incentivised through financialisation. See Helen Kennedy, *Post, Mine, Repeat. Social Media Data Mining Becomes Ordinary*, London: Palgrave Macmillan, 2016, 224.

accessible truth about trees, such as EcoMatcher, which states that with their tools, people will be able to ‘know everything about every tree planted’.¹²¹ Others emphasise how blockchain makes possible better and faster ways of verifying information about individual trees.

Verification is a recurring theme in blockchain projects in general. The etymology of the word ‘verify’ is significant here: it comes from the Latin *verus* (true) and *facere* (to make). Therefore, to verify means to make something true or to make a truth. Truth is not found in some initial state of nature, but rather it is that which is verified and is rendered thus on these blockchains. The blockchain forest protection project Veritree owes its name to this idea of truth-making (see Figure 7).

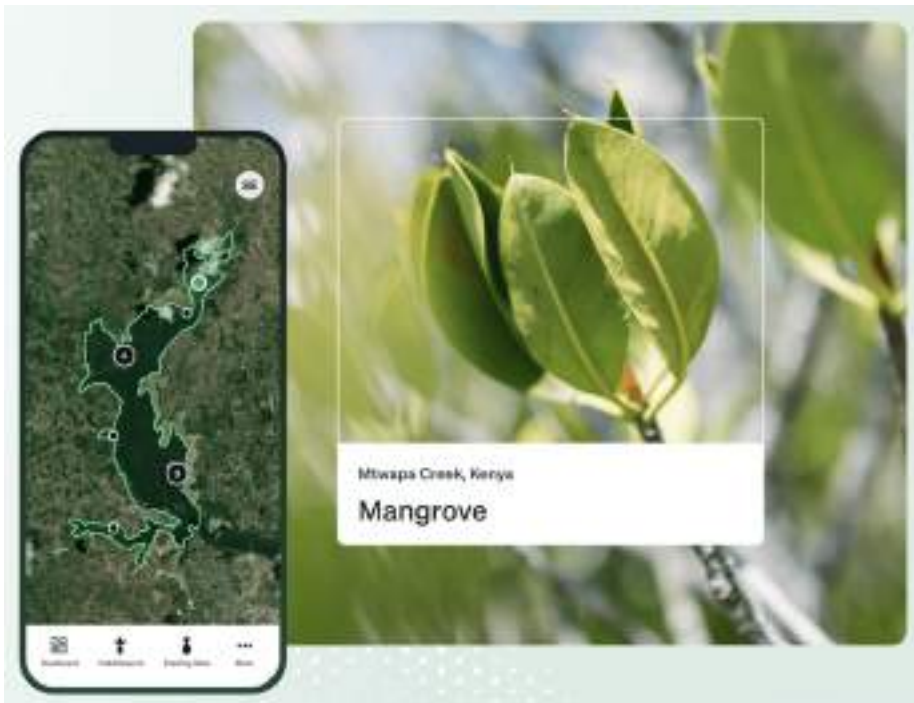


Fig. 7. Screenshot of Veritree's website.¹²²

This process of establishing categories and then using these to determine what is true about the world follows a colonial logic of ‘circularly verifying’ non-neutral categories.¹²³ For example, the subjugation of colonial Others first required the establishment of category pairs such as human and animal, or rational and irrational, and of the understanding of Black people as the ‘missing link’ between irrational animality and rational humanity in order to verify their

121 EcoMatcher, <https://www.ecomatcher.com/>.

122 Veritree, <https://www.veritree.com/>.

123 Wynter, ‘Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,’ 276.

inferiority and their need to be 'saved'.¹²⁴ Veritree's trees are token-certified 'true trees' based on predetermined categories in service of commodification. In effect, other trees that are not connected to a blockchain are rendered less-than-true through their invisibility in the system.

Validating Nature

The voice-over of Open Forest Protocol's video states that '[a] network of validators will be ready to validate your truth. Everyone in the world will be able to see the value in your forest'.¹²⁵ Such validation is a crucial step in any blockchain transaction, signifying the moment in which verified updates are pushed to the whole network and established as the new status quo: the new truth. To validate also has Latin roots: *validus* means strong, effective, powerful, active. Tokenised trees on a blockchain are verified and validated—made true and effective. They are both counted trees and trees that *count*, rather than the weak, passive, ineffective, and powerless trees that remain less-than-true, invisible, and uncounted around them.

Philosopher of science Isabelle Stengers¹²⁶ reflects on similar claims of powerful, valid truthfulness in the wider framework of the hegemony of scientific rationality. In particular, she focusses on the seemingly innocuous word 'really' as it is used to distinguish between what are called animist beliefs (e.g., the moon as an active and illuminating influence on life on earth) and what is 'really' happening as established through scientific rational objectivity (e.g., the moon as a giant object in space that has a gravitational pull). Claiming so authoritatively that something 'really' exists in effect stages an attack against any other understanding. It attempts to disprove other knowledges and make them inferior. Veritree and other projects that practice the 'making of effective truths' through verification and validation continue the legacy of perceived authority claimed by those adhering to scientific rationality 'to decide what objectively exists'.¹²⁷ The added feature is that 'what objectively exists' here is made immutable, transparent, and tradable.

This exertion of power is something that is characteristic of Western rationalism. Stengers continues: 'I would guess that those who are categorized as animists have no word for 'really,' for insisting that they are right and others are victims of illusions'.¹²⁸ On this point decolonial scholar Aníbal Quijano concurs:

[o]utside the 'West,' virtually in all known cultures, every cosmic vision, every image, all systematic production of knowledge is associated with a perspective of totality. But in those cultures, the perspective of totality in knowledge includes the acknowledgment of the heterogeneity of all reality.¹²⁹

124 Wynter, 'Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,' 301.

125 Open Forest Protocol, *Welcome to the Open Forest Protocol*.

126 Isabelle Stengers, 'Reclaiming Animism', *e-flux journal* 36 (2012): 1-10.

127 Stengers, 'Reclaiming Animism,' 4.

128 Stengers, 'Reclaiming Animism,' 4.

129 Quijano, 'Coloniality and Modernity/Rationality,' 117.

Outside of modernity/coloniality, the idea of totality includes the idea of difference, and those ‘differences are not necessarily the basis of domination’.¹³⁰ Instead, nature that is financialised through tokenisation *does* in effect dominate other parts of nature, continuing the logics of modernity/coloniality. That is, the tokenisation of nature aims for what decolonial scholar Walter Mignolo calls ‘a totalitarian notion of Totality’:¹³¹ the negation of the right to exist of other ways of knowing and other conceptions of the world than that which is dominant. This is the topic of the next section. To do so, I bring media theorist Alexander Galloway’s notion of ‘protocological control’¹³² to bear on tokenisation and show that it affords new medium-specific variations of such control.

Tokenisation as Protocols for Rationalist Totalitarian Totalities

The tokenisation of nature aims for universal coverage and, as a result, its processes of financialisation also centralise power. Through their universalising processes, tokenisation affords the establishment of an order of nature that has an effect on the kinds of nature that may be able to flourish in the future through the financialisation of only tokenised nature. That is, the tokenisation of nature affords what decolonial scholar Sylvia Wynter calls ‘language-capacitated form[s] of life’¹³³ in rationalist and financially biased ways. While blockchain forest protection projects may attempt to capture as much as they can of nature, they will always omit parts, resulting in the simplification that is inherent in the abstractions of tokenisation. Parts of nature that are tokenised are financialised and capable of receiving support, while other parts suffer from lack of visibility and receive inadequate support. Tokenised nature might thrive if these projects fulfil their promises, however, on the flipside, nature as an ecology might just as well wither in such a totalitarian totality.

Covering the World Through Tokenisation

Tokenised claims of totality reproduce coloniality in distributed systems. Here, the inclusion of the word ‘protocol’ in OPF’s name is revealing. While each in their own way, all the projects covered in this chapter can be understood (to aim) to institute a protocol of sorts that translates physical reality into tokens, Open Forest Protocol chose to name its project in explicit reference to this technological phenomenon. To understand the significance of tokenisation protocols in the context of coloniality, I will first discuss how protocols allow for control in the distributed network of the internet before bringing these insights to bear on the tokenisation of nature.

Computational protocols are ‘standards governing the implementation of specific technologies’¹³⁴ and internet protocols—which will serve as a basis for my analysis of tokenisation protocols—are, by their very nature, distributed. While on the internet anyone can in theory

130 Quijano, ‘Coloniality and Modernity/Rationality,’ 117.

131 Mignolo, ‘Delinking,’ 451.

132 Alexander R Galloway, *Protocol: How Control Exists After Decentralization*, Cambridge, MA: MIT press, 2004, 8.

133 Wynter, ‘Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,’ 270.

134 Galloway, *Protocol*, 7.

connect with anyone—it is non-hierarchical and distributed—to facilitate these connections within a global and diverse network, technical standards (protocols) are necessary. Although the internet is not centralised in the same way older media are, this does not mean it is inherently free from control, and protocols are way in which this new form of control occurs in distributed systems.¹³⁵ Protocols are necessary for distributed technology to function and represent a form of ‘voluntary regulation’¹³⁶ and subjection to control.

There are several layers of nested protocols governing internet-based connections and interactions, and one such protocol is especially interesting here. To establish a bridge between machine-readable codes and human-readable website addresses, the Domain Name System routes human users of the internet that input a URL in their browser through a tree-like system to find the location of the data they requested. Different institutions are situated at the branches of this tree-like system, and effectively represent hierarchies of control to whatever is further along the branch from them. This is one way in which control takes shape within the distributed system of the internet.¹³⁷

Blockchain technology also consists of a nested structure of protocols, for example the various algorithmic consensus protocols that set the rules for adding entries onto a blockchain ledger and different token protocols that define the way tokens function. The Ethereum blockchain is structured in such a way that it can act as a host to any token as long as it adheres to its token protocols, for example, ERC20 and ERC721.¹³⁸ The projects developing these tokens may then themselves define rules for their specific functionality. A project’s token design—its tokenomics—sets the rules for what a token in its purview represents, such as the rules of the process of tokenisation and the actions a token is capable of facilitating. Here again, a tree-like structure can be recognised in which each project represents a hierarchical point of control over the definition of what its token represents and how it functions. Because all the technology needed to launch a token is open source, theoretically anyone can create their own token with their own rules, however, practically speaking, not everyone will do so due to various factors related to, for instance, the cost of setting up and maintaining the required infrastructure, the technical literacy involved in doing this, and the time and effort needed to accumulate a critical mass of users. This is especially the case for blockchain forest protection projects that feature the tokenisation of physical phenomena, such as trees, that need to be measured off-chain through various additional technological infrastructures.

Protocols both universalise, ‘produc[ing] sameness or consistency where originally there existed arbitrariness’, and aim for universality—that is, to include everything within themselves.¹³⁹ Datafication features a similar drive toward totality and standardisation. By claiming comprehensiveness, data-based representation accumulates in value, and datafying systems thus aim toward “‘covering’ the world’ in its entirety through homogenising data-based repre-

135 Galloway, *Protocol*.

136 Galloway, *Protocol*, 7.

137 Galloway, *Protocol*.

138 ERC20 defines fungible tokens that are generally used for cryptocurrencies, and ERC721 defines non-fungible tokens that represent unique assets, such as art, real-estate, or indeed, individual trees.

139 Galloway, *Protocol*, 50.

sentations.¹⁴⁰ Blockchain climate projects often explicitly reference their aims toward totality. For example, in OPF's video, the project envisions their system as a 'new world standard. It will scale every forest value in a permanent, open, system update of this world'.¹⁴¹ Single.Earth's 'digital twin' of the planet echoes similar visions of universality when they confirm that they want to 'tokeniz[e] the whole planet'¹⁴² and 'assess and quantify nature as a whole'.¹⁴³

Through its universalising protocological control, tokenised nature is naturalised as nature itself. Olivier Jutel writes that this drive to universality is inherent in many blockchain projects because they function by creating 'a singular platform which mediates all data and the social world' and—specifically in the case of many blockchain forest protection projects—have 'developing world resources as [their] prized objects'.¹⁴⁴ For blockchain forest protection projects, it is not enough to capture data about some trees or forests. Instead, to really be effective and to understand nature enough to commodify it, *all* of nature—as defined by their categories—needs to be tokenised within a single system. Like with the emergence of big data, visions emerge of whole populations captured and represented one-to-one instead of having to use statistical samples in the analysis of society. Blockchain forest protection projects manage and regulate how things get to be included in their universality through standardisation to form a sort of 'second nature' that forms 'a 'layer' that has been folded back on itself such that it is simultaneously its core self and its own patina'.¹⁴⁵

Claims of totality in knowledge are not in themselves problematic until they deny the legitimacy of other knowledges.¹⁴⁶ Historical Western rationalism established itself as the only 'real' knowledge *at the expense of* other knowledges. The development of this rationalist paradigm—in which nature is other to humanity and can be isolated in scientific facts—was an integral part of the emergence of colonial power relations, in which racialised humans were biologically othered and dominated through the hierarchies established in the sciences. These hierarchies of cultures and peoples were thought of as part of a totality in which each had their role to play. It was in this view that the self-proclaimed intellectual and civilised West seemed to naturally embody authority.¹⁴⁷ Scientific truths about the world that claim superiority over

140 Couldry and Mejias, *The Cost of Connection*, 203.

141 Open Forest Protocol, *Welcome to the Open Forest Protocol*.

142 John Koetsier, 'The Stock Exchange of Nature? A Startup is Tokenizing the Planet to Save it', *Forbes*, 2021, <https://www.forbes.com/sites/johnkoetsier/2021/01/30/the-stock-exchange-of-nature-a-startup-is-tokenizing-the-planet-to-save-it/?sh=719fc8385aba>, n.p.

143 Single.Earth, <https://web.archive.org/web/20230606190446/https://www.single.earth/>. Digital twins are 'real-time digital representations of real-life physical entities or systems' that 'substitute their physical twin in knowledge production and decision-making processes'. See Paulan Korenhof, Else Giesbers, and Janita Sanderse, 'Contextualizing Realism: An Analysis of Acts of Seeing and Recording in Digital Twin Datafication', *Big Data & Society* 10.1 (2023), 1. They further abstract, isolate, and automate data-based knowledge and decision-making about reality that claim totality while obscuring those aspects of reality that they do not represent.

144 Jutel, 'Blockchain Imperialism in the Pacific,' 3.

145 Galloway, *Protocol*, 87, 92.

146 Quijano, 'Coloniality and Modernity/Rationality'.

147 Quijano, 'Coloniality and Modernity/Rationality'.

other ways of knowing reproduce the social dynamics of modernity/coloniality. This is their purpose: they are ‘truth[is]-for’ the reproduction of dominant sociality.¹⁴⁸

Establishing a Universal Order of Nature

The tokenisation of nature in these projects represents a universal order of nature, established through verification, validation, and standardisation. Blockchain forest protection projects occupy a privileged position with regard to the protocological question of how nature can be known and what is obscured or disregarded through their systems. Their ‘vision of *totality*’¹⁴⁹ is key in tokenisation’s coloniality. Rather than just showing reality, tokenisation *constructs* reality by continuing colonial ‘distortions of knowledge through power’¹⁵⁰ that attempt to universalise a rational understanding of nature.

Throughout modernity, rationalist scientific knowledge, secularism, and Western civilisational ideals came to define the understanding of humanity’s separation from the realm of animals and nature. Notions about what constitutes being human—especially in the form of hierarchies between humanity and animality, rationality and irrationality, culture and nature, civilisation and barbarity—supported the establishment of colonial relations. These notions are presented as scientific truths and operate under the guise of neutrality, while they rationalise the superiority and dominance of White Europeans over others that are categorised as less-than-human, irrational savages.¹⁵¹ Over time and through several iterations, this knowledge system established biological, cognitive, and economic markers of what it means to be human that overrepresent the West’s ‘genre of being human’¹⁵² as the universal essence of humanity itself.

Blockchain forest protection projects envision a world-encompassing system that becomes the world itself: a genre that stands in for all genres, a ‘digital twin’ that becomes the ‘single earth.’ Tokenised nature represents a particular ‘genre of nature’ as the essence of nature that excludes other genres through the system’s universalising and financialised power. Tokenised nature is nature-made-effective-and-truthful. The universalising visions of blockchain forest protection projects aim to turn tokenised nature into powerful truths. In Western rationalism, the natural is ‘that which Science will eventually explain’,¹⁵³ obfuscating other understandings that already exist. In blockchain forest protection projects, nature, trees, and forests are something that tokenisation will eventually define while obfuscating whatever it does not capture.

The imaginary of the truthful and total tokenisation of nature has onto-epistemological implications. All cultures and their respective ways of knowing and explaining the world exist as

148 Wynter, ‘Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,’ 270.

149 Couldry and Mejias, *The Cost of Connection*, 202 emphasis in original.

150 Nick Couldry and Ulises A. Mejias, ‘The Decolonial Turn in Data and Technology Research: What is At Stake and Where is it Heading’, *Information, Communication & Society* 26.4 (2023), 795.

151 Wynter, ‘Unsettling the Coloniality of Being/ Power/ Truth/ Freedom’.

152 Wynter, ‘Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,’ 281.

153 Stengers, ‘Reclaiming Animism’.

'language-capacitated form[s] of life'¹⁵⁴ in which life is a material-discursive phenomenon. The dominant, tokenised 'genre of being' defines what being constitutes ontologically through the operationalisation of its system of knowledge on a blockchain. Epistemic categories that are used for *knowing* nature in totalising blockchain forest protection projects also constitute what nature *is*, at least within the confines of their own aspirational totality.

Conclusion: A colonial Onto-Epistemology of Tokenised Nature

The tokenisation of nature constitutes a rationalist imaginary that hides the myths it is built on in order to gain power: tokenised nature is naturalised as nature itself and blockchain is naturalised as *the* solution to climate change. At the same time, its biases and the politics of its logics are obscured.

Where blockchain forest protection projects 'touch the ground'—take ownership of land, implement tokenisation procedures, sell tokens connected to nature, etc.—they perform a colonial environmentalism that reproduces the coloniality of power and knowledge. In the environmentalist blockchain imaginary, tokenisation provides a way to better manage trees and forests through rationalist methods in order to achieve sustainability. Blockchain forest protection projects present data about trees and other forms of natural life as an underexplored asset that—once connected to a blockchain system—can offer a sustainable route to economic growth: instead of more land and more logs, they produce more data about nature while the actual trees remain rooted in the ground and continue sequestering carbon and harbouring biodiversity. This seemingly innocuous pitch is supported by an imaginary that reproduces colonial logics of power and knowledge.

The tokenisation of nature should be understood as part of a long history of environmentalism that perpetuates colonial myths, power structures, and knowledge logics. The imaginary of blockchain as a truth machine of nature relies on the belief in the objectivity and virtue of tokenised data that atomises and disintermediates nature. It reproduces the Modernist myth of the neutrality of scientific rationalism and its attendant rhetoric of justified violence and domination over non-Western Others in order to 'emancipate' them to what its data substantiate as more sustainable lifestyles. As such, it reproduces the coloniality of power. While blockchain forest protection projects claim that through their system, more people will have more direct agency to protect trees and forests, the logics of tokenisation preclude the participation of already marginalised communities that live in those forests. The tokenisation of nature affords the creation of an off-world from which participation is not possible, but which is structured to facilitate the extraction of value to those in power over the system.

By making particular kinds of nature—kinds that are observable through scientific and technological methods—effective and truthful on a blockchain, blockchain forest protection projects afford the coloniality of knowledge in financialised ways. Tokenised truths are 'truths-for' the purpose of supporting existing power dynamics. Tokenised nature is different from nature and obscures it through its economic and epistemic power. Tokens represent

154 Wynter, 'Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,' 270.

a truth-for the universalisation of scientific rationality and the obfuscation of other ways of knowing. Blockchain forest protection projects continue the Western rationalist episteme by striving for totalising systems of tokenisation. Ways of knowing that are immutably stored on a blockchain stabilise their own epistemology because of their implication in financialisation. Token-based extractivism reinforces the value of its own system of knowledge over other forms of knowledge in a continuous financialised feedback loop, and as such presents a colonial onto-epistemological phenomenon.

Thus far, despite their imaginary of total and truthful tokenised nature, blockchain forest protection projects do not (yet) fulfil their totalitarian colonial affordances. Although in specific cases, blockchain forest protection projects have detrimental colonial consequences in actual lived reality, their logics of tokenisation do not, as yet, produce widespread protocological control. Life, in most cases, is still possible outside of their definitions. My analysis in this chapter has skewed toward the material-discursive imaginary that is constructed by rather dominant actors that reproduce dominant Western logics of thought. As such, the colonial affordances of an imaginary of tokenised nature warrant alertness: they are situated in power. However, for now, questions remain about the possibility of an imaginary of tokenisation that emerges from different ways of being, knowing, and relating.

Although as an artistic project it does not fit the selection criteria of this chapter, an analysis of the tokenisation of nature is not complete without a reference to blockchain artwork Terra0.¹⁵⁵ Terra0 is particularly relevant in terms of the questions I raise in this last paragraph: it invites reflection on such questions. The central concept of the project is that it seeks to give a forest the agency to govern itself by connecting it to a blockchain and sensor technology. If the sensors indicate that the forest is healthy, it may decide to sell some of its logs and slowly accumulate enough wealth in its blockchain wallet to buy new land to expand as a forest. In essence, instead of distributing ownership and agency to DAO-members, NFT- and crypto-holders—as the projects analysed in this chapter do—Terra0 appropriates capitalist logics for the growth of nature by giving nature agency to govern itself. Although ultimately, human developers decide how health or growth is defined technically, the project raises urgent questions with regard to non-human agency in decisions about nature within capitalist logics. The artists behind Terra0 invite reflection on the pros and cons of accelerationism for nature, about how non-humans may or may not be able to express their needs and intentions in systems designed by humans, how tokenisation might isolate elements from entangled ecosystems, and what remains of nature once it is rendered visible through technological and financial proxies. Terra0 does not answer these questions, but as an ongoing artwork, has continued to provide ways to, à la Haraway,¹⁵⁶ stay with the troubled intersection of technology, nature, finance, and sustainability without totalising tendencies.

155 Paul Seidler, Max Hampshire, and Paul Kolling, *Terra0*, 2016. <https://terra0.org/>.

156 Donna J Haraway, *Staying With the Trouble: Making Kin in the Chthulucene*, Durham: Duke University Press, 2016.

BLOCKCHAIN'S DIVINE DESIGN: BELIEF IN BLOCKCHAIN AS A TRUTH MACHINE

Back in 2015, *The Economist* drew attention to the then-new capacities of blockchain for usecases beyond cryptocurrencies in an article titled 'The Great Chain of Being Sure About Things'.¹ The title itself reduces a seemingly irreducibly complex technology to a simple promise: certainty. Within the article, blockchain is presented as a tool for securing trust where it is otherwise absent—covering examples ranging from ownership records to the tracing of precious stones and the management of personal identity. To reinforce this framing of certainty, the article included an image of interlocking, gear-like components coiling into what looks like a Fibonacci spiral, a shape long associated with natural harmony and divine order. In this visualisation, blockchain is less an administrative tool than a principle of cosmic precision, something humans can rely on but never fully comprehend. This blending of calculability and belief—the presentation of truths that can be verified alongside to-be-believed destinies that cannot—is precisely the tension that I examine in this chapter.

The element of belief that accompanies blockchain culture's rationalism is both ubiquitous and regularly repudiated.² In fact, technology culture is intimately entangled with technomystical, religious, and spiritual beliefs³ and these phenomena are 'crucial to understanding the sociotechnical imaginaries animating many technologies'.⁴ The cases I analyse showcase that such beliefs take many forms, ranging from the genuine conviction that a predestined future will appear, to the ideological use of religious references in order to spread belief in a particular teleology for selfish gains, or from the mystical portrayal of the technology as the saviour of society, to the magical obfuscation of its inner workings.

Belief and rationalism are commonly framed as if they are in opposition with each other. In the context of blockchain, however, they operate less as antagonists than as partners: both hinge on ideals of perfection and both rely on forms of concealment. Together, perfection and obfuscation sustain the widespread sense that the technology is resistant to full comprehension, reinforcing its aura of impenetrable complexity. In the previous chapter, we saw that the rationalism of blockchain culture relays the technological specificities of tokenisation in an idealised vision of universally valid systems of truthfulness, neutrality, and objectivity. The myths that support rationalism were actively kept out of view by blockchain

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- 1 'The Great Chain of Being Sure About Things', *The Economist*, 31 October 2015, <https://www.economist.com/briefing/2015/10/31/the-great-chain-of-being-sure-about-things>.
 - 2 From personal experience being immersed in blockchain culture I have gathered that mentioning the element of belief that pervades the culture to its proponents is likely to yield either puzzled faces or agitated denials.
 - 3 Steven Derk Aupers and Lars de Wildt, 'Digital Religion', in D. Rohlinger, and S. Sobieraj (eds) *The Oxford Handbook of Sociology and Digital Media*, Oxford: Oxford University Press, 2022, 159-77.
 - 4 Damien Patrick Williams, *Belief, Values, Bias, and Agency: Development of and Entanglement With 'Artificial Intelligence'*, PhD diss., Virginia Polytechnic Institute and State University, 2022, 5.

forest protection projects. They flattened nature's inherent complexity in order to offer what seemed like a universal solution to deforestation within the logic of rationalism. In this chapter, I build on these findings to show that the various forms of spiritual and secular belief in blockchain culture provide a second abutment of this same idealised vision. The combination of familiar narratives, inscrutability, and the perception of technological autonomy naturalise the imaginary of blockchain as a truth machine as an indisputable, god-given fact. The spiritual connotations of perfection aid rationalism in obfuscating the real-world frictions and inequalities that it denies exists.

Technology and belief have long been intertwined with each other and with power in Western societies. On the one hand, technology is integrated into the divine through a historical quest for perfection through philosophical, scientific, and religious means. On the other hand, rituals that emerge around technologies have throughout history shaped systems of belief. Additionally, technologies are also bound with non-religious forms of belief in ways that obfuscate not only their inner workings, but also their entanglement with power. Such technological mysticism relies, on the one hand, on a widespread modernist belief that technologies are beneficial to society, and on the other hand, on the mystification of how those technologies actually work. Together, these various forms of belief provide a coherent sense of futurity that supports the idea of technology as the unfathomably perfect solution to societal issues. Specifically, the idea of blockchain as the techno-rational saviour of society emerges from the dynamic of perfection and obfuscation that the mix of rationalism, algorithmic logics, religiosity, and mysticism in blockchain culture produces.

To substantiate this claim, I analyse formative episodes in blockchain's (pre-)history where the technology is cast as possessing a near-divine authority over knowledge. I begin by addressing how blockchain's consensus algorithm and its enigmatic inventor have been the object of reverence—phenomena for which the breeding ground can be found prior to the emergence of blockchain itself. I observe that the protocol for establishing truth on a blockchain is equated with a distanced and disinterested god who mediates exchange with an ultimate level of objectivity and neutrality. However, in this godly shape, algorithmic mediation is perceived to be as perfect as it is inscrutable. I show that blockchain takes on the characteristics of an all-knowing and impartial god that *needs* to be beyond human control in order to work.

Second, I turn to the point at which a new iteration of blockchain expanded its application beyond monetary exchange. Like we already saw a glimpse of in the opening paragraph of this chapter, in this moment, the technology is endowed with a form of sanctified truthfulness—it is framed as an object of unquestioning faith. In this new version, blockchain is presented as exceeding human capacities and understanding—something to be revered like a god or a portal to the divine. In the same way oracles supposedly speak a god's truth, the idea of blockchain as a truth machine—which we encounter literally in the community at hand here—is shaped by religious ideals. Here, accounting technology is understood to have spiritual significance for proving one's piousness and honesty to religious authorities. Blockchain provides what is thought of as a much-needed update to accounting technology in the face of the 2008 financial crash and the post-truth era.

Finally, I consider the ways in which Bitcoin is upheld as the best instantiation of the technology and celebrated for supposedly disclosing a universal logic. Within this discourse, self-identified Bitcoin Maximalists mobilise religious tropes to articulate their commitment, which leads them to envision the currency as carrying an almost prophetic promise of the future. Bitcoin Maximalists believe that the Bitcoin blockchain should and will take the place of all other currencies. Like rationalism, Bitcoin Maximalism is characterised by the idealisation of a natural, universal order that eradicates all other ways of knowing the world. By veiling Bitcoin's universal order in prophecies and belief, and by practicing animosity towards any kind of difference, Bitcoin Maximalism impedes critical reflection and contestation. It claims that Bitcoin possesses some kind of fundamental and universal truth that should guide all future actions, while it also presents the technology as a godlike force that is not to be technically understood, but rather faithfully followed.

Three Times 'Blockchain as a Truth Machine'

Before dealing with the element of belief, I will briefly outline the three components of the idea of truth in blockchain culture that feature in the cases I analyse: the idea that blockchain is neutral; the idea that blockchains can produce a new kind of truthfulness that has the capacity to overcome contemporary complexities; and the idea that a blockchain can represent universal truths about the nature of the world. While all these elements of the idea of truth in blockchain culture are present throughout the three cases I analyse, they each dominate the discussion in one case in particular, as I demonstrate below.

First, the notion of blockchain as an impartial technology mirrors Enlightenment rationalism's belief in its own capacity to deliver a universal and objective form of truth. Blockchain is thought of in these terms because of the idea that it is a 'trustless' technology. From a technical standpoint, the concept of 'trustlessness' denotes how blockchain's distributed design insulates the system from bad actors seeking to manipulate it for their own gain. On a social level, the term describes the possibility of transactions occurring without reliance on centralised intermediaries. In this idealised scenario, blockchain becomes a vehicle for an alternative form of social organisation, one defined—at least in theory—by radical decentralisation.⁵ For Satoshi, trustlessness was *the* innovation of blockchain technology. The Bitcoin whitepaper starts and ends with a reference to trust, the latter summing up the innovation of blockchain technology as 'a system for electronic transactions without relying on trust'.⁶ However, framing blockchain as a trustless system risks idealising it as 'neutral, objective and separate from the socio-political whims of humans and [the] "messiness of the world,"' even though, in practice, the technology remains deeply entangled with human agency and political contexts.⁷ Nevertheless, achieving neutrality remains a key driver in later blockchain developments as well. For example, Ethereum co-founder Gavin Wood writes that, in its blockchain, '[t]he incorruptibility of judgement, often difficult to find, comes naturally

5 Jaya Klara Brekke, *Disassembling the Trust Machine*, PhD diss., Durham University, Durham, 2019.

6 Nakamoto, Satoshi, 'Bitcoin: A Peer-to-peer Electronic Cash System', *Cryptography mailinglist*, 2008, <https://bitcoin.org/bitcoin.pdf>, 8.

7 Brekke, *Disassembling the Trust Machine*, 19.

from a disinterested algorithmic interpreter'.⁸ Blockchain's original consensus algorithm in particular is imagined as just such a 'disinterested interpreter' that, based on its rational calculations, reaches judgements that are unbiased, objective, and neutral. In the first part of my analysis, I focus on this algorithm, as well as on those who imagined and created it, to reveal the belief that surrounds it.

Second, in the subsequent section of my analysis, we encounter the phrase 'truth machine' literally. It is used in reference to the idea of blockchain's unprecedented capacities to produce truthful statements, particularly in the context of the post-truth era. The idea that society is going through a post-truth era exploded into the popular domain in 2016,⁹ around the same time that Ethereum broke into the mainstream.¹⁰ Its introduction expanded blockchain's functionality, enabling the technology to store and make accessible not just financial transactions but any form of information that can be rendered as digital data.¹¹ In the context of the uncertainty that the idea of the post-truth era connotes, 'blockchain as a truth machine' is used as a way to attempt to regain a shared sense of reality. However, blockchain's truth is very specific: it is an immutable, singular, incorruptible, and accessible form of truth.¹² These characteristics carry with them political implications for the way in which the world can be known through blockchains.

The final component of blockchain-based truth is the assumption that blockchain functions as a totalising system—one capable of holding and representing knowledge of everything. Although the idea is widespread in blockchain culture more generally, in the final part of my analysis I address this idea and its significance to a specific Bitcoin community known for its fervency on this point: Bitcoin Maximalists. As the first and largest¹³ blockchain, a loyal community has formed around Bitcoin over the years, and within it, some adherents maintain that no other blockchain—or even currency—should exist besides Bitcoin itself. The main difference with many subsequent blockchains is that Bitcoin is, first and foremost, imagined as a system for digital cash or some approximation of money without its traditional centralised controls,¹⁴ rather than a general purpose distributed ledger. In its imagined future as the world's singular currency, Bitcoin is portrayed as the idealisation of the free market,

8 Gavin Wood, 'Ethereum: A Secure Decentralised Generalised Transaction Ledger', *Ethereum*, 2014, <https://ethereum.github.io/yellowpaper/paper.pdf>, 1.

9 While the events surrounding the election of Donald Trump as president of the United States are key contributors to this explosion, the post-truth era did not start there. Rather, it emerged from a longer history, see e.g. Jayson Harsin, 'Regimes of Posttruth, Postpolitics, and Attention Economies', *Communication, Culture & Critique* 8.2 (2015): 327-33. The phrase 'blockchain as a truth machine' was used somewhat earlier as well, see e.g. "The Great Chain of Being Sure About Things".

10 see e.g. Steve Fuller, *Post-Truth: Knowledge as a Power Game*, London: Anthem Press, 2018.

11 The blockchain forest protection projects discussed in chapter 2 make use of this new capacity.

12 E. Johanna Hartelius, "'The Great Chain of Being Sure About Things': Blockchain, Truth, and a Trustless Network", *Review of Communication* 23.1 (2023): 21-37.

13 In terms of market capitalisation, not necessarily in terms of users. Notably, actual user counts are difficult to determine: What to count? All accounts ever created? Daily active accounts? Unique individuals with accounts? Nevertheless, Bitcoin is generally considered to be rivalled on this point by other major blockchains, chief among them Ethereum.

14 See Nakamoto, 'Bitcoin'.

technically designed to be outside of government control and representative of all economic interactions. Such a distributed economy is reminiscent of some of the work of economist of the Austrian School Friedrich Hayek, who considered the ‘discovery procedure’ of competition as the ideal organisational principle of society: distributed, competing parties uncovering all of economic reality through their actions on the unregulated market alone. In his vision, the market is the ideal informational system via which to know society.¹⁵ In the imagined scenario in which Bitcoin mediates all economic exchange, it would represent an all-encompassing informational system.¹⁶

The current section provides the backbone for my analysis of the relationship between the imaginary of ‘blockchain as a truth machine’ and the dynamics of faith and conviction that shape blockchain culture. The materials that I analyse range from books, mailing lists, and online articles, to memes, social media posts, online videos, and marketing material. In each section, I start from one of the conceptualisations of truth laid out above and show how blockchain is conceived as a rationalist machine in religious and mystical ways. The following two sections present the relationship between technology on the one hand, and various forms of belief on the other, to lay the foundation for these analyses.

Perfection, or the Entangled History of Religion and Technology in the West

The pursuit of technological and scientific perfection has long been entangled with both Western religious traditions and alternative spiritual currents. Enlightenment thought elevated mathematics and abstraction as markers of rationality while also casting them as channels to a divine, all-knowing truth. Today’s computational and algorithmic systems continue this logic, sustaining the belief in universal and transcendent truths that continues to inform how digital perfection is imagined. Furthermore, the element of belief seems to inherently accompany (the idea of) complex technology itself. Although over time particular technologies might lose

15 Friedrich A Hayek, ‘Competition as a Discovery Procedure’, *Quarterly Journal of Austrian Economics* 5.3 (2002 [1968]): 9-23. Although many consider Hayek a forefather of Bitcoin, I do not wish to draw too close a connection between the two. Hayek argued for the denationalisation of money and would have liked to see a competitive system of privately organised currencies. This is the part of Hayekian thought that leads many to consider his ideas to be foundational for cryptocurrencies. However, Hayek’s line of thought is not quite as compatible with Bitcoin’s reality as it may at first seem. For example, Hayek thought that in a system of multiple competing currencies, consumer choices would eventually weed out ‘bad money’ and promote ‘good money.’ What precisely ‘good money’ is, would be determined by its acceptability and its price stability. At the time of writing, Bitcoin is still experiencing problems with regard to both these requirements for ‘good money.’ See Luca Fantacci, ‘Cryptocurrencies and the Denationalization of Money’, *International Journal of Political Economy* 48.2 (2019): 105-26. <https://doi.org/10.1080/08911916.2019.1624319>; David Sanz Bas, ‘Hayek and the Cryptocurrency Revolution’, *Iberian Journal of the History of Economic Thought* 7.1 (2020): 15-28.

16 Although Hayek considered himself an anti-rationalist, his notion of rationalism merges distinct and sometimes ill-fitting ideas, causing some to dispute this self-characterisation, see e.g. Mats Lundström, ‘Is Anti-rationalism Rational? The Case of F. A. Hayek’, *Scandinavian Political Studies* 15.3 (1992): 235-48. Bitcoin Maximalists place their ideals of a free market within a broader framework of the universal laws of nature that echo the legacy of Enlightenment rationalism.

their mystifying sheen of novelty and become somewhat earthlier and mundane, human-made tools that exceed human capacities continue to provide a vision of perfection.

Conversely, technological changes can also co-shape religion and spirituality. New technologies afford changes within religious practices and contribute to evolving notions of the divine. Despite the predominantly Christian references in blockchain culture, I treat these as the expression, not of a pre-existing religion, but rather of thus far uninstitutionalised expressions of religiosity—i.e., a personal experience of belief—that have the capacity to shape the future of religion.

The Religion of Technology

Throughout history, the development of Western technology has repeatedly been driven by ‘spiritual yearnings for supernatural redemption’.¹⁷ In Europe’s early Middle Ages, this orientation was accompanied by a transformation in how tools themselves were understood and the kinds of significance attached to them. Once tied to the mundane existence of peasants engaged in manual work, tools came to be reimagined as offering a potential route to spiritual transcendence.¹⁸ Historian of technology David F. Noble characterises this shift as distinctly Christian: while many religions imagine humanity as created in the likeness of a divine being, Christianity uniquely proposes that this likeness, lost through the fall from grace, can be recovered through persistent human effort. Over the centuries, this pursuit of perfection became deeply entwined with technological practice.¹⁹ Technology came to be seen as the very instrument through which humanity might ‘restore [fallen man] to his prelapsarian condition’.²⁰ Noble argues that this fusion of faith and invention gave rise to what he calls ‘the religion of technology’,²¹ a worldview that endured for centuries. The myth of progress, distinctions between nature and culture, and the idea of development through science and technology emerged out of these histories and, remembering the theories touched upon in chapter 2, has significant connections to colonialism and coloniality.²²

The pursuit of a religious kind of perfection through technology on the one hand, and the rise to perceived epistemic superiority of abstraction and calculable logic became closely linked in the historical currents that laid the groundwork for modern computation. Descartes’

17 David F. Noble, *The Religion of Technology: The Divinity of Man and the Spirit of Invention*, New York: Penguin Group, 1997, 3.

18 Noble, *The Religion of Technology*; Erik Davis, *Techgnosis: Myth, Magic and Mysticism in the Age of Information*, London: Serpent’s Tail, 2004.

19 Noble, *The Religion of Technology*.

20 Elspeth Whitney, ‘Paradise Restored: The Mechanical Arts From Antiquity Through the Thirteenth Century’, *Transactions of the American Philosophical Society* 80.1 (1990): <https://doi.org/10.2307/1006521>, 82.

21 Noble, *The Religion of Technology*, vii.

22 On a related note, Walter Benjamin argues that the history of Christianity is so entangled with that of capitalism that it is essentially indistinguishable from it. Moreover, he writes that ‘capitalism is a pure cult religion’ that provides an inescapable and ubiquitous understanding of meaning, value, purpose, and responsibility, which is always in service of its own reproduction. Walter Benjamin, *Toward the Critique of Violence: A Critical Edition*, Stanford: Stanford University Press, 2021 [1921], 90–91.

attempts at pure thought free from ‘corporeal distortion’²³ featured an idealisation of the truly disembodied knowledge of god. Such pure, disembodied thought could, according to Descartes, be approached in the abstractions of mathematics, geometry, and arithmetic. A subsequent significant Enlightenment figure is Leibniz, who we already encountered in chapter 1 because of his foundational contribution to modern-day computation. Leibniz pursued a totalised mathematical understanding of the world through the development of a binary code system, which he thought would constitute a view of life that approached that of god.²⁴ Nineteenth-century mathematician George Boole continued the development of this binary logic to produce calculable true statements. Boole was a devout Christian and his pursuit of ‘the ultimate laws of thought’²⁵ through binary algebra ‘is impossible to separate [from his] religious beliefs’.²⁶ Boolean algebra is still a central logic at work in contemporary algorithmic processes.²⁷

What can be recognised throughout the history of the religion of technology is the ‘belie[f] that human thought was mankind’s link with the divine and that a mathematical description of human mental processes was therefore at the same time a revelation of the mind of God’.²⁸ Whereas Noble is particularly interested in the convergence of Christian belief in transcendence through technological perfection, Erik Davis’ work complements this with examples from less institutionalised belief-systems, including the occult, New Age, and esoteric beliefs. Davis shows not only that technology is embedded in everyday spiritual practices of diverse kinds, but also that the idealisation of the spiritual realm continues to shape the perfection of knowledge for which information technologies strive, ‘provid[ing] launching pads for transcendence’ in ways that do not always follow strict doctrines.²⁹

The Technologies of Religion

Besides the role of religion and spirituality in the development of technology, technology also influences the development of religions. Philosopher and historian Brian Rotman shows the significance of the history of writing technologies for developments in Judaeo-Christian monotheisms. For example, the emergence of the scrolled format of early Jewish Torahs historically relies on the invention of writing technology, such as the preparation of animal skins into kosher parchment. This process produced long scrolls that afford a linear reading practice, and still influences the practice of Jewish Torah reading in Synagogues to this day. Later, Christian proselytism was supported by the emergence of bound books, which made religious texts more portable and specific passages more easily retrievable.³⁰ Protestantism’s focus

23 Noble, *The Religion of Technology*, 147.

24 Noble, *The Religion of Technology*.

25 George Boole, *An Investigation of the Laws of Thought*, London: Project Gutenberg, 2005 [1854], 317.

26 Desmond MacHale, *The Life and Work of George Boole: A Prelude to the Digital Age*, Cork: Cork University Press, 2014, 222.

27 Noble, *The Religion of Technology*.

28 Noble, *The Religion of Technology*, 146.

29 Davis, *Techgnosis*, 9.

30 Brian Rotman, *Becoming Beside Ourselves: The Alphabet, Ghosts, and Distributed Human Being*, Durham: Duke University Press, 2008.

on the accessibility of religious scripture for common people was subsequently facilitated by the invention of the printing press and the ‘universal, directly accessible privatization of the word it made possible’.³¹ In other words, the affordances of writing technologies throughout the ages in combination with their changing cultural contexts made possible new forms of religion, new spiritual practices, and new conceptions of god. Over time, the automation and standardisation of writing and publishing practices has contributed to more uniform religions that spread across large geographic distances and remain stable over long periods of time.³² These examples show that, in a very real way, emerging technologies ‘[invite] the creation of new gods’.³³ Of course, the establishment of a religion and the development of a new conception of god is a slow process and difficult to decipher in (close to) real-time. In the context of blockchain, this is something upon which future research will have to reflect. However, in this chapter, I do already point to some practices that could be understood in this light.

During the early years of the twentieth century, sociologist Georg Simmel made a distinction between religion and religiosity that is relevant to the contemporary entanglement of technology and (systems of) belief. He thought of religion as the objective institutional form and religiosity as the internal, subjective content: the relatively stabilised rules and organisational shape versus the personal experience of belief. He wrote that ‘religion does not create religiosity, but religiosity creates religion’.³⁴ Rotman’s religious technological practices can be understood as religiosity that over time created new religions. Significantly, Simmel saw that religiosity moved outside the strict confines of religion—something that became even more pronounced in the free-floating techno-spirituality of post-World War II Western countercultures described by Erik Davis.

Debates surrounding the definition of religion have occupied sociologists and anthropologists for decades, each ‘emphasiz[ing] different aspects of it, whether belief and ideas, or ritual, or feeling, or morality, or community’.³⁵ For the purposes of my analysis it is not necessary to go into more nuance than to say that I take religion to be a shared experience of sociality and purpose that is shaped by one or several more-than-human forces which, to some degree, remain inscrutable to humans. In the same way that specific religions attribute more-than-human force or religious purpose to (what from a Western perspective are thought of as) inanimate objects such as stones, statues, or celestial objects, technology joins their ranks

31 Rotman, *Becoming Beside Ourselves*, 136.

32 Rotman, *Becoming Beside Ourselves*.

33 David Porush, ‘Hacking the Brainstem: Postmodern Metaphysics and Stephenson’s *Snowcrash*’, *Configurations* 2.3 (1994), 554.

34 Georg Simmel, *Essays on Religion*, New Haven: Yale University Press, 1997 [1898-1918], 150. Simmel also reflected on the relationship between religiosity and science. Although he argues that religiosity and science do not create a conflict between each other—with which I would agree—he also argues that this is the case because they do not relate to each other at all. Each is entirely wrapped up in its own coherent worldview, and neither is capable of disassembling the other. See Simmel, *Essays on Religion*. This, I argue, is either beside the point—I am not concerned with one side establishing primacy over the other—or it raises questions: I wonder whether the experiences of belief in blockchain culture can be seen as forming a bridge between religiosity and scientific rationalism.

35 Jack David Eller, *Introducing Anthropology of Religion: Culture to the Ultimate*, New York: Routledge, 2021 [2007].

in these communities. While the cases I examine frequently draw on Christian imagery, I do not take this as proof of adherence to Christian faith. Rather, I interpret such references as expressions of religiosity more broadly—individual experiences of belief that may or may not align with institutionalized religion. Beyond their spiritual overtones, references to religious imagery and ritual serve a strategic purpose, helping to legitimise specific economic and political interests.³⁶

Theistic Conceptions of the Capacities of Artificial Intelligence and Blockchain

AI predates blockchain and hence more research has been done on its spiritual and religious framing. Rather than being a factual description of a technology, the term ‘artificial intelligence’ connotes an idea of the automated perfection of thought that exceeds human capacities, sometimes so much so that it approaches or even achieves the divine³⁷ as, for example, in the work of transhumanists Giulio Prisco³⁸ and Ray Kurzweil.³⁹ Such framings are an example of ‘theistic AI narratives’ in which ‘the deification of AI occurs within a scientific perspective on the universe and its contents’.⁴⁰ In this sense, AI is framed as the ‘God-like operator’ of the universe.⁴¹ The idea of theistic technology narratives serves as a counterpoint to common portrayals of technology as rational to the n^{th} degree, but does not address how that religiosity is scientifically, technologically, and rationally constituted, or vice versa. Shedding light on exactly how these seemingly separate realms overlap and co-constitute each other in the context of specific technologies reveals important aspects of the values and worldviews they support.⁴²

This is why the religiously informed rationalism of Descartes, Leibniz, and Boole is important. Among many other developments, their thought inspired the pursuit of a ‘thinking machine’ that would rise above human capacities, towards godly perfection.⁴³ Such an imagined artificial intelligence would not only expand human cognitive capacities, but also—fulfilling a truly Cartesian ideal—do away with humanity’s physical constraints and its earthliness. Rather,

36 David Pucheu, ‘Network, Cryptography, and the Next World to Come: An Everlasting American Techno Millenarianism’, *Zygon: Journal of Religion and Science* 59.1 (2024): 258–79.

37 Nicolas Spatola and Karolina Urbanska, ‘God-Like Robots: The Semantic Overlap Between Representation of Divine and Artificial Entities’, *AI & Society* 35.2 (2020): 329–41.

38 Giulio Prisco, *Tales of the Turing Church: Hacking Religion, Enlightening Science, Awakening Technology*, Self-published, 2020.

39 Ray Kurzweil, *The Age of Spiritual Machines: When Computers Exceed Human Intelligence*, Middlesex: Viking Penguin, 1999.

40 Beth Singler, ‘Blessed By the Algorithm’: Theistic Conceptions of Artificial Intelligence in Online Discourse’, *AI & Society* 35.4 (2020), 954, 952.

41 Transhumanist Giulio Prisco quoted in Singler, ‘Blessed By the Algorithm’, 953. Although slightly outside my discussion here, it should be clear that the spiritual utopianism of technology visionaries like Kurzweil hides its politics, such as the way in which AI is an ‘instrument of capital’. See Nick Dyer-Witford, Atle Mikkola Kjøsen, and James Steinhoff, *Inhuman Power: Artificial Intelligence and the Future of Capitalism*, London: Pluto Press, 2019, 3.

42 R. M. Geraci, ‘Apocalyptic AI: Religion and the Promise of Artificial Intelligence’, *Journal of the American Academy of Religion* 76.1 (2008): 138–66. <https://doi.org/10.1093/jaare/lfm101>.

43 Noble, *The Religion of Technology*, 148.

this intelligence would be immortal and consist of pure thought.⁴⁴ The echoes of the Cartesian thinking machine in AI are clear. In addition, I argue that this line of thought also influences the imaginary of ‘blockchain as a truth machine’ because of its concern with disembodied, universal knowledge. Although blockchain technology is not generally conceived of as a ‘thinking machine,’ it does pursue truth through rationalist processes that are imbued with religiosity. Just like AI, claims about the potential of blockchain technology are tied up with its capacity to handle information in new ways. Instead of being framed in terms of intelligence and featured in debates about the degree to which that intelligence is capable of replicating or even surpassing human intelligence in a way that is godlike, blockchain is instead framed as a source of truth that takes on religious dimensions.

Blockchain is both technically and culturally connected to religious ideas and practices and builds on the religiosity of the communities from which it emerged. Before blockchain appeared, groups such as the cypherpunks and extropians promoted crypto-anarchist visions grounded in what David Pucheu identifies as ‘techno-millennialism’ and ‘eschatological expectations.’⁴⁵ Cryptography, in their view, offered a means of transcendence and liberation from human and institutional constraint. These imaginaries continue to shape Bitcoin culture. Sociologist Nigel Dodd notes that many of its proponents defend the currency ‘with quasi-religious zeal,’ seeing in the technology beneath it something distinctly ‘godlike.’⁴⁶ This godlike character rests upon blockchain’s capacity to ‘remember every discrete event.’⁴⁷ Moreover, ‘its memory is *infallible*’—i.e., a blockchain is all-knowing and its memory eternal like that of a god.⁴⁸ Significantly, blockchain transforms the centralised gods of monotheistic religions into a ‘distributed god’⁴⁹ that often takes on the personification of the mysterious character of Satoshi Nakamoto.⁵⁰

Artificial intelligence also informs the continuing evolution of religion and religious practices. For example, the explosive growth in the capabilities of AI in recent years means it is now capable of writing sermons, raising serious ethical questions regarding the implications of artificially produced teachings.⁵¹ AI is also already used in spiritual robots, which make possible new forms of religious interactions and practices.⁵²

44 Noble, *The Religion of Technology*. AI is anything but immaterial, from the rare earth minerals used in hardware and the architecture of server farms, to the devices used by end-users and the energy used per query posed. See Kate Crawford, *The Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*, New Haven: Yale University Press, 2021.

45 Pucheu, ‘Network, Cryptography, and the Next World to Come: An Everlasting American Techno Millenarianism,’ 16.

46 Nigel Dodd, ‘The Social Life of Bitcoin,’ *Theory, Culture & Society* 35.3 (2018), 49.

47 Dodd, ‘The Social Life of Bitcoin,’ 49.

48 Dodd, ‘The Social Life of Bitcoin,’ 49, emphasis in original.

49 Dodd, ‘The Social Life of Bitcoin,’ 49.

50 Mariam Humayun and Russell W. Belk, ‘“Satoshi is Dead. Long Live Satoshi”: The Curious Case of Bitcoin’s Creator,’ *Consumer Culture Theory* 19.1 (2018): 19–35. <https://doi.org/10.1108/s0885-211120180000019002>.

51 Justin E. Lane, *Understanding Religion Through Artificial Intelligence: Bonding and Belief*, London: Bloomsbury Academic, 2021.

52 Pauline Hope Cheong, ‘Religion, Robots and Rectitude: Communicative Affordances for Spiritual

Blockchain's quasi-religious quality shows itself in the rituals that have emerged around it. Such ritualised practices contribute to a collective faith regarding the capacity of the technology to be the harbinger of a better world.⁵³ Recurring events such as Bitcoin Pizza Day—an annual celebration in remembrance of the day in 2010 on which an early adopter spent 10,000 bitcoins on two pizzas, an amount that would today equate well over 600 million euros⁵⁴—repeated procedures that are thought to bring good fortune performed by crypto-traders, and community gatherings such as meet-ups and conferences offer ceremonial rituals and places of worship that replicate religious boundary work and community building exercises.⁵⁵

Obfuscation, or How Technology Thrives on Belief

Besides the religious dimensions of perfection, the element of belief in blockchain culture also contributes to obfuscation, particularly as part of the entanglement of technological developments with non-religious and non-spiritual forms of belief. Such technological mysticism relies on a widespread belief that technologies are beneficial to society on the one hand, and on the mystification of how those technologies actually work on the other hand. This mysticism positions technology as the ultimate, unquestioned remedy for social problems. The process of mystification unfolds across multiple dimensions: at the technical level, through opaque infrastructures; at the discursive level, through language that casts technological progress in magical or transcendent terms; at the cultural level, through the elevation of technology to an object of fascination and desire; and at the economic level, through speculative investment practices that valorise ambition and reward the promise of the impossible.

The mysticism that surrounds (big) data provides an insightful background to my analysis of blockchain mysticism because of its particular concern with objectivity. Dataism is the belief in the virtue and objectivity of data,⁵⁶ and it can be understood as a precursor to the belief in blockchain as the techno-rational saviour of society. Although dataism is generally thought of as a secular belief, the combination of this section and that which precedes it will provide a framework to understand the religiosity of mysticism in blockchain culture.

Knowledge and Community', *Applied Artificial Intelligence* 34.5 (2020): 412-31.

53 Sandra Faustino, Inês Faria, and Rafael Marques, 'The Myths and Legends of King Satoshi and the Knights of Blockchain', *Journal of Cultural Economy* 15.1 (2021): 1-14. <https://doi.org/10.1080/17530350.2021.1921830>.

54 This is based on the exchange rate on 29 March 2024.

55 Faustino, Faria, and Marques, 'The Myths and Legends of King Satoshi and the Knights of Blockchain'; Seung Cheol Lee, 'Magical Capitalism, Gambler Subjects: South Korea's Bitcoin Investment Frenzy', *Cultural Studies* 36.1 (2020), 1-3.

56 José van Dijck, 'Datafication, Dataism and Dataveillance: Big Data Between Scientific Paradigm and Ideology', *Surveillance & Society* 12.2 (2014): 197-208.

Technological Mysticism: The Element of Belief in Technology Culture

Technological mysticism is a ‘belief in the universal efficacy of technology’ and functions through the obfuscation of the mechanisms through which technology operates.⁵⁷ Technological mysticism rests on the belief that technologies possess a kind of autonomy—that they function beyond human control and thus hold the potential to accomplish remarkable feats. This notion tends to surface in three distinct forms. The first frames technology as neutral and objective, suggesting that it stands apart from the human intentions and contexts that shape its use. This is exemplified by the common rhetoric used in the US gun lobby that ‘guns don’t kill people, people kill people.’ A second variation is the belief that social advancement is propelled by technological innovation. This reflects a technological determinist perspective, closely tied to the discourse of progress that characterised modernity. According to this version of technological mysticism—despite the fact that technologies are human-made—they are what causes change or, as we saw above, transcendence. A third perspective depicts technology as something that escapes human control. Linked to Romantic thought, this view is epitomized by Mary Shelley’s *Frankenstein*, where a human invention acquires autonomy and turns against its creator. Such narratives resonate with dystopian anxieties and fears about the kinds of futures that emerging technologies might bring about.⁵⁸

These three strands of technological mysticism continue to circulate widely in contemporary technology culture. The first—technology imagined as neutral—appears in ‘the promise of algorithmic objectivity,’ a discourse common among digital media corporations that present themselves as impartial carriers of information rather than as institutions exercising editorial power.⁵⁹ It persists in the language used by search engine companies regarding the results they show for racist search queries or by social media companies regarding the neutrality of the selection of posts shown in their feeds, especially during election cycles.⁶⁰ Second, the notion of progress still underpins narratives that cast technology as the primary engine of social transformation. For example, the previous chapter featured rhetoric on the emancipatory effects of the tokenisation of nature. The broader culture of technological solutionism that dominates contemporary Western cultures is part of this type of mysticism.⁶¹ Finally, the surge of enthusiasm around generative artificial intelligence in 2023–24 revived long-standing anxieties about AI escaping human control and becoming a force that might one day threaten

57 William A. Stahl, *God and the Chip: Religion and the Culture of Technology*, Waterloo: Wilfrid Laurier University Press, 1999, 125–26.

58 Stahl, *God and the Chip*.

59 Tarleton Gillespie, ‘The Relevance of Algorithms’, in Tarleton Gillespie, Pablo J. Boczkowski, and Kirsten A. Foot (eds) *Media Technologies: Essays on Communication, Materiality, and Society*, Cambridge, MA: MIT Press, 2014, 168.

60 Safiya Umoja Noble, *Algorithms of Oppression: How Search Engines Reinforce Racism*, New York: New York University Press, 2018; Tarleton Gillespie, *Custodians of the Internet: Platforms, Content Moderation, and the Hidden Decisions That Shape Social Media*, New Haven: Yale University Press, 2018.

61 Evgeny Morozov, *To Save Everything, Click Here: The Folly of Technological Solutionism*, New York: PublicAffairs, 2014.

humanity's very survival.⁶² This is a contemporary example of the Romantic view on technology as out of human control.

Emerging technologies are imbued with the capacity to fundamentally change socio-cultural systems through processes that are unfathomable to humans.⁶³ The mystification of technology rests in part on its fetishisation: 'the habit humans have of endowing real or imagined objects or entities with self-contained, mysterious, and even magical powers to move and shape the world in distinctive ways'.⁶⁴ Emerging technologies often appear to operate as if by magic—generating outcomes that obscure the link between cause and effect—yet are nonetheless regarded as rational despite this opacity.⁶⁵ Media theorist Wendy Hui Kyong Chun describes the seemingly magical results that appear from clicking a button in a software interface as 'sourcery'.⁶⁶ The button itself is just a collection of pixels on a screen, yet, as users of software, we imbue it with the capacity to produce the results we intended. The interface hides the layers of computational work that go on in software and hardware beneath the visible surface. Chun thinks of '[c]ode as a medium in the full sense of the word. As a medium, it channels the ghost that we imagine runs the machine'.⁶⁷ This process of mystification depoliticises technologies, obfuscating their social and material implications.⁶⁸

Blockchain is similarly often imbued with mysticism and portrayed as in some way beyond the range of human capacities or understanding—a practice characterised by the 'substitution of technical, ideological, and socioeconomic discussion of financial technologies for a foregrounding of their fantastic, magical, unknowable qualities'.⁶⁹ Rather than the political acts of particular humans, blockchain mysticism cements the technology as an unstoppable force, always just beyond our human reach.⁷⁰ On the one hand, this mysticism features in

62 Timnit Gebru and Émile P. Torres, 'The Tescreal Bundle: Eugenics and the Promise of Utopia Through Artificial General Intelligence', *First Monday* 29.4 (2024): n.p.

63 Braden Allenby, 'The Industrial Ecology of Emerging Technologies: Complexity and the Reconstruction of the World', *Journal of Industrial Ecology* 13.2 (2009): 168–83.

64 David Harvey, 'The Fetish of Technology: Causes and Consequences', *Macalester International* 13.1 (2003), 3.

65 Alfred Gell, 'The Technology of Enchantment and the Enchantment of Technology', in Jeremy Coote and Anthony Shelton (eds) *Anthropology, Art and Aesthetics*, Oxford: Oxford University Press, 1992, 40–64; Suzanne L. Thomas, Dawn Nafus, and Jamie Sherman, 'Algorithms as Fetish: Faith and Possibility in Algorithmic Work', *Big Data & Society* 5.1 (2018): 1–11. <https://doi.org/10.1177/2053951717751552>. Frankfurt School philosopher Herbert Marcuse connects rationality and mystification in another way. He writes that rational technical production processes are the embodiment of the dominant ideology of society, and as such rationality 'becomes the most effective vehicle of mystification'. See Herbert Marcuse, *One-Dimensional Man: Studies in the Ideology of Advanced Industrial Society*, London: Routledge, 2002 [1964], 194. The tokenisation of nature discussed in chapter 2 is an example of the way in which ideology functions through rational technology while it is portrayed as objective and neutral.

66 Wendy Hui Kyong Chun, 'On 'Sourcery,' or Code as Fetish', *Configurations* 16.3 (2008), 299.

67 Chun, 'On 'Sourcery,' or Code as Fetish', 310.

68 Chun, 'On 'Sourcery,' or Code as Fetish'.

69 Martin Zeilinger, 'Contemporary Art Between Algomysticism and Fintech Activism', in Ine Gloerich, Geert Lovink, and Patricia de Vries (eds) *Moneylab Reader #2: Overcoming the Hype*, Amsterdam: Institute of Network Cultures, 2018, 76.

70 Zeilinger, 'Contemporary Art Between Algomysticism and Fintech Activism'.

memetic bottom-up portrayals, like the Bitcoin Wizard meme on Reddit that solidified the unofficial slogan ‘magic internet money’ for the cryptocurrency, or the portrayal of Ethereum founder Vitalik Buterin as a ‘wealth-conjuring sage’.⁷¹ On the other hand, it also characterises the practices of more institutional or powerful actors, for example the promises of mythical proportions made by blockchain start-ups,⁷² for example Highvibe Network, which promised that their system facilitates the expansion of human consciousness,⁷³ or the portrayals of the technology as having superhuman qualities on major cryptocurrency-themed news platforms such as Coin Telegraph.

The current system of financing innovation virtually ensures the persistence of technological mysticism. Emerging technologies are routinely depicted as pushing beyond existing limits and as cures for society’s problems.⁷⁴ A narrow yet powerful network of billionaire tech philanthropists, together with the profit-driven logic of venture capital, plays a central role in keeping these narratives alive.⁷⁵ Portraying a new technology as the solution to a real societal problem can be a way to convince funders they will see good returns on their investments.⁷⁶ Claims of mythical dimensions have the potential to yield equally impressive funds. Such widespread mystical portrayals support a societal belief in the power of technology to produce radically different (utopian or dystopian) worlds—e.g., the contemporary unquestionable faith in the societal benefits of algorithmic rationality.⁷⁷ What currently passes for the magical or mystical sets the horizon toward which technological development aims.⁷⁸ Sustaining this orientation is a mythic, idealized image of technology that projects perfection and, in doing so, continues to propel innovation.

Beyond the opaqueness that such fetishism produces, many corporately owned systems are also characterised by a technical opaqueness that results from their ‘blackboxed’ nature. These technologies are, in a very real sense, ‘mysterious to the user’⁷⁹ and often ‘evok[e] mystic metaphors’.⁸⁰ Through the increasing digitalisation of everyday life, blackboxes perform

71 Zeilinger, ‘Contemporary Art Between Algomysticism and Fintech Activism,’ 79.

72 Zeilinger, ‘Contemporary Art Between Algomysticism and Fintech Activism’.

73 Highvibe Network, <https://www.highvibe.network/>.

74 Morozov, *To Save Everything, Click Here*.

75 Sangeetha Chandrashekeran and Jathan Sadowski, ‘Post-Capitalist Techno-Futures: Beyond Instrumental Utopianism’, in Samuel Alexander, Sangeetha Chandrashekeran, and Brendan Gleeson (eds) *Post-Capitalist Futures: Paradigms, Politics, and Prospects*, Singapore: Palgrave Macmillan, 2022, 107-16.

76 Janet Haven and danah boyd, ‘Philanthropy’s Techno-Solutionism Problem’, *Knight Foundation*, 2020, <https://knightfoundation.org/philanthropys-techno-solutionism-problem/>; Thomas, Nafus, and Sherman, ‘Algorithms as Fetish: Faith and Possibility in Algorithmic Work’; David Graeber, ‘Fetishism as Social Creativity: Or, Fetishes Are Gods in the Process of Construction’, *Anthropological Theory* 5.4 (2005): 407-38.

77 Lee Worth Bailey, *The Enchantments of Technology*, Urbana: University of Illinois Press, 2005.

78 Alfred Gell, ‘Technology and Magic’, *Anthropology Today* 4.2 (1988): 6-9. <https://doi.org/10.2307/3033230>.

79 Adrienne Mayor, *Gods and Robots: Myths, Machines, and Ancient Dreams of Technology*, Princeton: Princeton University Press, 2020, 220.

80 Patricia de Vries, *Algorithmic Anxiety in Contemporary Art: A Kierkegaardian Inquiry Into the Imaginary of Possibility*, PhD diss., Erasmus University Rotterdam, Rotterdam, 2020, 123.

their mysterious functions in more and more aspects of social life. Due to their perceived complexity and the marketing strategies used to promote them, which focus on results while obscuring inner workings,⁸¹ blockchains—although technically open source and designed for transparency—can in practice function as blackboxes.

Belief in the Objectivity and Societal Benefits of Data

Their inner workings may be unverifiable due to complex algorithmic processes, yet information technologies are celebrated for what is perceived as unmatched scientific precision and fetishised for seeming to operate ‘outside the scope of present scientific knowledge’.⁸² Media scholar José van Dijck defines such dataism as ‘a belief in a new gold standard of knowledge about human behavior’,⁸³ a worldview grounded in the assumption that quantification itself guarantees objectivity.⁸⁴ Such conviction underpins both the expansion of datafication and the rise of so-called big data.⁸⁵ Through pervasive mythologies, data come to appear natural and politically neutral—‘higher form of intelligence and knowledge that can generate insights that were previously impossible, with the aura of truth, objectivity, and accuracy’.⁸⁶ Campolo and Crawford identify this dynamic as ‘enchanted determinism,’ warning that such technology-based knowledge imaginaries generate ‘power without responsibility’.⁸⁷

Dataism not only consists of a belief in the objectivity of data, but also of a belief in the virtue of datafication—i.e., its capacity to ‘fulfil human beings’ collective potential in some tran-

81 Wenyu (Derek) Du, et al., ‘Affordances, Experimentation and Actualization of Fintech: A Blockchain Implementation Study’, *The Journal of Strategic Information Systems* 28.1 (2019): 50-65. <https://doi.org/10.1016/j.jsis.2018.10.002>.

82 Alexander Campolo and Kate Crawford, ‘Enchanted Determinism: Power Without Responsibility in Artificial Intelligence’, *Engaging Science, Technology, and Society* 6.1 (2020): <https://doi.org/10.17351/ests2020.277>, 3.

83 van Dijck, ‘Datafication, Dataism and Dataveillance,’ 201.

84 The term was coined by New York Times columnist David Brooks, who described it as the belief ‘that everything that can be measured should be measured; that data is a transparent and reliable lens that allows us to filter out emotionalism and ideology; that data will help us do remarkable things’. See David Brooks, ‘The Philosophy of Data’, *The New York Times*, 2013, <https://www.nytimes.com/2013/02/05/opinion/brooks-the-philosophy-of-data.html>, 23. In 2018, author and public intellectual Yuval Noah Harari popularised the term when he presented about it at the World Economic Forum in Davos. See Yuval Noah Harari, *Will the Future be Human?*, 2018. <https://www.youtube.com/watch?v=hL9uk4hKyg4>. Philosopher Miriam Rasch later added to this the idea that the goal of dataism is to banish friction. Friction signifies unpredictability and as such undermines the belief in the capacity of data to measure well and do all the things it is believed to be capable of. See Miriam Rasch, *Fricție: Ethiek in Tijden Van Dataïsme*, Amsterdam: De Bezige Bij, 2020.

85 van Dijck, ‘Datafication, Dataism and Dataveillance’.

86 danah boyd and Kate Crawford, ‘Critical Questions for Big Data: Provocations for a Cultural, Technological, and Scholarly Phenomenon’, *Information, Communication & Society* 15.5 (2012): <https://doi.org/10.1080/1369118X.2012.678878>, 663; Roland Barthes, *Mythologies*, New York: Hill and Wang, 2013 [1957]; Os Keyes, Zoë Hitzig, and Mwenza Blell, ‘Truth From the Machine: Artificial Intelligence and the Materialization of Identity’, *Interdisciplinary Science Reviews* 46.1-2 (2021): 158-75.

87 Campolo and Crawford, ‘Enchanted Determinism: Power Without Responsibility in Artificial Intelligence,’ 1.

scendental way',⁸⁸ and features in the enchanted determinism of data-based solutionism.⁸⁹ By framing it as capable of bringing about future utopias, dataism justifies both the ongoing application and the technological expansion of big data and the depoliticisation of the epistemology such data establishes.⁹⁰ Beliefs in the inherent objectivity of data and the social virtues of datafication obscure the politics of the (design) choices that are made in the process of developing, implementing, and continuously updating datafying technologies.

Although the concept of dataism is about belief, that belief is generally understood to be neither religious nor spiritual. Van Dijck calls dataism a 'secular belief'⁹¹ that centres an unproven mythical story that nonetheless informs actions and thus has the capacity to shape futures, but does not invoke deities or spiritual forces. Whereas a lack of proof and the possibility for doubt are inherent in religious belief, dataism performs a logical somersault in its belief system. Dataism is the belief in the objectivity and virtue of data, which is provable by data. The belief in the objectivity of data thus makes the necessity of belief obsolete. Proponents of datafication do not frame the need for data in terms of belief, and likely do not conceive of it as such. Consequently, the belief in the virtue and objectivity of data is often not readily perceptible as a belief.

In the beginning was the Protocol, and the Protocol was with God

Religious narratives about blockchain began before it was even invented. Key foundations for blockchain mysticism and religiosity were laid by the Cypherpunks in the 1990s. One striking example can be found in the work of Cypherpunk Nick Szabo. In his influential essay, the mechanisms imagined for digital currency are cast in terms that evoke the instruments of a divine, all-knowing power. Such a godly tool *needs* to be out of human control or conception in order to work. This pre-history provides a foundation for the subsequent framing of blockchain as a holy force for good and the portrayal of its pseudonymous inventor Satoshi as a neutral, all-knowing god.

The God Protocols: A Prehistory of Blockchain

A major factor in the emergence of blockchain technology is the influential work of a group of individuals known as cypherpunks:⁹² about 700 people organised on the Cypherpunks mailing

88 Nick Couldry and Ulises A. Mejias, *The Costs of Connection: How Data is Colonizing Human Life and Appropriating it for Capitalism*, Stanford: Stanford University Press, 2020, 17.

89 Campolo and Crawford, 'Enchanted Determinism: Power Without Responsibility in Artificial Intelligence'; Christian Fuchs, 'Beyond Big Data Capitalism, Towards Dialectical Digital Modernity: Reflections on David Chandler's Chapter', in David Chandler and Christian Fuchs (eds) *Digital Objects, Digital Subjects: Interdisciplinary Perspectives on Capitalism, Labour and Politics in the Age of Big Data*, London: University of Westminster Press, 2019, 43-51.

90 Christopher Lee Adamczyk, 'Communicating Dataism', *Review of Communication* 23.1 (2023): 4-20; Daniel Faltesek, 'Big Argumentation', *tripleC: Communication, Capitalism & Critique* 11.2 (2013): 402-11.

91 van Dijck, 'Datafication, Dataism and Dataveillance,' 197.

92 Lana Swartz, 'What Was Bitcoin, What Will it be? The Techno-Economic Imaginaries of a New Money

list from the early 1990s onwards. Although many participants posted anonymously through the list's remailing functionality or used 'nyms'⁹³—pseudonyms—the identities of some of its most central contributors are known and some are famous for their influence on wider internet culture. These include Intel engineer and author of an early cypherpunk FAQ Timothy May, computer scientist researching digital currencies Nick Szabo, computer scientist and receiver of Satoshi's first-ever Bitcoin transaction Hal Finney, author of *A Cypherpunk's Manifesto* Eric Hughes, founder of the Electronic Frontier Foundation John Gilmore, and WikiLeaks founder Julian Assange.⁹⁴ The Cypherpunks' mailing list would later give rise to several spin-off mailing lists that hosted overlapping communities. Perhaps the most famous spin-off is the Cryptography mailing list. This is where Satoshi would post the Bitcoin whitepaper on October 31, 2008.

Rather than organising society through institutions and laws, the cypherpunks were adamant that technological systems—especially digital technology that deals with information, interaction, and exchange—can be designed to change the character of sociopolitical reality. As such, cypherpunks do not limit themselves to discussions on a mailing list, but consider that they have an obligation to also actively write code to pursue the futures for which they argue.⁹⁵ The cypherpunks' interests ranged far beyond a digital currency that supports the privacy of individuals using it, but in the pursuit of a social life free from government control and surveillance, such a currency was deemed to be of the utmost importance.⁹⁶

Of the cypherpunks listed above, Nick Szabo is one of the most influential figures for the development of a digital currency. Szabo's work is clearly recognisable as foundation for the Bitcoin whitepaper, and rumours that Szabo is actually Satoshi continue to resurface throughout the years, despite Szabo's refutations. In a 1997 text titled 'The God Protocols,' Szabo outlined how an anonymous system of exchange might operate without relying on a central authority. He introduced it in the following terms:

Imagine the ideal protocol. It would have the most trustworthy third party imaginable—a diety [sic] who is on everybody's side. All the parties would send their inputs to God. God would reliably determine the results and return the outputs. God being the ultimate in confessional discretion, no party would learn anything more about the other parties' inputs than they could learn from their own inputs and the output.⁹⁷

Technology', *Cultural Studies* 32.4 (2018): 623-50.

93 Craig Jarvis, 'Cypherpunk Ideology: Objectives, Profiles, and Influences (1992–1998)', *Internet Histories* 6.3 (2022): 315-42.

94 Adam Hayes, 'The Socio-Technological Lives of Bitcoin', *Theory, Culture & Society* 36.4 (2019): 49-72. <https://doi.org/10.1177/0263276419826218>; Eric Hughes, *A Cypherpunk's Manifesto*, Satoshi Nakamoto Institute, 1993.

95 Timothy C. May, *The Cyphernomicon: Cypherpunks FAQ and More*, Version 0.666, Satoshi Nakamoto Insitute, 1994.

96 Finn Brunton, *Digital Cash: The Unknown History of the Anarchists, Utopians, and Technologists Who Created Cryptocurrency*, Princeton: Princeton University Press, 2020.

97 Nick Szabo, *The God Protocols*, Satoshi Nakamoto Institute, 1997.

The central difficulty preventing the actual emergence of digital cash was how to confirm that digital currency was not being used twice—when exactly its digital form made duplication trivial—while keeping users unidentifiable. This is what is known as the double spending problem. In ‘The God Protocols,’ Szabo sketched a speculative fix to overcome it, and more than a decade later, Satoshi Nakamoto introduced the Proof of Work consensus mechanism, finally making fraud-proof, decentralised and anonymous digital cash feasible.⁹⁸ By doing so, the cypherpunks’ dream of anonymous digital cash moved from speculation to practice. Szabo’s text can be seen as an important precursor that both paved the way for Bitcoin and helped foster the atmosphere of reverence that has since grown around blockchain technologies.

In 1997, the god protocols were still only a thought experiment, and Szabo expressed clear unease about having to continue to depend on fallible people and the institutions they build. He portrays this dependence as a kind of misplaced faith: ‘Alas, in the our [sic] temporal world we deal with humans rather than deities. Yet, too often we are forced to treat people in a nearly theological manner, because our infrastructure lacks the security needed to protect ourselves.’⁹⁹ Szabo frames belief itself as unavoidable: one must either ascribe divine-like integrity to other people without evidence or entrust technology with the role of an all-seeing, incorruptible arbiter. This rhetorical move positions blockchain not just as a technical fix but as a substitute deity—an imagined guarantor of purity beyond human corruption. In doing so, Szabo contributes to a broader mythologising of technology, where machines are cast as more trustworthy than people and should be used to order their messy social relations.

Saints and Sacred Texts in an Emerging Techno-Religion

For many, Satoshi’s Bitcoin whitepaper holds near-scriptural status and is celebrated as a blueprint for cryptocurrencies and as a vision for reorganising society as a whole.¹⁰⁰ For example, the Church of Bitcoin¹⁰¹ is an online community for those that want ‘to free the world from the oppression that is currently enabled by government and central bank control over exchange’.¹⁰² The church states that their ‘scripture is short and simple. The whitepaper written by our prophet Satoshi Nakamoto is our core document. In cryptography we trust. We embrace open-source philosophy and freedom of information’.¹⁰³

98 In practice, Bitcoin no longer resembles key aspects of Satoshi’s vision. The network shows clear signs of centralisation, user anonymity is frequently compromised, and its extreme volatility—alongside its marginal use in ordinary retail—keeps it from functioning as a digital version of cash.

99 Szabo, *The God Protocols*.

100 Tobias A. Huber, and Didier Sornette, ‘Boom, Bust, and Bitcoin: Bitcoin-Bubbles as Innovation Accelerators’, *Journal of Economic Issues* 56.1 (2022): 113–36. <https://doi.org/10.1080/00213624.2022.2020023>.

101 The Church of Bitcoin was founded in 2017 in Vermont, the United States, and lists 106 members on its website. As with any registered religious organisation in the United States, donations to the Church of Bitcoin are tax deductible Church of Bitcoin, <https://churchofbitcoin.org/>.

102 Church of Bitcoin, <https://churchofbitcoin.org/>.

103 Church of Bitcoin, <https://churchofbitcoin.org/>.

Images that portray Satoshi as a holy figure abound in the online Bitcoin community. For example, NFT artist Robness's work PEPESHINICHI portrays mathematician Shinichi Mochizuki—who is one of the many people rumoured to be Satoshi—as a religious icon.¹⁰⁴ The Mochizuki-as-Satoshi-as-Jesus figure takes on the visual language of Orthodox Christianity: a blessing hand, a sacred book. Yet, the book bears the Bitcoin logo rather than the words Holy Bible. Through this imagery, Satoshi is cast as a messianic teacher, with the whitepaper elevated to the status of scripture for a community of believers. The reference to Eastern Orthodoxy is significant: Bitcoin Maximalism—which will feature in more detail in the next section—in particular is often thought of in terms of orthodoxy and tradition because its adherents advocate for remaining as true as possible to the original technical set-up of Bitcoin.¹⁰⁵ In Figure 8, we see a meme portraying Satoshi as Jesus Christ battling a devil figure labeled with the term fiat money—the state-controlled form of money that Satoshi intended to circumvent with Bitcoin. In the meme in Figure 9, Satoshi is portrayed in the biblical scene where Jesus expels the merchants from the temple, who he accuses of turning it into a 'den of thieves' (Matthew 21: 12–13). In the meme, Satoshi takes on the role of Jesus, while the thieving merchants are turned into bankers from Citi, Wells Fargo, JP Morgan Chase, Goldman Sachs, Dexia, and Barclays—some of the most powerful financial institutions, many of whom were key actors in the 2008 financial crash. These memes depict Satoshi as a divine saviour, a strong hero that makes society better, while the icon—of which other versions with other rumoured Satoshis also exist—portrays Satoshi as the keeper of wisdom and blessing of followers.

The mythology around Satoshi is heightened by his disappearance. The Bitcoin wallet attributed to Satoshi has remained untouched since the cryptocurrency's early days, and the creator's identity is still unknown. This withdrawal echoes the Christian god who, after creating the world, receded from direct human contact. In the vacuum left behind, others have taken on quasi-prophetic roles. Tech entrepreneur Andreas Antonopoulos and investor Roger Ver, for example, have each been labelled 'Bitcoin Jesus' for their efforts to spread the message of Bitcoin.¹⁰⁶ This kind of missionary zeal expresses confidence in Bitcoin's promise to improve the world but also the financial stakes of those whose fortunes are tied to its survival.¹⁰⁷

104 Robness, *Pepeshinichi*, 2022. <https://pepe.wtf/asset/PEPESHINICHI>. The NFT is a contribution to a series of cards called Rare Pepes that iterate on the memetic Pepe the frog character, hence Mochizuki's green skin colour. Although the character of Pepe was famously politicised as the symbol of the alt-right ideology culminating around Donald Trump's 2016 election as president of the United States, in blockchain culture references to Pepe seem to have a more general meaning that is not overtly political.

105 Bruno Campos Cardoso, 'Towards Hyperbitcoinization: Bitcoin Maximalism as Speculative Fiction', in Matan Shapiro (ed) *Crypto Crowds: Singularities and Multiplicities on the Blockchain*, New York: Berghahn Books, 2024, 23–39.

106 Faustino, Faria, and Marques, 'The Myths and Legends of King Satoshi and the Knights of Blockchain'; Harjani, Ansuya, 'Meet 'Bitcoin Jesus,' a Virtual Currency Millionaire', *CNBC*, 2013, <https://www.cnn.com/2013/12/02/meet-bitcoin-jesus-a-virtual-currency-millionaire.html>.

107 David Golumbia, 'Zealots of the Blockchain', *The Baffler* 38 (2018): 102–11. <https://doi.org/10.2307/26406449>.



Fig. 8. Satoshi portrayed as a Christ-like figure

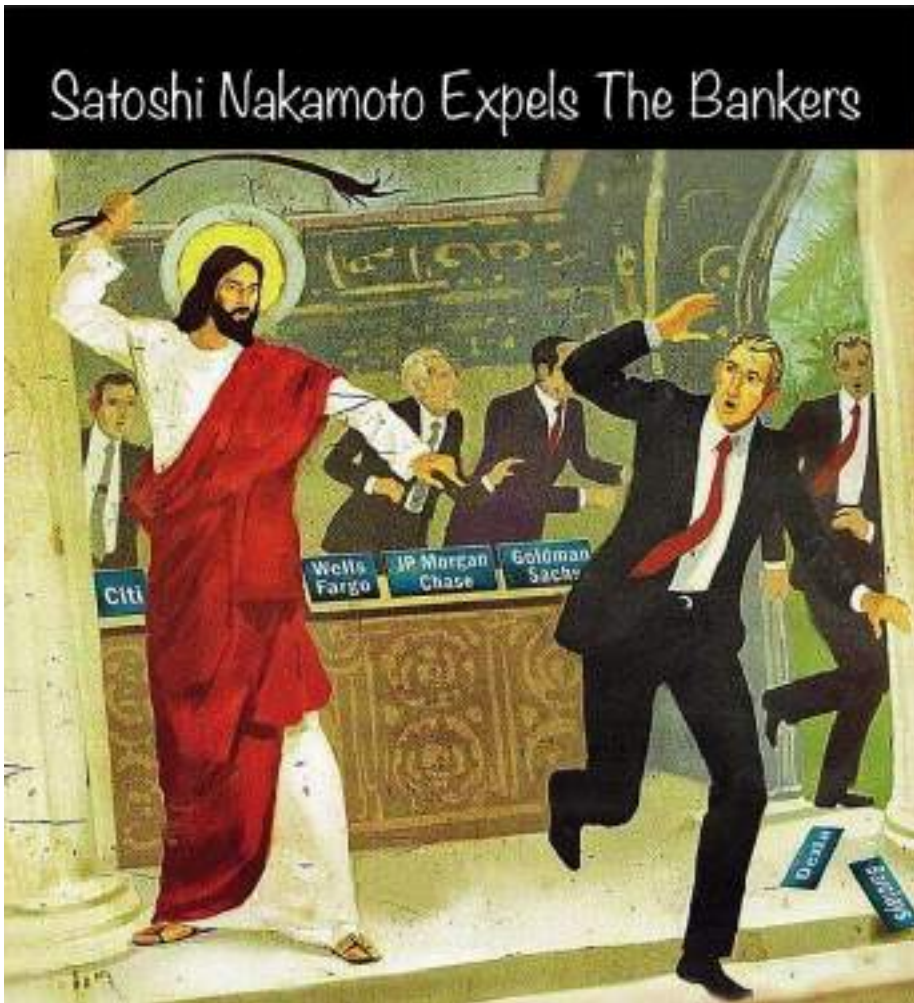


Fig. 9. Satoshi as Jesus expelling the merchants from the temple.

As with any religious scripture, various corners of the Bitcoin community have debated since Satoshi's disappearance what is the true meaning of the whitepaper and whether changing times warrant updated interpretations and new code, or whether the original interpretation of the text should be protected. One of the largest ongoing debates in the Bitcoin community is about the size of the blocks of its blockchain, and it constitutes a key site for expressions of reverence for Satoshi and of the sanctification of Bitcoin.¹⁰⁸ The question is: should the original Bitcoin blockchain as coded by Satoshi, with its quite limited block-size, be adhered to even as the user-base of Bitcoin grows? If the answer to this question is yes, this would in

108 Humayun and Belk, "Satoshi is Dead. Long Live Satoshi": The Curious Case of Bitcoin's Creator'; Faustino, Faria, and Marques, 'The Myths and Legends of King Satoshi and the Knights of Blockchain'.

practice mean that only a very limited number of transactions can be verified per block, ultimately relegating Bitcoin to function more like a store of value that people aim to accumulate than a medium of exchange that people use on a day-to-day basis. If the answer is no, and the block-size is to be expanded in order to allow for more transactions per block, this would mean that more individuals would be capable of transacting their bitcoins in real-time, even for small everyday purchases. Some argue that this answer would ultimately ‘fulfil[...] the original promise of Bitcoin as “Peer-to-Peer Electronic Cash”¹⁰⁹ as Satoshi envisioned in the whitepaper, even if it means making changes to the code Satoshi wrote. The latter interpretation of Satoshi’s true meaning—following the whitepaper rather than the code—has led to several hard forks in Bitcoin’s history, the most popular of which is Bitcoin Cash.¹¹⁰ Over time, different communities with irreconcilable interpretations of the Bitcoin whitepaper have emerged, each building their own ‘forks of the original Bitcoin source code, which represent the sacred object of Bitcoin’.¹¹¹

Key aspects of Bitcoin culture are performed through a religious lens: a hidden creator, revered messengers, and a sacred text. Whether these comparisons are intended literally or not is less important than the effect they produce. Such enactments of religiosity give the community a theistic origin story that frames Bitcoin as both divinely authored and technologically sanctified, shaping its cultural norms regardless of the sincerity of belief. It is thus not surprising that the fundamental change to the technology that the launch of Ethereum represented caused quite a shake-up in the community. This change is the topic of the next section, and the Bitcoin’s community’s response will follow subsequently.

Blockchains as Machines of a Religious Kind of Truth

When Ethereum added the possibility to execute smart contracts via its blockchain,¹¹² the technology burst into countless new domains. Talk of a blockchain revolution dominated blockchain-related articles and books between 2016–2018.¹¹³ A recurring theme amidst visions of blockchain revolution is the technology’s capacity for the establishment of truth about much more than just currencies. Like in other parts of blockchain culture, here too, belief and rationalism co-constitute the technology.

109 BitcoinCash, <https://bitcoincash.org/>.

110 At the time of writing on April 1, 2024, Bitcoin Cash is listed as the fourteenth largest cryptocurrency in terms of market capitalisation, while Bitcoin itself is listed at number one CoinMarketCap, <https://coinmarketcap.com/>.

111 Huber and Sornette, ‘Boom, Bust, and Bitcoin,’ 123.

112 Smart contracts are a key feature of contemporary blockchain that is also based on the work of Nick Szabo. Szabo described smart contracts as ‘computerized transaction protocol[s] that execut[e] the terms of a contract.’ See Nick Szabo, *Smart Contracts*, Satoshi Nakamoto Institute, 1994. In even less technical terms, smart contracts are agreements between two or more parties that are automatically enforced.

113 See e.g. Don Tapscott and Alex Tapscott, *Blockchain Revolution: How the Technology Behind Bitcoin and Other Cryptocurrencies is Changing the World*, London: Penguin, 2016.

Ethereum: A Blockchain Made of Aether and Oracles

Several mystical references lie at the core of Ethereum itself, such as the language and visual references that founder Vitalik Buterin and his Ethereum organisation use to represent the technology, which frames it either as a god or a portal to god. This new kind of blockchain is discursively constructed as something to be revered, outside of human capacities and understanding, autonomous and mystical.

Ethereum founder Vitalik Buterin chose its name because he liked that it referred to ether¹¹⁴—the mythical substance that ancient philosophers thought filled the universe. In ancient Greek mythology, Aether is the god of the upper sky and represents the purest kind of air, the kind that only gods breathe. This reference posits Ethereum as an invisible and pure infrastructure—a foundation that does not draw attention to itself, yet upon which everything else relies.

Oracles are another ancient reference used to signify a particular aspect of Ethereum and other blockchains like it. Projects that aim to represent an off-chain state of affairs—for example the location of a physical good or other data that originates outside of the ledger of a blockchain, such as the blockchain forest protection projects discussed in chapter 2—have to rely on oracles to collect that off-chain information and log it onto a blockchain ledger. I should stress that this off-chain data comes without the trustless, tamperproof, immutable affordances that are central to blockchain's appeal, and represents major weak spots where corruption might creep in for any system that uses them.

Oracles¹¹⁵ were portals through which it was thought gods spoke directly to people. Although often voiced in the form of ambiguous riddles, the predictions uttered by oracles were considered ultimately truthful because they came directly from a god. The ambiguity of oracles is exemplified by a famous ancient prediction regarding war, reading: 'Ibis redibis nunquam per bella peribis,' which is translated from Latin as 'you will go you will return never in war will you perish.' In the absence of punctuation, the prediction can be interpreted as that one will either die in battle or return from battle alive. Either outcome will thus retroactively prove the prediction. This ambiguous prediction gets to the heart of a central issue for blockchain oracles: the lack of trustlessness in the origin of information and the effective truthfulness¹¹⁶ of the data once they are logged on a blockchain.

In an often quoted post, Buterin, who is an avid blogger on the Ethereum Foundation Blog, defined blockchain technology as 'a magic computer'.¹¹⁷ (n.p.). Why specifically he feels it to be magic remains unclear, nonetheless, this more-than-rational framing of Ethereum persists.

114 Nick Paumgarten, 'The Prophets of Cryptocurrency Survey the Boom and Bust', *The New Yorker*, 2018, <https://www.newyorker.com/magazine/2018/10/22/the-prophets-of-cryptocurrency-survey-the-boom-and-bust>.

115 From the Latin 'orare,' meaning 'to speak.'

116 Effective in terms of their factual existence as facts on a blockchain, but also in terms of their capacity to produce results as financialised truths. I went into more depth about effective truths in chapter 2.

117 Vitalik Buterin, 'Visions, Part 1: The Value of Blockchain Technology', *Ethereum Foundation Blog*, 13 april 2015, <https://blog.ethereum.org/2015/04/13/visions-part-1-the-value-of-blockchain-technology>, n.p.

For example, the Ethereum logo is often depicted in mystical ways. The logo is made up of two pyramid shapes mirroring each other at their square side and it is often pictured as if it is magically floating in space, as seen in Figure 10, where a group of multi-species characters is huddled around the logo. Some have their hands together as if in prayer, while others seem to stand in awe of the logo, watching expectantly or listening carefully. Here it is Ethereum itself that is presented as an oracle, offering mystical, guiding insights to those engaging with it.



Fig. 10. Illustration featuring the Ethereum logo taken from the official Ethereum website.

A Machine with a Sacred Logic of Truth

In *The Truth Machine: The Blockchain and the Future of Everything*, Paul Vigna and Michael J. Casey¹¹⁸ describe blockchain as a system that restores moral and epistemic order to a corrupted world. Referring to it as ‘a record-keeping method that brings us to a commonly accepted version of the truth that’s more reliable than any truth we’ve ever seen,’¹¹⁹ they frame accounting itself as sacred, insisting that ‘[w]ell-maintained and clear accounting is sacrosanct.’¹²⁰ This religious tone situates blockchain not merely as a financial or technical

118 Both are influential in the blockchain media space, (regularly) appearing in such outlets as the MIT Media Lab’s Digital Currency Initiative, The Wall Street Journal, The Washington Post, Wired, CNN, CNBC, MSNBC, Fox, and PBS.

119 Paul Vigna and Michael J. Casey, *The Truth Machine: The Blockchain and the Future of Everything*, New York: St. Martin’s Press, 2018, 20.

120 Vigna and Casey, *The Truth Machine*, 25.

innovation but as a moral and even spiritual force—a divine mechanism for producing certainty in uncertain times.

Vigna and Casey trace blockchain's genealogy to fifteenth-century bookkeeping practices, particularly highlighting Franciscan friar Luca Pacioli. Pacioli conveyed a method known as double-entry bookkeeping, where credits and debits are recorded in parallel to ensure accountability and transparency. The way Vigna and Casey understand it, this technique exemplified both commercial rationality and moral legitimacy: 'Pacioli's method helped the merchants overcome the church's disdain for usury. The merchants had to prove to the church that their businesses were not, in fact, sinful, that they provided a benefit to mankind.'¹²¹ In this reading, accounting served a dual purpose: it organised trade and offered merchants spiritual redemption by demonstrating honesty before God.

Historical scholarship tells a more pragmatic story. As Sangster notes, double-entry bookkeeping was initially developed to manage credit between businesses, not as a theological practice. It became a closed, verifiable system only later, when the Church had already softened its position on usury.¹²² Yet this discrepancy is less important for Vigna and Casey's argument than the moral symbolism accounting provides. For them, the ledger—whether divine or digital—represents a universal framework for truth.

They continue arguing for bookkeeping's sanctity by connecting it directly to scripture. Discussing the Day of Judgement described in the Book of Revelation, they write: 'the dead stand before God and open their book. Then God opens his book. The second book. You might call this, oh, *double bookkeeping*.'¹²³ Through this analogy, accounting becomes not only a technical practice but a divine act of judgement, a means by which truth itself is recorded and verified.

Moving into the 21st century, Vigna and Casey identify the 2008 financial crisis as the collapse of this moral and epistemic order. 'Most (though certainly not all) people today have a hard time seeing the Bible as literal truth; but they had no trouble seeing Lehman Brothers' books as literal truth,'¹²⁴ they write. For them, the once sacred modern accounting had become corrupt and unreliable. 'Even when done honestly,' they argue, it is 'little more than an educated guess' that 'has become so convoluted that it is virtually useless.'¹²⁵ The crisis revealed, in their view, that faith in traditional ledgers made society 'vulnerable to fraud.'¹²⁶ In this context, blockchain appears as salvation: a technological reform that restores purity and trust to the practice of shared truth-making.

121 Vigna and Casey, *The Truth Machine*, 26-27.

122 Alan Sangster, 'Pacioli's Lens: God, Humanism, Euclid, and the Rhetoric of Double Entry', *The Accounting Review* 93.2 (2018): 299-314. <https://doi.org/10.2308/accr-51850>.

123 Vigna and Casey, *The Truth Machine*, 27, emphasis in original.

124 Vigna and Casey, *The Truth Machine*, 28.

125 Vigna and Casey, *The Truth Machine*, 28.

126 Vigna and Casey, *The Truth Machine*, 28.

They describe blockchain as ‘essentially a digitized, objective rendering of the truth’¹²⁷ that is ‘mathematically secured.’¹²⁸ The system promises a new epistemology grounded in computation and eliminating the need for human interpretation. For Vigna and Casey, consensus—the collective verification of information across the network—constitutes this new form of truth. ‘What’s been agreed upon as the truth *is* the truth. There is no room for debate.’¹²⁹ Once consensus is achieved, truth becomes immutable and timeless. In contrast to older systems of knowledge, which they regard as unstable or corruptible, blockchain introduces permanence through automation.

Significantly, rather than understanding consensus as a social process of ‘participation, deliberation, and competition,’¹³⁰ the authors portray it as an asocial, computational operation. Through algorithms, blockchain transforms human agreement into mathematical security—a ‘truth-discovery process’ that removes ambiguity, dissent, and interpretation. The result is a mechanised vision of truth as singular and universal, collapsing complexity into verifiable certainty.

In this framework, blockchain assumes divine attributes. Like a god, it is imagined as omniscient, incorruptible, and distant from human fallibility. Vigna and Casey thus follow Szabo’s earlier logic, even naming the chapter in which most of this is laid out after Szabo’s God Protocols article: they place their faith in technology over human judgement, seeing in blockchain a perfect, self-regulating order. The technology’s cryptographic operations are portrayed as unlocking a truth worthy of belief and capable of redeeming society from the moral failures of the financial system and post-truth dynamics.

Ultimately, Vigna and Casey’s account fuses theological and technological imaginaries. They construct blockchain as both the heir to double-entry bookkeeping’s sacred rationality and its transcendence: a divine apparatus that automates truth, eliminates human error, and restores faith in systems of exchange. In doing so, they transform a financial infrastructure into a cosmology—an enchanted determinism in which computation itself becomes a godlike source of order and moral clarity.

Preparing for the Rapture with Bitcoin

At this point, my attention turns back to Bitcoin culture, specifically the faction known as Bitcoin Maximalists, who define themselves through an uncompromising belief in the superiority of Bitcoin above all other blockchains. Their devotion reflects a broader cultural pattern in which the original whitepaper is elevated to the status of scripture and Bitcoin itself is treated with religious reverence. The religious discourse and practices of this community support a techno-mystical belief in the universal efficacy of Bitcoin and pushes for its absolute

127 Vigna and Casey, *The Truth Machine*, 30.

128 Vigna and Casey, *The Truth Machine*, 29.

129 Vigna and Casey, *The Truth Machine*, 65, emphasis in original.

130 S Jankowski, ‘Making Consensus Sensible: The Transition of a Democratic Ideal Into Wikipedia’s Interface’, *Journal of Peer Production* 15 (2022), 4.

dominance over other currencies. Because Bitcoin is understood as the saviour of the world, full dedication to the cryptocurrency is demanded in this faithful community. These unforgiving attitudes have only sharpened in reaction to the rise of competing cryptocurrencies and the enthusiasm for blockchain applications beyond money. Marked by rigidity and intolerance, Bitcoin Maximalism have responded by casting dissenters as heretics and making their opinions loudly heard in the broader community.

Bitcoin Maximalism is an expression of the kind of technological mysticism that emerges from modernity ideals of progress and rationality. Bitcoin is seen as the expression of perfect rationalist order and truth. Like the divine logics for which Enlightenment rationalist searched, Bitcoin is presented as revealing the order of nature. It is thought of as the universal, everlasting truth, meaning that it should be defended against any belief in other currencies: if people sell their bitcoins, the network will lose strength, and the Bitcoin truth will break down. By using references to saints, rituals, and sacred objects to make sense of Bitcoin and the role of individuals that are key for its development, the community places them in a familiar religious narrative. This narrative almost automatically transfers visions of an unavoidable rapture onto the future trajectory of Bitcoin and supports a culture of unwavering commitment to the cryptocurrency, despite its volatility. It presents the technology as a divine force that is not to be technically understood, but rather faithfully followed.

The Book of Bitcoin Maximalism

The introductory chapter of this book featured a reference to a prayer to Satoshi written by Hass McCook, who is an prominent advocate of a religious dedication to Bitcoin and is also known as friar Hass. Here we take another look at this prayer, this time in a bit more detail:

As always, we commence by sending thanks and good tidings to Satoshi Nakamoto—The Creator and First of the Believers, General of the Byzantines, Breaker of Banks and Fighter of Fiat—and to Hal Finney and The Apostles and The Disciples thereafter, and to The Stoic and Patient True Believers, who keep their tithe holy, and stack sats for their salvation. Oh you who believe, fear the day of economic reckoning, and do not face the Angel of Hyperbitcoinization as a nocoiner, for punishment awaits them in the hereafter.¹³¹

As the prayer exemplifies, Bitcoin Maximalist discourse borrows heavily from religious language. It includes references to specific beliefs—i.e., that banks should be broken and fiat should be fought, that there will be a future event called hyperbitcoinisation, holy figures such as Satoshi, Hal Finney, and the apostles, and sacred rituals such as keeping tithe holy and stacking sats—and is a good entrypoint to understanding the dimensions of the religiosity of Bitcoin Maximalism. ‘Hal Finney and The Apostles’ refers to a few central figures in the early development of Bitcoin during the first years after Satoshi published the whitepaper.¹³² The

131 Hass McCook, ‘The Passion of the Believers’, *The Bitcoin Times*, 2019, <https://bitcointimes.io/the-passion-of-the-believers/>.

132 Andrew Fenton, ‘Is Bitcoin a Religion? If Not, it Soon Could be’, *Magazine by Cointelegraph*, 2021, <https://cointelegraph.com/magazine/is-bitcoin-a-religion-if-not-it-soon-could-be>.

phrase ‘stacking sats’ operates as the equivalent of paying tithe: rather than selling them, one demonstrates faith by accumulating and holding bitcoins. A ‘sat’ (short for Satoshi) is the theoretically smallest divisible unit of Bitcoin,¹³³ with 100 million sats equalling a single coin. Small, repeated purchases of sats are thus framed as acts of commitment to the belief system. This ritual practice looks forward to ‘hyperbitcoinisation,’ imagined as an millenarian-like end-times scenario in which Bitcoin overtakes all other currencies and its adherents are ultimately rewarded. This speculative future moment is treated as an unavoidable reality by Bitcoin Maximalists, who prepare for it by exchanging as much fiat money as possible into bitcoins before hyperbitcoinisation inevitably causes an explosion of their value.

Bitcoin Maximalism does not have a defined set of rules or practices beyond a total commitment to Bitcoin over any other currency and a belief in its future world dominance. Many adherents signify their commitment by putting ‘Bitcoin Maxi’ in their social media bios, but there is no central organisation. However, Bitcoin Maximalist Max Keiser put together a publication called *The Book of Max: Relentless Optimism in the Age of Bitcoin*,¹³⁴ which can be read as a guide to some of Bitcoin Maximalism’s core characteristics and beliefs.¹³⁵ Featuring as a headline speaker at many Bitcoin conferences, Keiser is a prominent figure in the Bitcoin Maximalist community. For instance, at the Bitcoin 2021 conference,¹³⁶ he was introduced as the ‘high priest of Bitcoin’.¹³⁷ Keiser’s fame in the Bitcoin Maximalist community builds on his previous media experience, contributing to outlets such as Russia Today, Al Jazeera, and BBC World News. His public persona is flamboyant, energetic, and brazen. He is an effortless entertainer, has the stage presence to excite a crowd, and is not afraid to offer expletives and insults.¹³⁸ Keiser is a media-savvy performer who knows the power of quotable soundbites. While his statements might be designed for traction rather than nuance or authenticity, they do outline his religiously framed understanding of the importance of Bitcoin.

Most of Keiser’s book is filled with aphorisms and memes based on statements made by Keiser since the early days of Bitcoin. Particularly relevant to my analysis are the book’s introduction and conclusion, which provide more detail with regard to the coherence of the views expressed in the other pages. Although the publication itself seems to be little discussed in the Bitcoin Maximalist community, it draws upon Keiser’s unapologetically enthusiastic views of Bitcoin

cointelegraph.com/magazine/is-bitcoin-a-religion-if-not-it-soon-will-be/.

133 This limit to the divisibility is of course just an agreement, and Bitcoins could in the future be coded differently.

134 Max Keiser, *The Book of Max: Relentless Optimism in the Age of Bitcoin*, Calabasas: Swan Bitcoin, 2021.

135 The book was published by Swan Bitcoin, a company to whom Keiser acts as an advisor.

136 The conference, which took place during the Covid-19 pandemic, boasts being the largest Bitcoin event in history, and is reported to have had 12,000 attendees. The YouTube video of Keiser’s contribution has been viewed well over 100,000 times in two years. See Michael Saylor and Max Keiser, ‘Bitcoin 2021: Fireside: Michael Saylor and Max Keiser’, *YouTube video*, 2021. https://www.youtube.com/watch?v=_uJwvlmNns.

137 Saylor and Keiser, ‘Bitcoin’.

138 For example, he called United States Senator Elizabeth Warren ‘Lucifer’s mistress’ because of her critical stance on Bitcoin. See Keiser qtd. in Teuta Franjkovic, ‘Warren Criticizes Crypto’s Role in Sanction Evasion: Max Keiser’s Response Ruffles Regulatory Feathers’, *CCN*, 2024, <https://www.ccn.com/news/warren-criticizes-cryptos-role-sanction-evasion-max-keisers-response/>.

and his charismatic public performances in many major Bitcoin-themed events, and *those* have certainly had a wide reach. In what follows, I will relay key parts of the contents of *The Book of Max* that exemplify how Bitcoin Maximalism connects technology and belief.

The publication's very title echoes the format of Judeo-Christian scripture, recalling works like The Book of Genesis or The Book of Leviticus, and in doing so positions itself as a modern counterpart to these sacred texts. In an interview on the *What Is Money?* podcast, Keiser explains that he made the publication as a way to make his understanding of the 'metaphysical' significance of Bitcoin accessible to the community.¹³⁹ He explains that becoming a Bitcoin Maximalist necessarily implies an embrace of the metaphysical:¹⁴⁰

People who get into Bitcoin, eventually, they become metaphysicists [sic], they get into the metaphysical aspect of Bitcoin. [...] Unless you take the dark side, unless you become an altcoiner, or a 'shitcoiner.' Then you go down a different path toward hell, essentially. Those are the two paths on offer. You either ascend to the metaphysical heaven, or you descend into the analog basement of reality, somewhere in Satan's playpen.¹⁴¹

At the core of Keiser's book is a vision of Bitcoin as divinely ordered. The text elevates Satoshi to an all-knowing, godlike status, casts Bitcoin itself as the redeemer of society, and interprets its underlying structure as evidence of a divine design. Through these themes, Bitcoin is ultimately presented as a perfect and natural foundation for a prosperous future.

For Keiser, Bitcoin embodies ideals of reason and truth that echo the philosophical logics of Leibniz and Boole. He treats the blockchain as something eternal—already present within the fabric of the world and only recently uncovered. 'Bitcoin was a discovery, not an invention. We needed it all along. We finally found it. Now it's gonna save us',¹⁴² he writes. This framing casts Bitcoin as both pure and universal, 'No more illusions. Only the truth. Each and every block'.¹⁴³ Keiser envisions a future singularity in which Bitcoin's adoption grows uncontrollably, surpassing any human or regulatory authority and culminating in hyperbitcoinisation. In this vision, Bitcoin becomes the apolitical saviour of humanity, offering a perfect order for society.

The biblical story of Adam and Eve provides Keiser with a framework for his analogy. Before eating the forbidden fruit, Adam and Eve stood naked before god—innocent, pure, and fully transparent to divine insight. Once they sinned, however, their newly acquired awareness of their own nakedness drove them to hide. Keiser (2021) reworks this narrative to describe

139 Robert Breedlove and Max Keiser, 'The Art of Bitcoin With Max Keiser (Wim264)', *YouTube video*, 2023. <https://www.youtube.com/watch?v=5n9FSG5a8S8>.

140 Literally, Keiser uses the term 'Bitcoin' rather than 'Bitcoin Maximalism.' I treat these terms as the same when expressed by a Bitcoin Maximalist, because inherent to being a Bitcoin Maximalist is the belief that all other attitudes to Bitcoin are wrong and should not exist. To Bitcoin Maximalists, the culture of Bitcoin is or should be the same as Bitcoin Maximalism.

141 Breedlove and Keiser, *The Art of Bitcoin With Max Keiser*.

142 Keiser, *The Book of Max*, n.p.

143 Keiser, *The Book of Max*, n.p.

Bitcoin, claiming it places us '[n]aked before Satoshi'.¹⁴⁴ In his view, Bitcoin makes possible a return to that prelapsarian state: no concealment, no falsehoods—only perfect truthfulness that is guaranteed by an all-seeing, godlike Satoshi.

An Orthodoxy of Toxicity

Bitcoin Maximalists, known for their uncompromising attitudes, have been described as a toxic and sect-like.¹⁴⁵ Max Keiser gives perhaps the most dramatic defence of their reputation for toxicity. He likens their uncompromising stance to Michelangelo's chisel: 'The marble thought Michelangelo's chisel was 'toxic.' The result was a masterpiece. Bitcoin maximalists' toxicity is chiseling a masterpiece of perfect money from a lifeless slab of global, fiat money, central bank junk'.¹⁴⁶ Bitcoin Maximalists believe that the world is so utterly in disrepair that being toxic is the only honest and correct thing to do as it demonstrates pious dedication to the essence of Bitcoin and its prophesied future of hyperbitcoinisation. This rhetorical move reframes what outsiders dismiss as hostility into a badge of artistry, skill, and perfection, and Maximalists themselves embrace the label as integral to their identity.

The 'toxicity' of Bitcoin Maximalists is especially pronounced when dealing with what they call 'shitcoins.' In the eyes of Bitcoin Maximalists, any cryptocurrency that is not Bitcoin is a shitcoin—i.e., not worth consideration, actively hurting the potential of Bitcoin, and doomed to fail. This even applies to Ether, the cryptocurrency that is native to the Ethereum blockchain and is the long-time second to Bitcoin in terms of market capitalisation.¹⁴⁷

A set of tweets posted by the outspoken Bitcoin Maximalist Kevin Pham exemplifies the sub-culture's antagonism against members of the Ethereum community (Figure 11). Each tweet features two pictures side by side, the left featuring a person that embodies something quint-essential to the Bitcoin community, and the right doing the same for the Ethereum community. On the Bitcoin side, the pictures in the tweets feature many of the same stereotypes that some of the most fervent US-based right-wing communities have come to be famous for: pride in traditionalism, vehement defence and celebration of (militaristic) gun culture, and caps with political statements on them. One of the Bitcoin pictures features a man with a face covering displaying the Bitcoin logo, ensuring his anonymity. The man proudly holds up a machete and an axe, and wears a cap that reads 'UASF'.¹⁴⁸ The other Bitcoin picture features a man with a large firearm that appears to have been shot in the direction of the camera just a fraction of a second ago. He is wearing a cap that, in a direct reference to former US president Donald Trump's 2016 campaign slogan, reads 'Make Bitcoin Great Again' expressing discontent with

144 Keiser, *The Book of Max*, n.p.

145 See e.g. Huber and Sornette, 'Boom, Bust, and Bitcoin'; *Boomer*, 'A Newcomer's Perspective on Toxic Bitcoin Maximalism', *Bitcoin Magazine*, 2022, <https://bitcoinmagazine.com/culture/what-toxic-bitcoin-maximalism-means-to-onlookers>; Diwaker Gupta, 'The Bitcoin Maximalist Movement Has Gotten Toxic: Bitcoin Optimism is a Better Way', *Built In*, 2022, <https://builtin.com/blockchain/bitcoin-maximalist>.

146 Keiser, *The Book of Max*, n.p.

147 CoinMarketCap, <https://coinmarketcap.com/>.

148 UASF stands for 'user-activated soft fork,' which is an attitude associated with Bitcoin Maximalists in the block-size debate.

changes to the Bitcoin orthodoxy. In contrast, Ethereum culture is presented as playful and cute, featuring rainbows, unicorn hats, and a cat-shaped bag carried by Ethereum founder Vitalik Buterin. These tweets present the Bitcoin community as earnest, dedicated, and willing to fight for what they believe is right, while the Ethereum community is basically a bunch of kids playing around, unaware of what is really at stake.



Fig. 11. Screenshots of tweets by Bitcoin Maximalist Kevin Pham portraying Vitalik Buterin and other unknown individuals.

Michael Saylor is another prominent Bitcoin Maximalist who describes the community as ‘a swarm of cyber hornets’ in a tweet (Figure 12), likening it to the wasp commonly thought of as aggressive and, when a nest is disturbed, liable to retaliate as a swarm, each wasp capable of stinging multiple times.¹⁴⁹ On stage with Max Keiser at the Bitcoin Conference in 2021, Saylor stated that he considers these cyber hornets as a motley crew of Bitcoin defenders on X, Bitcoin-supportive journalists in the media, those that fight for the continuing existence of Bitcoin by mining the cryptocurrency, and those that build ‘altars to Satoshi’ by running a Bitcoin node.¹⁵⁰ Being a Bitcoin Maximalist is like being a proselytising member of a religion, convinced of its truthfulness and working hard to support its proliferation although often derided by outsiders.

Bitcoin Maximalism, as outlined across these subsections, presents itself as both mystical and absolute. Bitcoin is imagined as a divine technology, flawless yet inscrutable, whose defence must be uncompromising. However, this sacralisation comes with costs: it discourages critical examination, suppresses dissent, and undermines accountability.

149 Michael Saylor, <https://twitter.com/saylor/status/1307029562321231873>.

150 Saylor quoted in Saylor and Keiser, *Bitcoin* 2021.



Fig. 12. Screenshot of Bitcoin Maximalist Michael Saylor's tweet comparing the community to 'cyber hornets.' Saylor's avatar features his portrait with laser beams for eyes. This is a Bitcoin Maximalism reference, indicating his belief that one bitcoin will one day reach the value of 100,000 USD.

Conclusion: The Religiosity of 'Blockchain as a Truth Machine'

Religious metaphors have surfaced repeatedly in efforts to explain blockchain's role and significance. They appear at its very origins with the algorithm that made cryptocurrency possible, resurface when the technology expanded beyond monetary use into global hype, and re-emerge in the rhetoric of Bitcoin's staunchest defenders as they reacted to threats to its primacy. Yet these narratives do not clarify how blockchain actually works. Instead, they frame it as a vision of perfection—an ideal that humanity, fallible and corruptible, can approach but never fully comprehend.

Belief is a powerful element in the socio-techno-cultural imaginary of blockchain as a truth machine. Rationalism in blockchain culture is supported by overt religious and non-religious forms of belief. This is a departure from the findings of the previous chapter: there, the imaginary of blockchain as a truth machine gained strength by hiding the myth that sustains it, whereas in the current chapter, belief and religious references are overt and ubiquitous, and give blockchain a sense of perfection and inscrutability. Together, both chapters show that technological rationalism and more-than-rational belief prop up the same imaginary of blockchain as an objective, neutral, and universal truth machine. In this chapter, I have shown that religious narratives and technological mysticism do work in the world, offering blockchain directionality and purpose while sustaining its perception as too complex to understand.

Bitcoin's Proof-of-Work algorithm is frequently likened to a divine mediator—impartial, detached, and able to oversee exchanges with ultimate neutrality. Its perfection is emphasised precisely through its inscrutability, echoing the logic by which gods are imagined as both unknowable and authoritative. More broadly, blockchain's legitimacy depends on this sense of autonomy from corruptible human influence. Although such technological mysticism is not always religious, the god analogy crystallises how the technology comes to be trusted as objective and beyond reproach.

As blockchain moved out of its currency niche, its truthfulness became equated with a religious form of virtue. In the same way oracles speak god's truth, the idea of blockchain as a truth machine is shaped by religious logics. Blockchain is framed as an update to the sanctity of accounting capable of smoothing out the messiness of the contemporary post-truth era. In this framing, it grants access to a form of truth imagined as surpassing all previous human achievements with regards to establishing truth. As such, it is presented as an ideal of perfection that invites faith rather than doubt.

While the views of the Bitcoin Maximalist community cannot be taken as representative of other parts of blockchain culture, they do exemplify the preoccupation with universality and the role of power in the Western genealogies of rationalism and religion. Like rationalism, Bitcoin Maximalism is characterised by an aim to uncover a natural, universal order that eradicates all other ways of knowing the world. By veiling Bitcoin's universal order in religious predestination, by practicing animosity towards any kind of difference, and supporting unquestioning devotion rather than critical analysis, Bitcoin Maximalism impedes critical reflection and contestation and exemplifies enchanted determinism.

When Bitcoin Maximalist Kevin Pham tweeted about the differences between the Bitcoin and blockchain communities, he linked the comparative images with captions in which he makes the same spelling mistake twice, accidentally turning their meaning upside down. Instead of writing that the communities cannot be any more different, he writes: 'I don't think the Bitcoin and Ethereum community can be anymore different'¹⁵¹ and 'Bitcoin and Ethereum community can't be anymore different Pt. 2'¹⁵²—i.e., they can no longer be different, or they can no longer be regarded as different.¹⁵³ Furthermore, the previous paragraphs show that there are many overlaps between the different communities in terms of their regard for objectivity, neutrality, and truth. By using similar religious and mystical references—albeit with different emphases, concerns, and contexts—the imaginary of blockchain as a truth machine is aligned with a more or less cohesive sense of the future in which blockchain constitutes the foundation of interactions in an untrustworthy social world.

Like the blockchain forest protection projects of the previous chapters, Bitcoin Maximalists aim for a universal and universalising order—a way of knowing the world that is superior to other ways of knowing and that should ultimately eliminate them. Although the other communities discussed in this chapter do not explicitly use such language, the use of religious beliefs that portray blockchain as part of a cohesive mythology or an animating narrative does support a similar sense of universality. Religious narratives offer an explanation of the totality of the world, and by using such narratives to understand blockchain, the technology itself seems capable of addressing that totality as well. Historically and today, religious narratives are used to support unequal power relations. Combined with technologies such as blockchain

151 Kevin Pham (@_Kevin_Pham), 'I Don't Think the Bitcoin and Ethereum Community Can be Anymore Different', *Twitter*, 2017, https://x.com/_Kevin_Pham/status/927792297365848064.

152 Kevin Pham (@_Kevin_Pham), 'Bitcoin and Ethereum Community Can't be Anymore Different Pt. 2', *Twitter*, 2017, https://x.com/_Kevin_Pham/status/927795191985012736.

153 I cannot claim the eagle-eyedness to spot these meaningfully missing spaces myself: my co-supervisor Martin Zeilinger pointed them out to me in one of our conversations.

that animate the collective imagination, they deserve to be taken seriously rather than brushed off as inconsequential or insincere.

SURFING THE WAVES OF VOLATILITY: ASTROLOGICAL AND MEMETIC RESPONSES TO UNCERTAINTY IN THE CRYPTO MARKET

This chapter signals a shift in my focus. Until now, I have dealt with imaginaries that—in true rationalist fashion—seem to emerge from everywhere and nowhere at the same time, emphasising the potentiality of structures in the abstract over their embeddedness in cultures and lived reality. They were rationalist in terms of their emphasis on the supposed superiority of universal blockchain-based truths and the obfuscation of the situatedness of their operationalisation of truth in favour of the pretence of a view from nowhere. Despite the grand hopes that blockchain can regain the truth that was lost amid alternative facts in post-truth times, the lived experiences of blockchain communities, discussed in this and the next chapter, reveal that reality remains ambiguous, volatile, and chaotic. These two chapters form a counterweight of specificity and situatedness to the claimed universality of the previous two: the lived practices that form in the cracks of big dreams—never predetermined and always co-constituted in unexpected ways with their socio-cultural and historical context.

Moving beyond the dominant imaginaries represented in prominent texts, the situated communities I discuss in this chapter practice imagination differently. Additionally, a shift in topic takes shape. In the current chapter, for example, rather than the abstract promises of blockchain technology in general, the financial cryptocurrency market takes centre stage. Taking my cue from sociologist Aris Komporozos-Athanasiou, I do not simply critique how ‘finance hollows out contemporary social life’,¹ of which chapter 2 is an example, but I also contribute to a better understanding of how it affords ‘its own way of imagining—not just the future but our puzzlingly uncertain present too’²—by ‘mov[ing] beyond the structural perspective of technology companies and consider[ing] the dazzling spectacle of speculative communities from the perspective of their lay users’.³ Komporozos-Athanasiou notices that new forms of sociality and subjectivity emerge in the context of contemporary sociopolitical uncertainty and financial volatility permeated by the logic of speculation.⁴ I start by providing a theoretical foundation through which to understand speculation as an imaginative practice within and outside of the financial market in order to bring to the fore two speculative crypto imaginaries in subsequent sections.

My aim with this chapter is to show how the context of the volatile crypto market impacts the rationalism that characterised the imaginary of blockchain as a truth machine that I discussed

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- 1 Aris Komporozos-Athanasiou, *Speculative Communities: Living With Uncertainty in a Financialized World*, Chicago: The University of Chicago Press, 2022, 29.
 - 2 Komporozos-Athanasiou, *Speculative Communities*, 29.
 - 3 Komporozos-Athanasiou, *Speculative Communities*, 73.
 - 4 Komporozos-Athanasiou, *Speculative Communities*.

thus far. I show two different approaches from within the cryptocurrency community that attempt to grapple with the un(der)regulated chaos of crypto markets, oscillating between rationality and the more-than-rational, between calculation and speculation. The uncertainty that characterises contemporary societies—in terms of, for example, political ruptures, financial instability, and the disintegration of social cohesion—causes some to try to rationalise and attempt to regain a lost sense of control and future stability. However, the rationalist myth of order and control through universal and calculable logics is hard to uphold amid unpredictable chaos, causing others to dive in at the deep end of radical uncertainty. They surf on the high waves of present volatility without wishing for the waters to calm down in the future.

Whereas Komporozos-Athanasίου sees potential for the appropriation of speculative imagination for countercultural cultures and empowering relationality,⁵ I show that the two case studies in this chapter instead reinforce the status quo. The first case study focusses on crypto astrology, that is, the application of astrology to the cryptocurrency market in order to predict the performance of particular coins and adjust investment strategies accordingly. Although historically, financial astrology—which has existed for at least a century, if not longer—occupied an indeterminate position between calculation and divination, my analysis of prominent crypto astrologer Maren Altman shows that blockchain's rationalising promises have a strong pull. While astrology and finance both deal in uncertainty, Altman hopes combining them with blockchain technology will make the future legible and calculable in order to secure financial gains and stable, prosperous futures.

My second case study—which features a memetic crypto imaginary—makes use of the reinforcing connection between imaginaries and the performative affordances of memes. Through the use of vernacular language and subcultural references, memes contribute to establishing community boundaries, the values and norms that govern the crypto community, and its particular form of speculative imagination. Memes operate as normative performative language objects that supercharge the power of imaginaries through their memetic affordances.

Crypto memes specifically contribute to an imaginary that doubles down on uncertainty in order to thrive in the speculative present. Taking memetic phrases such as 'HODL' and 'do your own research' as my guide through a database of approximately 14,700 memes, I show how crypto memes reinforce the ambiguity of knowledge and the untrustworthiness of others, instead revelling in unknowability and uncertainty. On the crypto market, promises are not meant to be fulfilled at some future date, but rather offer an opportunity to take advantage of the volatility of the market in the present. Crypto memes mock the idea of ever being able to predict a market as ruthlessly unpredictable and susceptible to random outside influences as that of a cryptocurrency. Crypto speculators seem to 'knowingly descend into chaos'⁶ while memes spread their vernacular imaginary of speculative survival in radical uncertainty.

5 Komporozos-Athanasίου, *Speculative Communities*.

6 Komporozos-Athanasίου, *Speculative Communities*, 21.

Speculative Imagination in Volatile Times

As I move from imaginaries that appear from nowhere to imaginaries from a specific somewhere, the subject that does the imagining also comes into clearer view. The *imaginary* is always constituted by the *imaginative practices* of communities of specific people with specific degrees of power in society.⁷ To understand the relationship between imagination and society, I present a brief overview of key sociological and philosophical positions on imagination, guided by the work of, among others, Aris Komporozos-Athanasiou, Max Haiven, and Cornelius Castoriadis. Despite its historically negative, passive, and/or fictional connotations, imagination constitutes worlds and makes change possible. Significantly, it does so outside of the confines of rational calculability. In the context of increasingly widespread financial logics, speculation can be understood as ‘an *imaginative act*’⁸ that can bring about futures. Over time, speculation has evolved from an elite practice to a dominant logic in contemporary societies, and its political potential demands critical attention.

The Role of Imagination in (Financial) Capitalism

Academic discussions of the concept of imagination almost invariably reference Plato and Aristotle: the first dismissing imagination as the illusion of reality—like the shadows inside his famous cave that the prisoners mistake for reality—while reality itself is located in the world of the mind in the form of truth and reason; the second taking a different stance, arguing for the importance of imagination for understanding the world around you, as it produces narratives that create coherence, support the establishment of communities, and motivate action in the real world.⁹ These ancient differences in understanding and appreciation of imagination between Plato and Aristotle have remained recognisable throughout history.

Modernity also played host to significant reflections on imagination. Descartes, for example, believed the human pursuit of knowledge involved the intellect as well as the imagination and other faculties spanning both mind and body. He argued that animals do not have the capacity to imagine, but rather function like automata and are thus fundamentally different from humanity.¹⁰ However, ultimately, for Descartes, a god’s disembodied pure and rational

7 The materials analysed in the previous chapters were of course written by specific people, many of which are publicly known. Contrastingly, the memes studied in this chapter are anonymous by their very nature. What I mean to address with the idea of ‘imaginaries from somewhere’ has more to do with the story they tell and the imaginary they are representative of than the specific person producing them. The imaginaries in previous chapters proclaimed universality while hiding their positionality. The imaginaries in this chapter and the next are specific in their positionality and do not claim universality.

8 Komporozos-Athanasiou, *Speculative Communities*, 16, emphasis in original.

9 Max Haiven, *Crises of Imagination, Crises of Power: Capitalism, Creativity and the Commons*, London: Zed Books Ltd., 2014.

10 Gary Steiner, ‘Descartes on the Moral Status of Animals’, *Archiv für Geschichte der Philosophie* 80.3 (1998): 268–91. <https://doi.org/10.1515/agph.1998.80.3.268/html>. Rationalist ideas of imagination also play a role in modernity/coloniality: colonial Others were deemed not only too imaginative because of their non-Christian belief systems, which were perceived by Europeans as figments of the imagination, but also not imaginative enough because of their perceived incapacity to develop and create visions for social change, seeming—at least to the Western ‘Enlightened’ eye—to be stuck in the past. See Haiven,

knowledge consists only of the highest faculty of pure thought: the intellect. Later, in the context of German Romanticism, imagination was considered ‘humanity’s ‘divine spark’—that quality of being that set humans apart from animals¹¹ and was ‘intrinsically associated with liberal philosophy’s ethos of individual freedom’.¹² While, for Descartes, it was imagination that set us apart from animals—albeit simultaneously signifying an imperfection in humanity that separated it from the divine—for the Romanticists, imagination was a sign of divinity itself.

It is against the latter in particular that Karl Marx’ and Max Weber’s stance on imagination in relation to capitalism takes shape. For Marx, rather than a feature of Romantic *individualism*, imagination is part of *social* collaboration and transformation, and provides the road to an anticapitalist future.¹³ However, ‘Capital’s success is to interrupt in this process [of social organisation], offering its material manifestations (money, commodities, institutional hierarchies) as the means through which social relations are imagined and negotiated’.¹⁴ While the labourer’s power to imagine is ‘never fully suppressed’,¹⁵ imagination takes the shape of ‘a dark shadow’ and ‘a destructive force, undergirding the parasitic function of speculative capital and the alienating power of money ideology’ in Marx’ work.¹⁶

Weber’s understanding of capitalism differs from that of Marx. Rather than primarily based on material interests, Weber argued that the market took shape under the influence of shared cultural ideas and values. Specifically, Weber links the emergence of capitalism to the rise of Protestantism and its understanding of work as a divine calling.¹⁷ Under the dominant logic of rationalisation and efficiency, capital gained the ‘devastating power [...] to both conscript and stifle the human imagination,’ uniting ‘markets and publics under the aegis of the iron cage’ of bureaucratisation.¹⁸ Weber’s theory contains two key elements—individual sacrifice for future redemption through dedication to work, and the dominance of rationalisation and bureaucracy—that are particularly recognisable in more recent neoliberal logics of efficiency and its redemptive promise of future reward to those who work hard.

While for Marx and Weber, the social potential of imagination was suppressed by, or redirected to, capitalism, for philosopher Cornelius Castoriadis, it remains laden with alternative potential. Castoriadis argued that imagination has been misunderstood as fictional, fantastical, or otherwise ‘not real’ throughout the dominant history of Western philosophy, going back to both

Crises of Imagination, Crises of Power.

11 Haiven, *Crises of Imagination, Crises of Power*, 221.

12 Komporezos-Athanasiou, *Speculative Communities*, 55.

13 Max Haiven, ‘Finance as Capital’s Imagination? Reimagining Value and Culture in an Age of Fictitious Capital and Crisis’, *Social Text* 29.3 (2011): 93–124.

14 Haiven, ‘Finance as Capital’s Imagination?’, 103. A key example of this is commodity fetishism as the reification of value as inherent to commodities rather than the result of labour. In this context, labour is organised as subservient to the exchange value of commodities. See: Karl Marx, *Capital: A Critique of Political Economy, Volume 1*, New York: Penguin Books, 1976 [1867].

15 Komporezos-Athanasiou, *Speculative Communities*, 52.

16 Komporezos-Athanasiou, *Speculative Communities*, 51.

17 Max Weber, *The Protestant Ethic and the Spirit of Capitalism*, New York: Routledge, 2013 [1904].

18 Komporezos-Athanasiou, *Speculative Communities*, 53.

Plato and Aristotle.¹⁹ This history loaded it with connotations of the irrational, unachievable, and utopian. Another misconception plaguing the history of the concept of imagination is its association with passivity: imagination is often thought of as a mirror of reality, such as in Plato's metaphor of the cave. Against such misconceptions, Castoriadis proposes that imagination has the world-changing capacity to create: society is made and continuously remade via our imagination, and as such it is both real and significant. Moreover, in contrast to Marx' determinism, for Castoriadis imagination has *radical* capacities for change. Even if sociohistorical conditions make it possible, that which the imagination creates cannot be expected, calculated, or determined by what already existed: the imagination can create something radically new. Arguing that Marxism fundamentally does not break with the capitalist urge to rationally control nature and society, Castoriadis instead offers the imagination as the site for radical potentiality and fundamental change.²⁰

Speculative Imagination amid Broken Promises

Significantly, taking shape against the backdrop of Marx' and Weber's theories is the spread and subsequent dominance of finance capitalism and its logic of speculation, at first particularly in the United States. While Marx already critiqued what he called 'fictitious capital'²¹—stocks and other financial instruments, the value of which is determined by *expectations* (or, imaginations) of what the future might be like—technological breakthroughs in the 1870s and beyond saw this imaginative logic of financial speculation move beyond the strict confines of the stock market, where it had already existed for centuries. On a socio-political level, the popularisation of speculation should be seen in its historically chaotic context of the preceding Civil War and ongoing unrest, such as the emerging labour struggles of the time. In this context, promises about future reward for current asceticism lost believability. Technological developments influenced how this popularisation of speculation took shape. As the telegraph network expanded, and information could for the first time be transmitted over large distances without significant delay,²² speculation was no longer reserved to those privileged enough to have access to official markets.²³ Unofficial trading locations emerged, called 'bucket shops,' where 'anyone with a few pennies in their pocket [could] participate in the 'mass communion' of speculation'.²⁴

19 Cornelius Castoriadis, *The Imaginary Institution of Society*, Cambridge, MA: MIT Press, 1998 [1975].

20 Cornelius Castoriadis, *World in Fragments: Writings on Politics, Society, Psychoanalysis, and the Imagination*, Stanford: Stanford University Press, 1997 [1990]; Castoriadis, *The Imaginary Institution of Society*.

21 Karl Marx, *Capital: A Critique of Political Economy, Volume 3*, London: Penguin, 1992 [1894].

22 At least, for the standards of the time. In the context of twenty-first-century high-frequency trading technologies, the telegraph does constitute a terribly slow medium.

23 Komporez-Athanasiou, *Speculative Communities*; Haiven, 'Finance as Capital's Imagination?'. The privilege to access official stock markets was by and large reserved for rich, white men. See: Komporez-Athanasiou, *Speculative Communities*.

24 Komporez-Athanasiou, *Speculative Communities*, 43. In Europe, bucket shops emerged a bit later than in the United States, which was situated in its own fin de siècle social unrest and, after the turn of the century, a looming war.

Speculation occupies a curious position that seems to 'overcom[er] the historical bifurcation of rationality and irrationality'.²⁵ This is exemplified by another technology that was key to the popular uptake of bucket shops: the ticker tape. Ticker tape instantly transformed telegraphed Morse code into human readable information printed on a long strip. On the one hand, ticker tape provided a rational view of finance as a calculable, predictable whole. However, on the other hand, it was also central to its mystification, functioning as a sort of oracle spitting out truths, and constituted a popular medium of financial divination.²⁶

Speculation is often mistakenly confused with gambling. The difference between the two emerges from the difference between risk and uncertainty, where speculation is the 'engagement in investment activities that are *inherently uncertain* for the purpose of profit'.²⁷ Perhaps the first to deal with the distinction between risk and uncertainty is economist Frank Knight, who stated: 'It will appear that a measurable uncertainty, or 'risk' proper, [...] is so far different from an immeasurable one that it is not in effect an uncertainty at all'.²⁸ The famous economist John Maynard Keynes later specified:

By 'uncertain' knowledge [...] I do not mean merely to distinguish what is known for certain from what is only probable. The game of roulette is not subject, in this sense, to uncertainty [...]. The sense in which I am using the term is that in which the prospect of a European war is uncertain, or the price of copper and the rate of interest twenty years hence, or the obsolescence of a new invention, or the position of private wealth-owners in the social system in 1970. About these matters there is no scientific basis on which to form any calculable probability whatever. We simply do not know.²⁹

More recently, the sociologist Ulrich Beck recognised a contemporary tendency to attempt to rationalise and control uncertainty into becoming like risk.³⁰ For example, the systems of surveillance and security that permeate contemporary society—e.g., the ubiquity of CCTV cameras with increasing analytic capacities, crowd control technology, the continued military presence in major cities under frameworks such as the French Opération Sentinelle, strict airport security protocols and technologies, etc.—emerged in response to the unpredictable devastation caused by terrorist attacks such as that on 9/11. While the actual occurrence of future attacks will always remain fundamentally uncertain according to Keynes' definition, assessment strategies such as country risk profiles, travel safety advisories, and the technologies mentioned above attempt to produce risk from uncertainty while simultaneously claiming to manage it.³¹

25 Komporozos-Athanasiou, *Speculative Communities*, 51.

26 Alex Preda, 'Socio-Technical Agency in Financial Markets: The Case of the Stock Ticker', *Social Studies of Science* 36.5 (2006): 753-82; Peter Knight, 'Reading the Ticker Tape in the Late Nineteenth-Century American Market', *Journal of Cultural Economy* 6.1 (2013): 45-62.

27 Komporozos-Athanasiou, *Speculative Communities*, ix, emphasis added.

28 Frank H. Knight, *Risk, Uncertainty, and Profit*, New York: Houghton Mifflin Company, 1921, 20.

29 John Maynard Keynes, 'The General Theory of Employment', *The Quarterly Journal of Economics* 51.2 (1937): <https://doi.org/10.2307/1882087>, 213-14.

30 Ulrich Beck, *World At Risk*, Cambridge, UK: Polity Press, 2009.

31 Beck, *World At Risk*.

Beck sees potential for what he calls ‘enforced enlightenment’³² in the contemporary globalised media space. Fundamentally, his idea is that by knowing more about the world through globalised and distributed media production processes, people—especially those in privileged positions—might make better decisions. However, not only is his hope here ‘ultimately wedded to a belief in the liberating promise of (a type of) rationality’,³³ but his critique of the logic of risk is also ultimately tied to ‘a ‘self-transgression’ of instrumental reason that omens the possibility of a more self-aware and responsible modernity’.³⁴ For Beck, these responses to uncertainty produce a ‘world risk society’³⁵ that both limits social freedom in the present for the promise of future well-being and global solidarity. This is all imagined under dominant logics of rationality and asceticism in service of a regained sense of future security and stability.

Yet, for many people

the future often no longer harbors the promise of redemption that justifies sacrifice in the present (the logic of austerity)—rather, it signifies increasingly a time-space of repudiation. Speculation, then, can be understood as the very act of endorsing a failed promise or, put another way, the act of knowingly entering into a broken contract.³⁶

Despite the attempts at rationalisation highlighted by Beck, in times of uncertainty speculation thrives, as the emergence of cryptocurrency markets in the aftermath of the 2008 financial crash exemplifies. After a period of mid-twentieth-century stability and optimism, a context of uncertain social, political, and economic circumstances re-emerged and now dominates contemporary societies.³⁷ In recent decades, this is recognisable, for example, in disintegrating social cohesion and shared sense of reality, the rise of populist politics, and the 2008 financial crash.³⁸

In the period after the 2008 crash, practices of speculation that revel in the unknown and incalculable once again seeped into the mainstream, albeit in new ways and via new technologies. This time, speculation is not confined to the financial sphere, but shapes more and more aspects of social life through the inscrutable forces of algorithms. Contemporary technologies—among others social media, trading algorithms, location-based apps, news services, and AI—both make new connections possible and hide the ways in which their logics structure those possibilities: they both reveal *and* obscure complexity.³⁹ Such ‘com-

32 Beck, *World At Risk*, 57.

33 Komporez-Athanasίου, *Speculative Communities*, 130.

34 Klaus Rasborg, “(World) Risk Society” or “New Rationalities of Risk”? A Critical Discussion of Ulrich Beck’s Theory of Reflexive Modernity, *Thesis Eleven* 108.1 (2012), 17.

35 Beck, *World At Risk*, 9.

36 Komporez-Athanasίου, *Speculative Communities*, 19.

37 Neoliberal policies such as US President Richard Nixon’s end of the gold standard in 1971 and broader cultural phenomena such as the punk slogan of ‘no future’ show the early signs of the re-emergence of uncertainty. See Franco ‘Bifo’ Berardi, *After the Future*, Oakland: AK Press, 2011.

38 Komporez-Athanasίου, *Speculative Communities*.

39 Komporez-Athanasίου, *Speculative Communities*.

plex machines and opaque algorithms [do not] make the world more legible or its impending crises more predictable'.⁴⁰ Instead, they 'nurse a speculative mode of being 'in the present,' a disbelief in capitalism's future promises, and an invitation of the unknown into all aspects of life'.⁴¹ Phenomena as diverse as the rise of freelancing and gig working, growing opportunities for debt-relations in buy-now-pay-later schemes, increasing housing insecurity, and political polarisation are all inseparable from inscrutable algorithmic systems that impact people's capacity to act in and influence social systems, affording speculative engagements with uncertainty.⁴²

In this context, speculation should be understood as 'an *imaginative* act that confronts uncertainty by way of both hedging and wagering'⁴³ by creating fictions about what could happen in the future, and playing the field of possibilities until that future does or does not become real. Mediated by speculative algorithmic technologies, uncertainty-endorsing speculative imagination permeates contemporary society.⁴⁴ As such, speculation 'not only [is] fiction,⁴⁵ but also perform[s] its own fictionality' into existence.⁴⁶

Moving beyond the dichotomy of the rational and the irrational that characterised 'homo economicus,' a new subject emerges in this context: '*homo speculans* lingers in the space of fictions, not irrationally, but because fictions become the most reliable ordering principle in a protean reality'.⁴⁷ For *homo speculans*, volatility and uncertainty offer a space of imagination and the performance of possible futures in the present. Fact and fiction no longer oppose each other, but can be put to use for speculative ends by *homo speculans* 'throw[ing] itself in the whirlpool of life's chaos without the desire to master its contingencies,' but rather the desire to use its potential in the present.⁴⁸

Komporozos-Athanasiou is particularly interested in the spread of speculative imagination beyond the confines of the financial market in the context of algorithmic technologies and their obscuring and connective capacities and uses.⁴⁹ However, my focus differs in three ways. First, in this chapter, I stay within the realm of the financial market, albeit the cryptocurrency market that is characterised by its immersion in everyday life. In this way, the space of the crypto market echoes the newfound accessibility and popularity of speculation that the *fin de siècle* bucket shops also represented. Second, blockchain technology does not exactly fit the parameters of algorithmic technologies that Komporozos-Athanasiou uses. Blockchain is at its core open source and the transparency of blockchain-based data and algorithms

40 Komporozos-Athanasiou, *Speculative Communities*, 64.

41 Komporozos-Athanasiou, *Speculative Communities*, 9.

42 Komporozos-Athanasiou, *Speculative Communities*.

43 Komporozos-Athanasiou, *Speculative Communities*, 16, emphasis in original.

44 Komporozos-Athanasiou, *Speculative Communities*.

45 As in Marx' notion of the 'fictitious capital,' which is speculated with in financial markets. See Marx, *Capital Volume 3*, 525.

46 Urs Stäheli, *Spectacular Speculation: Thrills, the Economy, and Popular Discourse*, Stanford: Stanford University Press, 2013, 70.

47 Komporozos-Athanasiou, *Speculative Communities*, 26, emphasis in original.

48 Komporozos-Athanasiou, *Speculative Communities*, 14.

49 Komporozos-Athanasiou, *Speculative Communities*.

are central tenets of its popularity, compared with the corporately owned, blackboxed social media algorithms and apps that Komporezos-Athanasiou labels ‘speculative technologies [...] that both represent and occlude the uncertain conditions of everyday social life’.⁵⁰ However, as I have argued in chapter 3, blockchain technology is inscrutable *in practice* due to its perceived complexity, as well as, in the case of the cryptocurrency market, a culture of profound volatility. Blockchain and cryptocurrencies make visible and obscure everyday life in ways that can reinforce uncertainty. Third, I treat astrology and memes as technologies that have the capacity to contribute to speculative imagination. Both serve to reveal and obscure in their own ways: astrology occupies a middle ground between calculation and divination—something Komporezos-Athanasiou also contends,⁵¹ although he refers to it historically and my analysis looks at the significance of astrology in the present of the cryptocurrency market—and memes are particularly suited to understanding the collectivisation of felt uncertainty in a highly individualising context.

Disintermediating the Stars: Rationalisation in Crypto Astrology

In the context of the imaginary of blockchain as a truth machine, financial astrology on the cryptocurrency market—although it possesses the potential to function as speculative imagination—takes on the character of rationalist logic striving for control, proof, and predictability. Historically, financial astrology was a popular practice in nineteenth- and early twentieth-century financial culture. Astrology owes at least part of its spread in dominant Western culture to its blend of rationality and irrationality, combining factual measures and calculations with mystical divinatory practices. In part because of this ambiguity, astrology is often considered inconsequential. Nevertheless, it simultaneously represents a powerful way in which norms are performed and spread throughout societies. In the context of the contemporary loss of belief in future rewards for present sacrifices and increasingly widespread speculative practices, the popular revival of astrology can be understood as speculative imagination doubling down on ambiguity, riding the wave of uncertainty for all its present potentiality.⁵² Superficially, Maren Altman’s crypto astrology seems to be a straightforward continuation of this phenomenon, echoing a historical burst in the mainstreaming of financial speculation amid technological innovation. However, a closer look at Altman’s conceptualisation of blockchain and astrology reveals significant discontinuities with this past. Rather than participating in speculative imagination, Altman attempts to rationalise astrology by putting it on a par with logical processes of pattern recognition, technical analysis, and computation. Furthermore, she proposes the use of blockchain technology to further pull astrology out of the imaginative and into the rational by logging planetary positions and interpretations of their worldly significance on a blockchain, making them back-testable, statistical, and provable. As such, Altman’s crypto astrology, faced with the profound uncertainties of contemporary life and the crypto market, grasps at the final straws of rationalism in an attempt to regain control.

50 Komporezos-Athanasiou, *Speculative Communities*, x.

51 Komporezos-Athanasiou, *Speculative Communities*.

52 Komporezos-Athanasiou, *Speculative Communities*.

A Revival of Financial Astrology amid Profound Uncertainty

The application of astrology to financial trading on cryptocurrency markets represents an iteration of a historical phenomenon. Financial astrology was ‘not fringe, but mainstream’ in the late nineteenth-century culture of finance.⁵³ Traders and market analysts such as W. D. Gann and Charles Dow are known as early pioneers, while investment banker J. P. Morgan was ‘a devout follower of celebrity astrologer Evangeline Adams’.⁵⁴ The influence of astrology beyond an elite set of famous bankers is evidenced by several ‘how-to’ books published throughout the last century, for example *Astro Cycles and Speculative Markets*,⁵⁵ which was published in the 1930s and became ‘an instant sensation on Wall Street and around the country’.⁵⁶ Other examples of such books include *The Lost Science: Esoteric Math and Astrology Techniques for the Market Trader*,⁵⁷ *A Trader’s Guide to Financial Astrology: Forecasting Market Cycles using Planetary and Lunar Movements*,⁵⁸ and *Financial Astrology: How to Forecast Business and the Stock Market*.⁵⁹

Astrology was able to gain large mainstream popularity compared with the relative marginality of other forms of non-religious spirituality—at least in a Western context—because its predictions were (seemingly) substantiated by actual celestial positions and movements also described by scientific astronomers as objective facts.⁶⁰ At the same time, the predictive relationship that is described between these celestial facts and life circumstances is not verifiable. Adorno described this combination as the ‘pseudo-rationality’⁶¹ of astrology. This ambiguity between rationality and belief allowed astrology to establish itself in societies characterised by the dominance of rational thinking.

By casting personal, financial, and world events in a predestined light, astrology puts reality at a distance, pushing it beyond human agency or critical reflection.⁶² Adorno found that the predictions published in the 1950s LA Times astrology column reproduced a normative, gendered, and capitalist lifestyle, including advice about when to work, when to do housework, or when it was time for leisure activities in the context of the nuclear family.⁶³ Appearing

53 Komporez-Athanasios, *Speculative Communities*, 77.

54 Komporez-Athanasios, *Speculative Communities*, 77.

55 L.J. Jensen, *Astro Cycles and Speculative Markets*, Pomeroy: Lambert Gann Publishing Company, 1985.

56 Komporez-Athanasios, *Speculative Communities*, 77.

57 M.G. Bucholtz, *The Lost Science: Esoteric Math and Astrology Techniques for the Market Trader*, Bloomington: iUniverse, Inc, 2013.

58 Larry Pesavento and Shane Smoleny, *A Trader’s Guide to Financial Astrology: Forecasting Market Cycles Using Planetary and Lunar Movements*, Hoboken: Wiley, 2015.

59 David Williams, *Financial Astrology: How to Forecast Business and the Stock Market*, Tempe: American Federation of Astrologers, 1982.

60 Theodor Adorno, *The Stars Down to Earth and Other Essays on the Irrational in Culture*, London: Routledge, 2002 [1975].

61 Adorno, *The Stars Down to Earth*, 46.

62 Moira Weigel, ‘The Authoritarian Personality 2.0’, *Polity* 54.1 (2022): 146-80.

63 Adorno, *The Stars Down to Earth*. For example, ‘[i]n P.M. arrange cleaning, laundry, clothing, furnishing, diet problems. (November 1952, Virgo)’ or ‘A.M. finds your own desires conflicting with those of family member. Don’t argue, cooperate to prevent lasting resentment. (2 January 1953, Taurus).’ Carroll

in a 'conservative newspaper leaning far to the right wing of the Republican Party'⁶⁴ these predictions perform the social dynamics of 1950s American conservatism and present unequal gender norms as natural facts written in the stars. What Adorno's analysis shows is that astrological predictions are consequential in their performance of particular imaginaries, while flying under the radar of scrutiny because of their association with the frivolity of gendered pseudo-rationality.

In the context of contemporary uncertainty, broken promises, and widespread speculative practices, the popular revival of astrology doubles down on this ambiguity between 'the realistic and the irrational'.⁶⁵ Rather than searching for concrete answers to regain certainty, 'the poetic ambivalence of astrology'⁶⁶ forms a practice that is responsive to the deep sense of uncertainty that pervades contemporary everyday financialised life. Astrology performs speculative imagination about the future that may or may not come true, offering a way of being in the present that does not depend on hopes and dreams that no longer seem viable or attainable. Komporez-Athanasίου's analyses contemporary astrological practices beyond the confines of the financial market proper, for example in the use of astrology in dating apps, and shows that the speculative logic that astrology exemplifies influences everyday life in a profound way.⁶⁷ I add to this by arguing that crypto culture represents the spread of finance into everyday life in its own ways, and the emergence of astrology in this context offers a distinctive phenomenon worthy of scrutiny.

Social media influencer Maren Altman operates exactly at the intersection where financial speculation with cryptocurrencies and astrology meet. Among astrology videos with predictions related to world events (wars, politics,⁶⁸ etc.), Altman's YouTube channel also features a playlist called 'crypto astro' in which she uses the birth charts of Bitcoin and other cryptocurrencies⁶⁹ to predict their upcoming performance. In these predictions, she combines astrology with technical analysis in order to base her trading strategies on all the available information. Like any crypto influencer, she is adamant that her videos are not to be taken as financial advice, although in reality that does seem to be their main take-away.⁷⁰

Righter's LA Times column Astrological Forecasts qtd in Adorno, *The Stars Down to Earth*, 88, 93.

64 Adorno, *The Stars Down to Earth*, 56.

65 Adorno, *The Stars Down to Earth*, 157.

66 Komporez-Athanasίου, *Speculative Communities*, 94.

67 Komporez-Athanasίου, *Speculative Communities*.

68 Altman's fame rose to new levels after she predicted the 2020 US presidential election outcome correctly—a call she describes as being very easy to make if you look at the astrology. See Luke Martin Crypto, 'Predicting Crypto Cycles Based on the Moon?? (Specific Dates Explained!)', *YouTube video*, 2022, <https://www.youtube.com/watch?v=E5wRW2D0Stw>.

69 Just like with birth charts of people, cryptocurrency birth charts are made by analysing the celestial situation at the moment a cryptocurrency was 'born'—i.e., when it was first logged on a blockchain.

70 Maren Altman is the most prominent crypto astrologer in the community. She has over 200,000 subscribers on YouTube and 1.4 million followers on TikTok, and has been invited to share her approaches in popular crypto-themed media. See the podcasts analysed in this section and e.g., Jilian Godsil, 'Astrology Charts and Technical Analysis: Maren Altman is a Star', *Magazine by Cointelegraph*, 2022, <https://cointelegraph.com/magazine/astrology-charts-beat-technical-analysis-maren-altman-star/>. The presence of other crypto astrology accounts attests to the spread of the phenomenon beyond Altman's popularity. For example, @WatersAbove is a YouTube channel that combines crypto trading

Altman presents herself as ‘an astrologer and social media personality and entrepreneur focusing on finance and mostly crypto.’⁷¹ Besides YouTube, she uses several other social media platforms and has substantial followings on each.⁷² Altman is young,⁷³ energetic, brazen, and speaks in a vernacular that displays her immersion in crypto and internet subcultures. Before her introduction to crypto, she studied for a philosophy BA, and her interest in philosophical topics such as the nature of reality and the possibility of free will is noticeable in her discussions of crypto and astrology.

Altman records many of her videos from her bedroom featuring red atmospheric lighting. Altman herself always appears in red, black, or white clothing, with the dramatic lighting and make-up adding to the goth aesthetic. She overlays her videos with the relevant astrological and financial charts as talks through her analyses to substantiate her predictions, pointing out the patterns upon which she is basing them (see e.g., Figure 13). For example, in a video from January 25, 2022, entitled ‘FEBRUARY 2022 CRYPTO MARKET PREDICTIONS: BTFD or GTFO?’ she predicts how Bitcoin’s price will fluctuate as it goes into February:

the new moon on the 1st, I will say, is right on the United States’ North node, which indicates increase or ambition. So, I expect there to be a striking pump into the new moon on the first and a dump after.⁷⁴

A little later in the same video, she goes on to say that ‘on the 19th to the 20th, we have the Sun sextile Bitcoin’s Pluto, that’s great for conviction and price action and just overall confidence in the market’.⁷⁵

with gematria, numerology, and astrology (50,000+ subscribers, nearly 2,500 Patreons); whereas @AstroCryptoGuru (also known as Crypto Damus) has 31,000 followers on Twitter. Other smaller accounts such as @ProfitWithThePlanets, @ellavictoriaalexandra, and @astrologynewsBtcpredictions follow the same overall format. Compared to other crypto astrologers, Altman is the most outspoken and visible on the crypto scene. She is an astrology enthusiast and a crypto enthusiast, whereas some other creators only occasionally examine these topics together, sometimes, it seems, simply because their audiences are interested in their overlaps.

71 REKT Podcast, ‘Interview 103: Maren Altman’, *YouTube video*, 2021, <https://www.youtube.com/watch?v=TEM91H16aXY>.

72 Expanding on her work-life balance during her interview on the Bankless podcast, Altman states: ‘I work pretty non-stop, so in a certain way, if I’m doing [host interrupts unintelligibly] yeah! Because I love what I do, I love, um, people ask me ‘how do you put out so much content,’ I’m like ‘actually I would put out more if I didn’t have to add captions and text and the hashtags or whatever the hell I have to add.’ So, I can say a lot more, I don’t shut up’. Bankless, ‘The Astrological Crypto Maven With Maren Altman - Layer Zero’, *YouTube video*, 2022, <https://www.youtube.com/watch?v=aVVnrvJvSok>.

73 In several of the included videos she mentions being either 23 or 24 years old.

74 Maren Altman, ‘February 2022 Crypto Market Predictions: BTFD or GTFO?’, *YouTube video*, 2022, <https://www.youtube.com/watch?v=gLTPPdafTc4>. The title of this video exemplifies the advisory character of the Altman’s videos, despite her adamant claims to the contrary. BTFD means ‘buy the fucking dip’—i.e., the practice of buying cryptocurrency when their prices are low in order to have maximum yield when their popularity picks up again—and GTFO means ‘get the fuck out’—i.e., sell a particular cryptocurrency quickly before prices plummet.

75 Altman, ‘February 2022 Crypto Market Predictions’.



US stock market correction, BTC ETF, DOGE taking over the world: weekly crypto astrology 3/7

Figure 13: Screenshot detail of one of Maren Altman's crypto astrology videos. This one features the astrological chart for the Dogecoin cryptocurrency.⁷⁶

Although there is a wealth of content in her crypto astro playlist,⁷⁷ this is not the best place to analyse Altman's understanding of the connection between astrology and crypto trading. Although the crypto astro videos are filled with actual analyses and predictions, they do not contain explanations of the how and why of these things. Altman's various appearances in crypto-themed media outlets provide a better source for in-depth descriptions of her work. Many of the hosts of these outlets are either unaware or indeed seem downright sceptical of astrology and thus they provide a useful baseline against which Altman can respond with clarity and within the frame of reference of crypto culture. REKT Podcast, Bankless Shows, Luke Martin Crypto, and Gokhshtein Media are all YouTube channels focussed on crypto, with subscribers ranging from 377 (REKT Podcast, which only publishes audio and is not very YouTube-oriented) to the 192,000 follower Bankless Shows (which also publishes its content on other platforms). The host of the Unchained Podcast, Laura Shin, is an interesting exception on the list as the only female interviewer and a crypto journalist who would later publish *The Cryptopians: Idealism, Greed, Lies, and the Making of the First Big Cryptocurrency Craze*.⁷⁸ Shin explains that having been a yoga instructor herself, she has interacted with astrology in the past due to the overlap between the two communities.⁷⁹ Her interview includes

76 Maren Altman, 'US Stock Market Correction, BTC ETF, DOGE Taking Over the World: Weekly Crypto Astrology 3/7, 2021', YouTube video, 2021, <https://www.youtube.com/watch?v=vF6boUK61-Y>.

77 Forty-eight videos of varying lengths between a couple of minutes to half an hour (at the time of writing 27 January 2023).

78 Laura Shin, *The Cryptopians: Idealism, Greed, Lies, and the Making of the First Big Cryptocurrency Craze*, New York: PublicAffairs, 2022.

79 Unchained, 'Why Astrologer Maren Altman Believes 2023 Will Bring an Innovative 'Cyber Pandemic' – Ep. 205', YouTube video, 2021, <https://www.youtube.com/watch?v=Fd88sCH40Fk>.

a more in-depth discussion of some of the more technical aspects of astrology. Another outlier on the list was posted on the account of Altman’s new business venture: Astrace Official. This is a short promotional interview in which Altman presents her new company, which puts astrology itself on a blockchain. Although the nature of this interview is different from Altman’s other media appearances—it is a much more one-dimensional promotional piece, whereas the other interviews serve promotional purposes while at the same time offering entertainment and information—it has been included in the analysis because it provides some useful additional insights on the intricacies of merging astrology and blockchain. A full list of the audio analysed is listed in Table 1 below.

Date	Duration	Title	Channel	Host(s)
2021-02-12	1:04:03	Interview 103: Maren Altman	REKT Podcast	‘Bunchu’ and ‘Chamber’
2021-11-26	25:25	Why Astrologer Maren Altman Believes 2023 Will Bring an Innovative ‘Cyber Pandemic’ - Ep. 205	Unchained Podcast	Laura Shin
2022-05-17	59:33	The Astrological Crypto Maven with Maren Altman Layer Zero	Bankless Shows	David Hoffman
2022-08-28	3:24	Maren Altman: Inside Look at the Beginning of Astrace	Astrace Official	[unnamed]
2022-09-25	23:35	Predicting Crypto Cycles Based on the Moon?? (Specific Dates Explained!)	Luke Martin Crypto	Luke Martin
2022-11-09	55:54	Gokhshtein Unfiltered EP15 w/ Maren Altman Astrology & Cryptocurrency YouTuber @ MarenAltman	Gokhshtein Media	David Gokhshtein

Table 1: The interviews with Maren Altman included in the analysis. These metadata come from YouTube, but many are posted on (several) other platforms as well.

Astrology: A technical Analysis for all of Reality

Like the nineteenth-century financial astrologers that came before her, Maren Altman ‘blur[s] the lines between prediction and divination’⁸⁰ when she describes astrology as ‘a mirror for reality or the clock on the wall that’s telling you it’s 3 PM. It’s not making it 3 PM’.⁸¹ She explains that she is ‘being very deliberate about saying this is correlation not causation. I’m not someone that claims any magical abilities. In fact, I’m pretty, like, grounded, like, I say I’m like a human computer’.⁸² These statements include three underlying themes that are present throughout Altman’s interviews: the stated abilities of astrology in relation to reality and crypto; the rationalisation of a particular kind of astrology and market prediction; and the alignment of divinatory practices with computational processes—more specifically, blockchain’s affordances. I will use the rest of this section to discuss each of these themes in more detail.

Altman repeats throughout all her interviews that crypto astrology is about recognising correlation, not about finding direct causation. This nuance in her approach signifies an important alignment with the technical analyses practiced by crypto traders, who look for historical patterns to predict future market movements, yet do not frame such an approach deterministically. However, correlation does not preclude predestination—the movement of celestial bodies in the future is after all highly dependent on their past—and in fact this is a loophole that Altman implicitly uses to square astrology with rationality. She elaborates that ‘when people say things like ‘it was not the planets, it was this black swan event,’⁸³ I’m like ‘exactly! The planets were *informing* us that this was going to happen’;⁸⁴ and ‘I like to think of it as a glitch in the architecture of reality that we can look at, where we’re getting a sneak peek into a point of view that we wouldn’t normally have’.⁸⁵ The stars are like a map containing information about the fate of reality on earth that can be read by those who are skilled enough. To Altman, there is indeed such a thing as fate, the natural, predestined occurrence of events. Although she explains that interpreting the positions of the stars requires awareness of socio-political contexts on earth, Altman’s crypto predictions represent the crypto market’s shifts as outside of human agency: the stars already map out what will happen, and it is up to an astrologer’s interpretative capacities to unveil these fates.

Nevertheless, the complex contradictions of free will and cosmic predestination are not resolved in Altman’s view. For example, she considers the ethics of her own performative influence on the market when she explains that

there is a certain amount of basic responsibility with a very low cap project, or something, where I’m not going to speak about it, but this has kept me from speaking on

80 Komporozos-Athanasiou, *Speculative Communities*, 77.

81 Bankless, ‘The Astrological Crypto Maven’.

82 Bankless, ‘The Astrological Crypto Maven’.

83 A black swan event is an unexpected event with big consequences.

84 Bankless, ‘The Astrological Crypto Maven’. Emphasis added.

85 Bankless, ‘The Astrological Crypto Maven’.

any large cap alts⁸⁶ as well, just because I do feel like it hurts that community and it hurts the team.⁸⁷

At the same time, Altman is very determined to set herself apart from other astrologers who, in her view, diverge from logic and rationality. For example, Ophira Edut—an astrologer for ELLE magazine who has elaborated on the influence various astrological signs have on commercial, financial, or technical undertakings⁸⁸—describes astrology as the analysis of ‘[h]ow the stars *shape* the world and the future’,⁸⁹ and in doing so attributes agency to the stars in determining what life on earth looks like. Altman fundamentally disagrees with such conceptualisations of astrology, which attribute what she considers to be magical abilities to celestial bodies. Speaking about such approaches, Altman states:

they can literally bring in some weird random technique and it’s like [sarcastically] ‘okay there, okay all right sir, like, have fun with that.’ Like, they can bring in some really weird things like speeds and proximity to the Sun and very, like, more exotic things, which, um, I don’t have experience with.⁹⁰

According to Altman, besides her own, there is another kind of astrology which is weird, random, or magical. Here, she seems to echo some of the scepticism that her interviewers voice in the interviews, and aligns herself with their (uninitiated) perception of mainstream astrology when, in fact, she argues her own version of ‘astrology is TA’⁹¹ for all of reality⁹² and that these two practises complement each other:

I found that tracking finance and tracking candlestick charts is very similar to tracking planets in the way that it’s symbolism and patterns and angles, even, like, shapes, angles, knowing how that translates into psychological and world event significations. It’s basically the same.⁹³

86 Altcoins (non-Bitcoin cryptocurrencies) with a large market capitalisation—i.e., the total value in USD of a project’s tokens at a given time.

87 Gokhstein Media, Gokhshtein Unfiltered Ep15 W/ Maren Altman - Astrology & Cryptocurrency Youtuber @marenaltman, YouTube video, 2022. <https://www.youtube.com/watch?v=TQ0jkuwJK4Y>.

88 For example, she explains that with ‘Uranus, which rules technology, making its transit through Taurus, the sign of money’ between 2017 and 2026, the stars were set for the world to experience a digital finance revolution. To give some historical data showing the significance of the relation between planets and signs, she explains that Pluto was in Sagittarius (which is good for publishing) in the years the internet (and internet-based publishing innovations such as blogs and social media) became mainstream between 1995 and 2008, after which it moved into Capricorn, which is reflected in the emergence of big Web 2.0 corporations—Capricorn being the sign of business and corporations. See AstroTwins, ‘What Astrology Can Teach Us About NFTs’, *Astrostyle*, 2022, <https://astrostyle.com/what-astrology-can-teach-us-about-nfts/>.

89 AstroTwins, ‘What Astrology Can Teach Us About NFTs’. Emphasis added.

90 Luke Martin Crypto, ‘Predicting Crypto Cycles Based on the Moon??’.

91 TA stands for Technical Analysis and refers to the practice of analysing historical data about a cryptocurrency to make predictions about its future market performance.

92 Unchained, ‘Why Astrologer Maren Altman Believes 2023 Will Bring an Innovative ‘Cyber Pandemic’.

93 Luke Martin Crypto, ‘Predicting Crypto Cycles Based on the Moon??’.

In the same way astrology consists of finding patterns in celestial movements and correlating them with earthly events in order to predict future circumstances, performing a technical analysis consists of finding historical patterns in financial data, correlating them with current price movements, and predicting the future fluctuations of a coin's price on that basis. While astrology is often associated with the inscrutable skill of the individual astrologer, by describing it in similar terms as technical analysis, Altman's astrology gains a pseudo-rational aura of facticity and logic. Conversely, as she attempts to rationalise astrology, Altman also exposes the pseudo-rationality of technical analysis. She often emphasises the similarities between astrological and technical analyses and explains how she uses both in tandem because they have such similar structures.

She proposes that thinking of astrology only as magical is a narrow view, and instead offers her own, supposedly very different version of astrology. This grounded version of astrology, she argues, is comparable to algorithmic processes, logics, and the precision of data:

It's essentially looking at the angles between planets and how that changes as planets move relative to a stagnant chart. So we're I think looking at a whole lot of angles and it's very... it reminds me of logic class or logical equations where this in this relation to this equals this. And over time when you multiply all of that data up, it gives us a really good indicator of what is to come.⁹⁴

It's a ton of, basically, data collection over time that is refined into certain alignments [that] represent certain world events.⁹⁵

Generally, there's a really strong consensus [among astrologers] because astrology is basically first order logic, like, if-this-then-that statements, pretty clearly.⁹⁶

Altman presents astrological prediction for crypto trading as a three-part process that starts with data collection—meaning, in this case, the past performances of coins, their birth charts, and ongoing data about planetary movements. Then, patterns are discerned in these data, particularly through the analysis of the angles and shapes between two charts. These patterns are then turned into logical statements producing a prediction: if <X planetary movement over a specific coin's birth chart at a specific moment in time> then <predict Y future market condition for this coin>.

Framing astrology as functioning under these computational logics can be understood as a tactic to make it palatable for an audience of crypto traders that is accustomed to proof and transparency on-chain:⁹⁷

94 Unchained, 'Why Astrologer Maren Altman Believes 2023 Will Bring an Innovative 'Cyber Pandemic''.

95 Bankless, 'The Astrological Crypto Maven'.

96 Luke Martin Crypto, 'Predicting Crypto Cycles Based on the Moon??'.

97 Komporozos-Athanasiou recognises a similar strategy among early financial astrologers 'appeal[ing] to scientific rationality' to increase 'their appeal in the eyes of financiers'. See Komporozos-Athanasiou, *Speculative Communities*, 77.

I want to be more legitimizing with something that I think talks about the nature of reality and the framework of existence. It's not woo-woo.⁹⁸ It's literally here and now, we can see these patterns. I want to put it on chain for everyone to see transparently.⁹⁹

By logging astrological crypto predictions on a blockchain—the proposition of Altman's new business venture called Astrace—Altman attempts to overcome the opacity of the crypto market. With Astrace, Altman is attempting to lift astrology to the same standards of verification, proof, and immutability that the rationalist imaginary of blockchain as a truth machine boasts. Whereas astrological predictions previously depended in part on the contextual knowledge of the astrologer for the correct interpretation of the stars, on a blockchain, it can now be freed from human error as well as an excess amount of more-than-rational mysticism:

putting [astrology] into something that can take my error out of it, of trade execution, or of... saying the moon instead of mercury or [turning] 90 degrees into 40 degrees. Like, I've done that before many times, where I just... human error comes in.¹⁰⁰

there's no large scale meta-analysis or study to quantify this that is acceptable to me. So I think what's missing is something that I'm building, which is combining these two things into an astrological algorithm that will power a trading vault.¹⁰¹

Altman discusses astrology in terms that are more reminiscent of a technology business than the occult. To her, astrology is quantitative and is produced through algorithmic calculation based on metrics. It can be verified through back-testing, because it can—through Astrace—make use of an immutable blockchain ledger logging past data transparently. All of these ways of framing crypto astrology serve to present it as rational, calculable, and predictable. The future of the market can finally be understood with something approaching certainty through logics and clear processes. By making astrology back-testable, Altman uses blockchain to disintermediate the stars, eliminating human error and the subjectivity of interpretation once and for all. Although Altman engages with the logic of speculation, she continuously attempts to regain lost capacities for control and the feeling of a future full of promised profitability, while holding on to the myth of order and predictability attached to computational processes in general and blockchains specifically. The second half of this chapter—featuring crypto memes—succeeds in shaking of this myth.

98 'Woo-woo' according to the Urban Dictionary is 'descriptive of an event or person espousing New Age theories such as energy work, crystal magic, Reiki, bizarrely restrictive diets, or supernatural/paranormal/psychic occurrences'. See Lowe, Jennifer, 'Woo-Woo', *Urban Dictionary*, 2005, <https://www.urbandictionary.com/define.php?term=woo-woo>.

99 Astrace Official, 'Maren Altman: Inside Look At the Beginning of Astrace', *YouTube video*, 2022, <https://www.youtube.com/watch?v=ndzSo3jTmKg>.

100 Bankless, 'The Astrological Crypto Maven'.

101 Astrace Official, 'Maren Altman'.

The Performativity of Memetic, Vernacular, Speculative Imaginaries

Before I move to my analysis of crypto memes, some more theoretical foundations are needed, particularly on the connection between the concepts of performativity, vernaculars, and memes. Memes are particularly suitable for an analysis that is concerned with performativity, because of their inherent characteristics as circulating media objects that imitate, iterate, and transform meaning in a distributed manner. As such, memes function like language: they get their meaning not solely or primarily based on the intention of a speaker or sender, but in the sociocultural contexts in which they operate; they are subject to multiple iterations via ongoing citational expressions that carry forward norms and ideals about society and everyday life. Through the vernacular use of language and subcultural references, memes contribute to establishing community boundaries, the values and norms of which they consist, and their particular form of speculative imagination. Memes are normative performative language objects that supercharge imaginaries through their memetic affordances.

Because memes are such distributed and decentralised phenomena, capturing them for analysis is not a straightforward process. Therefore, I also detail the methods I used for collecting and selecting memes for analysis. Using digital methods, I develop both a way to collect a large set of memes *and* ready them for the closer analysis.

The Performativity of Memes and Online Vernaculars

The term ‘meme’ was first coined by Richard Dawkins in 1976, well before computers and the internet became accessible to everyday users. Reflecting on the expansions of mass media and popular culture of the time, Dawkins wrote that a meme is a ‘unit of cultural transmission’,¹⁰² for example:

tunes, ideas, catch-phrases, clothes fashions, ways of making pots or of building arches. Just as genes propagate themselves in the gene pool by leaping from body to body via sperms or eggs, so memes propagate themselves in the meme pool by leaping from brain to brain via a process which, in the broad sense, can be called imitation.¹⁰³

While for a period, image macros were their most ubiquitous and recognisable form, memes are more than just images with text on them¹⁰⁴ encompassing ideas ‘presented as a written text, image, language ‘move,’ or some other unit of cultural ‘stuff’’.¹⁰⁵ This ‘cultural stuff’ is not inconsequential, in fact, it ‘shape[s] and reflect[s] general social mindsets’.¹⁰⁶

102 Richard Dawkins, *The Selfish Gene*, Oxford: Oxford University Press, 2016, 249.

103 Dawkins, *The Selfish Gene*, 249.

104 Limor Shifman, *Memes in Digital Culture*, Cambridge, MA: MIT Press, 2013; Ryan M Milner, *The World Made Meme: Public Conversations and Participatory Media*, Cambridge, MA: MIT Press, 2018.

105 Michele Knobel and Colin Lankshear (eds) *A New Literacies Sampler*, New York: Peter Lang, 2007, 202.

106 Shifman, *Memes in Digital Culture*, 4.

Yet, Dawkins' description of memes has in recent years been criticised for its disregard of the role humans play in their spread, portraying memes as viruses that have agency and strive for Darwinian survival through self-replication.¹⁰⁷ Most prominently, Limor Shifman argues that instead of viruses, each user sharing and changing memes contributes to a larger cultural memetic phenomenon. She writes that a meme is '(a) a *group of digital items sharing common characteristics* of content, form, and/or stance; (b) that were created *with awareness of each other*; and (c) were circulated, imitated, and/or transformed *via the Internet by many users*'.¹⁰⁸ This means that, due to the networked and evolving nature of the object, a meme can contain a multitude of individual contributions differing in their stance on the same subject matter.¹⁰⁹ As such, a meme's meaning can be taken in different directions by different individuals, subcultures, or political groups, showing that 'memetic media afford polyvocality'.¹¹⁰ Additionally, memes often feature irony in abundance, making any intended meaning hard to determine. This is especially true in anonymous spaces such as 4chan, in which many memes are thought to originate.¹¹¹

As circulating, imitating, and transforming cultural objects, memes are performative by definition. Instead of searching for intended meaning, I will use the concept of performativity to understand what the memes *do* in their broader sociocultural context. As I discussed in chapter 1, performativity refers to the capacity of language to have an effect in the world that emerges from its embeddedness in a social system of conventions, repetition, and iteration,¹¹² or what is known as 'citationality'.¹¹³ Citationality is relevant to the study of the performativity of memes for three reasons. First, it means that they '*can and must be able to function in the radical absence of the sender, the receiver and the context of production*'.¹¹⁴ Second, while memes are repeatable, they are not the very same, identical thing in two different occasions: they are not impermeable units isolated from their environment, but rather get their meaning in relation to their context. Third, this context is not tied to the meme, and rather travels with it into its future iterations as a trace.¹¹⁵ Meaning, therefore, exists partly¹¹⁶ outside of the intention of any meme-maker or sender. Additionally, individuals cannot express themselves without referencing, reiterating, reflecting on expressions made previously—a phenomenon Deleuze

107 Kate M. Miltner, 'Internet Memes', in J. Burgess, A. Marwick, and T. Poell (eds) *The Sage Handbook of Social Media*, London: SAGE Publications Ltd, 2018, 412-28.

108 Shifman, *Memes in Digital Culture*, 7-8, emphasis in original.

109 Shifman, *Memes in Digital Culture*.

110 Milner, *The World Made Meme*, 6.

111 Daniël de Zeeuw and Marc Tuters, 'Teh Internet is Serious Business: On the Deep Vernacular Web and Its Discontents', *Cultural Politics* 16.2 (2020): 214-32.

112 James Loxley, *Performativity*, Oxon: Routledge, 2007; Jillian R. Cavanaugh, 'Performativity', Oxford Bibliographies (2015): <https://doi.org/10.1093/obo/9780199766567-0114>.

113 Jacques Derrida, *Limited Inc*, Evanston: Northwestern University Press, 1988, 12.

114 Derrida, *Limited Inc*, 48, emphasis in original.

115 Derrida, *Limited Inc*.

116 Derrida does not do away with intentionality all together, because if this logic would be followed to its extreme, language would start to function like an automaton of societal conventions. Rather, meaning is the result of the combination of a particular speaker and the contexts in which something is, and was previously, spoken. See Jacques Derrida, *Without Alibi*, Stanford: Stanford University Press, 2002.

and Guattari refer to as the indirectness of discourse¹¹⁷—and so, expressions are not individual, but have a ‘necessarily social character’ carrying forward ‘implicit presuppositions’.¹¹⁸ Memes exist by virtue of their repetition, citation, and iteration and as such reaffirm and reshape acceptability and social norms within their vernacular language communities.

Memes spread, replicate, and shape-shift through time and space as ‘order-words’¹¹⁹ that influence what we know about the world and how to act in accordance with this knowledge. In the process, they contribute to the ‘patterning of unconscious obedience [...] to the imagined desires of the vernacular language community’.¹²⁰ The notion of order-words highlight how memes carry within them assumptions about the world and how each individual memetic speech act exists as part of, is the result of, and performs a ‘collective assemblage of enunciation’ casting society in particular frames and dimensions.¹²¹

As contributors to the establishment of imaginaries, memes are simultaneously a tool for the negotiation, as well as the establishment of norms.¹²² Each created or shared meme contributes to the continuous performance of societal norms and, as such, plays a part in creating those norms. In the repeated performance of these norms, memes play a role in establishing boundaries, for example of acceptability, normalcy, or even what is considered to be natural under the influence of powerful societal institutions.¹²³

Memes are ‘socially constructed public discourses’¹²⁴ and ‘play an important role in the construction of group identity and social boundaries’.¹²⁵ This means that to understand the social relevance of particular memes as well as their dense layers of subcultural references, intimate familiarity with the community from which they originate is necessary.¹²⁶ Indeed, ‘knowing *what* to say as well as *how* to say it is key to signal (sub)cultural belonging’ in online

117 Gilles Deleuze and Félix Guattari, *A Thousand Plateaus*, London: Bloomsbury Academic, 2021 [1980].

118 Deleuze and Guattari, *A Thousand Plateaus*, 91–92.

119 Deleuze and Guattari, *A Thousand Plateaus*, 92.

120 Marc Tuters, ‘Why Meme Magic is Real But Memes Are Not: On Order Words, Refrains and the Deep Vernacular Web’, in Mike Watson, and Jerry Galle (eds) *Memnesia*, Rotterdam: V2_Publishing, 2021, 51–52.

121 Deleuze and Guattari, *A Thousand Plateaus*, 91. An infamous example from the world of alt-right memes showcases this performativity. The election victory that saw Donald Trump become president in 2017 was produced in part through what is known in alt-right circles as ‘meme magic’: the alt-right meme army brought into being a real-life version of their imagined homeland Kekistan by sharing Pepe the Frog memes, and are thought to have helped ‘meme Trump into the presidency’. See Hannah Barton, ‘Memes, Magic and What it All Means: On Explaining Current Research When Your Findings Leave You Lost for Words’, *Dandelion* 8.1 (2017): 1–6.; See also S. Peeters et al., ‘On the Vernacular Language Games of an Antagonistic Online Subculture’, *Frontiers in Big Data* 4.1 (2021): 1–15. <https://doi.org/10.3389/fdata.2021.718368>.

122 Noam Gal, Limor Shifman, and Zohar Kampf, ‘“It Gets Better”: Internet Memes and the Construction of Collective Identity’, *New Media & Society* 18.8 (2016): 1698–714.

123 Judith Butler, ‘Performative Acts and Gender Constitution: An Essay in Phenomenology and Feminist Theory’, *Theatre Journal* 40.4 (1988): 519–31.; Gal, Shifman, and Kampf, ‘“It Gets Better”’.

124 Shifman, *Memes in Digital Culture*, 8.

125 Shifman, *Memes in Digital Culture*, 100.

126 Shifman, *Memes in Digital Culture*; Milner, *The World Made Meme*.

pseudonymous communities, and the continuous emergence of neologisms plays a significant role in this process.¹²⁷ Being able to understand a meme may in this sense be seen as a form of linguistic cultural capital.¹²⁸ Generally, this is associated with institutionalised languages like those of law, education, and medicine, and their normative capacities. However, slang—such as subcultural memetic language—although spoken by a dominated class, also carries within it an element of oppositional power.¹²⁹ As such, in-group slang can act as ‘a quasi-aristocratic expression of distinction’ and mastery of subcultural language is embedded in power dynamics, ideology, and the establishment of group boundaries.¹³⁰

Language in memes constitutes subcultural vernaculars, emerging from the ‘deep vernacular web’ of anonymous platforms such as 4chan.¹³¹ This deep vernacular web is a place characterised by the inability to convey one’s identity in any other way than through continuous use of visual and textual vernacular cues. An online vernacular is ‘a group-based discourse, or sociolect, [featuring] highly innovative forms of expression that are often intentionally obscure to those outside the in-group’.¹³² Speakers of such a vernacular are part of ‘subcultural language communities’ connected through shared ideas on linguistic and social norms.¹³³ Similar to Benedict Anderson’s imagined communities,¹³⁴ subcultural language communities consist of groups of people that do not know each other, yet feel some sense of belonging to a shared and finite grouping. Technologies such as the printing press played a significant role in the historical emergence of these imagined communities because they facilitated the sharing of information in simultaneous and standardised ways and capitalised on vernacular languages.¹³⁵ Internet culture, making use of emerging technologies, has given new impetus to the speed with which language communities take shape and spread.¹³⁶ Memes represent a technology that is particularly suitable for the establishment of imagined communities: the speed with which they travel between individuals gives the appearance of simultaneity and

127 Sal Hagen and Daniël de Zeeuw, ‘Based and Confused: Tracing the Political Connotations of a Memetic Phrase Across the Web’, *Big Data & Society* 10.1 (2023), emphasis in original. Memes that are known because of their repeated phrases or syntax, such as lolspeak or doge’s thoughts, can even turn into their own language variety. See Piia Varis and Jan Blommaert, ‘Conviviality and Collectives on Social Media: Virality, Memes, and New Social Structures’, *Multilingual Margins: A Journal of Multilingualism from the Periphery* 2.1 (2015): 31–45.

128 Asaf Nissenbaum and Limor Shifman, ‘Internet Memes as Contested Cultural Capital: The Case of 4chan’s/b/board’, *New Media & Society* 19.4 (2017): 483–501.

129 Pierre Bourdieu, *Language and Symbolic Power*, Cambridge: Polity Press, 1991.

130 Marc Tuters and Sal Hagen, ‘(((they))) Rule: Memetic Antagonism and Nebulous Othering on 4chan’, *New Media & Society* 22.12 (2020), 2225.

131 de Zeeuw and Tuters, ‘Teh Internet is Serious Business,’ 214.

132 de Zeeuw and Tuters, ‘Teh Internet is Serious Business,’ 216.

133 Peeters, ‘On the Vernacular Language Games of an Antagonistic Online Subculture,’ 2.

134 Benedict Anderson, *Imagined Communities: Reflections on the Origin and Spread of Nationalism*, London: Verso, 1983.

135 Anderson, *Imagined Communities*.

136 Holmes David, *Communication Theory: Media, Technology, Society*, London: Sage Publications, 2005; Craig Calhoun, ‘Indirect Relationships and Imagined Communities: Large-Scale Social Integration and the Transformation of Everyday Life’, in Pierre Bourdieu and James Samuel Coleman (eds) *Social Theory for a Changing Society*, Boulder: Westview Press, 1991, 95–120.

puts the focus on present shared experiences, and the logics of memetics provide a level of standardisation within continuously shifting subcultural vernaculars.¹³⁷

Komporozos-Athanasiou elaborates on the spread of financial jargon through the ‘social vernacular of speculation’ of contemporary culture in which promises of a given future have lost their power to convince and ‘shorting and hedging the unknown becomes the zeitgeist of contemporary financialised societies’.¹³⁸ Such vernaculars contribute to the establishment of narratives and myths through which the world can be understood and imagined. The crypto community uses memes to share notions about how the world works and who belongs to their group, without knowing each of its own members. Rather than just reflections of societal dynamics, memes *co-create* them and in the context of financialisation are a medium for the ‘speculative imagination’.¹³⁹

In using memes as an object of analysis for the study of imaginaries it is important to take into account that humour plays an important role in memes and they can be understood as a ‘funhouse mirror for culture and society’,¹⁴⁰ meaning that while they do reflect reality, at the same time they also distort it. Therefore, it is safe to assume that the memes represent exaggerated and stereotypical versions of different proclivities in crypto culture, and real people and their practices will most likely exist between such extremes, shifting positions over time.

Collecting and Selecting Memes for Analysis

I employed two separate modes of collecting memes, the first being informal and based on my own experience in crypto culture, and the second being structured and more distanced. After gaining my initial interest in memes as a conveyer of blockchain imaginaries in 2021, I

137 Much research has been (and continues to be) done on the memetic qualities of memes, focusing particularly on the way they spread and change. See Maria Carolina Zanette, Izidoro Blikstein, and Luca M. Visconti, ‘Intertextual Virality and Vernacular Repertoires: Internet Memes as Objects Connecting Different Online Worlds’, *Revista de Administração de Empresas* 59.1 (2019): 157–69. Others, however, critique this approach because it ‘dismiss[es] the medium’ as an inert channel’ and focus instead on, for example, the influence of platforms such as Reddit and 4Chan on meme culture. See Tony D. Sampson, *Virality: Contagion Theory in the Age of Networks*, Minneapolis, MN: University of Minnesota Press, 2012, 72; and e.g., Adrienne Massanari, ‘#gamergate and the Fappening: How Reddit’s Algorithm, Governance, and Culture Support Toxic Technocultures’, *New Media & Society* 19.3 (2017): 329–46; Ioana Literat and Sarah van den Berg, ‘Buy Memes Low, Sell Memes High: Vernacular Criticism and Collective Negotiations of Value on Reddit’s Memeconomy’, *Information, Communication & Society* 22.2 (2019): 232–49. This perspective can be understood in terms of a McLuhanian focus on the effect of media on the message, but also has its roots in linguistic pragmatics, which looks at languages and oral cultures in their context of use: ‘internet vernaculars are shaped by the affordances of their milieus as well as how they are transcoded between them’. See Tuters, ‘Why Meme Magic is Real But Memes Are Not,’ 47. For example, the study of platform vernaculars focusses on use practices and the affordances of both the platform and the technology of memes themselves. See Martin Gibbs et al., ‘# Funeral and Instagram: Death, Social Media, and Platform Vernacular’, *Information, Communication & Society* 18.3 (2015): 255–68.

138 Komporozos-Athanasiou, *Speculative Communities*, 9, 2.

139 Komporozos-Athanasiou, *Speculative Communities*, 66.

140 Miltner, ‘Internet Memes,’ 413.

committed to downloading memes that sparked my interest as I came across them. These encounters occurred in various places, such as newsletters (e.g., Bankless), subreddits (e.g., r/WallStreetBetsCrypto), and on Twitter. Seeing an interesting meme format or vernacular term, I then also searched for more examples or iterations through dedicated meme databases (e.g., Know Your Meme or Meme Generator) as well as general searches on Google (e.g., using a search query such as ‘HODL meme’). This process lasted several months and resulted in 138 memes. The purpose of this process was mostly exploratory, based on, and extending, my own knowledge of the culture that I studied. It served to familiarise me with what is being expressed in the culture and some of these memes ended up being used in my analysis. Reading the memes broadly allowed me to map the scope of their content and better target my exploration of a larger dataset. This also informed my choice to focus specifically on cryptocurrency-related memes, as these were the overwhelming majority of the memes I collected thus far.

For the second part of my collection process, I worked in a more formal manner. While many memes can be found in newsletters and on Twitter, the platform that offers the most consistent stream of memetic content, while also lending itself well to automated collection, is Reddit. After spending time reading through several active and well-populated subreddits (e.g., r/Bitcoin, r/CryptoCurrency, r/NFT), I picked r/CryptoCurrencymemes as the subject of my analysis because it combines a topical interest that is broad enough (not just a particular cryptocurrency or only NFTs) with a commitment to memetic content. The content of the subreddit was collected from the moment it started on November 8, 2017, until March 29, 2022.¹⁴¹ After cleaning the data, the final number of memes downloaded was 14,767.

Having such a large dataset can be a great resource, but it can also be stifling. Before any qualitative analysis can be done, it needs to be made accessible in a meaningful format. In order to allow for textual querying, the memes were put through an Optical Character Recognition (OCR) process that extracted the text in the meme.¹⁴² As my analysis is focussed primarily on the words and phrases in the memes, this step was fundamental for my ability to parse their meaning across multiple memes. I queried the dataset for words and acronyms particular to the vernacular of the cryptocurrency community, such as HODL, DYOR, pump, or rekt. The resulting memes were stored in a separate folder and examined to gain an initial

141 The code for the scraping process was written and executed by Gaurika Chaturvedi, who was a student on the RMA Media Studies at the University of Amsterdam at the time. The code can be found at <https://github.com/acidicgaseouscloud/Scraping/blob/main/Reddit/Pushshift>. It functions by querying the Pushshift.io API using Python. See Jason Baumgartner et al., ‘The Pushshift Reddit Dataset’, *Proceedings of the International AAAI Conference on Web and Social Media*. Online, June 8-11, 2020, 2020. <https://doi.org/10.1609/icwsm.v14i1.7347>. Chaturvedi also performed the downloading and OCR process described below. The data captured for each post consists of its URL, its subject line, and the URL to the media file of the meme posted in it. Then, the GNU Wget software package was used to download the media file located at those latter URLs. Due to some noise (e.g., posts without memes, but other visual content such as screenshots or embedded Tweets), Wget was set to only download from URLs containing ‘imgur’ or ‘redd.it’. Some memes were no longer accessible due to removal by a moderator.

142 This process made use of the Google Vision AI via 4CAT, a tool developed by OILab and the Digital Methods Initiative at the University of Amsterdam. The tool can be found at <https://4cat.oilab.nl/>.

understanding as well as to find new terms with which I was not yet familiar. Additionally, I selected several random samples from the dataset for close examination. The list of queries grew over time, snowballing from term to term. Finally, seventy-five such queries were made. During this process I got to know the dataset, allowing me to untangle the imaginary to which they contribute.

The next section focusses on crypto memes as cultural objects that shape and reflect the imaginary of the cryptocurrency community. Due to the platform and manner of collection, the selection of memes will consist of image macros, gifs, or other visual formats and the repeated phrases in them, for example the memetic sentence that a coin will go ‘to the moon,’ indicating that it is predicted to skyrocket in price. I focus particularly on memes that relate to ideas on knowledge, truth, volatility, and uncertainty in order to reflect on the imaginary of blockchain as a truth machine. Each of the four subsections features memes using key phrases from the crypto vernacular. I outline the meaning of these terms outside and inside the crypto community before interpreting the meaning of the memes in which they feature. Taken together, the memes substantiate a crypto imaginary that doubles down on radical uncertainty and frames speculation as a means of survival.

Fanning the Flames of Radical Uncertainty: Speculative Imagination in Crypto Memes

As a space that, besides genuine, cooperative, and benevolent behaviour, is also home to everything from opportunistic money-grabs to collectively orchestrated market manipulation, the language spoken in crypto memes sometimes comes across as a riddle that is difficult to solve completely. There is always a layer of context that remains inaccessible, especially since the speaker is anonymous or pseudonymous. The crypto vernacular is not to be taken at face value, yet, it *is* indicative of the imaginary that characterises the crypto community: instead of blockchain as a truth machine, the imaginary that takes shape here can be summarised as ‘crypto as an uncertainty machine.’ Rather than the immutable or universal truths that previous analyses in this book highlighted, here the truths that crypto churns out are so dizzyingly abundant, so ambiguous, and so rapidly turned upside down that instead of relying on their veracity, what counts is their immediate effectiveness to those that know how to revel in uncertainty.

Before I present my analysis, a quick word on the sexism and racism that is rampant in crypto memes. In their use of vernacular language and other subcultural cues, memes contribute to the establishment of community boundaries.¹⁴³ In terms of who gets to be included in this crypto community, the memes paint a very clear picture: white, heterosexual men.¹⁴⁴ When included at all, people of colour are cast in stereotypical, racist roles. While (white) women are more prevalent, more often than not they appear while pleasing or tempting men sexually. Sometimes the goal of crypto trading itself is compared to penetrative heterosexual sex. When women do appear at the focus of a meme, their perceived sexiness *is* their success. Generally,

143 Shifman, *Memes in Digital Culture*.

144 This also reflects the demographics of memers in general Milner, *The World Made Meme*.

however, success is portrayed as masculine, muscular, and aggressive. Stereotypical white heterosexual men are those envisioned in this imaginary, to the detriment of all others. Often, a particular idea about crypto is expressed in different memes, some of them racist or sexist, others not. When I have the choice between these options, I will choose not to reproduce harmful or hurtful imagery even if this is a significant part of the character of the community as represented in the memes. In total, I have chosen not to include eight memes in this chapter despite their suitability to illustrate the crypto imaginary. By discussing their character in general in this paragraph, I intend to convey this part of the culture without reproducing it.

Influencing in Uncertainty

The memes featured here portray the perspective of those trying to find a way through what seems like an inscrutably volatile market, looking to influencers for help. Such influencers are a ubiquitous source of information that is at the same time barely illuminating. Through advice that may not be named advice, and the ever-present possibility of ulterior motives, the impact of influencers is defined less by the truthfulness of the information they share than by its potential to have an effect on the market.

In the context of internet culture in general, influencers are understood as social media users who, through (self-)branding, gain a large following that is monetised through advertorials, paid endorsements, and other collaborations with larger brands.¹⁴⁵ Crypto influencers share in-depth market analyses as well as personal or paid opinions on the potential of particular coins. There is a certain performativity to these predictions, as the large following an influencer may attract might in itself influence the trajectory of a coin's value. Just like many other social media influencers, crypto influencers often make a living out of their public status by establishing partnerships with companies.¹⁴⁶

The ubiquity of influencers is perhaps unsurprising, as amid the ever expanding list of possible crypto projects to buy into, some kind of guidance or curation is a welcome aid to anyone not themselves an expert in the field. Yet, the memes generally portray influencers in a sceptical light. For example, Figure 14 refers to influencers who state that the information they share is not to be taken as financial advice in order to avoid legal liability in case they inadvertently promote a fraudulent or crashing project. However, such statements are so ubiquitous that they have turned into a running joke: everyone says it, even if the content of their message does look and sound very much like financial advice and seems, very often, to be treated as such. The assessment of the usefulness of advice is made additionally difficult by the inability to identify the success or longevity of an influencer's own activity on the anonymous market (see Figure 15). In combination with the fact that influencers often monetise their

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- 145 Crystal Abidin, "‘Aren't These Just Young, Rich Women Doing Vain Things Online?’: Influencer Selfies as Subversive Frivolity", *Social Media + Society* 2.2 (2016): 1-17. <https://doi.org/10.1177/2056305116641342>; Susie Khamis Lawrence Ang, and Raymond Welling, 'Self-Branding, 'Micro-celebrity' and the Rise of Social Media Influencers', *Celebrity Studies* 8.2 (2017): 191-208.
- 146 Alberto Cossu, 'The Digital Traces of Crypto-Finance', in E. Armano, M. Briziarelli, and E. Risi (eds) *Data Traces, Digital Platforms, and Algorithmic Self*, London: University of Westminster Press, 2022, 123-34.

audiences by partnering with businesses, advice generally takes on a somewhat dubious nature in blockchain culture.

The cryptocurrency market is also swayed by the influence of celebrities (see e.g., Figure 16). Celebrities influencing crypto markets range from community insiders, such as Vitalik Buterin, to mainstream celebrities such as Donald Trump and Elon Musk. This latter group is made up of people who, by virtue of their power in general society, cannot help but influence the crypto market. Elon Musk—who is known for commenting on cryptocurrencies—is by far the most mentioned celebrity influencer in the memes analysed. Celebrity influence is often portrayed as unwanted and unpredictable, representing an unwelcome relic from the centralised society crypto is supposed to dismantle.

What the mention of crypto influencers in the memes reveals about the world of which they are part is that, while they are important players shaping the market, they should not be trusted. The information they share offers a potentially valuable way to weed through the sprawl of crypto projects and find the right one to invest in, but it can just as easily be disruptive. Amid such widespread uncertainty, ‘speculation itself becomes the primary terrain of power conflicts’:¹⁴⁷ those who are able to operationalise ambiguity and uncertainty will learn to amplify them, causing the differences between ‘speculators and those who are speculated on’¹⁴⁸ to continue to grow.

Although the etymology of the term is not entirely clear, it seems the expression ‘shilling’ emerged as a common practice in the early twentieth century. The 2023 edition of the Oxford English Dictionary defines a ‘shill’ as ‘an accomplice of a confidence trickster or swindler who poses as a genuine customer to entice or encourage others’ or ‘a person who pretends to give an impartial endorsement of something in which they themselves have an interest.’ In the contemporary context of online auction sites such as eBay, shill bidding is a practice where a seller places fake bids on their own items in order to attempt to artificially increase the final price they will be paid for it. It is generally considered an undesirable practice, yet because eBay financially benefits from higher prices as well, they are incentivised to not enforce a ban against it.¹⁴⁹ Shill bidding has also been recorded in NFT marketplaces.¹⁵⁰

Descriptions of shilling in memes are mostly connected to the behaviour of influencers that guide people through the chaos of the crypto market (see e.g., Figure 17) and is reinforced by the difficulty to distinguish on social media between remarks made professionally and content shared as a personal opinion.¹⁵¹ As is the case in the general use of the word, it is

147 Komporozos-Athanasiou, *Speculative Communities*, 21.

148 Komporozos-Athanasiou, *Speculative Communities*, 21.

149 Joseph Engelberg and Jared Williams, ‘Ebay’s Proxy Bidding: A License to Shill’, *Journal of Economic Behavior & Organization* 72.1 (2009): 509–26; Nazia Majadi et al., ‘Real-Time Detection of Shill Bidding in Online Auctions: A Literature Review’, *Computer Science Review* 25 (August 2017): 1–18.

150 Dipanjan Das et al., ‘Understanding Security Issues in the NFT Ecosystem’, *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security*, Los Angeles, USA, 7–11 November 2022. <https://doi.org/10.1145/3548606.3559342>.

151 Robert Mark Simpson and Eliot Michaelson, ‘The Big Shill’, *Ratio* 33.4 (2020): 269–80. <https://doi.org/10.1017/S0034423320000000>.

understood that shilling in the crypto community is something done, at least partly, out of self-interest. Often, coins are proclaimed by influencers to be going ‘to the moon’ (i.e., shoot upwards in terms of their price). The phrase is used to create hype around a particular coin, but can also be a sign of a shill that is too good to be true, as in the memes in Figure 18. Not only is it difficult to spot someone’s motivations for speaking highly of a particular project, the practice of creating hype is also so constant, and the revolving list of projects so long, that there is hardly any stable information to hold on to within crypto culture: it is a culture of broken promises and uncertainty.

“I am not a financial advisor.”



Figure 14: Meme making fun of influencers pretending not to give financial advice.



Figure 15: Meme emphasising how an influencer might have poor trading skills themselves.



Figure 16: Memes referring to Elon Musk's influence over the crypto market.



Figure 17: Meme emphasising the incomprehensibility of the crypto market and the role of influencers making sense of it, the genuineness of which is ridiculed here.



Figure 18: Memes about extreme promises of future rewards.

A Context of Relentless Scamming

Through widespread practices such as so-called ‘rug pulls’ and ‘pump and dump schemes,’ untrustworthy information takes on an organised form. While such practices should be understood quite straightforwardly as scams, they are not regarded by the community as negative or disposable per se. Rather, the memes present scams that emphasise the ambiguity of information—initiations that people have to go through in order to be part of the crypto community.

The term ‘rug pull’ is a derivation of the saying ‘to pull the rug out from under someone.’ The nominalised term generally refers to a trading practice that predates the emergence of the crypto market.¹⁵² A rug pull in the crypto-specific meaning occurs when the developers or owners of a project suddenly abandon it while taking all the money investors put into the project with them.¹⁵³ Crypto rug pulls are associated with DeFi projects,¹⁵⁴ which rose to prominence in 2020,¹⁵⁵ and can be understood as an iteration on what became known as an ICO scam during 2017–2018.¹⁵⁶ A rug pull occurs when, with or without the knowledge of influencers endorsing a DeFi project, its owners suddenly take all the value out of the project, leaving the investors with worthless and illiquid stakes: the rug has been pulled out from under them.

Rug pull memes often refer to the promises of influencers that are too good to be true, such as in Figure 19: 3500% annual percentage yield. This strategy of making claims that are quite easily spotted as untenable by insiders, but tempting to less informed newcomers, is reminiscent of the gullible character of victims of 419 email scams commonly known as Nigerian Prince scams.¹⁵⁷ Here difficult to interpret information and volatility are operationalised as a community boundary mechanism. Those who know enough would never fall for the scams, but to people just entering the crypto market the extreme ups and downs of the market graphs may seem to warrant such outrageous promises.

152 Bruno Mazorra, Victor Adan, and Vanesa Daza, ‘Do Not Rug on Me: Leveraging Machine Learning Techniques for Automated Scam Detection’, *Mathematics* 10.6 (2022): 949. <https://doi.org/10.3390/math10060949>.

153 Pengcheng Xia et al., ‘Trade or Trick? Detecting and Characterizing Scam Tokens on Uniswap Decentralized Exchange’, *Proceedings of the ACM on Measurement and Analysis of Computing Systems* 5.3 (2021): 1–26.

154 Decentralised Finance, indicating crypto projects that aim to minimise centralisation, for example by using decentralised exchanges such as Uniswap.

155 Valerio Puggioni, ‘Crypto Rug Pulls: What is a Rug Pull in Crypto and 6 Ways to Spot it’, *CoinTelegraph*, 2022, <https://cointelegraph.com/explained/crypto-rug-pulls-what-is-a-rug-pull-in-crypto-and-6-ways-to-spot-it>; Robert Stevens, ‘The 2020 Year in Review: DeFi’, *Decrypt*, 2020, <https://decrypt.co/52298/the-2020-year-in-review-defi>.

156 An Initial Coin Offering is the crypto variant of the Initial Public Offering in traditional finance. An ICO is a way for a crypto project to generate funds to further develop its product and at the same time gives early investors the potential of eventual high returns on their investments. Like in a rug pull, developers disappear with investors’ money in an ICO scam.

157 Richard G Brody, Sara Kern, and Kehinde Ogunade, ‘An Insider’s Look At the Rise of Nigerian 419 Scams’, *Journal of Financial Crime* 29.1 (2020): 202–14. <https://doi.org/10.1108/JFC-12-2019-0162>.

Despite their negative connotation in the memes described above, rug pulls are also portrayed as something you just have to go through in order to become part of the crypto community. Figure 20 shows an example of a meme in which an innocent-looking, wide-eyed character, after experiencing a rug pull, turns into an extremely weathered and beaten yet focussed trader. Like the ‘uncertain and anxious pursuit’ amid widespread speculative logics that ‘may not chase after security but is nonetheless productive of new social connections’,¹⁵⁸ going through a rug pull is a sort of *rite de passage* to enter the crypto community: it will make you less gullible, less focused on the future, and instead able to weather the chaotic now of crypto speculation.

So-called ‘pump and dump’ schemes are defined as ‘a type of fraud in which the offenders accumulate a commodity over a period, then artificially inflate the price through means of spreading misinformation (pumping), before selling off what they bought to unsuspecting buyers at the higher price (dumping)’.¹⁵⁹ Such schemes appeared in financial markets as early as the eighteenth century, when fraudsters spread false rumours of the profitability of South Sea Bubble stock in order to inflate its price and popularity.¹⁶⁰ After the invention of the telephone, pump and dumpers would cold call their victims in order to get them to invest.¹⁶¹ When email use increased, the pump and dump practice transformed again and grew exponentially: it is estimated that in 2008, 15% of spam emails were ‘stock touts’—i.e., emails that spread falsely positive messages about a stock and invite their many recipients to invest in it.¹⁶² Because major financial markets have regulated against pump and dump schemes and stock touting, much of it takes place in markets with low regulations such as today’s cryptocurrency markets.¹⁶³ Crypto-based pump and dump schemes are often organised through semi-private social media, such as Telegram chats or Discord channels, where countdowns are posted well in advance so that everyone involved knows exactly when to buy a coin. This creates a sudden increase in that coin’s price, which is likely to be picked up by outsiders as a sign of its profitability. As soon as this happens, the orchestrators of the scheme then collectively sell their tokens for a much higher price than they bought them for, leaving the outsiders with a deflated asset.¹⁶⁴

158 Komporozos-Athanasiou, *Speculative Communities*, 9-10.

159 Josh Kamps and Bennett Kleinberg, ‘To the Moon: Defining and Detecting Cryptocurrency Pump-and-dumps’, *Crime Science* 7.1 (2018), 2.

160 Kevin C Bartels, ‘Click Here to Buy the Next Microsoft: The Penny Stock Rules, Online Microcap Fraud, and the Unwary Investor’, *Indiana Law Journal* 75.1 (2000): 353-77.

161 Adam J O’Donnell, ‘The Evolutionary Microcosm of Stock Spam’, *IEEE Security & Privacy* 5.1 (2007): 70-75.

162 Laura Frieder and Jonathan Zittrain, ‘Spam Works: Evidence From Stock Touts and Corresponding Market Activity’, *SSRN Electronic Journal* (2007): 1-28.; Michael Siering, ‘The Economics of Stock Touting During Internet-Based Pump and Dump Campaigns’, *Information Systems Journal* 29.2 (2019): 456-83. <https://doi.org/10.1111/isj.12216>.

163 Siering, ‘The Economics of Stock Touting During Internet-Based Pump and Dump Campaigns’; Kamps, and Kleinberg, ‘To the Moon’.

164 Friedhelm Victor, and Tanja Hagemann, ‘Cryptocurrency Pump and Dump Schemes: Quantification and Detection’, *International Conference on Data Mining Workshops*, Beijing, China, 8-11 November, 2019, 2019. <http://doi.org/10.1109/icdmw.2019.00045>.

Even if they are known scams, memers' attitudes towards pump and dump schemes are not outright dismissive: sometimes they joke about the quick turnover between pumps and dumps or even revel in the thrill of a pump. However, there are also negative portrayals of pump and dump schemes, particularly in association with meme coins and celebrity influencers. In such cases, pump and dump schemes are portrayed as negative outside influences that wreak havoc on the market. Taken together, the memes, such as those in Figure 21, indicate an ambiguous relationship to pump and dump schemes, on the one hand dismissing them as unwanted elements stemming from outside of crypto culture itself, but on the other hand engaging with them, thriving on them, and taking them as a fact of life within crypto culture.



Figure 19: Memes showing the attraction of rug pulls.



Figure 20: Meme presenting rug pulls as an initiation.



Figure 21: Examples of memes referring to pump and dump schemes either positively or negatively.

The Meaninglessness of Information

The notion of information loses all meaning in memes that feature phrases such as FUD, DYOR, and technical analysis. While these phrases all concern practices that are meant to separate reality from fiction, at the same time, these memes indulge in the chaotic uncertainty they signify. In a hostile and volatile world, the only way to survive is to be sceptical of anything others say. This is the anti-social world that is imagined through speculation on the crypto market.

The term 'FUD' (Fear, Uncertainty, and Doubt) was first used in the context of marketing and technology in 1975 by Gene Amdahl, who, after leaving his executive job at IBM, started his own competing company. In response, IBM launched a marketing campaign focused on invoking feelings of fear, uncertainty, and doubt in customers in relation to the Amdahl company, ultimately aiming to stifle the enterprise at its outset.¹⁶⁵ In later years Microsoft became a prominent alleged practitioner of FUD marketing, especially against open-source software.¹⁶⁶ However, FUD strategies are not just used by multinationals. 'How-to' sales books such as *The Complete Sales Letters Book* describe FUD as a common practice between competitors, which includes the use of negative press and the dissemination of comparative charts.¹⁶⁷ In the context of crypto, FUD, often in the form of negative press, has been shown to produce 'market uncertainty' and dropping prices.¹⁶⁸

Just like rug pulls, FUD is often portrayed in the memes as part of an initiation into crypto culture (see Figure 22). Those who are not experienced enough to recognise that such market uncertainty offers profitable waves upon which to surf, fall for the FUD, and are portrayed as a more or less dispensable Other. Conversely, those who know better than to follow the FUD are portrayed as smart and heroic, as in Figure 23, where they are portrayed as David pitted against a Goliath-sized wave of FUD, using the dip in prices to stock up with more coins at a low price.

FUD joins shills, rug pulls, and pump and dump schemes on the list of inauthentic information floating around the crypto community. This supports a culture in which anyone discussing any aspect of blockchain might genuinely believe what they are saying, but it could just as well be that they are simply trying to pull you in for their own ulterior profit motives. FUD, shills, and all the other forms of information that are potentially less than true are not seen as problems to regulate against, but inherent parts of the free market. To thrive in this society, individuals should learn to deal with this lack of solid ground.

165 Bryan Pfaffenberger, 'The Rhetoric of Dread: Fear, Uncertainty, and Doubt (Fud) in Information Technology Marketing', *Knowledge, Technology & Policy* 13.3 (2000): 78-92.

166 Eric S Raymond, 'Why Microsoft Smears-and Fears-Open Source', *IEEE Spectrum* 38.8 (2001): 14-15.; Pfaffenberger, 'The Rhetoric of Dread'.

167 Rhonda Harris and Ann McIntyre, *The Complete Sales Letter Book: Model Letters for Every Selling Situation*, Armonk: Sharpe Professional, 1998.

168 Jamil Civitarese and Layla Mendes, 'Bad News, Technical Development and Cryptocurrencies Stability', *SSRN Electronic Journal* (2018), 1.

It is not clear where exactly the acronym DYOR (Do Your Own Research) first appeared, but it is often used in conspiratorial communities. Specifically, its use is linked to conspiracies around the Covid-19 pandemic and the 4chan-based anonymous Q, who in 2017 instigated the QAnon conspiracy. While it consists of many diverging claims, at its core the conspiracy warns against a supposed global political and societal elite of paedophiles, cannibals, and sex traffickers that lists many high-profile politicians among its members. Q's mode of operation was to 'drop' little bits of information to be reassembled into meaningful narratives by QAnons, leading to their use of the term DYOR. Collaboratively working through these drops to decipher their meaning, QAnons' research is mutually reinforcing.¹⁶⁹ In the context of the pandemic, DYOR was often used in two ways by those suspicious of vaccines or other Covid-19-related public health measures: on the one hand, those sharing conspiratorial information would call for the audience to verify a conclusion they made, seemingly opening up to critical reflection, but actually relying on confirmation bias through which the audience would reach the same conclusions they did; or, on the other hand, as a measure against difficult questions, meaning that if the person asking such a question would have done their own research well, they would not have posed such a question in the first place.¹⁷⁰

Referring to the French sociologist Émile Durkheim, researcher of the QAnon phenomenon Matthew Hughey (2021) writes that "DYOR" reflects a 'cult of the individual'; a patina of independent thinking covering a reactionary collective consciousness'.¹⁷¹ Writing around the turn of the twentieth century, Durkheim analysed the decline of Christianity as the context of the rise of the cult of the individual: 'a new religion whose sacred object is the rational and autonomous individual as enshrined in democratic rights discourses'.¹⁷² The cult of the individual constituted a new route to social cohesion based on autonomy and the shared human drive towards connection.¹⁷³ This idea took on new dimensions in times of social distancing and dissipating feelings of belonging during the Covid-19 pandemic. Conspiracy theories provided a retreat into alternative communities, establishing a subversive kind of collectivism in the face of a supposed oppressive authority and lack of social cohesion,¹⁷⁴ all while disintegrating the notion of truth.

Just like in conspiracy cultures, DYOR is repeated like a mantra in crypto communities (see e.g., Figure 24). Tweets shilling crypto projects include it as a hashtag almost by default, and some companies and influencers include it in their profile bio as a blanket disclaimer, broadly meaning that anything mentioned there is not financial advice, and they will not be

169 Matthew W Hughey, 'The Who and Why of Qanon's Rapid Rise', *New Labor Forum* 30.3 (2021): 76-87.

170 B. Hughes, et al., 'Development of a Codebook of Online Anti-Vaccination Rhetoric to Manage Covid-19 Vaccine Misinformation', *International Journal of Environmental Research and Public Health* 18.14 (2021): 1-18. <https://doi.org/10.3390/ijerph18147556>.

171 Hughey, 'The Who and Why of Qanon's Rapid Rise,' 81.

172 Paul Carls, 'Modern Democracy as the Cult of the Individual: Durkheim on Religious Coexistence and Conflict', *Critical Research on Religion* 7.3 (2019): <https://doi.org/10.1177/2050303218823069>, 293.

173 Charles E Marske, 'Durkheim's 'Cult of the Individual' and the Moral Reconstitution of Society', *Sociological Theory* 5.1 (1987): 1-14.

174 Madison Wesenberg, 'Covid-19 and the Rise of the Conspiracy: Examination of Covid Related Conspiracies Using Durkheimian Concepts', *Crossing Borders: Student Reflections on Global Social Issues* 3.1 (2021): 1-4.

accountable for any losses someone might make following their suggestions—i.e., people are responsible for basing their decisions on their own research. Yet, as the practice of shilling and the presence of influencers attest, crypto culture is full of people sharing their trading and investment strategies. Amid the chaos of market volatility, scams, and a never-ending wave of new crypto projects, the line between doing your own research and watching someone else's video supposedly honestly reporting on their own research is not always as easy to draw as it may seem. The readily available resources that others share can be tempting, especially when you are new to the game. On top of everything, there is the ever-present influence of celebrities and hype that might—or even should—overrule the most diligent research at times, as shown in Figure 25.

The DYOR strategy seemingly prevents people from falling for short-lived hypes, shills, rug pulls, and pump and dump schemes. When newcomers ask for help determining which crypto projects to invest in among this wealth of dubious information, the only reasonable answer is 'DYOR.' In a hostile and volatile world, the only way to survive is to be sceptical of anything others say. Advising someone to DYOR is better advice than any practical investment strategy anyone else might share.

Doing your own research consists of, among other research practices, technical market analysis. In traditional finance, the terms 'bull' or 'bear' markets are often used to describe market behaviour, and they are also ubiquitous in crypto culture. In the case of a bull market, prices are going up from a low point, whereas in a bear market, they are going down. The term bull market, or an investor being 'bullish,' is thought to refer to the energetic and eager charge of the bull, exemplified by the famous statue of a charging bull on Wall Street. Conversely, a bear market, or 'bearish' investor, is thought to refer to the old English proverb 'don't sell the bear's skin before you've caught it,' which is used to caution against relying on anticipated favourable future outcomes before they are certain. Instead of waiting for such a future to become reality or not, bearish investors anticipate unfavourable futures by selling their holdings when they expect the market to crash, feeding the crash as they do so.¹⁷⁵

While these terms abound in memes, there are also other metaphors used in market analyses that are worth highlighting here. For example, due to the high levels of volatility and unpredictability of crypto markets, other animals have been used to describe their graphs as well, such as the spiky stegosaurus seen in Figure 26. As references become more random, such as recognising the shape of Bart Simpson's head in a graph or the naming of ever smaller sections of a chart in order to predict what will come next, the concept of technical market analysis starts to disintegrate under the weight of that volatility, reaching a low point in the crude, sexist, or immature variant where any pretence of usability has dissipated (see e.g., Figure 27). These memes seem to mock the idea of ever being able to predict a market as ruthlessly unpredictable and susceptible to random outside influences as that of a cryptocur-

175 Maria Lojko, 'Animals as a Source Domain for Metaphorical Expressions in English Economic Discourse', *Language and Literary Studies of Warsaw* 8.1 (2018): 107-98.

rency. Crypto speculators seem to ‘knowingly descend into chaos’¹⁷⁶ while memes standardise their vernacular imaginary of speculative survival in radical uncertainty.



Figure 22: Memes portraying FUD as an initiation.

176 Komporozos-Athanasiou, *Speculative Communities*, 21.



Figure 23: Memes portraying those who know to disregard FUD.



Figure 24: Memes showing people giving others the advice to DYOR.



Figure 25: Memes showing that hypes can be tempting to overrule one's own research.



Figure 26: Meme showing an example of an animal-themed market analysis.



Figure 27: Memes showing the disintegration of technical analysis.

Surfing the Waves of Volatility

Finally, I move to the perspective of three significant figures in the crypto community: HODLers, day traders, and degens. Either by being uninterested in, hyper focused on, or recklessly reinforcing the uncertainty of the market, these crypto insiders surf the waves of extreme volatility. Despite their fundamentally different attitudes, HODLers, day traders, and degens share a capacity to thrive while speculating in uncertainty.

The term HODL is among the most prevalent in the crypto community. Usually written in all-caps, it refers to the action of holding on to your crypto assets in the face of dropping prices because of the belief that eventually those prices will rise again.¹⁷⁷ It is often used in reference to holding bitcoins in particular, and has an additional, deeper meaning relating to staying strong in the belief of the prosperous and revolutionary future of the cryptocurrency and its benefits after political or societal collapse,¹⁷⁸ as discussed in relation to Bitcoin Maximalism in chapter 3. The term HODL originates from the online Bitcoin Talk forum, where in 2013, someone explained that they were going to hold on to their coins, but—having had ‘some whiskey’—accidentally wrote that they were ‘hodling’ them instead.¹⁷⁹ The typo soon caught on in true memetic fashion and is still used abundantly today. Besides this origin, the term has also been reverse-engineered as the acronym for ‘Hold On for Dear Life’, which again reflects the practice of buying and keeping coins in the face of any ensuing market chaos.

HODL memes emphasise two main points about crypto’s speculative imagination. First, as shown in Figure 28, HODLers are not perturbed by volatility and chaos. They hold fast in their beliefs despite the profound uncertainty that characterises the crypto market. Second, as shown in Figure 29, being able to HODL is portrayed as a sort of superpower that mere mortals do not possess. It reflects a disregard for practices that are often associated with stability and future reward, such as spreading one’s assets across various investments for risk management and holding on to real estate as a stable store of value. All of this presents HODLers as special individuals who are not influenced by current volatility and have a unique capacity that goes against received knowledge and allows them to live and speculate through the chaotic ‘now’ unscathed.

Day trading can be understood as the polar opposite of HODLing: it is defined as buying a financial product and then selling it again within the same day, whether that is on the traditional stock market or on the crypto market.¹⁸⁰ Especially during the dot-com bubble of the 1990s, day trading gained prominence because of the potential for high turnovers amid the hype. How-to and tell-all style books were published on the topic, for example *The Day Trader*:

177 Claus Dierksmeier, ‘Just HODL? On the Moral Claims of Bitcoin and Ripple Users’, *Humanistic Management Journal* 3.1 (2018): 127–31.

178 Megan L Knittel and Rick Wash, ‘How ‘True Bitcoiners’ Work on Reddit to Maintain Bitcoin’, *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. Glasgow, UK, 4–9 May 2019. <https://doi.org/10.1145/3290607.3312969>.

179 GameKyuubi, <https://bitcointalk.org/index.php?topic=375643.0>, n.p.

180 Brad M Barber et al., ‘The Cross-Section of Speculator Skill: Evidence From Day Trading’, *Journal of Financial Markets* 18.1 (2014): 1–24.

From the Pit to the PC¹⁸¹ and *Dumb Money: Adventures of a Day Trader*.¹⁸² Nevertheless, it has been shown that many day traders lose money.¹⁸³ Furthermore, while the traditional stock market regulates the practices of day trading with various safeguards, such rules do not exist in the cryptocurrency market. Hence, here day trading is accessible to many more people.



Figure 28: Memes of HODLers as people who are unperturbed by volatility and chaos.

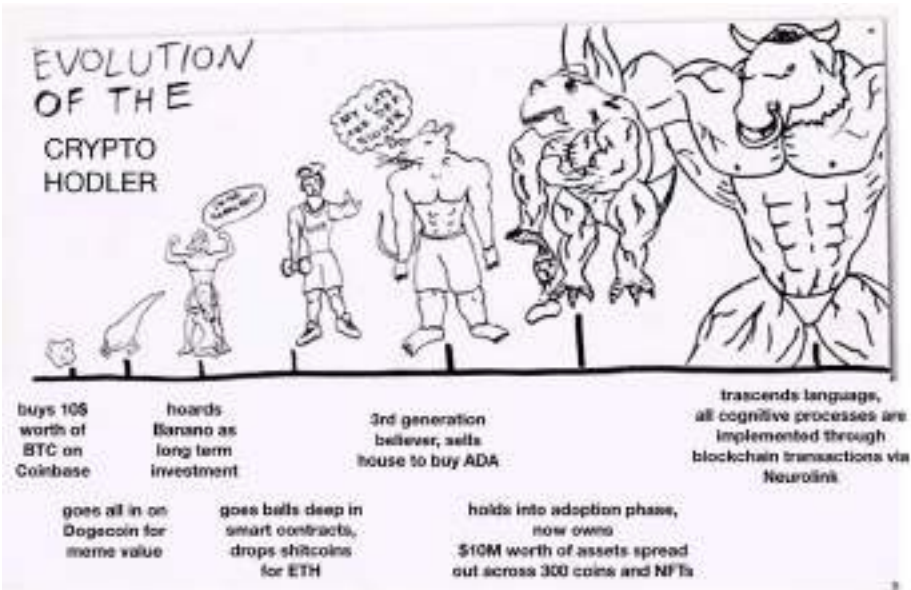


Figure 29: Memes of the HODLer as transcending human capacities.

181 Lewis Borsellino, *The Day Trader: From the Pit to the PC*, Hoboken: John Wiley & Sons, 1999.

182 Joey Anuff and Gary Wolf, *Dumb Money: Adventures of a Day Trader*, New York: Random House, 2000.

183 Barber, 'The Cross-Section of Speculator Skill'.

While HODLers and day traders alike invest in cryptocurrencies because they believe it will bring them financial gains, how that translates into day-to-day practice differs radically. The difference between HODLers and day traders is clearly represented in Figure 30. Instead of being distanced from the everyday experience of the market, day traders are always in the midst of it, analysing it, hardly having time to shower in case they lose out on a trading opportunity. Rather than being indifferent to the chaos of the market, they actively stimulate it. This attitude is made possible by, and at the same time reinforces, the extreme volatility of the crypto market. Day traders are agnostic to the way in which they make money—whether a particular coin or the cryptocurrency market as a whole ultimately survives or dissolves. While uncertainty is like background noise to a HODLer, a day trader can think of nothing else. Indeed, a stable market is useless to the day trader, as volatility provides the very possibility for the day trader's actions.

Another figure mentioned in the memes that thrives on uncertainty is the 'degen.' The origin of the term 'degen' (which is short for 'degenerate') is difficult to trace, although a possible connection can be made with the use of the term in gambling communities.¹⁸⁴ As outlined earlier in this chapter, gambling and financial speculation are separate practices, the first being associated with risk, and the latter with uncertainty.¹⁸⁵ However, the distinction between gambling and speculation seems to blur in the degenerate behaviour in gambling communities, where degens are individuals that 'exhibit unwise betting behaviour,' while at the same time being considered capable and confident players.¹⁸⁶ Similarly, in crypto memes (see e.g., Figure 31), the term 'degen' refers to individuals who recklessly invest in anything, including known scams such as Ponzirand.¹⁸⁷ Rather than the quiet and humble genius,¹⁸⁸ the degen (Figure 32)¹⁸⁹ is shown in a positive light while raging through his analyses and revelling in anything that shows off his wealth. Degens are inexplicably successful, and their existence reinforces the ideal of reckless behaviour and mystifies the inner workings of the crypto market.

184 Paul Delfabbro et al., 'Cryptocurrency Trading, Gambling and Problem Gambling', *Addictive Behaviors* 122 (November 2021): 1-6.

185 Stuart Banner, *Speculation: A History of the Fine Line Between Gambling and Investing*, Oxford: Oxford University Press, 2017; Komporozos-Athanasios, *Speculative Communities*.

186 Breigh Radburn and Rachel Horsley, 'Gamblers, Grinders, and Mavericks: The Use of Membership Categorisation to Manage Identity By Professional Poker Players', *Journal of Gambling Issues* 26 (2011), 44.

187 Ponzirand is a crypto project that announces itself as a scam in order to comment on the widespread fraud among memecoins. See Ponzirand, <https://ponzirand.wordpress.com/>.

188 Pictured in the meme is Benjamin Cowen, who has a PhD in nuclear engineering and is known as a serious financial analyst in the crypto community. His YouTube channel has more than 800,000 subscribers and features him analysing market trends from a famously sparsely decorated room. See e.g. Benjamin Cowen, 'Ethereum Falls Below Its Bull Market Support Band', *YouTube video*, 2024, <https://www.youtube.com/watch?v=DRjLAed8Qzc>.

189 Here pictured is Christopher Jaszczynski, a German crypto analyst and influencer with approximately 570,000 subscribers. Jaszczynski is known for his exhibitionist style, often shooting videos on the go and featuring constant signs of wealth, such as private pools, expensive cars, and luxury apartments. See e.g. Christopher Jaszczynski, 'Bitcoin: Horrible News!!!!!!', *YouTube video*, 2024, <https://www.youtube.com/watch?v=gT-lkjaSC3I>.

Some memes ridicule those who are not capable of riding the waves of volatility like HODLers, day traders, and degens. For example, there are memes that make fun of so-called ‘virgin’ traders who try yet do not have the nerves of steel needed to make it in the crypto market. Other memes feature those that ‘panic sell’—i.e., get rid of assets as soon as disaster strikes, exacerbating volatility¹⁹⁰—or those who have ‘paper hands’ and avoid risk and volatility altogether. Together, these negative portrayals reinforce the idea that without the right kind of uncertainty-endorsing attitude, people will not make it in the chaos of the crypto market.



Figure 30: Memes portraying day traders amid volatility.

190 Amin Zokaei Ashtiani, Marc Oliver Rieger, and David Stutz, ‘Nudging Against Panic Selling: Making Use of the Ikea Effect’, *Journal of Behavioral and Experimental Finance* 30.1 (2021): 1-11.



Figure 31: Memes showing degens investing recklessly.



Figure 32: Meme showing the difference between a degenerate and a genius.

Conclusion: Speculative Survival on the Crypto Market

The context of the volatile crypto market has a profound impact on the rationalism that characterised the imaginary of blockchain as a truth machine discussed in the previous chapters. The uncertainty that characterises contemporary societies causes some to try to rationalise and attempt to regain a lost sense of control and future stability, while others surf on the high waves of the present without wishing for calmer waters.

Although historically, astrology functioned in support of speculative financial practices that overcome the binary distinction between rationality and irrationality, in the case of Maren Altman's crypto astrology, it instead functions in support of the rationalist imaginary of blockchain as a truth machine. Maren Altman's descriptions of crypto astrology frame it as operating through computational processes and rational calculations. Astrology and cryptocurrencies both represent large historical datasets, which, with the right contextual knowledge, can reveal the future with a renewed sense of certainty. By logging astrological data and interpretations on a blockchain, Altman uses the technology to disintermediate the stars, and thus eliminate human error and subjectivity once and for all. Astrology and rationality become one on the blockchain.

Contrastingly, the affordances of memes supercharge an imaginary of crypto as an uncertainty machine. Rather than the immutable or universal truths that previous analyses in this book highlighted, here the 'truths' that crypto churns out are so dizzyingly abundant, so ambiguous, and so rapidly turned upside down, that instead of relying on their veracity, their immediate effectiveness to those who know how to revel in uncertainty is what counts. The memes show that indulging in chaotic uncertainty is the only way to survive in the anti-social world that is imagined through speculation on the crypto market. Although Komporozos-Athanasiou sees potential for the appropriation of speculative imagination for countercultures,¹⁹¹ in crypto memes, speculation 'functions as capital's imagination',¹⁹² reinforcing boundaries and power dynamics between an elite and wealthy in-group and struggling outsiders.

Although crypto astrology and crypto memes seem in many ways to showcase diametrically opposed approaches to the volatility of the crypto market, if we take a more distant perspective, it becomes clear that they do not fully negate each other. Altman and crypto memers fundamentally agree that everyone should enter the crypto market because, if you know how to handle it, it can make you rich. Moreover, Altman lets slip that even though she describes astrology in rationalist terms, it always involves human interpretation and is thus never fully rational. She presents her business Astrace as a solution to this problem, but here I am bound to ask: Does Astrace really change the subjective nature of astrology, or is it just a way to make astrology palatable to crypto traders through the lure of its pseudorationalism? It is indeed likely that by making astrological data and predictions verifiable on a blockchain, Astrace makes an astrologer's track record easier to evaluate by looking back through this data. However, does Astrace's data really make a difference when attempting to predict the future? Considering the fundamental uncertainty of social, political, and economic future events, their influence on the market, and the volatility-inducing practices of memers and others that revel in the chaos of the crypto market, future astrological readings cannot help but engage in uncertainty. Astrology does not equate to the position of the planets, but rather exists as the speculative engagement with their uncertain yet nonetheless possible relation to earthly events. Like surveillance measures attempt to transform the uncertainty of future attacks into risk measures, Altman's approach to crypto astrology attempts to turn the future of the market into a calculable prediction. However, ultimately, neither is possible. Both the memes discussed in this chapter and Maren Altman's approach to crypto astrology confirm that in times of uncertainty, speculation certainly thrives

191 Komporozos-Athanasiou, *Speculative Communities*.

192 Haiven, 'Finance as Capital's Imagination?', 93.

A REALITY MACHINE FOR MUTABLE AND PLURAL WORLDS: THE DELIBERATE IMAGINARIES OF PREFIGURATIVE DAOS

In this chapter, I engage with blockchain projects that contribute to imaginaries of radical sociocultural change and I look at how ‘imaginaries have the ability to embed and embody prefigurative realities capable of producing change’.¹ Previous chapters have demonstrated that, while claims about blockchain as a force for good abound in the culture that surround it, extractivist, individualist, and universalising logics and ideals often lurk not far beneath the surface. ‘Blockchain-for-good’ narratives can be understood as rhetorical devices that seem to be committed to sociocultural change—the sector proclaims that it wants to do good, to make the world a better place, i.e., to change it somehow—while in actuality these systems perpetuate existing hierarchies and power dynamics. My findings in chapter 2 as well as findings regarding other parts of the blockchain-for-good sector by other researchers² substantiate this. Projects that are part of the blockchain-for-good sector are not the kind of use of blockchain that fits the parameters of the current chapter.

While I continue along the situated trajectory upon which I embarked in the previous chapter, I also make a further significant shift in order to address such imaginative work. Up to this point, I have used the concept of imaginaries to unravel the socio-techno-cultural ideals, assumptions, and commitments that shaped engagement with blockchain. The use of imaginaries provided a theoretical lens via which to make sense of communities that did not explicitly frame their engagement with the technology in such terms. Instead, this chapter deals with a part of blockchain culture that *deliberately* tries to establish new imaginaries around the technology. Here imaginaries are used as a tool towards future realities that are different from the status quo.

Social movements may be understood as ‘convocations of the radical imagination’³ in which ‘counter-imaginaries’⁴ take shape and provide a shared sense of direction away from the status quo. However, the emergence of these imaginaries may be ‘sometimes intentional,

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- 1 Stefania Milan and Emiliano Treré, ‘Big Data From the South(s): Beyond Data Universalism’, *Television & New Media* 20.4 (2019), 329.
 - 2 See e.g., Silvia Semenzin, ‘“Blockchain for Good”: Exploring the Notion of Social Good Inside the Blockchain Scene’, *Big Data & Society* 10.2 (2023): 1-14; Mirca Madianou, ‘The Biometric Assemblage: Surveillance, Experimentation, Profit, and the Measuring of Refugee Bodies’, *Television & New Media* 20.6 (2019): 581-99. <https://doi.org/10.1177/1527476419857682>.
 - 3 Alex Khasnabish and Max Haiven, *The Radical Imagination: Social Movement Research in the Age of Austerity*, London: Zed Books, 2014, 7, emphasis in original.
 - 4 Becky Kazansky and Stefania Milan, ‘“Bodies Not Templates”: Contesting Dominant Algorithmic Imaginaries’, *New Media & Society* 23.2 (2021), 364.

sometimes incidental'.⁵ To address the deliberate use of imaginaries specifically, I zoom in on the intersection of artistic and activist engagement with blockchain. Here, imaginative work take centre stage in attempts to break with established imaginary horizons of use for the technology. Such deliberate artistic imaginaries can be understood as a kind of activism that uses the performativity of imaginaries as a chisel to carve out a changed reality. My focus is on the ways in which such a shift from imaginaries as a scholarly concept, to imaginaries as a deliberative tool, might take shape in this context.

Two examples give an insight into the emergence of this deliberate use of imaginaries in the early years of the creative and artistic engagement with blockchain. In March 2014, the Amsterdam-based Institute of Network Cultures organised a conference entitled *MoneyLab: Coining Alternatives*.⁶ Much of the conference dealt with the emerging field of cryptocurrencies and other financial technologies with the explicit intention to find and support creative and artistic alternatives to their financialising uses. The description of one session at the conference explained how it would dive 'into today's social, technical, and art practices in relation to money, [inquiring] into what we are missing in today's landscape. Who designs alternatives, and for which constituencies? *Can we see a collective imagination arising here?*'.⁷ In the ten years since then, subsequent MoneyLab conferences and publications have been a space in which this kind of imagination has featured prominently. In 2017, the publisher Torque Editions collaborated with then London-based arts organisation Furtherfield for an edited publication titled *Artists Re: Thinking the Blockchain*.⁸ The work explicitly addresses the need to imagine blockchain *differently* through artistic work. Co-editor Ruth Catlow explains that artists play a key role in attempts to establish new imaginaries because

they know that a way to get to know something that doesn't yet exist is to collaborate with its possibilities and to do something/anything with it or about it. And by doing so they materialize and shape what it will be.⁹

Such artistic engagement with blockchain imaginaries contributes to an opening up of the horizon of the technological landscape—to explore that which is not yet, but could in the future become, real. Of course, numerous other publications, artworks, conferences, workshops, and exhibitions—too many to list here—have contributed to this work of 'rethinking' blockchain in artistic and activist ways in the past ten to fifteen years.

5 Khasnabish, and Haiven, *The Radical Imagination*, 7.

6 I joined the Institute of Network Cultures team in September 2014, a few months after this first MoneyLab conference. Soon after joining, I collaborated with the team on the MoneyLab project, which is focussed on bridging criticism of the power dynamics and extractivist logics of financial technologies with creative and grassroots experimentation, with the purpose of democratising the digital economy. Although since 2020, I have stepped back from day-to-day involvement in the core MoneyLab team, it significantly shapes my understanding of the field. Being part of the wider MoneyLab network continues to be a source of much inspiration, and I hope to give back to the community in some way with this book.

7 Geert Lovink et al., *Moneylab: Coining Alternatives*, Amsterdam: Institute of Network Cultures, 2014, n.p., emphasis added.

8 Ruth Catlow et al. (eds) *Artists Re:thinking the Blockchain*, Liverpool: Torque Editions, 2017.

9 Catlow, *Artists Re:thinking the Blockchain*, 22.

A particularly well-developed strand among such rethinking practices consists of a move from framing blockchain as a technology with primarily financial, monetary, or administrative capacities to casting it in a social light. This move is particularly noticeable in work on Decentralised Autonomous Organisations (DAOs), which—as the name suggests—are concerned with the fundamentally social phenomenon of organisation. This chapter, therefore, focusses specifically on the imaginaries that surround DAOs and the way they are imagined to contribute to the prefiguration of different sociocultural realities.

In what follows, I propose two concepts that are useful in developing a more nuanced understanding of deliberately developed imaginaries, particularly those at the crossroads of art and activism. First, to account for the element of the technical in these DAO practices, I develop the notion of radically reimagined affordances in which the active and creative capacity of the imagination is put to use in order to institute new potentialities for blockchain. Second, building on decolonial epistemological critiques,¹⁰ I apply Patricia de Vries' notion of artistic plot-work¹¹ to unravel how DAO-based storytelling practices may world new worlds.¹² However, I start by providing a brief history of DAOs with a specific focus on the role that politics play in the perception and mainstream use of DAOs.

DAO Cultures Between Credible Neutrality and Covert Politics

A key moment in the emergence of DAOs is the shift from blockchains as primarily distributed networks to the use of blockchains as platforms to deploy applications—like DAOs—such as was signified by the emergence of Ethereum. Blockchain as a platform carries connotations of neutrality that do not do justice to the many ways in which its design and use constitute political choices. Below, my analysis features a discussion of two DAOs: MolochDAO and RaidGuild. While prominent DAO thinker and maker Kei Kreutler thinks of such DAOs as constituting a covert kind of political practice that embodies political change by virtue of its technical features,¹³ this view is built on a technological determinist view of how change is achieved, pre-emptively justifying the application of the technology while preventing deeper political engagement.¹⁴

10 Sylvia Wynter, 'Novel and History, Plot and Plantation', *Savacou* 5.1 (1971): 95–102; Sylvia Wynter, 'Unsettling the Coloniality of Being/ Power/ Truth/ Freedom: Towards the Human, After Man, Its Overrepresentation—an Argument', *The New Centennial Review* 3.3 (2003): 257–337. <https://doi.org/10.2307/41949874>; Sara C. Motta, 'Notes Towards Prefigurative Epistemologies', in Sara C. Motta, and Alf Gunvald Nilsen (eds) *Social Movements in the Global South: Dispossession, Development and Resistance*, New York: Palgrave Macmillan, 2011, 178–99.

11 Patricia de Vries, *Plot Work as an Artistic Praxis in Today's Cityscapes: An Introduction to the Lectorate Art & Spatial Praxis / the City*, Amsterdam: Gerrit Rietveld Academy, 2022.

12 Donna J Haraway, *Staying With the Trouble: Making Kin in the Chthulucene*, Durham: Duke University Press, 2016.

13 Kei Kreutler, 'A Prehistory of DAOs: Cooperatives, Gaming Guilds, and the Networks to Come', *Gnosis Guild*, 2021, <https://gnosisguild.mirror.xyz/t4F5rltMw4-mlpLZf5JQhElbDfQ2JRVKAZepanyxW1Q>.

14 Sally Wyatt, 'Technological Determinism is Dead; Long Live Technological Determinism', in E. Hackett, et al. (eds) *Handbook of Science and Technology Studies*, Cambridge, MA: MIT Press, 2008,

Blockchain as a Platform upon Which to Build DAO Worlds

The launch of Ethereum in 2015 impacted blockchain technology and culture significantly: from merely functioning as a system for the exchange of tokens, it began to function more like a platform upon which others could build. As Tarleton Gillespie's framing of the politics of social media as platforms indicates,¹⁵ this new form of blockchain technology is not just a computational shift, but also carries with it political connotations and implications. In the narrow computational sense, these blockchains function as platforms because they facilitate outside actors to build and deploy applications, such as smart contracts and DAOs, on top of their existing systems. However, platforms pre-exist digital technology, and older meanings of the word highlight the important political dimensions that also permeate digital platforms. Indeed, the very term 'platform' gives the sense of a foundation that has no politics itself, but neutrally allows for others to express themselves with greater reach.¹⁶ Despite this supposed a-political appearance, social media platforms institute rules and community guidelines that shape the kinds of expressions that are facilitated through their systems, and they are in reality anything but 'neutral conduits'.¹⁷ Similarly, in the case of blockchains, '[e]very line of code reflects a policy choice about the blockchain system as a whole [...] and technical choices about how to best reflect the policy mandate in code'.¹⁸ Despite the framing Ethereum founder Vitalik Buterin gave of his and other blockchains as 'credibly neutral'¹⁹—echoing the rhetoric of social media platforms—blockchains and the applications built on top of them are political by their very nature.

As platforms upon which to build applications, these kinds of blockchains opened up a new space for experimentation and organisation. For example, using smart contracts—i.e., pieces of code that automatically make changes to the data stored on a blockchain based on updated inputs and previously agreed terms—various kinds of 'decentralised autonomous entities'²⁰ such as DAOs may be implemented on Ethereum by individuals, communities, or corporations outside of the Ethereum organisation.²¹ Even before Ethereum went live in 2015, this new space of opportunity captured imaginations in blockchain culture, with Buterin stating that 'all in all, it is safe to say that 'DAOism' is well on its way to becoming a quasi-cyber-religion'.²² In general, DAOs can be understood as a particular way of using blockchain technology to

15 Tarleton Gillespie, 'The Politics of "Platforms"', *New Media & Society* 12.3 (2010): 347-64; Tarleton Gillespie, *Custodians of the Internet: Platforms, Content Moderation, and the Hidden Decisions That Shape Social Media*, New Haven: Yale University Press, 2018.

16 Gillespie, 'The Politics of "Platforms"'.
 17 Gillespie, *Custodians of the Internet*, 14.

18 Angela Walch, 'Deconstructing "Decentralization": Exploring the Core Claim of Crypto Systems', in Chris Brummer (ed) *Cryptoassets: Legal, Regulatory, and Monetary Perspectives*, Oxford: Oxford University Press, 2019, 52.

19 Vitalik Buterin, 'Credible Neutrality as a Guiding Principle', *Nakamoto.com*, 2020, n.p. <https://nakamoto.com/credible-neutrality/>.

20 Vitalik Buterin, 'DAOs, DACs, DAs and More: An Incomplete Terminology Guide', *Ethereum Foundation Blog*, 6 May 2014, <https://blog.ethereum.org/2014/05/06/DAOs-dacs-das-and-more-an-incomplete-terminology-guide>.

21 Ethereum is not the only blockchain that has this functionality, but it was the first blockchain offering it.

22 Buterin, 'DAOs, DACs, DAs and More,' n.p.

achieve some kind of organisational purpose, whether that be commercial, activist, or personal. DAOs automate certain interactions between various stakeholders according to the goals and rules of the DAO, which are sometimes defined in collective and iterative design processes, but other times are part of a corporate top-down implementation.

As with blockchain technology itself, there is no clear-cut technical definition that describes DAOs once and for all. In general terms, a DAO may be thought of as ‘a blockchain-based system that enables people to coordinate and govern themselves mediated by a set of self-executing rules deployed on a public blockchain, and whose governance is decentralised’.²³ Martin Zeilinger states that DAOs are ‘computationally enforceable and socially enacted’ and ‘(more or less) blockchain-reliant organisations’, emphasising that DAO code does not determine the social realities that surround them, but rather DAOs are always co-constituted both by code, and by the groups of people around them.²⁴ Echoing this emphasis on the social, a popular saying in the field describes DAOs as ‘group chats with a shared bank account.’ This common understanding of DAOs as to some degree social phenomena—group chats, organisations, etc.—rather than just the protocols of financial exchange that first-generation blockchains are generally thought to be, contributes to their popularity in the context of marketing trends such as the notion of ‘brand communities’ that attempt to frame commercial entities in terms of their embeddedness in culture, lifestyles, identities, or social groups. Since there is no clear delineation of DAOs, words that emphasise sociality, such as ‘collective’ and ‘community,’ can be used rhetorically to hide ‘decentralised venture capitalism and fractionalised ownership’ under the superficial appearance of such embeddedness.²⁵

This latter element of DAO culture is exemplified by the history of a decentralised autonomous organisation (presumptuously) called ‘The DAO.’ It was the first DAO in the history of Ethereum, and it was funded through what was at the time hailed as one of the largest crowdfunding campaigns in history. The DAO was designed as a system to automate pay-outs to projects vying for investor funds.²⁶ However, it was hacked in June 2016: due to a technical loophole, a user was able to transfer the equivalent of 50 million USD to their own wallet without the consent of the original owners of those funds. While in the realm of regulated property or currencies this would be called theft, a staunchly anarcho-capitalist part of the Ethereum community expressed that they saw this as a fair game: the hacker simply exploited the agreed upon system to the fullest for their own benefit. Others adhered to more mainstream understandings, indeed calling the hack a theft. This rift caused a ‘hard fork’²⁷ in which what

23 Samer Hassan and Primavera de Filippi, ‘DAO (Decentralized Autonomous Organization)_’, in Valeria Ferrari et al. (eds) *Log Out_ a Glossary of Technological Resistance and Decentralization_*, Amsterdam: Institute of Network Cultures, 2023, 109.

24 Martin Zeilinger, *Structures of Belonging*, Ljubljana: Aksioma – Institute for Contemporary Art, 2023, 12–13.

25 Zeilinger, *Structures of Belonging*, 13.

26 Cade Metz, ‘The Biggest Crowdfunding Project Ever – the DAO – is Kind of a Mess’, *Wired*, 2016, <https://www.wired.com/2016/06/biggest-crowdfunding-project-ever-dao-mess/>.

27 A hard fork happens when the protocol that is in use in a particular blockchain is changed, making it no longer backward compatible. As a result, two versions of the network exist at the same time, one with the change implemented, and one without. Especially when dealing with cryptocurrencies, implementing hard forks is undesirable: the value of cryptocurrency coins depends on their

would become Ethereum Classic (in which the theft was left unaltered) split from Ethereum, which restored the stolen funds.²⁸

It is in this context that Kei Kreutler, a prominent thinker and developer who is concerned with the social and imaginative potential of blockchain infrastructures, writes that MolochDAO, which launched in 2019, 'rekindled the fire of decentralized governance after The DAO hack'²⁹ (n.p.) had caused it to die down. MolochDAO is still one of the most prominent examples in the DAO field, as its services consist of distributing funding for software development that contributes to the strengthening of Ethereum as a public infrastructure. MolochDAO frames these activities as contributing to a fight for a shared social good, not in the business-like way the blockchain forest protection projects did in chapter 2, but in a rather more imaginative way. Despite its practical focus on software development, MolochDAO's website, whitepaper, and other documentation presents the DAO as if it is caught in a battle of epic proportions. In a large font, the landing page of the website reads:

Whom Do You Serve?

The future of humanity requires the sacrifice of your shallow desires, but its reward is the head of Moloch himself.

This demon god of coordination failure, who consumes our future potential for perverse immediate gain, will be slain. Pledge your oath to his demise, or go down with him. [...]

DEMON BLOOD LOOKS GOOD ON YOU³⁰

The name Moloch is a reference to the ancient Canaanite god of the same name, who convinced adherents that if they sacrificed some of their children during times of war, this would increase their chances of victory. Consequently, individuals who did not adhere to this divine logic were seen as a threat to the entire society, although the sacrifice of too many children could in reality just as well constitute the same. MolochDAO writes that they make this reference because 'problems associated with collective action, where individual incentives are misaligned with globally optimal outcomes' are important issues that they want to address in the context of DAOs.³¹ The worldview they present in their documentation seems split between a fundamental critique of the status quo, in which individualism trumps collaboration for a common good, and being stuck in the logics that contribute to this status quo, for example,

cryptographically secured scarcity and duplicating the network means duplicating the coins. Hard forks are seen as a last resort measure to solve community disagreements, such as the one surrounding the DAO hack.

28 John Flood and Lachlan Robb, 'Trust, Anarcho-Capitalism, Blockchain and Initial Coin Offerings', *SSRN Electronic Journal* (November, 2017): 1-27. <https://doi.org/10.2139/ssrn.3074263>.

29 Kreutler, 'A Prehistory of DAOs,' n.p.

30 MolochDAO, <https://molochdao.com/>.

31 Ameen Soleimani et al., *The Moloch DAO: Beating the Tragedy of the Commons Using Decentralized Autonomous Organizations*, Moloch DAO, n.d.

competition. They explain that they aim for individuals to retain or regain the freedom to act in their self-interest in a free market, but also that this ‘game’ must run itself into the ground—a thought that echoes accelerationist critiques of capitalism—in order to create a different yet unspecified ‘altruistic game’.³² Although it is ideologically unclear, in its naming, copy, and visuals, MolochDAO invites participants to step into another world. This story gives meaning to interactions and purpose to actions: Moloch urges participants to band together against power, even if that power remains ill-defined.



Fig. 33. Screenshot of RaidGuild’s website.

Another DAO that presents itself as if fighting a battle against a big villain is RaidGuild. Similar to MolochDAO, the services they offer are rather practical, consisting of consultation, design sprints, software development, and marketing.³³ Yet, the landing page of their website features an illustration of a group of small characters fighting a giant dragon-like figure together (see Figure 33). RaidGuild calls itself a ‘decentralized collective of mercenaries’³⁴ in which workers band together to gain power, functioning like a ‘DAO-based worker collective’.³⁵ RaidGuild references this epic battleground setting throughout their documentation. For example, like many other DAOs, they use the term ‘summoning’ to mean the forming of a new DAO or to gather the members for a new DAO. This is likely a reference to the act of summoning someone in a video game, meaning to pull a character into your immediate area from a distant location, usually in order to aid you in a battle. Summoning in DAOs implies the need for a like-minded community when starting a project of such seemingly mythological proportions, and can be understood as an alternative that carries more social connotations than some

32 MolochDAO, ‘Whitepaper Postscript’, *MolochDAO*, n.d., <https://molochdao.com/docs/vision-and-values/supplement-to-whitepaper/>.

33 RaidGuild, ‘Raidguild: A Decentralized Collective of Mercenaries Ready to Slay Your Web3 Product Demons’, *RaidGuild*, n.d., <https://www.raidguild.org/>.

34 RaidGuild, ‘Raidguild’.

35 Ruth Catlow and Penny Rafferty, ‘Introduction: What is Radical Friendship Made of?’, in Ruth Catlow and Penny Rafferty (eds) *Radical Friends: Decentralised Autonomous Organisations and the Arts*, Liverpool: Torque Editions, 2022, 29.

of the ‘more utilitarian terms like “create,” “start,” or “fund”’ that are used in the venture capitalist start up sector.³⁶

The Limits of Covert Politics in DAOs

Kei Kreutler³⁷ traces more significant overlaps between gaming culture and DAO culture. She is specifically interested in the social forms explored in Massively Multiplayer Online Gaming guilds as they self-organise collaboration between strangers in order to achieve a common goal that would be unattainable by individual players, for example slaying a large dragon. The loot collected from these wins has to be distributed fairly between the members of the guild in order to maintain its future coherence. However, there are usually not enough pieces to go around. This conundrum—collective decision-making over the distribution of shared yet scarce assets—reminds Kreutler of decision-making processes in (some) DAOs. Joshua Citarella,³⁸ an artist and writer working on the topic of online political subcultures, describes such gaming guilds as examples of market socialism in which publicly owned resources are distributed through market processes. Kreutler agrees and writes that ‘DAO tooling often tries to technically solve problems that gaming guilds have already culturally refined for several decades, and it may be time for DAOs and gaming guilds to merge their practical knowledge more closely’.³⁹

Kreutler also writes that ‘a DAO *could* be described as a voluntary association with the operating principles of digital cooperativism’,⁴⁰ although she concurs that not everyone might agree with this definition. However, ultimately, she finds that cooperativism is a feature of the technology itself and it does not really matter whether one agrees or disagrees with its politics:

Although many DAOs would not embrace the label of digital cooperative, one could say DAOs embrace cooperativism as a protocol, meaning an evolving set of relational practices that are distinct from traditional corporate structures or decentralized autonomous corporations, because they prioritize member ownership.⁴¹

The two prehistories that Kreutler identifies for DAO culture—cooperativism and gaming guilds—are examples of alternatives to the free market in which workers own the means of production and have agency over the distribution of value. In this framing, the villains DAOs fight can be understood as capitalists, their extractive models, and centralised logics of ownership. Seen from Kreutler’s perspective, both MolochDAO and RaidGuild seem capable of offering a space in which to practice the social relations that could exist outside of capitalism.

36 CoinMarketCap, ‘DAO Summoning’, CoinMarketCap Glossary, n.d., <https://coinmarketcap.com/academy/glossary/dao-summoning>.

37 Kreutler, ‘A Prehistory of DAOs’.

38 Joshua Citarella, ‘DKP is Market Socialism’, *D/S.art*, 2022, <https://dis-art.libproxy.mit.edu/dkp-is-market-socialism>.

39 Kreutler, ‘A Prehistory of DAOs’.

40 Kreutler, ‘A Prehistory of DAOs,’ emphasis in original.

41 Kreutler, ‘A Prehistory of DAOs,’ emphasis in original.

Many members of gaming guilds would likely not associate themselves with the market socialism recognisable in their practices. However, these forms of organising have grown organically to be the most popular in games such as World of Warcraft, their popularity resting, at least in part, on this vagueness.⁴² A similar vagueness is also recognisable in the guild- or coop-like structures and mythologies of DAOs such as those mentioned above. Kreutler describes this vagueness an advantage: DAOs repackaging politics in an unrecognisable shape that gives them a ‘powerful ambiguity’ to ‘build surreptitious solidarity across ideological divides’.⁴³ By telling epic stories that entice people to practice different ways of relating to others, Kreutler argues, these DAOs stealthily and alluringly explore alternative understandings of life and living together.

However, there are significant limitations to this understanding, especially concerning its reliance on technological determinist foundations and a covert form of politics. When Kreutler writes that ‘DAOs embrace cooperativism as a *protocol*’, she injects them with the capacity to establish particular social outcomes through technology: an ‘evol[ed] set of relational practices’ brought about by the structures of DAOs that ‘prioritize member ownership’.⁴⁴ Such a framing leans on a techno-deterministic belief in the power of technology to change society’s ills and deemphasises the social and cultural work that is required to achieve and maintain such shifts. The covert politics of these DAOs—the supposed embodiment of cooperativism by design without using the discourse usually associated with these politics—might indeed allow people that would otherwise never engage with cooperativism to experience its social forms, but does little to politicise participants and encourage practical political engagement with the social reality of which DAOs are part.

Artistic-Activist Engagement with DAOs

Notably, on other occasions Kreutler seems to contradict the idea of DAOs as covert politics by referring to them as manifestos⁴⁵—a term others reiterate.⁴⁶ Political and artistic manifestos alike have the ‘intention to trigger a collective rupture, and [...] assume the voice of some collective ‘we’’.⁴⁷ Rather than covert politics, manifestos overtly state their social and political commitments and, as manifestos, DAOs might be thought of as explicit guidelines for a form of engagement that is starkly different from that which dominates the present. Instead of aiming for Buterin’s credible neutrality⁴⁸ or Kreutler’s own earlier covert politics, thinking of DAOs as manifestos puts politics front and centre in imagining what they may be used for and

42 Citarella, ‘DKP is Market Socialism’.

43 Kreutler, ‘A Prehistory of DAOs’.

44 Kreutler, ‘A Prehistory of DAOs,’ emphasis in original.

45 See e.g., Kreutler quoted in Pete Gomes, ‘The Blockchain: Change Everything Forever’, Vimeo video, 2016, <https://vimeo.com/184677819>; “DAOs in the Art World” ‘DAOs in the Art World’, Right Click Save (2022):

46 See e.g. Catlow and Rafferty, ‘Introduction’.

47 Hans Ulrich Obrist, ‘Manifestos for the Future’, *E-flux Journal* 12 (2010): 1-8.

48 Media scholar focussing on blockchain, social coordination, and economics Nathan Schneider writes on this point: ‘Neutrality is an implied refusal, a missed opportunity, a failure of imagination’. See Nathan Schneider, ‘How We Can Encode Human Rights in the Blockchain’, *Noema*, 2022, <https://www.noemamag.com/how-we-can-encode-human-rights-in-the-blockchain/>.

defining the practices that they could consist of. It is in this light that DAOs have sparked the interest of initiatives that attempt to *overtly* harness the technology for societal or activist ends.

Using DAOs to Support Communities, Interdependency, and Social Coordination

In artistic-activist circles that engage with DAOs,⁴⁹ blockchain technology is often imagined in the context of its potential to support ways of being together with (human and non-human) Others that are different from the logics that structure dominant geopolitical and societal relations. For example, building on the idea of DAOs as manifestos, Ruth Catlow and Penny Rafferty, who are both prominent voices in the artistic-activist blockchain space and together edited the book *Radical Friends: Decentralised Autonomous Organisations and the Arts*,⁵⁰ consider DAOs with particular attention to their potential in achieving socio-technical-cultural futures that diverge radically from the status quo. In the book, Catlow is concerned with 'translocal belonging and cultural cooperation'⁵¹ and sees in DAOs the potential to overcome the limitations of borders and nation states to such collectivities. She argues that DAOs can support self-organisation based on mutual interdependence between individuals, communities, and more-than-human ecologies.⁵² Framing DAOs in primarily social terms, Catlow writes that she wants to 'explor[e] how the governance and coordination tools of DAOs might help forge new paths to belonging (to each other and to this earth)'.⁵³ She continues:

For plural, interdependent cultures to thrive at a planetary scale the future fortunes of all societies must be understood as connected and emplaced. DAOs provide a useful tool for thinking through translocal cultural cooperation and coordination.⁵⁴

Instead of being defined by nationalisms, borders, and separations, Catlow understands contemporary human experiences as fundamentally translocal, 'as a result of globalisation,

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- 49 It is not the aim of this chapter to develop an airtight definition of this space of artistic-activist engagements with DAOs, however, I use this term to broadly indicate approaches to DAOs that are engaged with social, political, and cultural issues using imaginative, creative, and artistic approaches. It refers to a field that I understand as building on the ongoing legacies of media and technology art, critical technical practices and, for example, the theories and practices of cyberfeminism and tactical media. See Philip E. Agre, 'Toward a Critical Technical Practice. Lessons Learned in Trying to Reform AI', in Geoffrey Bowker et al. (eds) *Social Science, Technical Systems, and Cooperative Work: Beyond the Great Divide*, New York: Psychology Press, 1998, 131-58; Mindy Seu (ed) *Cyberfeminism Index*, Los Angeles: Inventory Press, 2022; Geert Lovink, *Dark Fiber: Tracking Critical Internet Culture*, Cambridge, MA: MIT Press, 2003. This space is where previous assumptions about the affordances of blockchain technology may be questioned and where new horizons may open up.
- 50 Ruth Catlow and Penny Rafferty (eds) *Radical Friends: Decentralised Autonomous Organisations and the Arts*, Liverpool: Torque Editions, 2022.
- 51 Ruth Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain – a Citizen Sci-Fi', in Ruth Catlow, and Penny Rafferty (eds) *Radical Friends: Decentralised Autonomous Organisations and the Arts*, Liverpool: Torque Editions, 2022, 173.
- 52 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain'.
- 53 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain,' 188.
- 54 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain,' 189.

migration and hyperconnectivity'.⁵⁵ She writes that 'most of us now feel that we belong to many communities in multiple places at once', leading to 'the political complexity associated with the question of where we come from [...] becoming more apparent to us all, along with questions about territory, who rules us, and by which systems of governance'.⁵⁶ Here, the work of activist scholar David Graeber helps further contextualise Catlow's framing. Graeber argued that anarchist movements are deeply globalist, albeit in a profoundly different way to that in which the term is often used. Theirs is not a globalism that stands for free trade and the easy import and export of capital and commodities, but rather one characterised by a focus on the freedom of people to move around the world, the dissolving of national borders, and the global accessibility of knowledge.⁵⁷ This is the kind of 'translocal' community that Catlow describes.⁵⁸

Although DAOs are often discussed in terms of their capacity to *autonomously* operate technical procedures—i.e., without direct human instructions—to follow Catlow, the 'A' in 'DAO' instead stands for a political kind of autonomy. That is, the autonomy to organise beyond the purview of state power, to do so with a self-chosen community, and to use this autonomy to work towards the social relations that one wants to see in the future. She stresses that such translocal autonomous communities should be understood as more-than-human entanglements that are committed to ecological well-being. DAOs are thus imagined to offer a way to do justice to, and to regain agency in, the context of the inextricable entanglement of all life, cultures, economies, and political realities of the contemporary world.

Although global support networks connecting activist movements pre-exist blockchain, as techno-financial systems that can operate (to a degree, temporarily) outside of the regulatory purview of governments, DAOs offer important opportunities for financial self-sufficiency through translocal interdependence. While managing resources in common with people that are geographically close can be done relatively straightforwardly without a blockchain-based ledger, doing so across localities, currencies, governments, and borders is much harder. Seen in this light, DAOs can consist of contemporary iterations of tactical media practices: they strive to circumvent established power clusters temporarily by appropriating media technology in tactical ways, in this case in the form of self-organised decentralised networks. As 'a set of dirty little practices, digital micropolitics',⁵⁹ which can carry monetary value across borders based on guidelines established by their communities, the tactical use of DAOs can sprawl into the cracks of geopolitical power to support its opposition, even if they are not likely to last for extended periods of time.

Artistic-activist DAOs can thus highlight social relations and embody and reinforce interdependency in the context of the dominant neoliberal ideals of individual responsibility. For example, the now discontinued Dual Power App DAO that the Black Socialists in America (BSA) envisioned can be understood in this light. Formed by a group of 'anti-capitalist, anti-authoritarian,

55 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain,' 174.

56 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain,' 174.

57 David Graeber, 'The New Anarchists', *New Left Review* 13.6 (2002): 61-73.

58 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain'.

59 Lovink, *Dark Fiber*, 267.

internationalist Black Americans who believe people should control the means of production in a directly democratic and decentralized way that allows symbiosis with non-human nature,⁶⁰ the BSA are particularly concerned with funding grassroots projects building interdependent networks of support among Black American communities and the need to make and keep critical knowledge accessible. Rooted in the work of radical Black scholars and activists, they hoped to support cooperative communities, mutual aid networks, and labour organisers through the non-hierarchical governance structures, self-defined distributions of value, and collective systems of ownership that the Dual Power App DAO would afford.⁶¹

Another example is the ReUnion Network, a speculative blockchain project initiated by Rotterdam-based artist and designer Yin Aiwen. ReUnion Network imagines how society might function if care was the central logic of social relations and organisation.⁶² It turns contemporary approaches to care as either invisibilised, domestic work, underpaid professional work, or a commodified sector upside down, instead focussing on 'trans-local care system for chosen families and mutual-aid communities'⁶³ that gain more financial support from social organisations as their commitment to each other and time spent caring for others grows. Although not explicitly framed as a DAO, the project imagines shared wallets with mutual funds and smart contracts that automate the release of funds for caring relationships, and as such can be understood as DAOs.⁶⁴ Building on this framework, Liquid Dependencies is a game version of the ReUnion Network in which players act out an alternate reality in which they are incentivised to respond in caring ways to each other's life events. Together, the players shape a vision for a future 'decentralised caring society'⁶⁵ and the role of DAOs in bringing such a society about.

Black Swan DAO, an artist collective based in Berlin of which Penny Rafferty is a member, also proposes new forms of social coordination. Black Swan is particularly interested in rethinking the logics at work in decentralised collective decision-making. Whereas in most DAOs, members are assigned voting power according to the number of tokens they hold and, as such, practice a form of decentralised decision-making that can have plutocratic tendencies, Black Swan operationalises voting processes in significantly different ways. Their 'quadratic' voting system consists of a process in which every member of the community gets the same amount of voting credits and gets to allocate them across proposals made by the community.

60 Black Socialists in America, <https://blacksocialists.us/about>.

61 Daisy Alioto, 'Join a Union – But Also Join a DAO', *The Nation*, 2021, <https://www.thenation.com/article/society/dao-crypto-union/>. On December 19, 2023, the Dual Power App team announced its project was paused indefinitely. See The Dual Power App, 'FYI: This Project is on Pause (Indefinitely) as @blacksocialists Reassesses Its Organizational Priorities. Everything We've Published Across Product, Design, and Engineering (Minus the Otd + Dpa Brands) is Open to Reuse and Modification for the Commons. <https://dualpower.app>', X, 2023, <https://x.com/DualPowerApp/status/1737187927279423738>.

62 ReUnion Network, <https://www.reunionnetwork.org/>.

63 Aiwen Yin and Mengyang Zhao, 'Playing 'Liquid Dependencies: What Does a Decentralised Caring Society Look Like' (2021 - Present)', *Arts of the Working Class*, 2023, <https://artsoftheworkingclass.org/text/playing-liquid-dependencies-what-does-a-decentralised-caring-society-look-like-2021-present>.

64 ReUnion Network, <https://www.reunionnetwork.org/>.

65 Yin and Zhao, 'Playing 'Liquid Dependencies,' n.p.

This way, voting turns from a binary decision into a more nuanced expression of value and interest. Importantly, distributing one's voting credits across multiple proposals instead of giving them all to a single one is incentivised by increasing the weight of those distributed votes.⁶⁶ In this way, Black Swan structures DAOs to function as tools against polarisation and for consensus-building, against zero-sum-games and for nuance. CultureStake, a project by Furtherfield, a London-based arts organisation of which Ruth Catlow is a co-founder and co-director, also uses quadratic voting to engage local communities in decisions about the distribution of cultural funding in projects in their neighbourhood.⁶⁷

Bridging Prefiguration, Affordances, and Imagination

The above examples show that, rather than just manifestos that consist of a set of guidelines for future (inter)actions that one can underwrite, but which do not do more than make these commitments textually explicit, these DAOs also offer structures to facilitate acting according to those guidelines—spaces in which to practice new forms of engagement. With this in mind, Catlow and Rafferty refer to DAOs as having a *prefigurative* capacity to practically and structurally embed sociocultural change.⁶⁸ Prefiguration consists of practices that aim to change social systems at their roots, 'transcending capitalism by tackling social reproduction, by embodying change, and by radically reshaping human needs, habits, and beliefs'.⁶⁹ To account for the element of technical engagement in these prefigurative DAO practices, I propose the notion of *radically reimagined affordances*. Such affordances put the active and creative capacity of the imagination to use in order to institute new socio-techno-cultural potentialities. Particularly, thinking of DAOs as 'structures of belonging' rather than 'property-oriented enclosure systems'⁷⁰ opens an imaginative horizon that moves beyond the logic of mere ownership to include something more social, for example, membership of a community or collective.

Prefigurative DAOs

Prefiguration is a form of activism that aims to enact revolutionary change in the here and now. While the term gained prominence in the 1970s, it can retroactively be used to understand debates and frictions between anarchists and Marxists as far back as the 1860s. While many associate prefiguration with anarchists and Marxists alike, differences do exist. Particularly, Marx's idea of revolution depended on the working class seizing power and, until that happens, to him, revolution could only play out in some undefined future. Anarchists, however, argued that 'revolutionary struggle must be consistent with its ends'.⁷¹ That is, movements

66 Black Swan, <https://trust.support/feed/black-swan>. Black Swan voting rounds are scheduled according to moon cycles in an attempt to free social organisation from the inflexible and ubiquitous logics of time reinforced by calendars and computational schedules.

67 CultureStake, <https://www.furtherfield.org/culturestake/>.

68 Catlow and Rafferty, 'Introduction'; Ruth Catlow, 'To Larp a DAO', in Ruth Catlow and Penny Rafferty (eds) *Radical Friends: Decentralised Autonomous Organisations and the Arts*, Liverpool: Torque Editions, 2022, 305-13.

69 Lara Monticelli, 'On the Necessity of Prefigurative Politics', *Thesis Eleven* 167.1 (2021), 102.

70 Zeilinger, *Structures of Belonging*, 5, 12.

71 Matthijs van de Sande, 'Prefiguration', *Krisis* 38.2 (2018), 133.

demanding change should already act out the change they desire: i.e., the movement should *prefigure* change .

Catlow and Rafferty are not alone in connecting blockchain with prefiguration, although some of the uses by others indicate that more clarity is needed on the implications of the term. On the one hand, Joshua Dávila, author of *Blockchain Radicals: How Capitalism Ruined Crypto and How to Fix It*,⁷² largely aligns with Catlow and Rafferty's politically particular understanding of prefiguration and writes that it is time to 'start prefiguring and designing the non-alienating systems of relations we would prefer to live under instead, by collectively wielding the technological tools we have at our disposal for pro-social ends' (14). On the other hand, Husain et al. emphasise the *inherently* political nature of designing systems that govern the social world. They write that 'blockchain projects personify 'prefigurative politics' *by design*'⁷³—whatever these politics may be—because they describe social relations in their code and in that way (the authors argue) prefigure the kind of relations they want to establish. This is a technologically deterministic view that is not compatible with a historical understanding of the term prefiguration as signifying specifically anti-capitalist struggles. Based on the breadth of political dynamics reinforced by blockchain projects, Husain et al. seem to equate the fact that blockchains are indeed political by their very nature—I agree that no technology is neutral—with a depoliticised understanding of prefiguration, stripping it of any necessarily anarchist, Marxist, or otherwise radical signification—a point upon which we fundamentally diverge.⁷⁴ For a more historically situated and political understanding of prefiguration, I turn to political sociologist Lara Monticelli, who argues that prefiguration *inherently* 'aims at transcending capitalism by tackling social reproduction, by embodying change, and by radically reshaping human needs, habits, and beliefs' by proposing a 'radical, emancipatory, and decolonized' vision for possible futures to act out in the here and now.⁷⁵ Prefiguration is thus characterised by material changes as well as shifts in how people understand themselves and their relation to the world of which they are part.

In order to apply this notion of prefiguration it is necessary to briefly unpack the adjectives Monticelli employs—radical, emancipatory, decolonised. The radicalness of prefigurative practices can be recognised in their aim to establish 'a profound rearrangement of the roots [...] of the existing order';⁷⁶ their emphasis on emancipation in the sense that they are 'directed towards the sustainable flourishing of human and non-human life on this planet';⁷⁷ and, as decolonised practices, they 'acknowledge the profound interdependence and interconnectedness between the so-called Global North and Global South at the economic, political,

72 Joshua Dávila, *Blockchain Radicals: How Capitalism Ruined Crypto and How to Fix it*, London: Repeater Books, 2023.

73 Syed Omer Husain, Alex Franklin, and Dirk Roep, 'The Political Imaginaries of Blockchain Projects: Discerning the Expressions of an Emerging Ecosystem', *Sustainability Science* 15.2 (2020): 380. <https://doi.org/10.1007/s11625-020-00786-x>. Emphasis added.

74 Husain, Franklin, and Roep, 'The Political Imaginaries of Blockchain Projects'.

75 Monticelli, 'On the Necessity of Prefigurative Politics,' 102.

76 Monticelli, 'On the Necessity of Prefigurative Politics,' 101.

77 Monticelli, 'On the Necessity of Prefigurative Politics,' 101.

and ecological level'⁷⁸ and support 'epistemological decolonisation' through the facilitation of 'multiple knowledges, multiple subjects of knowing and multiple practices of creating knowledge'.⁷⁹ Furthermore, in recognition of the interdependent nature of all aspects of life, 'prefigurative social movements emphasize the need to tackle, simultaneously, economic, social and environmental issues'⁸⁰ and take their own organisation as the experimental form in which alternative being and relating are exemplified.⁸¹

The discussions in chapters 2, 3, and 4 of this book show that in many cases, blockchain culture is anything but prefigurative according to this definition. Crypto memes and astrology ultimately conform to the status quo of widespread financialisation and speculation; blockchain mysticism limits emancipation by presenting the technology beyond human control; and the tokenisation of nature contributes to the ongoing reproduction of coloniality. Catlow engages with similar contexts when she writes:

Crucial to [the construction of the Web3 worlds we want⁸²] is an acknowledgement of the multiple layers of devastating losses that are the result of colonial extractivist petrocapiatalism upon which this webbed mechanosphere is built: the mass dispossession, destruction and loss of human lives, the loss of species biodiversity and habitats and the impoverishment of futurity that is its aftermath.⁸³

Catlow's notion of 'colonial extractivist petrocapiatalism'⁸⁴ calls to mind scholar and activist bell hooks' phrase 'imperialist white supremacist capitalist patriarchy',⁸⁵ and similarly serves the purpose of doing justice to the inextricable and mutually reinforcing nature of such systems of oppression and exploitation. Catlow echoes Monticelli's and Motta's understandings of the politics of prefiguration and the need to simultaneously address economic, social, and environmental issues. As the existence of books such as *Radical Friends* and *Blockchain Radicals* shows, blockchain sparks hopes of prefiguring radically different futures with certain activists and artists. As editors of *Radical Friends*, Catlow and Rafferty frame DAOs as 'powerful social

78 Monticelli, 'On the Necessity of Prefigurative Politics,' 101.

79 Sara C. Motta, 'Decolonising Critique: From Prophetic Negation to Prefigurative Affirmation', in Ana Cecilia Dinerstein (ed) *Social Sciences for an Other Politics: Women Theorizing Without Parachutes*, Cham: Palgrave Macmillan, 2016, 44, 45. Significantly, Sara Motta crystallises the importance of addressing the geopolitical coloniality of knowledge in any attempt at decolonisation through the idea of 'prefigurative epistemologies' (33)—i.e., knowing differently has ontological, prefigurative power. I will expand on the meaning of this notion in the context of DAOs and the imaginary of blockchain as a truth machine in a subsequent section.

80 Monticelli, 'On the Necessity of Prefigurative Politics,' 109.

81 Graeber, 'The New Anarchists'; van de Sande, 'Prefiguration'.

82 This phrasing is taken from a few sentences above on the same page as the quoted section. Catlow refers to prefigurative DAOs in the context of the move away from the centralised control and extractivist practices that dominate Web 2.0 and towards yet undefined 'Web3 worlds'. See Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain,' 177.

83 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain,' 177-78.

84 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain,' 178.

85 bell hooks, *Writing Beyond Race: Living Theory and Practice*, New York: Routledge, 2012, 4.

instrument[s]⁸⁶ that can be used to test and practice new worlds and different ways of being together that address these issues.

The DAOs mentioned above each echo elements of these descriptions of prefiguration in different ways. The Dual Power app by The Black Socialists in America is most clearly concerned with emancipation by establishing mutual aid networks between workers, but also emphasise that access to critical knowledge on ongoing coloniality is essential in this process. The ReUnion Network and the Liquid Dependencies game can be seen as tools for people to practice and make a radical shift in their understanding of how societies should or could be structured. Black Swan and CultureStake replaces competitive logics that dominate many processes of cultural funds allocation with emancipatory logics that aim to produce a flourishing community in which many initiatives may arise simultaneously.

Thus, while the previous chapters showed that blockchain affords the reproduction of extractive social relations, depoliticised understandings of technology, and hierarchical power dynamics, something different seems to occur in the artistic-activist DAOs discussed here. To understand this shift, I combine Castoriadis' notion of the radical imagination (which we encountered in the previous chapter)⁸⁷ with Nagy and Neff's concept of imagined affordances (which I discussed in chapter 1)⁸⁸ to argue that prefigurative engagements with blockchain open up new potential horizons by radically reimagining the technology's affordances. More generally, I propose that the concept of *radically reimagined affordances* could be understood as an important practice in prefiguration that features complex technologies dominated by imaginaries that reproduce the status quo.

Radically Reimagined Affordances

In 2019, blockchain researcher and practitioner Laura Lotti⁸⁹ wrote that despite the technology's overwhelming absorption into extractive capitalist practices, blockchain is still an emerging technology that has yet to assume a definite shape. Since then, it has anything but settled, with hype cycles around Web3, NFTs, and DAOs, each pushing and pulling the technology in different sociocultural directions. Lotti argues that, instead of taking the dominance of capitalist logics as a cue for those with other ambitions to steer clear of the technology, they should continue to reshape blockchain's affordances for different uses and attempt to redirect it towards supporting different sociocultural dynamics.⁹⁰ Zeilinger stresses the role of the imagination in such work, writing that this 'involves *reimagining* blockchains not as property-oriented enclosure systems, but instead as structures of belonging'.⁹¹ This is the space I want to address with the notion of radically reimagined affordances.

86 Catlow and Rafferty, 'Introduction,' 45.

87 Cornelius Castoriadis, *The Imaginary Institution of Society*, Cambridge, MA: MIT Press, 1998.

88 Peter Nagy and Gina Neff, 'Imagined Affordance: Reconstructing a Keyword for Communication Theory', *Social Media + Society* 1.2 (2015): 1-9.

89 Laura Lotti, 'The Art of Tokenization: Blockchain Affordances and the Invention of Future Milieus', *Media Theory* 3.1 (2019): 287-320. Lotti is, like Rafferty, part of the Black Swan DAO.

90 Lotti, 'The Art of Tokenization'.

91 Zeilinger, *Structures of Belonging*, 12, emphasis added.

On the one hand, this notion builds on Nagy and Neff's concept of 'imagined affordances',⁹² which emphasises that affordances do not emanate solely from a technological setup, but emerge in the interplay between materiality and socio-cultural factors, such as the expectations and intentions that designers and users of technology may have. Especially 'complex socio-technical systems that act with and sometimes without or despite us'⁹³ require a theory of affordances that includes the imagination. Complex technologies, like blockchains, always involve user engagement that is not based on technological capacities, but rather the affordances that users imagine those technologies to have. Furthermore, in an iterative and continuous process, 'imagined affordances [...] may *performatively* help to shape the platforms themselves'.⁹⁴ For example, in imaginaries of blockchain as a truth machine, blockchain is *imagined* to afford a new universal and immutable kind of truth and applications like those of the blockchain forest protection projects in chapter 2 are shaped according to this imagined affordance.

On the other hand, the notion of radically reimagined affordances builds on Cornelius Castoriadis' understanding of the imagination as a potentially radical phenomenon.⁹⁵ As discussed in more detail in chapter 4, the radical capacity of the imagination consists of its ability to institute fundamentally *new* things. Rather than thinking of the imagination as a passive faculty of an individual's brain, Castoriadis frames it as a social phenomenon that is consequential and powerful, rather than passive or reflexive. The radical imagination opens up the possibility to create different futures from the status quo that are not logically connected to it. Understanding radical imagination in this way shows its significance to the prefiguration that Monticelli seeks. The radical imagination can be understood as a key element in the 'profound rearrangement of the roots [...] of the existing order'⁹⁶ and 'a necessary step towards realizing the possibilities for articulating alternatives to today's financialized economy and society'.⁹⁷

Bringing prefiguration, imagined affordances, and radical imagination together—and keeping in mind the myriad ways in which blockchain affords the reproduction of the status quo—I argue that *radically reimagined affordances* are essential in making prefigurative blockchain practices possible at all. While imagined affordances per se may support any political or seemingly apolitical direction, and thus may or may not be part of prefiguration, affordances that are reimagined as part of radical prefigurative practices gain an intentionality. They are imagined in order to alter the very roots of society as we know it.

92 Nagy and Neff, 'Imagined Affordance,' 1.

93 Nagy and Neff, 'Imagined Affordance,' 2, emphasis added.

94 Taina Bucher and Anne Helmond, 'The Affordances of Social Media Platforms', in Jean Burgess, Thomas Poell, and Alice E. Marwick (eds) *The Sage Handbook of Social Media*, Thousand Oaks: Sage Publications Ltd., 2017, 241, emphasis added.

95 Castoriadis, *The Imaginary Institution of Society*.

96 Monticelli, 'On the Necessity of Prefigurative Politics,' 110.

97 Aris Komporozos-Athanasiou and Chiara Bottici, 'Rethinking Prefiguration: Between Radical Imagination and Imaginal Politics', in Lara Monticelli (ed) *The Future is Now: An Introduction to Prefigurative Politics*, Bristol: Bristol University Press, 2022, 65.

Catlow's framing of DAOs offers an insight into how this radical reimagination of blockchain's affordances might take shape. When she emphasises the ecological devastation that is tied up with the intertwined hierarchies of power and agency in colonial capitalism, her framing mirrors concerns that dominated chapter 2 of this book, but this time with the hope of using blockchain technology to undermine and negate the colonality that blockchain forest protection projects reproduce. For this to be possible, blockchain's affordances need to be radically reimagined. When Catlow frames DAOs as facilitating 'translocal belonging and cultural cooperation',⁹⁸ (173), as places to practice different ways of living together, and building networks of support, she emphasises social belonging instead of the financial, and in effect thinks of DAOs in terms of what Zeilinger calls structures of belonging.⁹⁹ This does not mean that the colonality of blockchain forest protection projects is prevented or called to a halt, but rather that it opens a new imaginative space of engagement with blockchains and DAOs that starts from radically different assumptions about its uses. Rather than focussing on how to establish universal, atomised, and immutable truths about nature through tokenisation, thinking about DAOs as structures of belonging in such a context may instead emphasise ecological interdependencies that are never reducible to individual tokenised representations.

The Dual Power App and ReUnion Network can also be understood as contributing to this reimagination of blockchains as structures of belonging that afford forms of community that are deprioritised, discouraged, or impeded in the dominant logics of contemporary societies. Imagining DAOs as social support structures, they attempt to prefigure worlds with caring, mutual aid, self-organised communities with strong, long-term, and interdependent relationships at their core. They imagine alternative forms of social coordination: the Dual Power App centred unionisation and mutual aid networks as key forms of organising workers across space, while ReUnion reimagines long-term and intimate commitments that break open and reimagine the family as a core unit of society.

Black Swan offers an interesting additional perspective on the idea of radically reimagining affordances as part of prefigurative practices. Its members say that they use 'blockchain and DAOs as tools for the imagination',¹⁰⁰ meaning that, rather than actually implementing ledgers, smart contracts, and DAOs, they use the *logics* of blockchain-based computation in order to experiment with more equitable forms of cultural funding without the need to code complete and secure technological systems. They call this approach 'blockchain thinking without the technology'.¹⁰¹ Without being tied to actually working technologies, the reimagination of blockchain's affordances gains a new level of flexibility to develop changes that address the very root of the logics that structure much of its application today. I will return to this practice in more detail in the concluding chapter.

98 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain,' 173.

99 Zeilinger, *Structures of Belonging*.

100 Laura Lotti et al., 'Simulating Equitable Art Worlds: DAOs Through a Role-Played Lens', *KW-Berlin*, n.d., <https://www.kw-berlin.de/en/black-swan-dao/>.

101 Lotti, 'Simulating Equitable Art Worlds'.

The examples discussed in this section exemplify how prefigurative engagement with DAOs works by radically reimagining the affordances of the technology as structures of belonging in social, collectivist, and caring ways. In doing so, they attempt to make unlikely worlds available for lived experiences, even if only temporarily and in limited ways. These are important prefigurative practices in the context of the pervasiveness of the logic of borders, financialisation, exclusion, polarisation, individualisation, and precarity. However, what remains unaddressed in these cases is how prefigurative DAOs may open up new horizons in the context of the imaginary of blockchain as a truth machine specifically.

Toward Prefigurative Truth Imaginaries

In order to radically reimagine blockchain as the producer of truth, knowledge, and a shared reality, what needs to be reimagined is how we may come to know through blockchain. In chapter 2, we encountered a version of truth that was based on modernity/coloniality. It aimed for universal coverage and afforded a limited group of people the capacity to create and act on knowledge. In chapter 3, truth appeared in the shape of predetermined facts, emerging from beyond the human capacity. Similarly continuing modernity/coloniality, it aimed to reduce the complexity of the world to a divinely ordained system of knowledge. To address these issues, it is necessary to ‘transform epistemology into a prefigurative practice of everyday life’.¹⁰² Such ‘prefigurative epistemologies’¹⁰³ offer a way of thinking about what truth and knowing the world constitute from a perspective that changes the roots of the universal logics of knowledge that dominate the imaginaries of blockchain as a truth machine. Rather than the controlled order of rationalism, Rafferty finds a different epistemological root in chaos, and takes this as the starting point for a new prefigurative imaginary for DAOs. In this imaginary, DAOs are a space for ritualised narrative practices through which communities are able to continuously re-establish themselves.¹⁰⁴

Setting Intentions for Plural Worlds

In the context of the control and order that characterises modernity/coloniality—as is visible in the use of blockchain in chapter 2—Rafferty turns this widespread tendency on its head to argue that chaos should be appreciated instead.¹⁰⁵ Chaos, she argues, is something that many people have learned to fear, hate, or be suspicious of because of the way their societies have conditioned them, rather than because it is something inherently bad, and that in the effort towards self-organisation it is paramount to learn to appreciate chaos, even if dominant stories about life have shown it in a negative light:

To love chaos would mean, for many, that we would have to hate control and that would mean a complete overhaul of the tyrannical society we live in, and so we learn

102 Motta, ‘Notes Towards Prefigurative Epistemologies,’ 182.

103 Motta, ‘Notes Towards Prefigurative Epistemologies,’ 178.

104 Penny Rafferty, ‘The Reappropriation of Life and the Living—a Cosmic Battleground’, in Ruth Catlow, and Penny Rafferty (eds) *Radical Friends: Decentralised Autonomous Organisations and the Arts*, Liverpool: Torque Editions, 2022, 102-14.

105 Rafferty, ‘The Reappropriation of Life and the Living’.

to reject chaos's offerings in favour of the comfortable shackles of mastery. To live alongside or to embrace chaos is to become ungovernable.¹⁰⁶

Rafferty argues that different stories about chaos can and should be told. To do this, she finds inspiration in the etymological history of the word itself: the Greek *khaos* means an abyss, a vast and empty chasm, a void. In ancient Greek myths about the origin of the universe, chaos refers to the void or gap at the start of creation: that moment when nothing *is* yet, but at the same time, everything is *possible*. She describes this mythological chaos as an 'early genesis hole, this empty yet full state [that] was once akin to a babbling spring, oozing life and creativity'.¹⁰⁷ She states that 'occultists, anarchists, artists, poets, philosophers and thinkers' create the 'generative-chaos' in which people may 'act with a playful sense of metanoia (going beyond thought) and a philosophy of flux, discarding dogma in favour of fluid mutation'.¹⁰⁸ Within this mythology of chaos, thinking and doing go hand in hand, continuously changing and iterating between experimental practices and thought experiments free from order or control.¹⁰⁹

However, learning to reappraise chaos is not easy:

For someone or a collective body to go against [the suspicion of chaos] and love chaos they must be agile, thoughtful and fluid. They must also believe they have access to resources that allow them to survive and grow in chaos. Chaos lovers are believers in both the abundance of change, and the abundance that flows from change.¹¹⁰

In chapter 4, we saw crypto traders embracing the chaos of a volatile market. They revelled in uncertainty and the unexpected, speculating on the sheer abundance of possibility. It was a way to survive in an otherwise precarious 'now,' always with the hope of a big break, that one time hitting x1000 yield, but even without achieving those, speculation on uncertainty was always already better than the option of not choosing to speculate at all and fall back on calculation and universal logics that ultimately feel anachronistic in the midst of such fundamental uncertainty.

Rafferty shares some premises with the crypto traders: she appreciates chaos and understands it as a (potential) source of abundance. Like the crypto traders that surf on the high waves of volatility, she argues that those who 'love chaos [...] must be agile, thoughtful and fluid'¹¹¹ in response to those waves. Yet, there is also a fundamental point of disagreement on what this chaotic potentiality is *for*. Against the individualism of the crypto traders, Rafferty is concerned with self-organised groups of people and communalism. Rather than profit, it is the potential for the autonomy to define the character of one's life in community with others

106 Rafferty, 'The Reappropriation of Life and the Living,' 103.

107 Rafferty, 'The Reappropriation of Life and the Living,' 103.

108 Rafferty, 'The Reappropriation of Life and the Living,' 106.

109 Rafferty, 'The Reappropriation of Life and the Living'.

110 Rafferty, 'The Reappropriation of Life and the Living,' 104.

111 Rafferty, 'The Reappropriation of Life and the Living,' 104.

that attracts Rafferty to the blockchain space. More specifically, she writes that ‘DAOs could allow us to collectively set up these void states *together*, and through the act of proposal making and voting, harness intention to regulate new reality making devices’.¹¹² For Rafferty, DAO-based practices and the new forms of social organisation they establish offer sites in which to become acquainted with chaos, and to learn to live in unstructured ecologies with others rather than in competition with them.

Besides the ancient Greek lineage of chaos, Rafferty is also inspired by a more recent yet equally mystical reference to chaos: Chaos Magick.¹¹³ Researcher of religion and magic Bernd-Christian Otto explains that Chaos Magick can be understood in relation to chaos theory. Both emerged around the same time in the 1970s, although they also diverge significantly. Chaos theory started with the notion of non-linear dynamics, made famous by the analogy of ‘the butterfly effect,’ which conveys the idea that small changes may have large and unpredictable effects in complex systems. Inspired by this theory, Chaos Magick then turned this non-determinism and uncertainty inward: to practitioners, individually held beliefs can have massive effects in the wider universe. They understand reality to be subject to these beliefs, and central to the practice is the active shaping of one’s beliefs as a tool to manifest different realities.¹¹⁴

Sigils are central in Chaos Magick: they are symbols, often circular in shape, which are charged with magical capacities to make things happen or appear. In more traditional forms of magic that existed before Chaos Magick, sigils were passed down through the ages and believed to be effective exactly because they endured over time. However, in the early years of the twentieth century a practice called sigilisation gained prominence, and later became a central tenet of Chaos Magick. Sigilisation turned sigils from eternal and universal into self-made symbols for individual purposes. In Chaos Magick, a practitioner may produce custom sigils to fit their own intentions, wishes, and desires. Subsequently, by reaching ritualised states of ecstasy, chaos magicians alter their beliefs and consequently, practitioners believe, reality is also altered.¹¹⁵ In Chaos Magick, “chaos’ is [...] promoted to a universal life force’¹¹⁶—similar to that bubbling origin of everything that so inspires Rafferty.¹¹⁷

112 Rafferty, ‘The Reappropriation of Life and the Living,’ 107, emphasis added.

113 Rafferty, ‘The Reappropriation of Life and the Living’.

114 Bernd-Christian Otto, ‘The Illuminates of Thanateros and the Institutionalisation of Religious Individualisation’, in Martin Fuchs et al. (eds) *Religious Individualisation: Historical Dimensions and Comparative Perspectives*, Berlin: De Gruyter, 2020, 759–96.

115 Otto, ‘The Illuminates of Thanateros and the Institutionalisation of Religious Individualisation’.

116 Otto, ‘The Illuminates of Thanateros and the Institutionalisation of Religious Individualisation,’ 765.

117 While Rafferty thinks of the potentiality of chaos as fundamentally pluralistic, it should be noted that Chaos Magick’s sigilisation principles have been used to understand extreme right-wing practices that strive for cultural homogeneity as well. See Egil Asprem, ‘The Magical Theory of Politics: Memes, Magic, and the Enchantment of Social Forces in the American Magic War’, *Nova Religio: The Journal of Alternative and Emergent Religions* 23.4 (2020): 15–42; Jacob Christiansen Senholt, ‘Radical Politics and Political Esotericism: The Adaptation of Esoteric Discourse Within the Radical Right’, in Egil Asprem and Kennet Granholm (eds) *Contemporary Esotericism*, London: Routledge, 2014, 244–64.

Emerging in the 1970s, Chaos Magicians were influenced by the development of postmodern theory and its critique of universal and a-historical truths. Chaos Magicians took ideas from postmodern theorists and pushed them to a relativist, individualised, and internalised extreme. Liberating themselves from dominant, ideological, universalist truth narratives, Chaos Magicians became convinced that shared truth in fact does not matter at all. Instead, their belief system operates through a utilitarian approach to knowledge: whether a god or some other higher cosmic force actually exists is unimportant if the sigil or prayer worked. This also means that established belief systems have no precedence over new ones or even self-made ones. In short, Chaos Magicians believe the things they themselves have tested and have experienced to be effective.¹¹⁸

Here again, the attitudes of the crypto traders that featured in chapter 4 are recognisable. Rather than searching for lasting and universal truths, the memes painted a picture of crypto traders that approach each bit of information in terms of the degree to which it is operationalisable on the market. What counts to crypto traders is not whether information reflects some shared reality in a truthful manner, but whether it moves the market.

Again, Rafferty shares some of these premises regarding the shortcomings of understanding ‘truth’ as universal and a-historical. Like Chaos Magicians, she is concerned with the power of beliefs, world views, and intentions in shaping reality, but unlike them, she does not address this capacity in a relativist and individualistic sense. Like the crypto traders, she is interested in the performativity of information on a blockchain, but unlike them, she is not concerned with spectacular, individual monetary gains. Instead, her critiques are part of a larger framework in support of community, subsistence, flexibility, and interdependence.¹¹⁹

This way of thinking about different practices of knowing as ways to make different realities come into view resonates with DAO projects in the field. For example, sigils and sigil-like imagery are a recurring presence in the wider DAO space. The artistic project *Claves Angelicae*¹²⁰ by Cullen Miller and Gabriel Dunne asks its audience members to create a sigil of their desires and intentions, and to inscribe it onto the Ethereum blockchain as an act of ritualised manifestation (see for example Figure 34). The sigil is surrounded by a circular rendering of the hash that represents the block on the Ethereum blockchain in which the sigilised intentions are encrypted, and the written intention itself is algorithmically rendered into a line drawing in the centre of the sigil.

Miller and Dunne write that they understand DAOs as ‘emergent entities formed by groups of magicians collectively constructing a shared, single thought’ and that such ‘decentralized systems allow for a plurality of deities to inhabit the Ether and entrust magicians to enact their Will without permission’.¹²¹ In less mystical terms, they understand DAOs as tools that collectives can use to find alignment among their members—think for example of the

118 Otto, ‘The Illuminates of Thanateros and the Institutionalisation of Religious Individualisation’.

119 Rafferty, ‘The Reappropriation of Life and the Living’.

120 Cullen Miller and Gabriel Dunne, *Claves Angelicae*, 2018. <https://claves-angelicae.github.io/dossier/>.

121 Miller and Dunne, *Claves Angelicae*.

quadratic voting used by Black Swan and CultureStake mentioned above—and, at the same time, offer the possibility for many collectives to do this at the same time without central oversight or regulation.

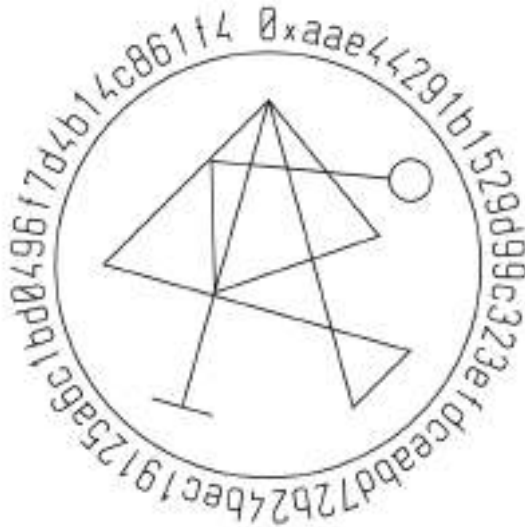


Fig. 34. Screenshot of a sigil featured on the *Claves Angelicæ* website.¹²²

Another example of sigil-like imagery in the DAO scene is the visual identity of Gnosis Guild and Zodiac, in which the aforementioned Kei Kreutler played a central role until 2022. Gnosis Guild is a group focussed on making DAO-building more accessible and modular, with the Zodiac DAO toolkit being a prominent output. Zodiac consists of various openly available tools and modules that other DAOs can use in their own systems, and to which anyone is free to contribute new items.¹²³ Each tool or module is recognisable by a unique icon, which is made up of a set of standardised design elements, each with their own listed meaning. Together, the design elements form a sigil-like icon seemingly imbued with ‘intentions’ for it to contribute to a DAOs strength, magic, dexterity, intelligence, or wisdom (see Figure 35).¹²⁴

¹²² Miller and Dunne, *Claves Angelicæ*.

¹²³ Kei Kreutler, ‘Zodiac: The Expansion Pack for DAOs’, *Gnosis Guild*, 2021, <https://gnosisguild.mirror.xyz/OuhG5s2X5uSVBx1EK4tKPhnUc91Wh9YM0fwSnC8UNcg>; Gnosis Guild, <https://www.gnosisguild.org/>.

¹²⁴ These terms are a nod to the terminology used in tabletop and digital role-playing games such as *Dungeons and Dragons*, and illustrate the influence of gaming culture on DAO culture that Kreutler highlights. See Kreutler, ‘A Prehistory of DAOs’.



Fig. 35. Sigil-like imagery used by Gnosis Guild and Zodiac. On the left a screenshot of the Zodiac design elements, and on the right a screenshot of the sigil-like Zodiac icon that is created with these elements, which features on the Gnosis Guild website.¹²⁵

Collectively Enacting Mutable Worlds

Based on her definition of chaos as a new root from which to engage with blockchain, Rafferty radically reimagines blockchain's affordances for establishing shared truths.¹²⁶ A key element in this is the transformation of blockchain's affordance of immutable data storage to its affordance of the support of mutable collectives. Immutability features prominently in the mainstream imaginaries of blockchain as a truth machine, from the verification processes in blockchain forest protection projects and the perfect predestined truths in religious narratives surrounding the technology discussed in previous chapters, to its wider application by organisations that trace the origins of fair-trade ingredients or by identity management systems. Instead, as part of Black Swan, she states that what makes DAOs so interesting to the collective is that they make organisations 'mutable, liable to change,' and are able to 'metamorphosise at the will of their members'.¹²⁷ More concretely, Rafferty reflects on the mutability of DAOs as follows in *Radical Friends*:

The mechanics of DAOs [...] don't exist in a fixed state and can be altered depending on propositions, progressions and failures, which allow them to socially evolve based on shared knowledge and learnt experiences—the lived life of the group, rather than taboo and untested tradition.¹²⁸

125 Kreutler, 'Zodiac'; *Gnosis Guild*, <https://www.gnosisguild.org/>.

126 Rafferty, 'The Reappropriation of Life and the Living'.

127 Rafferty in Laura Lotti et al., *CTM 2022: Prototyping Sonic Institutions*, 2022. <https://www.youtube.com/watch?v=DKb9za47twU>.

128 Rafferty, 'The Reappropriation of Life and the Living,' 107.

Rafferty proposes DAOs as ‘an experimental practice for moving towards a different way of living together’¹²⁹ that is never finished, but always in a state of becoming, mutating with the needs of the community.¹³⁰ Because they never settle into a final state, always remaining ‘forkable, mutable, [and] questionable’,¹³¹ prefigurative DAOs make possible new practices of knowing that destabilise mainstream understandings of truth as universal.

Such DAOs hold the potential for setting up structures that support many different ways of knowing and being together *simultaneously*. For Rafferty, the sigils mentioned above are a way ‘to collectively set up these void states together, and through the act of proposal making and voting, harness intention to regulate new reality making devices’ ‘in a kaleidoscope of ways’.¹³² Such ‘void’ states, as new beginnings, are made together with others: they are the result of bottom-up processes—proposals and votes that make desires and intentions explicit, questionable, and mutable over time. Furthermore, even though these processes are collective, those collectives do not have to stay cohesive: they can mutate, fork, and become plural.

An example that attempts to change the very basic assumptions about knowing shared reality that structure contemporary social relations is Dayra, a DAO project by a Palestinian art collective called The Question of Funding (QoF). Dayra was publicly presented for the first time in 2022 at the quinquennial contemporary art exhibition Documenta 15 in Kassel, Germany. It emerged in the precarious Palestinian context of the time, in which cultural funding depended on foreign donors and was, therefore, always constrained by politics.¹³³ Independent organisation was near impossible due to a general lack of money, and because access to basic resources was highly restricted by Israeli laws and infrastructures.¹³⁴ With Dayra, the collective radically reimagines cultural funding and wonders how to organise ‘in

129 Rafferty, ‘The Reappropriation of Life and the Living,’ 107.

130 Lotti, CTM 2022. Rafferty emphasised the her understanding of DAOs as mutable sociotechnical assemblages when I interviewed her in 2024: “For me the love of DAOs, the love of chaos, the love of change, is that they should be mutable, not just to different communities, but also over time. Different times need different forms of energy put into the community. There are moments where you can draw back and begin to tinker with the spreadsheet of community organisation, and then there are moments where you have to be on the street, with a sign of what that community means, expects, and demands.” See Rafferty quoted in Inte Gloerich (ed) *Artists, Activists, and Worldbuilders on Decentralised Autonomous Organisations: Conversations About Funding, Self-Organisation, and Reclaiming the Future*, Amsterdam: Institute of Network Cultures, 2025, 29. A DAO does not end at the spreadsheet, but also comes into being through ongoing, changing, community practices.

131 Rafferty, ‘The Reappropriation of Life and the Living,’ 108.

132 Rafferty, ‘The Reappropriation of Life and the Living,’ 107, 108.

133 Since the Documenta showcase, this context has radically changed. With Israel waging a genocidal war on Palestinians, the future of Dayra is uncertain. However, when I interviewed one of its core members, Yazan Khalili, in 2024, he explained that “Recently, Dayra has been moving slowly. People’s attention is elsewhere, understandably, and we did not reach the implementation level before the war started. This is unfortunate because we need a system like Dayra badly, especially in the context of the genocide! Currently, we are slowly starting things up again and we are rethinking what Dayra could be”. See Khalili quoted in Gloerich, *Artists, Activists, and Worldbuilders on Decentralised Autonomous Organisations*, 85.

134 Lara Khaldi, ‘Dayra and Communal Debt’, *Performing Borders 2 Rallying the Commons* (2022): n.p.

the absence of money'¹³⁵ through 'inverted resources'.¹³⁶ In the introductory video for Dayra, they state:

Not having a measuring tape to measure a table, does not mean the table does not exist. We only don't know its dimensions. Same goes for communal resources. Not having money to exchange them, does not mean that they don't exist, or that they don't have value.¹³⁷

Here, Dayra mirrors the onto-epistemological critiques of the tokenisation of nature presented in chapter 2: the system of measurement (tokens or money) does not constitute the thing that is measured (trees or communal resources), but often is treated as such and as financialised systems, blockchains were shown to reinforce their limited understanding subject to power and extractivism. More concretely, Dayra diverges from mainstream tokenisation projects by

measuring social relations rather than assets. So, if we go back to the metaphor of the table: Dayra is not about measuring the table itself, but about measuring the act of sharing the table. As an object, the table has no value or meaning until it is shared. Once that happens, the table enters into the Dayra system, and it is recognised as part of a communal wealth that takes the shape of practices of sharing.¹³⁸

Rather than using money—always conditioned, always tied up with geopolitical forces, always ready for financialisation—as a measuring tape for the resources of the community, Dayra chooses debt as its guiding logic. The Dayra DAO inverts the lack of money into a resource, circulating it as debt on its blockchain, and as such offers an economic space outside of the purview of dominating powers. Importantly, QoF understands debt not as a *lack* of economic means, or the requirement of future repayment in money, but as a *wealth* of social embeddedness, as an interpersonal indebtedness that casts communities into the future:

Dayra begins from a point of interdependencies. We are indebted to each other, when using Dayra we acknowledge this. We engage in a process of circulating debt. Debt becomes a communal resource; to be indebted to the community means you are part of it.¹³⁹

135 Khaldi, 'Dayra and Communal Debt,' n.p.

136 Yazan Khalili and Lara Khaldi, *Towards Counter-Hegemonic Infrastructures*, 2024. <https://www.youtube.com/watch?v=U0GrY8GJTJw&t=1088s>.

137 Dayra, 'Dayra - How it Works?', *Vimeo video*, 2022. <https://vimeo.com/721710848>.

138 Khalili quoted in Gloerich, *Artists, Activists, and Worldbuilders on Decentralised Autonomous Organisations*, 76-77.

139 Khaldi, 'Dayra and Communal Debt'.

Debt becomes a form of “communal wealth”¹⁴⁰ because “being indebted to your community is how that community is created and sustained. Mutual indebtedness builds communal relations that are dynamic and strong. It creates interdependence rather than independence.”¹⁴¹

In this way, Dayra radically reimagines blockchain’s affordances as a system that facilitates a growing network of entangled social ties, rather than a system that facilitates monetary exchange. Moving away from money as a universal unit of account, it imagines how communities might thrive on their continuous indebtedness to each other. In the absence of money, Dayra nurtures community values as a logic via which to shape societies in their engagement with DAO technology.

The Material-Discursive Character of Prefigurative Epistemologies

The examples above illustrate how prefigurative DAOs, rather than simply being overt manifestos or spaces to put those manifestos into practice, instead continuously merge processes of imagination and realisation. The notion of ‘prefigurative epistemologies’¹⁴² is key to understanding the prefigurative work that they undertake. Prefigurative epistemologies centre situated stories concerned with the creation of collectivities, feature reflexivity with regard to processes of knowledge creation, and facilitate plurality to reflect the diversity of social realities different communities want to create. With Sylvia Wynter’s notion of the plot¹⁴³—which is at the same time a space of lived realities, a story in which new meanings and futures are imagined, and a prefigurative plan towards change—the material-discursive dimension of prefigurative DAO practices comes into focus. In applying the notion of the plot to the artistic-activist DAO space, these practices can be understood as contributing to a move from blockchain as a truth machine to DAOs as ‘micro reality machines’¹⁴⁴ that deliberately make use of the performativity of material-discursive worlds in becoming.

Storytellingly Bringing About Different Worlds

Thinking of prefigurative DAOs through Wynter’s simultaneously literary and historic notion of the plot puts their material-discursive character into focus. The notion of the plot at the same time signifies a historically situated place in which resistance against colonisers could take root, a narrative concept that connects stories with practices and imagination with the emergence of spaces, and a practical scheme to institute change. Wynter explains that during the period of colonisation in the Americas, small areas of land—plots—were provided by (and for the benefit of) plantation owners to enslaved people ‘on which to grow food to feed

140 Khalili quoted in Gloerich, *Artists, Activists, and Worldbuilders on Decentralised Autonomous Organisations*, 77.

141 Khalili quoted in Gloerich, *Artists, Activists, and Worldbuilders on Decentralised Autonomous Organisations*, 77.

142 Motta, ‘Notes Towards Prefigurative Epistemologies,’ 178.

143 Wynter, ‘Novel and History, Plot and Plantation’.

144 Rafferty, ‘The Reappropriation of Life and the Living,’ 112.

themselves in order to maximise profits'.¹⁴⁵ These pieces of land were hard to cultivate and valueless to the plantation, and offered yet another layer of colonial exploitation in which enslaved people were effectively made to maintain their own status as an exploitable resource. However, this is not all the plots represent: enslaved people did not *just* use these plots as ground on which to grow provisions in the larger exploitative logic of the plantation. As they were often situated at the peripheries of plantations, working on the plot was also an opportunity to be out of sight of plantation owners, a rare moment in which to relate to oneself and others differently than was possible on the rest of the plantation.¹⁴⁶ This is why at the same time as being a cost-cutting measure, plots were a source of worry for plantation owners, 'for who knows what the enslaved were 'plotting' there'.¹⁴⁷

As a metaphor, the plantation represents the institutions that order and control society, their dominant knowledge politics, and their monolithic version of history.¹⁴⁸ The 'monological subject of knowing'¹⁴⁹ that emerged during modernity/coloniality instituted a seemingly universal subject that stands in for others, has the power to erase others, and is a-historical and singular. This Western subject is identified as the human that represents all of humanity, while reproducing the epistemic violence of coloniality that bars those who are Othered from speaking for themselves and knowing themselves outside of prescribed Western logics. This knowledge logic characterised the plantation and continues to influence contemporary societies through the ongoing coloniality of knowledge, as discussed in more depth in chapter 2. By universalising a particular, the monological knowing subject obscures historical changes, local contexts, and socio-cultural differences.¹⁵⁰

Against the monolithic and impersonal history as told by the all-knowing subject of modernity/coloniality, Wynter finds folk culture, suppressed memories, and the seeds of emancipatory futurity on the plot. More than just places to grow food for survival, plots offered space for ways of being together that were not possible on the plantation. The plot is where 'the roots of culture' can be found, where 'secretive histor[ies]'¹⁵¹ can be remembered, new futures imagined, and different ways of relating practiced. On plots, the values and traditions of cultures forcibly left behind could be reinvigorated and reimagined. As a metaphor, the plot represents both a place and a time to imagine and practice different ways of being and relating within larger systems of domination. By reflecting on the practices taking place on plantations and plots, Wynter shows that it is possible to create space for different social relations and different ways of knowing within larger contexts of exploitation and extraction. Just as coloniality is ongoing, plotting practices constitute an ongoing potential for the imagination and prefiguration of difference.¹⁵²

145 Wynter, 'Novel and History, Plot and Plantation,' 99.

146 Wynter, 'Novel and History, Plot and Plantation'; de Vries, *Plot Work as an Artistic Praxis in Today's Cityscapes*.

147 de Vries, *Plot Work as an Artistic Praxis in Today's Cityscapes*, 9.

148 Wynter, 'Novel and History, Plot and Plantation'.

149 Motta, 'Decolonising Critique,' 38.

150 Motta, 'Decolonising Critique'; Wynter, 'Unsettling the Coloniality of Being/ Power/ Truth/ Freedom'.

151 Wynter, 'Novel and History, Plot and Plantation,' 100, 101.

152 Wynter, 'Novel and History, Plot and Plantation'.

The literary connotation of the term 'plot' is significant in this context as well. Wynter thinks of humans as 'hybridly nature-culture beings'¹⁵³ that '*storytellingly invent themselves*'.¹⁵⁴ In chapter 2, I referred to Wynter to show that human ways of being and relating are 'language-capacitated',¹⁵⁵ whether they are prefigurative or not. Modernity/coloniality tells a story that hides its own partial identity under the powerful rhetoric and institutions of scientific rationalism. Its monological narrator produces a monolithic history told from a privileged Western perspective and universalises one 'genre of being human'¹⁵⁶ as the universal essence of humanity itself.

However, this universal subject and its monolithic history is but one story among many possible stories. Thinking of humans as 'hybridly nature-culture beings'¹⁵⁷ means that they can also *reinvent* themselves in radical, emancipatory, and decolonised ways by creating different mythologies, stories, and methods of meaning-making. As hybrid beings, humans have a 'uniquely auto-instituting mode of living being, *we humans cannot pre-exist our cosmogonies or origin myths/stories/narratives* anymore than a bee, at the purely biological level of life, can pre-exist its beehive'.¹⁵⁸ Radical imagination can create different stories to tell and retell in order to bring into being truly new ways of being together. Wynter writes: 'Human beings are magical. Bios and Logos. Words made flesh, muscle and bone animated by hope and desire, belief materialized in deeds, deeds which crystallize our actualities'.¹⁵⁹ Hopes, deeds, stories, and embodied experiences all work together in new mythologies that offer an alternative from the monolithic story of colonial capitalism. This is the material-discursive character of life that artistic-activist DAOs appropriate for their prefigurative practices.

The hybrid practices of the plot are prefigurative in many ways: They are based in past and present experiences, but forward-looking, small but radical, necessarily implicated in exchange relations, but choosing to foreground care, community, and interdependence. They question the truths that are handed down from history and imagine what knowing differently could do to support plural future worlds. Dominant history is a story told from positions of power, but different narratives based in secretive histories create the possibility for different futures. As radical imagination, such narratives of futurity do not logically follow out of the

153 Wynter, 'Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,' 295.

154 McKittrick in Sylvia Wynter and Katherine McKittrick, 'Unparalleled Catastrophe for Our Species? Or, to Give Humanness a Different Future: Conversations', in Katherine McKittrick (ed) *Sylvia Wynter: On Being Human as Praxis*, Durham: Duke University Press, 2015, 11, emphasis in original.

155 Wynter, 'Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,' 270.

156 Wynter, 'Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,' 281.

157 Wynter, 'Unsettling the Coloniality of Being/ Power/ Truth/ Freedom,' 295.

158 Sylvia Wynter, 'The Ceremony Found: Towards the Autopoietic Turn/ Overturn, Its Autonomy of Human Agency and Extraterritoriality of (Self-)cognition', in Jason R. Ambrose and Sabine Broeck (eds) *Black Knowledges/black Struggles: Essays in Critical Epistemology*, Liverpool: Liverpool University Press, 2015, 213, emphasis in original.

159 Sylvia Wynter, 'The Pope Must Have Been Drunk, the King of the Castile a Madman: Culture as Actuality, and the Caribbean Rethinking Modernity', in Alvina Ruprecht and Cecilia Taiana (eds) *The Reordering of Culture: Latin America, the Caribbean and Canada in the Hood*, Ottawa: Carleton University Press, 1995, 35.

dynamics that make up the present, but are instituted through the performative material-discursive practices that make up the plot.

To perform such a reinvention, prefigurative DAOs embrace collective, reflexive, and multiple approaches to sources of knowledge and ways of knowing. Rather than monologically speaking for, and representing those that are known through universal categories and logics, they can better be understood through the 'figure of the storyteller',¹⁶⁰ who is rooted in, and in dialogue with communities that collectively produce knowledge. Such a storyteller does not speak *for* the community, but rather consists *of* the community. Instead of attempting to create universal truths, this storyteller frames the notion of truth in relational and situated ways, actively supporting plurality. Importantly, such 'pluriversality is not cultural relativism, but the entanglement of several cosmologies',¹⁶¹ together constituting 'a world of truth in parentheses':¹⁶² an entangled patchwork of qualified and conditional truths that together make up the whole.¹⁶³

Because it does not attempt to create universal truths, such prefigurative storytelling is capable of a kind of reflexive openness about the politics of knowledge, and the genre of knowing and being it reproduces. It establishes and continuously rethinks ways of knowing in order to support community rather than power, and to be attuned to lived experiences rather than abstract logics.¹⁶⁴ When the prefigurative epistemologies of DAO-based plots align with these approaches, 'our epistemological horizons [may be] opened to dialogical grounds of becoming through multiple knowledges, multiple subjects of knowing and multiple practices of creating knowledge'.¹⁶⁵

Rafferty is concerned with the creation of new stories through which to imagine *different* future realities than the one handed down through the dominant version of history. She explains: 'This unincorporated reality sketch given to you by the colonial grandfather will always be a shadow in our imaginations and the only way to extinguish it is through an insurrectionary uprising in one's own mind'.¹⁶⁶ For Rafferty, DAOs are structures that are co-constituted by the discursive practices of collectives, in which communities can experiment with ways of knowing through processes of proposal-making and voting. Notably, these material-discursive processes are iterative, and their logics can mutate to accommodate changing values or new community members.¹⁶⁷ Making use of the interoperable and open-source nature of blockchain, many DAOs can exist next to each other while being connected via underlying infrastructures.

160 Motta, 'Decolonising Critique,' 39.

161 Walter Mignolo, 'Foreword: On Pluriversality and Multipolarity', in Bernd Reiter (ed) *Constructing the Pluriverse: The Geopolitics of Knowledge*, Durham: Duke University Press, 2018, x.

162 Mignolo, 'Foreword,' xii.

163 Mignolo, 'Foreword'.

164 Motta, 'Decolonising Critique'.

165 Motta, 'Decolonising Critique,' 45.

166 Rafferty, 'The Reappropriation of Life and the Living,' 110.

167 Rafferty, 'The Reappropriation of Life and the Living'.

It is important to point out that the (DAO-based) plots are not a place of reminiscence for a lost past or idle hope for a pure way of living that is untouched by the structures of domination and extraction: 'there is no question of going back'.¹⁶⁸ There is no way to make a clear and sudden break with the relations that define life as instituted through modernity/coloniality,¹⁶⁹ or to forget history ever happened and regain precolonial ways of living. Instead, the plot can provide a place to find 'a focus of criticism against the impossible reality in which we are enmeshed'.¹⁷⁰ Everyone is undeniably involved in the extractive and exploitative structures that feature in so many aspects of contemporary life, however, participating in a plot means that there is ambiguity and contrariness in that involvement. This is where resistance, however marginal, finds fertile soil.¹⁷¹ However, 'It is not plot *versus* plantation, but rather plot-and-plantation: they implicate each other. Plot and plantation are at odds with each other but inhabit the same locus'.¹⁷² Just like the historical plot is an imperfect place that remains implicated in the larger extractive structures of the plantation, creating prefigurative DAOs is an ongoing struggle for space in which difference from the financialisation that blockchain often implies can emerge. Nevertheless, new mythologies—new stories about worlds—create the possibility to prefigure radical, emancipatory, and decolonised futures with and through DAOs.

DAOs as Material-Discursive Micro Reality Machines

By way of Donna Haraway—whose work resonates in interesting ways with Wynter's—Catlow stresses the importance of recognising more-than-human entanglement in engagements with DAOs,¹⁷³ creating spaces in which it can be recognised that '[n]atures, cultures, subjects, and objects do not preexist their intertwined worldings'.¹⁷⁴ Worlds and myths, bees and beehives, humans and ways of knowing humanity: all are in a process of entangled becoming. Storytelling holds prefigurative potential in this intertwined becoming: stories 'propose and enact patterns for participants to inhabit, somehow, on a vulnerable and wounded earth'.¹⁷⁵ 'It matters what stories make worlds, what worlds make stories'¹⁷⁶ because prefigurative worlding 'affords the opportunity for the cessation of habitual temporalities and modes of being'.¹⁷⁷ In other words, to change reality, stories about life need to change. Radical, prefigurative imagination can institute new communities around new conceptions of life, and create new narrative self-conceptions for future generations to inhabit. To prefigure new worlds, new mythologies thus need to be told: new myths about life and sociality that allow for the practicing of different ways of being, living together, and relating to nature and more-than-human others. In these mythologies, communities might begin to establish different value

168 Wynter, 'Novel and History, Plot and Plantation,' 100.

169 In fact, creating such "aggressive cuts with the past" were the *modus operandi* of modernity/coloniality, and should not be repeated. See Khalili quoted in Gloerich, *Artists, Activists, and Worldbuilders on Decentralised Autonomous Organisations*, 84.

170 Wynter, 'Novel and History, Plot and Plantation,' 100.

171 Wynter, 'Novel and History, Plot and Plantation'.

172 de Vries, *Plot Work as an Artistic Praxis in Today's Cityscapes*, 9.

173 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain'.

174 Haraway, *Staying With the Trouble*, 12.

175 Haraway, *Staying With the Trouble*, 10.

176 Haraway, *Staying With the Trouble*, 12.

177 Helen Palmer, and Vicky Hunter, 'Worlding', *New Materialism Almanac* (2018), n.p.

systems in close relationship with more-than-human others and different forms of relating within ecological networks. Such mythologies stand at the base of new worlds.

The creative capacity of the radical imagination that characterises the plot inspires researcher Patricia de Vries to think of ‘plot-work as a transformative artistic practice’.¹⁷⁸ Plot-work, she writes,

interlaces ‘the material and the metaphoric,’ past, present and future, earth and earthlings. It is a conceptual tool and historic reality. It is figurative language and a challenge to current spatial arrangements. It is a verb and a narrative device. It shows us how art is ensnared in market logic and reminds us of other possibilities. The plot signifies practices in which other values are acted upon, constituted, grounded in space, and where dominant social orders and values are challenged and evaded.¹⁷⁹

De Vries thinks of artistic plot-work as ‘the ways by which the idea of the plot is socially enacted, embodied, narrativised, materialised, and spatialised in art practices’.¹⁸⁰ She asks, how does this ‘plot-work call [...] for an imaginative mind willing to *find*—or rather, to *create*—something different, something other in the endless repetition of the same, to *make* stories that make different places, to *seek them* out’?¹⁸¹ How can artistic work, implicated as it is in institutional and capitalist dynamics, perform plot-work to create space for relating ‘within and yet against the grain of the limits’ of these dynamics?¹⁸²

In Rafferty’s work, plotting means crafting new mythologies for an ungovernable way of life—origin stories that provide a new sense of futurity out of the endless possibilities of the void of chaos. It means taking radically different roads to creating knowledge beyond that which the origin story of modernity/coloniality provides. Rafferty uses DAOs as tools for the radical reimagination of life and its possibilities, as prefigurative spaces where different social relations and new conceptions of life emerging from ‘generative-chaos’ can be practiced.

The material-discursive interplay in plot-work means that such DAOs have the potential to be stories that are continuously becoming material worlds, always open to mutation and retellings, and always part of plural world-becomings. As such, DAOs become ‘micro reality machines’¹⁸³ to be used in tests, iterations, and experiments for new ways of living, knowing, and relating. Like sigils, DAOs set intentions based on shared new mythologies about life and living, and through the rituals of proposals and votes new worlds actualise out of those mythologies. Engaging in prefigurative ways with these micro reality machines generates plots to practice difference on in the here and now despite ongoing exploitations and extractivism.

178 de Vries, *Plot Work as an Artistic Praxis in Today’s Cityscapes*, 10.

179 de Vries, *Plot Work as an Artistic Praxis in Today’s Cityscapes*, 12, emphasis in original.

180 de Vries, *Plot Work as an Artistic Praxis in Today’s Cityscapes*, 11.

181 de Vries, *Plot Work as an Artistic Praxis in Today’s Cityscapes*, 14, emphasis in original.

182 Marsha Meskimmon, ‘From the Cosmos to the Polis: On Denizens, Art and Postmigration Worldmaking’, *Journal of Aesthetics & Culture* 9.2 (2017), 25.

183 Rafferty, ‘The Reappropriation of Life and the Living,’ 112.

The case of Dayra shows how this might take shape in practice. Dayra works by telling a different story about how the world works, how societies are organised, and what value is—i.e., debt does not signify a lack, but a measure of one's entanglement with their community. It attempts to establish a prefigurative epistemology that centres on ways to know social entanglement rather than service systems of control or extraction. Dayra participates in plot-work in multiple dimensions: as a space practicing different forms of engagement with others, different interactions, and exchanges; as a new mythology about life and the meaning of living together with others; and as a plan that operates beyond the view of the powers that be—a plan that offers a horizon for a new sense of futurity, one that offers hope rather than desperation.

Conclusion: Onto-Epistemologically Prefiguring New Worlds

My aim with this chapter was to propose conceptual tools to develop nuanced understandings of how deliberate engagement with imaginaries may take shape, particularly concerning practices in which art and activism are combined. First, to be able to recognise the profound imaginative work that is necessary to break with extractivist, individualist, and universalist tendencies that characterise much of the application of blockchain technology, I developed the notion of radically reimagined affordances. It builds on both the imaginative element that is part of affordances and the radical, creative capacities of the imagination, and occupies a space in which the material and the discursive are inextricably connected. Prefigurative engagement with blockchain technology that features radically reimagined affordances can contribute to emancipatory and decolonial imaginaries around the technology that performatively bring about change.

Certain DAOs represent prefigurative attempts to radically reimagine blockchain from its financial affordances toward the affordance of new kinds of sociality. These include the support of translocal communities, the reinforcement of interdependency, and the exploration of self-organised forms of social coordination. The (proposed) DAOs of the Black Socialists in America, ReUnion Network, and Black Swan all respond to elements of the prefigurative imaginary of belonging developed by Catlow¹⁸⁴ and Zeilinger.¹⁸⁵

Notably, both the logic of the plantation and that of the plot are possible within DAO-based structures of belonging. Belonging indicates both the capitalist logic of ownership and a socio-cultural sense of kinship or collectivity.¹⁸⁶ However, these examples show that 'if we begin to put cultures before structures'¹⁸⁷—i.e., if we radically reimagine blockchain's affordances *only* as a part of a commitment to community—then 'radically new worlds can appear'.¹⁸⁸ Significantly, the Black Swan DAO uses an approach they call 'blockchain thinking without the technology'¹⁸⁹ that takes this emphasis on communities and imagination over technological development to the extreme. This approach raises interesting questions about the degree

184 Catlow, 'Translocal Belonging and Cultural Cooperation After the Blockchain'.

185 Zeilinger, *Structures of Belonging*.

186 Zeilinger, *Structures of Belonging*.

187 Rafferty, 'The Reappropriation of Life and the Living,' 110.

188 Rafferty, 'The Reappropriation of Life and the Living,' 110.

189 Lotti, 'Simulating Equitable Art Worlds'.

to which artistic-activist engagements with blockchain technology require functional pieces of software and how other imaginaries of blockchain and other complex technologies might open up to radically new futures, if only they were not bogged down pre-emptively by the need for stable working systems.

DAO-based practices have the potential to constitute prefigurative epistemologies able to break open the limited views of rationalist imaginaries of blockchain as a truth machine in radical, emancipatory, and decolonised ways. For example, Rafferty develops an alternative understanding of what it means to know, addressing the root of the modernity/coloniality system of knowledge and emerging from what she calls 'generative-chaos'.¹⁹⁰ New worlds based on radically different mythologies that emerge from this chaos may be instituted through DAO practices. These practices can be recognised in the rituals through which DAO communities collectively discuss and decide their futures and develop radically different stories about what it means to live together with others.

It should be noted that prefiguration is not a process with a clearly defined end goal or measurable results. It is slow, iterative, and never truly finished.¹⁹¹ In the context of Castoriadis' radical imagination, it becomes clear that prefiguration is just as much about *practicing* difference as it is about *imagining* it. That is, prefiguration is constituted by both a practicing of different socio-political realities in the here and now, *and* the reimagining knowledge processes as collective, reflexive, and pluriversal. The 'onto-epistemic character of prefiguration'¹⁹² gives it the power to create alternatives to the imaginary of blockchain as a truth machine; to practice new ways of knowing and new ways of being together with others that iteratively establish themselves. This is the potential I have argued for through the use of radically reimagined affordances on prefigurative plot-work.

Thinking of artistic-activist DAOs that engage in prefigurative plot-work, I am not trying to determine if the DAOs I discuss are perfect examples, as no such perfection exists in the continued context of colonial-capitalist domination. Rather, my intention is to highlight the ways in which they break open established narratives about possible futures and create the potential for different horizons to emerge: to understand the radical, material-discursive, prefigurative, epistemological practices that contribute to this openness. Prefigurative DAOs are made up of stories that are continuously becoming material worlds. They are material constellations that are continuously narrativised and ritualised, always open to mutation and part of plural world-becomings.

Discourse and materiality are intrinsically entangled. Stories about life and the ongoing natural, biological, material becoming of life are inseparable. Stories have material significance.¹⁹³

190 Rafferty, 'The Reappropriation of Life and the Living,' 106.

191 Marianne Maeckelbergh, 'The Prefigurative Turn: The Time and Place of Social Movement Practice', in Ana Cecilia Dinerstein (ed) *Social Sciences for an Other Politics: Women Theorizing Without Parachutes*, Cham: Springer International Publishing, 2016, 121-34.

192 Arturo Escobar, 'Foreword', in Lara Monticelli (ed) *The Future is Now: An Introduction to Prefigurative Politics*, Bristol: Bristol University Press, 2022, xxv.

193 Motta, 'Notes Towards Prefigurative Epistemologies'; Motta, 'Decolonising Critique'; Wynter, 'Novel and

Artistic-activist DAOs engage in prefigurative plot-work in which the entanglement of the material and the discursive is actively put to use and pursued. Rafferty's work contributes to a prefigurative imaginary of blockchain as a reality machine for mutable and plural worlds. Like Rotman's religious technologies that contribute to shifting paradigms discussed in chapter 3,¹⁹⁴ the ritual of inscribing and re-inscribing a DAO collectively becomes 'a sorcerous operation' that is continuously involved in 'world-making and worldbuilding'¹⁹⁵ along what could be radical, emancipatory, and decolonised lines.

History, Plot and Plantation'.

194 Brian Rotman, *Becoming Beside Ourselves: The Alphabet, Ghosts, and Distributed Human Being*, Durham: Duke University Press, 2008.

195 Rafferty quoted in WAC Lab, 'Radical Friends: How DAOs Could Change the Art World', *WAC Weekly*, 2023, <https://medium.com/@wac-lab/radical-friends-how-DAOs-could-change-the-art-world-d4a8777d06d8>, n.p.

CONCLUSION: IMAGINING LIFE WITH BLOCKCHAIN

I began this book by introducing the imaginary of blockchain as a truth machine and by discussing the widespread hope within technology culture that blockchain would change the world for the better through its (supposedly) new knowledge logics. I observed that blockchain culture is characterised both by rationalism *and* the more-than-rational, and set out to understand not only how the notion of ‘truth’ takes shape in between these seemingly opposing forces, but also the socio-cultural relevance of these conceptions for how the world and entangled life within it may be known.

In this concluding chapter, I weave together the threads of my overarching arguments. First, throughout my research, I expanded the notion of imaginaries. I argued for moving from sociotechnical to socio-techno-cultural imaginaries to include a broader range of performative processes. I emphasised the need to understand and expose the situatedness of imaginaries within specific social, cultural, political, scientific, and ideological contexts that are crucial for understanding their embeddedness in power relations. I also argued for imaginaries to be considered as tools for artistic-activist engagement with technology, with which to forge new realities.

Second, through a critical reading of rationalism’s history, this study reveals two insights: the entanglement of rational and more-than-rational elements, and the importance of engaging with the politics these elements and their relationship. Despite the ongoing dominance of rationalist rhetoric in many aspects of contemporary Western societies, it is crucial to critically engage with moments in which these logics are problematised. My analyses exposed how different conceptions of truth shape different ways to understand the world through blockchain, based on its presumed ability to function as a ‘truth machine.’

Finally, I situated the contributions of this book in the context of the broader socio-cultural phenomenon of technological development and the progress rhetoric of ‘technology for good.’ Because of the temporal focus on what is new *now* in technology culture, it is important that research reveals continuities with historically entrenched logics, attitudes, and dynamics to make sense of the way in which power plays a role in claims of revolution. Additionally, for the topic of this book, the emergence of what has come to be known as the post-truth era is significant in understanding the hype around blockchain’s presumed truth affordances. In many ways, the imaginary of blockchain as a truth machine can be understood as a response to these developments. Similarly, to develop prefigurative truth imaginaries, it was and is necessary to address the assumptions and power relations that have led to this post-truth predicament.

Imaginaries as a Concept and a Tool

The Performativity of Culture

Imagining life with and around emerging technologies is an inherently material-discursive exercise. Affordances and performativity are methodological entry points to the study of imaginaries as material-discursive phenomena. The ways in which complex technologies develop depend on what users and designers imagine, hope for, or expect such technologies to afford. The way such imaginations, hopes, and expectations are performed—for example, in the shared myths that emerge around a given technology, but also of-the-cuff social media posts or meticulously worded business pitches and whitepapers—impact the ongoing materialisation of technologies. This book has been an interdisciplinary endeavour committed to bringing multiple sides of this complex process into view. The intersection of media studies and cultural analysis has proven to be a fruitful perspective from which to take on this research that is concerned with the phenomenon of blockchain technology as an evolving assemblage.

My research has indicated the importance of including culture in analyses of imaginaries. Expanding Jasanoff's concept of sociotechnical imaginaries¹ into socio-techno-cultural imaginaries brings into view a more comprehensive picture of how imaginaries are embedded into different kinds of engagements with technology. While sociotechnical imaginaries are effective in bringing to light dominant social visions of the future with and around technologies, with the inclusion of culture, different kinds of visions, with more diverse kinds of relationships to power, come into view.

My argument for this emphasis emerges from the widespread use of digital technologies such as blockchain and the resulting multifaceted processes of performativity. Digital technologies such as blockchain are used by numerous and diverse communities and, as a result, many imaginaries emerge across various cultural contexts, in contrast to the more centralised imaginaries that develop around technologies such as space travel or quantum computing.² Expanding sociotechnical imaginaries into socio-techno-cultural imaginaries not only broadens the range of cases that come into focus, but also acknowledges the performativity of everyday engagements with technology. By focussing on the way in which blockchain is situated within these different subcultural contexts, the evolving plurality of hopes, dreams, fears, and expectations that shape their diverse imaginaries can be appreciated. This approach recognises not only the performativity of 'languages of power',³ but also that of everyday practices, subcultural ideas, shared values, and cultural debates.

The case of cryptocurrency memes, analysed in chapter 4, is a concrete example of the effects of an emphasis on the socio-techno-cultural in research on imaginaries. It exemplifies

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- 1 Sheila Jasanoff, 'Future Imperfect: Science, Technology, and the Imaginations of Modernity', in Sang-Hyun Kim and Sheila Jasanoff (eds) *Dreamscapes of Modernity: Sociotechnical Imaginaries and the Fabrication of Power*, Chicago: University of Chicago Press, 2015, 1-33.
 - 2 Astrid Mager and Christian Katzenbach, 'Future Imaginaries in the Making and Governing of Digital Technology: Multiple, Contested, Commodified', *New Media & Society* 23.2 (2021): 223-36.
 - 3 Jasanoff, 'Future Imperfect,' 25.

the performative nature of mundane cultural objects, especially when they incorporate the distributed logics of memetics. Memes supercharge the imaginary of crypto traders because they put performativity in overdrive through a mix of memetic affordances, financial incentives, and a culture of individual responsibility amid profound uncertainty.

Situating Imaginaries

The second and related emphasis for which I argued concerns the situatedness of imaginaries. However, rather than a departure from earlier work on imaginaries—Jasanoff writes that sociotechnical imaginaries are ‘temporally situated and culturally particular’⁴—my contribution offers an expansion of the logic at work within them. A key interest for Jasanoff is the comparison between imaginaries from particular countries.⁵ As such, these imaginaries are understood as situated in particular socio-political contexts. With the introduction of culture in the analysis, the understanding of the situatedness of imaginaries is further nuanced and expanded. Rather than simply comparing imaginaries, it is important to recognise the ways in which particular imaginaries function to hide their cultural situatedness in favour of the appearance of universal relevance. In this book, in such instances rationalism features not only in the reliance on and reverence for logical processes as a way to establish universal truths, but also in the seeming universality of these imaginaries themselves. That is, who contributes to the production of such imaginaries, whose interests they serve, and the frictions involved in their real-world applications are obfuscated. Blockchain forest protection projects (chapter 2) present their solutions as neutral yet effective intermediaries intended to make the world a better place, but fail to acknowledge the inherent inequality of the outcome for those involved. Similarly, in religious narratives about blockchain’s superior truth capacities (chapter 3), thinking of the technology as ‘divinely designed’ imbues it with the connotation that it is simultaneously positioned outside our earthly realm while also permeating it, able to express all that is of value. This idealised, ‘infinite vision is an illusion, a god trick’.⁶ It is a rationalist, universalising story that hides its partiality.⁷

In the preceding chapters, the imaginaries discussed gradually became more specific, more overtly situated, and representative of particular people at particular times in particular places. While Maren Altman’s view of astrology as a cure to the uncertainties of the crypto market (chapter 4) still featured a view from nowhere and her solution appeared universally applicable, cryptocurrency memes (also chapter 4) complicate such a view. The imaginary featured in these memes emerges from a lived struggle with profound uncertainty in which the logics of control and universalisation appear ill-fitting and ineffective. Instead of guaranteed solutions, the memetic imaginary of blockchain as an uncertainty machine envisions an uneasy future of speculative survival in the perpetual ‘now’ of the crypto market. The prefigurative

4 Jasanoff, ‘Future Imperfect,’ 19.

5 Jasanoff, ‘Future Imperfect’.

6 Donna Haraway, ‘Situated Knowledges: The Science Question in Feminism and the Privilege of Partial Perspective’, *Feminist Studies* 14.3 (1988): <https://doi.org/10.2307/3178066>, 582.

7 Sylvia Wynter, ‘Unsettling the Coloniality of Being/ Power/ Truth/ Freedom: Towards the Human, After Man, Its Overrepresentation—an Argument’, *The New Centennial Review* 3.3 (2003): 257–337. <https://doi.org/10.2307/41949874>.

Decentralised Autonomous Organisations (DAOs) of chapter 5 completed the move to imaginaries that are explicitly from a specific place by putting ‘cultures before structures’⁸—i.e., making imagined blockchain futures dependent on what specific situated communities want or need. The prefigurative practices that shape DAOs do so in situated relationships to the dominant powers around them, and in specific and shared lived experiences.

With regard to my own focus on the rational and the more-than-rational in blockchain culture, such a situated socio-techno-cultural approach to imaginaries has proved indispensable because it connects the normativity of consciously performed social visions with the revealing discontinuities, discrepancies, and frictions that appear in everyday expressions. In societies in which the rhetoric of rationalism is strong, it can be expected that the dominant and commercial narratives of a technology’s social impact align with its values. Think about, for instance, the rhetoric of the blockchain-for-good sector that the cases in chapter 2 exemplified. Socio-techno-cultural imaginaries have the capacity to reveal the implications of this alignment from different perspectives.

Blockchain culture offers an interesting case here, because its internal ‘languages of power’⁹ range from the rationalist visions of technology companies in chapter 2, to the more-than-rational narratives in chapter 3. Yet, these more-than-rational aspects can appear irrelevant when one is purely seeking to understand how societies are imagined to take shape in the context of new technologies. Rather, they become revealing of the norms and values, the conceptions of life and relationality, social responsibility and roads to change when looking through the lens of socio-techno-cultural imaginaries.

Artistic-Activist Imagination with Technology

Based on my engagement with prefigurative DAO practices (chapter 5), I have argued that imaginaries not only offer a conceptual lens, but can also function as a practical tool. This notion of imaginaries is specific to the realm of artistic-activist engagements with technologies. Certainly, imaginaries that take shape in the context of wider technology culture are always instrumental to something as well: they are always political and they always *do* something. Yet, to think of them as tools that are designed for particular purposes attaches too much intentionality to most of them. In contrast, artistic-activist engagements with technology are intentionally political in their use of the radical imagination around technology because they aim to perform radically different worlds.

As with the use of imaginaries as a concept, affordances and performativity are key here too: affordances can be reimagined to bring into view different socio-techno-cultural futures that are performed through ongoing artistic-activist engagements. In chapter 5, I developed the notion of radically reimagined affordances to understand the ways in which artistic-activist

8 Penny Rafferty, ‘The Reappropriation of Life and the Living—a Cosmic Battleground’, in Ruth Catlow and Penny Rafferty (eds) *Radical Friends: Decentralised Autonomous Organisations and the Arts*, Liverpool: Torque Editions, 2022, 110.

9 Jasanoff, ‘Future Imperfect,’ 25.

DAO practices break open blockchain's perceived future potentialities. This notion combined Castoriadis' understanding of imagination as capable of being radical and creative (i.e., not necessarily logically connected to the status quo and capable of establishing fundamentally new realities)¹⁰ with Nagy and Neff's understanding of imagined affordances (i.e., the idea that the potential for action that complex technologies offer can be based on what users and designers expect and imagine the technology to be capable of).¹¹ By reimagining blockchain's affordances for social belonging instead of financial exchange, and by doing this in radical, creative, and consequential ways, prefigurative imaginaries emerge that can bring about new worlds.

Second, also in chapter 5, the notion of artistic plot-work¹² illuminated the power of storytelling in artistic-activist DAOs. Thinking of prefigurative DAOs as plots—which are simultaneously places, stories, and plans—brought into sharp focus both the material significance of the stories that are used to make sense of technologies, and the activist potential of performing different stories. I have shown that by telling and retelling different stories, by establishing new mythologies and developing new meaningful rituals within communities, prefigurative DAOs can iteratively institute radical, plural, emancipatory, and decolonial worlds within worlds.

The title of this conclusion can be interpreted in two ways and suggests one more dimension of the relationship between technology and imagination. First, 'imagining *life with blockchain*' refers to the different blockchain imaginaries that featured in this book, from rationalist truth machines to speculative uncertainty machines and prefigurative reality machines, and the way in which conceptions of sociocultural life change according to how technology is understood. In chapter 2, we saw a conception of life that reproduced age-old global hierarchies through the logics of tokenisation; in chapter 3, humanity was framed as incapable of influencing or fully understanding the perfect order that blockchain supposedly made available; in chapter 4, a starkly individualist notion of being emerged from the influence of the profound uncertainty of the cryptocurrency market; while in chapter 5, collective life gained a sense of mutability through the processes of collaboration in DAOs.

In the second meaning of the title, 'imagining *life with blockchain*,' blockchain might not be involved in the futures of life that are imagined at all. Instead, the technology primarily takes up a role in the process of imagining itself. This idea refers specifically to the work of the members of Black Swan DAO, who use 'blockchain and DAOs as tools for the imagination'¹³—that is, they use 'blockchain thinking without the technology'.¹⁴ Reflecting on these approaches, Rafferty explains that they were born out of circumstance—Black Swan emerged out of a

10 Cornelius Castoriadis, *The Imaginary Institution of Society*, Cambridge, MA: MIT Press, 1998.

11 Peter Nagy and Gina Neff, 'Imagined Affordance: Reconstructing a Keyword for Communication Theory', *Social Media + Society* 1.2 (2015): 1-9.

12 Sylvia Wynter, 'Novel and History, Plot and Plantation', *Savacou* 5.1 (1971): 95-102; Patricia de Vries, *Plot Work as an Artistic Praxis in Today's Cityscapes: An Introduction to the Lectorate Art & Spatial Praxis / the City*, Amsterdam: Gerrit Rietveld Academy, 2022.

13 Laura Lotti et al., 'Simulating Equitable Art Worlds: DAOs Through a Role-Played Lens', *KW-Berlin*, n.d., <https://www.kw-berlin.de/en/black-swan-dao/>.

14 Lotti, 'Simulating Equitable Art Worlds'.

speculative essay commissioned by Kunst-Werke Institute for Contemporary Art in Berlin—but also align with the conviction that engaging with and understanding the social, political, and cultural dimensions of complex emerging technologies should not require technical know-how, but rather should be as intuitive as child's play.¹⁵

Rather than using imaginaries as a concept to understand the significance of emerging technology cultures or using imaginaries as a tool to influence their direction, what I aim to convey with this second interpretation of the title of this chapter is the way in which blockchain itself—or, for that matter, any other emerging and complex technologies with a similar potential for sociocultural significance—may be used to reimagine what *life* itself constitutes, what it means, how it is entangled, and what practices it consists of. The algorithmic processes, ritualised practices, and imagined affordances of an emerging technology may trigger unexpected reflections on how different things could be, especially when they are the subject of ongoing community debates situated in real-life struggles.

Technology as a tool for the imagination might be understood as a kind of 'defamiliarisation'¹⁶ in which it is neither art nor design that brings to light critical aspects of taken-for-granted technological systems—as the term is often used—but rather the fictionalised, detechnologised logics of emerging technology that bring into view otherwise obscured or forgotten potentialities of life. Different technologies might spur the imagination differently and may open up visions for entangled life in becoming with others that would not have appeared otherwise. Using technology as a tool for the imagination repurposes the excitement for new and different ways of seeing things that emerging technologies can bring about—the excitement that technology culture often harnesses for the purpose of extraction and control—for profoundly changed conceptions of life. Yet, while technologies can inspire these changed conceptions, this does not mean that their implementation is required or even useful in the pursuit of those new imaginaries of life.

Multiple, Contesting, and Commodified Truth Imaginaries

Throughout this book, I have distinguished between the multiple ways in which truth is understood, and the—sometimes mutually contesting and sometimes commodified¹⁷—truth imaginaries that exist in blockchain culture. Below, I first relay how the rational and the more-than-rational are intertwined in dominant understandings of blockchain's truth affordances. While these different emphases may seem oppositional, they ultimately reinforce a rationalist

15 Inte Gloerich (ed) *Artists, Activists, and Worldbuilders on Decentralised Autonomous Organisations: Conversations About Funding, Self-Organisation, and Reclaiming the Future*, Amsterdam: Institute of Network Cultures, 2025.

16 c.f. Genevieve Bell, Mark Blythe, and Phoebe Sengers, 'Making By Making Strange: Defamiliarization and the Design of Domestic Technologies', *ACM Transactions on Computer-Human Interaction (TOCHI)* 12.2 (2005): 149-73; Luke Stark and Kate Crawford, 'The Work of Art in the Age of Artificial Intelligence: What Artists Can Teach Us About the Ethics of Data Practice', *Surveillance & Society* 17.3/4 (2019): 442-55; Elise Morrison, *Discipline and Desire: Surveillance Technologies in Performance*, Ann Arbor: University of Michigan Press, 2016.

17 Mager and Katzenbach, 'Future Imaginaries in the Making and Governing of Digital Technology'.

truth imaginary. In the second part of this section, I zoom in on parts of blockchain culture that problematise rationalism, either because the volatile nature of the cryptocurrency market eclipses any attempt at rational calculation, or because of activist commitments to plurality.

Blockchain Imaginaries Between the Rational and the More-than-Rational

This book proposes to regard rational and more-than-rational approaches to knowing not as irreconcilable, oppositional forces, but rather always in each other's presence, in different proportions or emphases, always also in particular relationships with power. The way in which the rational or the more-than-rational are present in different corners of blockchain culture and with which emphasis may not always be intentional, yet always has political repercussions. This book has not presented an argument for or against the rational, or indeed for or against the more-than-rational. Instead, two key arguments with regard to these elements have taken shape. First, this book constitutes an argument against the depoliticisation of either the rational or the more-than-rational. Second, it constitutes an argument for a historically, materially, culturally, and socially aware engagement with technology that frames its relation to the rational *and* the more-than-rational with particular, situated, and explicit goals in mind.

Mainstream, culturally powerful understandings of blockchain's truth affordances are overwhelmingly rationalist, but gain strength also because of their reliance on more-than-rational discursive framings of, and practices around, the technology. Even though these rationalist and more-than-rational narratives around blockchain technology appear to be incompatible, on a deeper level they fundamentally align. On the surface, blockchain forest protection projects that tokenise nature (chapter 2) and narratives that make sense of blockchain in religious and mystical ways (chapter 3) seem to represent irreconcilable understandings of why blockchain-based knowledge is so superior to other kinds of knowledge. The first emphasises verifiability, validity, quantification, and calculation. These projects imagine blockchain's truths as statements derived from the supposedly purely rational processes of tokenisation and financialisation, datafication and abstraction. By capturing nature on the blockchain, they argue that its protection would become rationally unavoidable simply because it makes calculable, verifiable, financial sense. The latter emphasises predestination, divine design, and inscrutability. By framing blockchain technology in religious and mystical narratives, it gains a teleology, a sense of destiny and unavoidability. Here, blockchain-based ways of knowing approach the distant and unbiased perspective of a god that referees the morality of worldly affairs. Framed this way, blockchain-based knowledge should be believed rather than verified, and revered rather than interrogated.

Yet, underneath these differences in appearance, chapter 2 and 3 fundamentally align: they both feature rationalist understandings of truth that are made possible by more-than-rational beliefs—from modernist myths to technological mysticism. The systems proposed by blockchain forest protection projects (chapter 2) are convincing because they create a myth that equates tokenised nature with nature itself. The logics of their processes of tokenisation determine which trees count, how much they count, and how they might be owned, traded, and financialised, while at the same time obscuring and devaluing the fact that these atomised bits of tokenised nature can only exist because of the irreducible ecologies

of which they are part. When belief in this myth dissolves, the idea of blockchain as a tool to protect the ecological value of these forests also starts to dissolve. Conversely, the more-than-rational discourse discussed in chapter 3 ultimately serves to institute a universal and logical blockchain-based epistemology. In these narratives, blockchain signifies divine design or some other mystical origin *because* it is understood to be so purely rational that it surpasses human capacities. In these narratives, the logics of blockchain-based knowledge production is understood as a road towards a god's ultimately rational and logical way of understanding the world—of the logic that makes up everything around and between us.

Ultimately, whether what is at stake is an overtly religiously framed rationalism or a rationalism that hides the more-than-rational myths by which it is supported, the imaginary illustrated by these two chapters builds on the history of rationalism with all its contradictions and tensions. Like for Descartes, Leibniz, and Boole and their religiously motivated rationalism, in this blockchain imaginary knowledge processes are designed to be as purely logical and calculable as possible in an attempt to transcend human capacities and overcome human fallibility. As seen in modernity/coloniality more widely, blockchain-based rationalism serves those in power by committing to the idea that a perfect and universally valid kind of rationality is possible and to material-discursive practices that obscure their own inner workings. Indeed, blockchain does not appear 'out of nowhere,' and framing it as a sterile, perfect 'truth machine' does not do justice to its social, cultural, and historical situatedness.

Furthermore, the medium-specificities of blockchain mean that tokenised knowledge is tied up with immediate financial valuation. The blockchain forest protection projects discussed in chapter 2 demonstrate that rationalist blockchain-based knowledge enhances power because it can be used to control people and phenomena on the one hand, and because it produces new forms of value that can be extracted by those with sufficient financial means on the other.

When Rationalism is Problematised in Blockchain Imaginaries

The dominant blockchain imaginary 'from nowhere' discussed above does not engage with, or indeed does not do justice to, the profound uncertainty that permeates economic, political, and environmental aspects of contemporary life in globalised and neoliberal societies. Uncertainty signifies the severe inability to anticipate future events and problematises the very foundational convictions that characterise rationalism with regard to the capacity of logical processes to 'account for every part of the material [...] social and mental worlds'.¹⁸ The cases in chapters 4 and 5 showed that rationalism fails to constitute a way out of the current precarious predicaments because it strives to control that which cannot be controlled.

In chapter 4, the pseudorationalism of crypto astrology emerges as a hybrid between rationalist control and more-than-rational divination. Crypto astrology appears as an attempt to rescue crypto traders from the unpredictable volatility of the market by rationalising financial decisions through the logical calculations that feature in astrology. Nevertheless, both

18 David Golumbia, *The Cultural Logic of Computation*, Cambridge, MA: Harvard University Press, 2009, 1.

astrology and the market are ultimately rendered through more-than-rational processes of interpretation, speculation, anticipation, and divination.

In memes about the crypto market (also chapter 4), speculation is instead presented as a way out of the detrimental effects of uncertainty, not by making things more certain or rationally calculable, but by using the more-than-rational processes of speculation to learn to thrive within it. The notion of truth that permeates the crypto market is one in which truth claims are judged based on their capacity to perform financially. The crypto market echoes the profound uncertainty of contemporary life, offering a place in which the more-than-rational practice of speculation becomes a survival mechanism. Repetitive speculation keeps the focus of information continuously in that moment immediately beyond the present. Never settling into a final state, memetic crypto culture's truth claims are 'unstable rather than fixed, simulated rather than real. They do not occupy a single, 'proper' place in knowledge',¹⁹ but shift continuously in response to the uncertainty of the market *and* in order to produce even more volatile waves. Here, performativity becomes the 'order-word' that structures social relations and personal interactions:²⁰ crypto traders inundate the culture with memes, X posts, reddit threads, videos, discord messages, blog posts, etc., because to survive the speculative present, they must 'Perform—or else!'²¹

In the realm of an artistic-activist engagement with DAOs (chapter 5), we saw that, for some, reckoning with uncertainty means learning to appreciate chaos and to act against the controlling and universalising tendencies of rationalism, instead exploring and appreciating different ways to know together with others, always in reference to specific contexts and always open to mutation. I have shown that the way in which artistic-activist engagements with DAOs actively address the role of affordances and performativity has the capacity to break open the imaginary of blockchain as a truth machine. Such DAOs may facilitate plural worlding practices based on diverse practices of knowing and relational, situated understandings of what can or should be known. At the same time, they may still implement logical, calculable decision-making processes. They may still tokenise aspects of the world around them, but to prefigure different worlds, these processes are always supplemented by active community discussions, embedded in their lived realities, and made to function within their specific characteristics without striving for universal coverage—in striking contrast with the immutable and universal tokenisation of nature that blockchain forest protection projects (chapter 2) instituted top-down. Instead, prefigurative DAOs do not negate rational approaches to knowing the world *per se*, but rather negate the power relations of rationalism by opening up the imaginary of blockchain as a truth machine to different ways of understanding what knowing means, the processes by which it is constituted, and the applicability it is supposed to have.

By framing blockchain in terms of social belonging rather than casting it in a financial light, and by rendering knowledge processes through the perspective of a prefigurative and collective storyteller, DAOs can stay committed to telling the story of their own community in a way that

19 Jon McKenzie, *Perform or Else: From Discipline to Performance*, London: Routledge, 2002, 18.

20 McKenzie, *Perform or Else*.

21 McKenzie, *Perform or Else*, 19.

makes sense to that community. Through ritualised practices such as community-based proposal-making and voting, communities continuously imagine and institute themselves, and may then reimagine and reinstitute themselves anew. Such a blockchain imaginary—one of mutable and plural worlds—is significant, not because it proposes a different, clear definition of what truth is, but because it renounces such permanent definitions. Instead, it breaks open the imaginary of blockchain as a permanent, immutable, universal truth machine to yet unknown understandings of what it means to know the world, the processes knowing consists of, and the implications of these ways of knowing for how living life together with others might be organised.

Future Research on Truth Imaginaries

Looking ahead, I outline three directions for future research. To start with, the existence of different notions of truth and their relationship with rationalism and the more-than-rational continues to be a relevant topic for research into blockchain culture as well as wider technology culture. Further work could expand on the potential of prefigurative truth imaginaries with blockchain. Second, it is also important to expand research to include a wider spectrum of blockchain cultures, especially perspectives beyond the dominant White, male, Western narratives. Third, understanding the intersection of blockchain truth imaginaries with other parts of technology culture (e.g., Effective Altruism [EA]) is vital for revealing the broader dynamics of contemporary rationalism and its entanglement with more-than-rational elements.

With regard to the first, even though the prefigurative imaginary of chapter 5 opened up new ways of thinking about what knowledge is and how it can be made together with others, what has remained out of view is how the plural worlds that emerge from such knowledge practices might actually interface with each other. Rather than initiating DAOs as disconnected islands, a pluriversal imaginary envisions entangled worlds making up a self-organised patchwork of reality. What happens when different knowledge processes come into contact with each other? In addition, how can prefigurative DAO tokens, meant for community-use, be protected from their wider context of extractivism, financialisation, and control? In other words, how to protect DAO-based plots from the overwhelming powers of the plantation?

The second direction addresses the necessary limitations of the scope of any book by introducing additional cases that may shed light on blockchain culture's internal differences with regard to truth imaginaries. With the exception of the examples discussed in chapter 5, the cases discussed in this book have represented overwhelmingly White, male, Western, and wealthy imaginaries. More research is needed into the truth imaginaries that take shape around blockchain technology from other perspectives—particularly, blockchain imaginaries 'from the South(s)'.²² How can blockchain's affordances be radically reimaged to offer functionalities in these contexts, and what situated plot-work may take shape there? Conversely, how does the superficial and inauthentic appropriation of indigenous values in '[n]

22 Stefania Milan and Emiliano Treré, 'Big Data From the South(s): Beyond Data Universalism', *Television & New Media* 20.4 (2019), 319.

arratives of techno-indigeneity'²³ by the Western tech sector interfere with genuine attempts to break open the horizon of possibilities for blockchain imaginaries?

With regard to the third direction, the truth imaginaries around artificial intelligence and the technology's intersection with popular and influential contemporary subcultures that feature rationalist tendencies demand more attention. In particular, the overlaps and discrepancies of blockchain culture with subcultures such as EA, Longtermism, and 'the Rationalists'²⁴ deserve attention. The latter congregates around an influential online blog called LessWrong, and offers resources to improve one's rational capacities because it is deemed the best way to know the world and make decisions. EA attempts to rationally calculate the most efficient ways of 'doing good,' and direct as much funding as possible to these causes, while the long-termists, who take EA's aims and project them onto astronomical time frames, disregard present concerns in favour of future hypothetical threats.²⁵ The core idea of the connection between blockchain culture and EA consists of the use of cryptocurrency profits to fund projects that are mathematically proven to be 'doing good.'²⁶ However, as EA and related subcultures have attracted more critical attention, it has become abundantly clear that their rational calculations of what 'good' constitutes are—unsurprisingly—intimately entwined with politics, most notably, the history of eugenics.²⁷ What is less clear, however, are the ways in which the rationalism that permeates these ideologies relies on more-than-rational elements. Understanding this dynamic may present further nuanced insights into the politics at play in the practices of these communities.

Ongoing Hype Cycles and the Relevance of Analysing 'Truth'

I opened this book with a reference to the widespread idea that blockchain revolutionised 'truth,' but I did not set out to test whether this hype was justified or what the practical applications of blockchain that make use of its capacity to establish shared truths could be. Instead, I set out to discover what kinds of ideals, values, social visions, and power dynamics constitute blockchain's truth imaginaries. I have discussed my findings in detail above and in each of the previous chapters. Here, instead, I reflect on the phenomenon of that hype itself, the enticing newness of emerging technologies in general, the practical kind of hope that they can engender, and the importance of a historicisation of technologies in this context.

Although this has not been a historical media studies book, I did use a historically situated approach to understanding the emergence of blockchain. I described the significance both

23 Olivier Jutel, 'Blockchain Imperialism in the Pacific', *Big Data & Society* 8.1 (2021), 6.

24 Timnit Gebru and Émile P. Torres, 'The Tescreal Bundle: Eugenics and the Promise of Utopia Through Artificial General Intelligence', *First Monday* 29.4 (2024), 6.

25 Gebru and Torres, 'The Tescreal Bundle: Eugenics and the Promise of Utopia Through Artificial General Intelligence'.

26 Until recently, Sam Bankman-Fried was a prominent proponent of EA. The now convicted crypto fraudster damaged the reputation of EA, although the ideas at the subculture's core continue to have impact.

27 Gebru and Torres, 'The Tescreal Bundle: Eugenics and the Promise of Utopia Through Artificial General Intelligence'.

of the immediate period before and after blockchain technology was first developed *and* its embeddedness in longer social, cultural, political, economic, religious, and scientific genealogies. Like other technologies that are hyped, blockchain is economically incentivised to present itself as a revolutionary, radical break with the past, as capable of doing what has never been done before, as the solution to the most pressing of contemporary global issues, or even as the saviour of humanity. Paradoxically, a technology hype that claims to be the harbinger of a revolution is itself ‘business-as-usual.’ This framing of revolution and profound newness is a continuation of the progress rhetoric that was established in Western Europe during modernity as part of scientific, religious, and geopolitical developments. Like modernity’s myth of universality and morality, contemporary technology culture’s claims of revolution and progress—often recognisable in projects that frame themselves under the header of ‘technology for good’—hide the politics of technology and the ways it serves those in power.

This book has presented ways in which this process takes shape and is problematised in blockchain culture specifically. It has put into focus that rhetorical alignment with the rational or the more-than-rational is no indication of the politics that lie beneath, but should be investigated with regard to who benefits from such an alignment. Investigation into other technology hypes such as digital twins and neurotechnology deserve similar critical historicisation, being attentive to the politics of the forms of both rationality and belief that shape their socio-techno-cultural imaginaries.

It is no coincidence that a book on the idea of a ‘truth machine’ was written during a historical period that has come to be known as the ‘post-truth’ era. The year 2016 marked both the period in which the events around Donald Trump’s election for president of the United States popularised this term and the beginning of the blockchain hype in which its then new usability beyond the confines of cryptocurrencies would cause widespread claims of a ‘blockchain revolution.’ The post-truth era is an important historical context for the popularity of the imaginary of blockchain as a truth machine. Post-truth refers not only to persisting lies about easily verifiable claims—for example about the size of a crowd at a presidential inauguration—but also signifies a wide-ranging collapse of trust in society’s institutions. For example, if the size of a crowd at a widely televised event is debatable, then certainly the amount of CO2 sequestered in a tree halfway across the world is equally debatable. It is in this climate of doubt that blockchain forest protection projects (chapter 2) aim to institute what they present as a newly trustworthy, a-political, and incorruptible truth that is instantly and continuously verifiable. They claim that they ‘disrupt’ the sector of sustainable forestation by offering more trustworthy information to financial donors and investors than non-blockchain-based—and therefore, so they claim, opaque, corruption- and error-prone, untrustworthy—companies. By reading this imaginary through the notion of the colonality of knowledge,²⁸ it loses its revolutionary sheen and instead the many ways in which it constitutes a continuation of the exclusionary and extractivist knowledge logics of modernity become clear. The previous

28 Ramón Grosfoguel, ‘Colonial Difference, Geopolitics of Knowledge, and Global Coloniality in the Modern/colonial Capitalist World-System’, *Utopian Thinking* 25.3 (2002): 203–24. <https://doi.org/10.2307/40241548>.

chapters demonstrate that attempting to regress to notions of truth and knowing from before the post-truth era—even if it appears new—does not address the assumptions and power relations that have led to this post-truth predicament.

This book has instead argued that making a real break with the power relations that characterise the knowledge logics of the past requires radical imagination. The historicisation of emerging technologies brings into focus how they are connected to forces that conserve or expand current power relations and wealth distribution. This knowledge is essential if there is to be any attempt to break with these dynamics. Combining this historical situatedness with the imagination—which via Castoriadis gained a forward-looking, radical, and creative capacity²⁹—is important in the context of technology culture’s connection to power. To break with the incentivisation of technological solutionism and a utopianism that claims revolution, but is actually in many ways a regression, different relationships between technology and the societies and cultures in which it is embedded need to be imagined. Chapter 5 shows practical ways to approach this material-discursive struggle to institute new worlds through radically reimagined affordances and artistic-activist plot-work. These tools can make historically situated yet radically reimagined entanglements ready for their prefigurative performance. By addressing technology culture’s relation to power through these and other possible artistic-activist tools of the radical imagination, unexpected and exciting relationships can take shape between the future and the economy, between technology and society, and between individuals and infrastructures. Rather than technology, it is ‘human beings [that] are magical’,³⁰ capable of creating the realities we live in by starting to imagine them. Tools such as the ones discussed here help to imagine new mythologies about life and new ways to know and appreciate what entangled life may be. They open up horizons and provide new ways to tell and retell stories about different worlds, and they allow collectives to perform and materialise those stories ‘in undared forms’.³¹

29 Castoriadis, *The Imaginary Institution of Society*.

30 Sylvia Wynter, ‘The Pope Must Have Been Drunk, the King of the Castile a Madman: Culture as Actuality, and the Caribbean Rethinking Modernity’, in Alvina Ruprecht and Cecilia Taiana (eds) *The Reordering of Culture: Latin America, the Caribbean and Canada in the Hood*, Ottawa: Carleton University Press, 1995, 35.

31 Wynter, ‘The Pope Must Have Been Drunk, the King of the Castile a Madman,’ 35.

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ABOUT THE AUTHOR

Inte Gloerich is a researcher who critically, creatively, and collaboratively reimagining emerging technology from decolonial, feminist, and ecological perspectives. She has been a researcher at the Institute of Network Cultures (Amsterdam University of Applied Sciences), a PhD researcher at Utrecht University, and a postdoc at the Visual Methodologies Collective (AUAS) and at the Antwerp Media in Society Centre (University of Antwerp).

Inte Gloerich is a teacher who fosters a critical stance on how technologies, methods, and ideologies affect society and culture. She has been a lecturer at the BA Media & Information and the MA New Media & Digital Culture at the University of Amsterdam, and currently teaches at the Thinker's Lab at AUAS.

Inte Gloerich has collaborated with cultural organisations like Aksioma, Furtherfield, NeMe, Baltan Laboratories, Amateur Cities, and Hackers and Designers by organising knowledge exchanges and creative workshops across disciplines. She is particularly interested in involving people in creatively imagining and prefiguring the future together with others.

Moreover, Inte Gloerich attempts to be a potter, pretends to be a sewer, dreams of being a carpenter, and throws dough as if she is a pizzaiola. She loves dogs whole-heartedly, respects spiders fearfully, invents words frequently, blushes easily, laughs generously, repeats jokes endlessly, and collects power tools avidly. She makes detailed plans to throw them out of the window once she has started, and she reads maps well but does not know where she is going.

